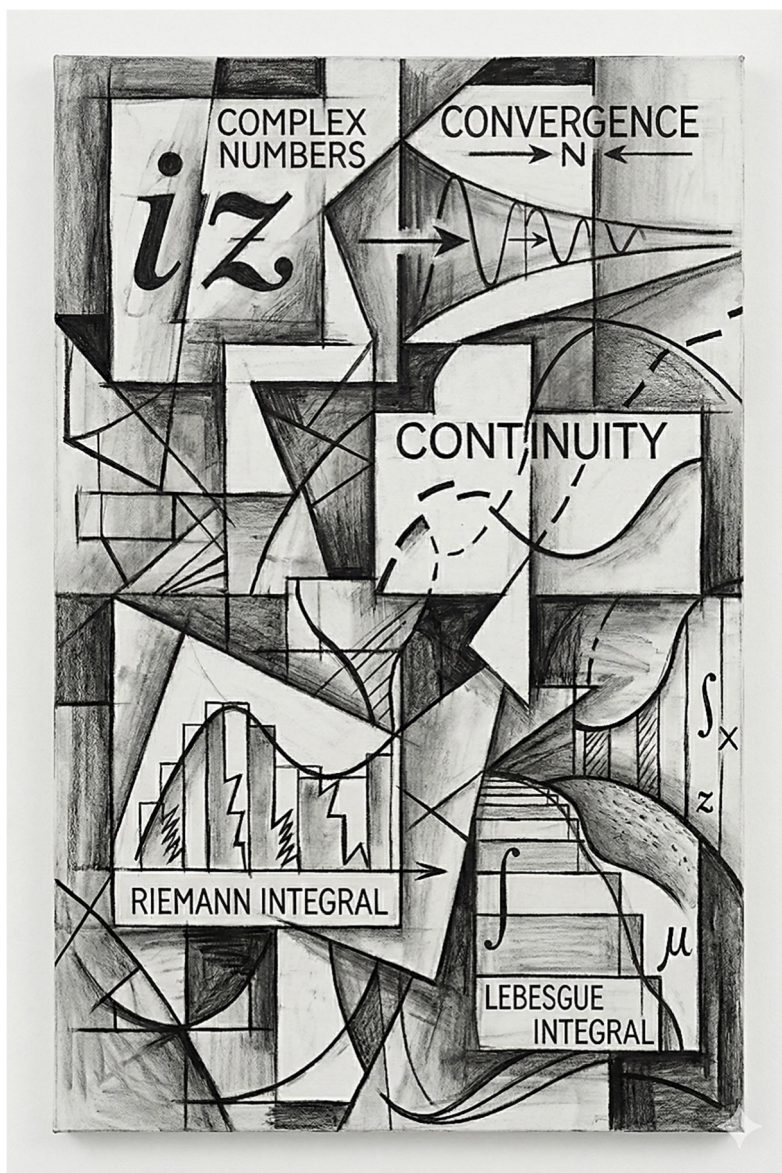


A Course of Modern Analysis Revisited

Complex Numbers, Theory of Convergence, Continuity, Riemann Integration, and the
Lebesgue Integral

Revisiting Whittaker & Watson with Modern Commentary



Stephen Fratini

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Preface

For many decades, *A Course of Modern Analysis* by E. T. Whittaker and G. N. Watson [1] has been regarded as one of the classic texts of mathematical analysis. First published in 1902, the book became widely known for its elegant and concise presentation of topics in complex analysis, special functions, and classical analysis. Generations of mathematicians encountered important ideas in analysis through its pages.

The style of the original work, however, reflects the mathematical culture of its time. Arguments are often presented in a highly compressed form, intermediate steps are frequently omitted, and certain logical transitions—especially those involving limits, rearrangements of series, or asymptotic reasoning—are left implicit. As a result, while the book remains mathematically rich and historically important, many readers today find parts of it difficult to follow without additional explanation.

The purpose of the present book is to revisit some of the material contained in Chapters I to IV of Part I of Whittaker and Watson and present it in a form accessible to contemporary readers. The approach taken here is to preserve the structure and much of the language of the original text while providing clarifications, explanatory commentary, and occasional modern perspectives where helpful.

In particular, passages that rely on implicit assumptions or compressed reasoning have been expanded to make the logical flow explicit. This includes:

- clarifying the grouping or rearrangement of terms in infinite series,
- making precise the meaning of partial sums and indexing conventions,
- supplying intermediate steps in asymptotic expansions and limiting arguments,
- and distinguishing clearly between finite identities and limiting processes.

The aim is not to alter the mathematics, but to make visible the reasoning that is often only implicit in the original exposition. In this sense, the present work may be viewed as a guided reading of Whittaker and Watson, in which the underlying ideas are unfolded in a manner consistent with modern pedagogical standards.

In expanding and commenting on Whittaker and Watson's exposition, the guiding principle has been that a reader should first understand what a theorem says, when it may be used, and why its hypotheses matter. In some cases, a detailed proof is essential and is therefore expanded carefully. In other cases, especially where the original proof is highly compressed or mainly technical, the emphasis is placed instead on the central idea, the correct interpretation of the result, and examples or counterexamples that show what can go wrong if the assumptions are not satisfied. The goal is not to replace rigor with intuition, but to make the structure and purpose of the mathematics visible before the reader is asked to follow every technical detail.

The topics covered in this volume include complex numbers, the theory of convergence of sequences and series, continuity and uniform convergence, and the theory of Riemann integration. These subjects form an important part of the classical foundation of analysis and serve as preparation for more advanced study in complex analysis and related areas.

In addition to revisiting the classical material, the book concludes with a brief appendix introducing the Lebesgue integral. The theory of Lebesgue integration, developed shortly after the publication

of Whittaker and Watson's text, represents one of the most significant advances in the development of modern analysis. While the original book naturally focuses on the Riemann integral, the inclusion of this appendix provides readers with a glimpse of how the theory of integration evolved in the twentieth century and why the Lebesgue approach plays such an important role in modern mathematics.

This book is intended for readers who have some familiarity with undergraduate mathematics, including calculus and elementary algebra, and who wish to deepen their understanding of classical analysis. It may be useful to students preparing to study complex analysis, to instructors seeking supplementary explanations of classical material, or to readers interested in the historical development of mathematical ideas.

In preparing this work, the author has made use of modern computational tools, including artificial intelligence, as an aid in transcription, verification, and exposition. The process has been a collaborative one: the original text provides the foundation, the author guides the mathematical interpretation and pedagogical direction, and AI assists in expanding intermediate steps, checking consistency, and exploring alternative explanations.

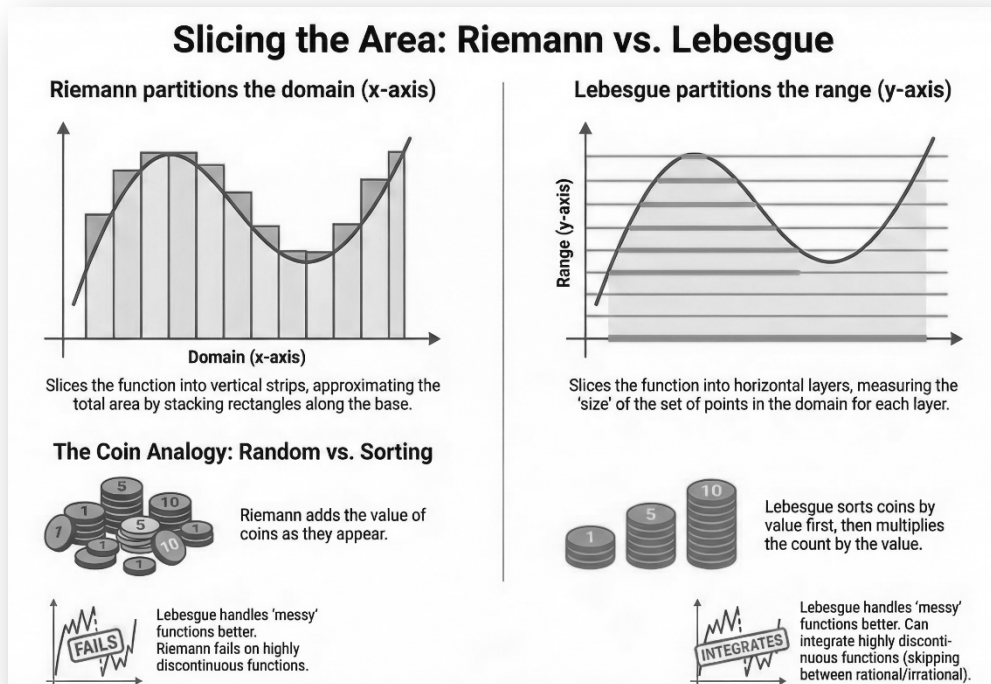
While these tools have proved valuable in clarifying and organizing the material, all mathematical content and editorial decisions have been reviewed and shaped to ensure accuracy, coherence, and fidelity to the original work. The aim throughout has been to combine the strengths of classical exposition with the capabilities of modern technology in service of clearer understanding.

The author hopes that this revisiting of Whittaker and Watson will help make a remarkable mathematical work more accessible to modern readers while preserving the clarity and elegance that have made the original text a classic.

Legal

Everything old is new again.

Originates from the song of the same name, written by Peter Allen and Carole Bayer Sager



Other books by the author:

- *The Art of Managing Things (2nd edition)*, self-published on Amazon, <https://www.amazon.com/Art-Managing-Things-Stephen-Fratini-ebook/dp/B07N4H4YWH/>, January 2019.
- *Mathematical Thinking: Exercises for the Mind (2nd Edition)*, self-published on Amazon, <https://www.amazon.com/dp/B0CL34FRP1>, October 2023.
- *Financial Mathematics (3rd Edition)*, self-published on Amazon, <https://www.amazon.com/dp/B0H2VM86Z9>, May 2026.
- *Math in Art, and Art in Math*, self-published on Amazon, <https://www.amazon.com/dp/B091D1F8MB>, March 2021.
- *Algebra through Discovery and Experimentation*, self-published on Amazon, <https://www.amazon.com/dp/B09B5L9WL5>, July 2021.
- *The Struggle Against Chaos*, self-published on Amazon, <https://www.amazon.com/dp/B09BLPQ86Q>, July 2021.
- *Mathematical Vignettes: Number theory, stochastic processes, game theory, cryptography, linear programming and more*, self-published on Amazon, <https://www.amazon.com/Mathematical-Vignettes-stochastic-cryptography-programming-ebook/dp/B0BBP1PBJQ/>, August 2022.
- *Learning Math through Puzzles: Number properties, counting, sequences and series, algebra, functions, and mathematical reasoning*, self-published on Amazon, <https://www.amazon.com/dp/B0BZFZP5B>, March 2023.
- *Mathematical Vignettes: Volume II: Topics from combinatorial design, magic squares, finite geometry, abstract algebra, error correcting codes, geometric packing problems and much more*, self-published on Amazon, <https://www.amazon.com/dp/B0CM1CLSK8>, October 2023.
- *Shape Up and Solve It!: Learn Geometry Through Puzzles*, self-published on Amazon, <https://www.amazon.com/dp/B0CRS7DRWF>, January 2024.
- *Mathematical Vignettes III (2nd edition): Introductions to non-Euclidean geometry, topology and complex analysis*, self-published on Amazon, <https://www.amazon.com/Mathematical-Vignettes-III-Introductions-non-Euclidean/dp/B0F1N6522B>, March 2025.
- *The Shape of Space: A Guided Tour of Vectors, Matrices, Tensors and Markov Chains*, self-published on Amazon, <https://www.amazon.com/dp/B0FQCGRNW1>, September 2025.

- *Beyond Limits: Infinitesimals, Abstract Algebra, Category Theory, Synthetic Differential Geometry, and Automatic Differentiation*, self-published on Amazon, <https://www.amazon.com/Beyond-Limits-Infinitesimals-Differential-Differentiation/dp/B0G7FXMQ6V>, January 2026.
- *A Symphony of Patterns: Endless Variations on a Theme*, self-published on Amazon, <https://www.amazon.com/dp/B0GRBSXRDT>, March 2026.

Electronic versions of my books are available (free of charge) at

https://github.com/sfratini33/art-of-managing-things-external/tree/master/free_books

My Video Series:

- *Linear Algebra*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UniSLwX82IFFqN1OsloWPK
- *Matrices - Some Less Common Properties*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_Umu0emwEcVtjDQs7O4ioZgr
- *Vector and Matrix Calculus*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UlI8xLnrTOQaNe-KX6v0EDp
- *Tensors*, https://www.youtube.com/playlist?list=PLSaIWQAK7_UltOIPJUX7nRHxIZ1RMYJXN
- *Non-negative matrices, Markov chains and Least Squares*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UlvuQ49JQUmV6GZGU7U1voD
- *Puzzles to Exercise the Mind*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UmCBhNtgSpmgss0XAJq392j
- *Non-Euclidean Geometry*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UnXtByWwyjg9L-0JcA81HK8
- *Topology*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_Uk62dBW9a6myMpmXrOIlR90
- *Complex Analysis*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UkA9q_2Ptfz9mK9yzmngTRe
- *Abstract Algebra*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UmjFOzc39_O5U5o953dJ22N
- *Number Theory Problems*, https://www.youtube.com/playlist?list=PLSaIWQAK7_UmEL-RoC5MqszhLIUGP1ft
- *Combinatorics*,
https://www.youtube.com/playlist?list=PLSaIWQAK7_UmH81dp3K9ApvFviltCaimW
- *Graph Theory*, https://www.youtube.com/playlist?list=PLSaIWQAK7_UICE6yqQeEYjl-P5zzkzb-d
- *Financial Mathematics*,
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If you are working from a paper copy of my book, it may be easier to visit my YouTube channel rather than typing in the long URLs above.

www.youtube.com/@sfratini

1 Introduction

The essence of mathematics lies in its freedom. — Georg Cantor

1.1 Purpose

This book aims to:

- revisit the classical material from Chapters I–IV of *A Course of Modern Analysis*, recast for the modern reader;
- preserve the elegance and structure of the original arguments while providing clarification where needed;
- explain intermediate steps and background ideas that are often omitted in the original exposition;
- introduce the foundational topics of classical analysis, including complex numbers, convergence, continuity, and Riemann integration;
- provide a brief introduction to the Lebesgue integral as a modern extension of the classical theory of integration (new material – not in *A Course of Modern Analysis*).

1.2 Intended Audience

This book is written for:

- undergraduate students who have completed an introductory course in calculus and wish to deepen their understanding of analysis;
- students preparing to study complex analysis or advanced real analysis;
- instructors who would like a supplementary text that explains classical arguments in greater detail;
- readers with an interest in the historical development of mathematical ideas.

While some familiarity with elementary algebra and calculus is assumed, the exposition attempts to provide sufficient explanation so that readers encountering these topics for the first time can follow the main ideas.

1.3 Prerequisites

To fully engage with this book, readers should have:

- a working knowledge of differential and integral calculus;
- familiarity with algebraic manipulation and basic mathematical notation;
- some exposure to limits, sequences, and functions.

No prior knowledge of complex analysis or measure theory is assumed.

1.4 Outline

The book is organized as follows.

Chapter 2 — Complex Numbers

This chapter introduces the extension of the number system from rational numbers to real numbers and then to complex numbers. The construction of complex numbers and their basic properties are developed, together with their geometric interpretation in the complex plane.

This material corresponds primarily to Part I, Chapter I of Whittaker and Watson.

Chapter 3 — The Theory of Convergence

This chapter develops the theory of convergence of sequences and infinite series. These ideas provide the foundation for handling infinite processes in analysis and are essential for the rigorous development of power series and analytic functions.

This material corresponds to Part I, Chapter II of Whittaker and Watson.

Chapter 4 — Continuous Functions and Uniform Convergence

This chapter introduces the concepts of continuity and uniform convergence, which play a central role in modern analysis. These ideas clarify how functions behave under limiting processes and provide the theoretical framework needed for many later developments in analysis.

This material corresponds to Part I, Chapter III of Whittaker and Watson.

Chapter 5 — The Theory of Riemann Integration

This chapter presents the classical theory of the Riemann integral and its basic properties. The Riemann integral provides a rigorous framework for defining the area under a curve and forms one of the central ideas of classical analysis.

This material corresponds to Part I, Chapter IV of Whittaker and Watson.

Finally, an **Appendix** provides a brief introduction to the Lebesgue integral. This material does not appear in *A Course of Modern Analysis* but is included here to illustrate an important development in the theory of integration that occurred shortly after the publication of Whittaker and Watson's work. The appendix offers a short overview of how the Lebesgue approach extends and generalizes the classical Riemann integral presented in Chapter 5.

1.5 Conventions

In preparing this book, the aim has been to preserve the mathematical content and spirit of *A Course of Modern Analysis* by Whittaker and Watson while presenting the material in a form more accessible to modern readers. In order to achieve this, certain editorial conventions have been adopted.

- **Simplifications and rewording.**
Where Whittaker and Watson's phrasing is unusually compressed, the main text has occasionally been reworded or simplified to improve readability, while preserving the mathematical content. Sections altered substantially, or added as new, are noted as such.
- **Section numbering.**
The original text uses a compressed decimal numbering system (for example §2.11 or §2.211) to indicate subsections. In this book these have been expanded into the more

familiar hierarchical form (e.g., §2.1.1 or §2.2.1.1) in order to make the logical structure of the material clearer. Further, because an Introduction is added as Section 1, each Section number in this book is one greater than the corresponding Chapter number in Whittaker and Watson.

- **Notation.**

- Some notation has been adjusted where doing so improves readability while preserving the meaning of the original text. In particular, standard modern grouping conventions are used in formulas: parentheses $()$ for ordinary algebraic grouping, square brackets $[]$ for larger groupings when needed, and braces $\{\}$ reserved for sets.
- Whittaker and Watson denote the real and imaginary parts of a complex number z by $R(z)$ and $I(z)$, respectively (corresponding to the modern notation $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$). This notation is retained here for consistency with the original text.
- Displayed equations are written without terminal punctuation to avoid ambiguity in mathematical notation.
- We consistently use σ to denote sums involving absolute values and S to denote sums of the original (possibly signed) series.
- The notation $\log x$ denotes the natural logarithm (that is, the logarithm to base e). This convention follows the usage in Whittaker and Watson; in other sources, $\ln x$ is often used for the natural logarithm.
- Expressions such as $\lim x_n$ are to be understood as limits taken as $n \rightarrow \infty$, unless otherwise indicated. This convention follows the usage in Whittaker and Watson, where the limiting variable is often left implicit when no ambiguity arises.
- Expressions such as $x \rightarrow 1 - 0$ are to be understood as one-sided limits, with x approaching 1 from below. In modern notation this is usually written $x \rightarrow 1^-$. More generally, Whittaker and Watson often use $a - 0$ and $a + 0$ to indicate approach to a from the left and from the right, respectively.
- We generally preserve Whittaker and Watson's use of x for real variables and z for complex variables. In some examples, the authors state a result for z but verify it by restricting z to real values. This is legitimate when proving non-uniform convergence near a point, since failure along the real axis already implies failure in any surrounding complex neighborhood.

- **Terminology.**

Terminology used in the original text has generally been retained. In a few places brief explanatory remarks are included where historical terminology may differ from that commonly used in modern texts.

- **References.**

In several places the references given in the original text of Whittaker and Watson are reproduced at the end of sections. These references are included primarily for historical context and therefore reflect the mathematical literature of the late nineteenth and early twentieth centuries.

- **Explanatory remarks.**

Short explanatory remarks have occasionally been added to clarify steps that are presented very briefly in the original exposition or to connect the classical presentation with modern terminology or interpretation. These remarks are distinguished typographically from the main text by indentation, italics, and a contrasting font.

- **Footnotes.**

The original text uses a variety of typographical symbols (such as *, †, ‡) for footnotes. In this book these have been replaced by standard numbered footnotes generated automatically by the word processor.

- **Limits superior and inferior.**

The original text denotes the limit superior and limit inferior using a bar above or below the symbol “lim.” In this book these are written using the modern notation $\limsup$ and $\liminf$, respectively.

- **Whittaker and Watson.** Within the text, there are cases where reference is made to the original book. In such cases, we refer to Whittaker and Watson as “the authors”.

2 Complex Numbers

Good math teachers don't teach students how to get an answer. They teach students how to think.

— Pranav Gupta

2.1 Rational numbers

The idea of a set of numbers is derived in the first instance from the consideration of the set of positive¹ integral numbers, or positive integers, that is to say, the numbers 1, 2, 3, 4, Positive integers have many properties, which will be found in treatises on the Theory of Integral Numbers, but at a very early stage in the development of mathematics it was found that the operations of Subtraction and Division could only be performed among them subject to inconvenient restrictions, and consequently, in elementary Arithmetic, classes of numbers are constructed such that the operations of subtraction and division can always be performed among them.

To obtain a class of numbers among which the operation of subtraction can be performed without restraint we construct the class of integers, which consists of the class of positive² integers (+1, +2, +3, ...) and of the class of negative integers (−1, −2, −3, ...) and the number 0.

To obtain a class of numbers among which the operations both of subtraction and of division can be performed freely³, we construct the class of rational numbers. Symbols which denote members of this class are $\frac{1}{2}, 3, 0, -\frac{1}{4}$.

We have thus introduced three classes of numbers, (i) the signless integers, (ii) the integers, (iii) the rational numbers.

It is not part of the scheme of this work to discuss the construction of the class of integers or the logical foundations of the theory of rational numbers⁴.

The extension of the idea of number, which has just been described, was not effected without some opposition from the more conservative mathematicians. In the latter half of the eighteenth century, Maseres (1731–1824) and Frend (1757–1841) published works on Algebra, Trigonometry, etc., in which the use of negative numbers was disallowed, although Descartes had used them unrestrictedly more than a hundred years before.

A rational number x may be represented to the eye in the following manner.

If, on a straight line, we take an origin O and a fixed segment OP_1 (P_1 being on the right of O), we can measure from O a length OP_x such that the ratio OP_x/OP_1 is equal to x , the point P_x is taken on the right or left of O according as the number x is positive or negative. We may regard either the point P_x or the displacement OP_x (which will be written $\overline{OP_x}$) as representing the number x .

The figure below illustrates the case where $x > 1$.

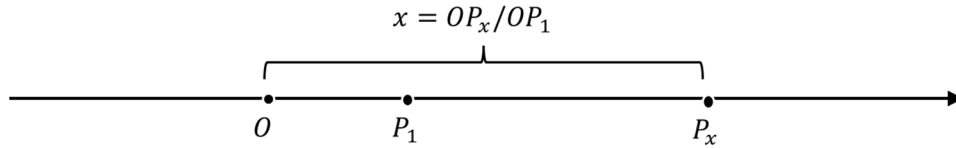
Here, the “fixed segment” OP_1 serves as the unit segment, representing the magnitude 1. All other real numbers are defined as ratios relative to this unit length.

¹ Strictly speaking, a more appropriate epithet would be, not positive, but signless.

² In the strict sense.

³ With the exception of division by the rational number 0.

⁴ Such a discussion, defining a rational number as an ordered number pair of integers in a similar manner to that in which a complex number is defined in §1.8 as an ordered number pair of real numbers, will be found in Hobson's *Functions of a Real Variable*, §§ 1–12.



All the rational numbers can thus be represented by points on the line, but the converse is not true. For if we measure off on the line a length OQ equal to the diagonal of a square of which OP_1 is one side, it can be proved that Q does not correspond to any rational number.

Points on the line which do not represent rational numbers may be said to represent irrational numbers, thus the point Q is said to represent the irrational number $\sqrt{2} \cong 1.414213$. But while such an explanation of the existence of irrational numbers satisfied the mathematicians of the eighteenth century and may still be sufficient for those whose interest lies in the applications of mathematics rather than in the logical upbuilding of the theory, yet from the logical standpoint it is improper to introduce geometrical intuitions to supply deficiencies in arithmetical arguments, and it was shown by Dedekind in 1868 that the theory of irrational numbers can be established on a purely arithmetical basis without any appeal to geometry.

2.2 Dedekind's theory of irrational numbers⁵

The geometrical property of points on a line which suggested the starting point of the arithmetical theory of irrationals was that, if all points of a line are separated into two classes such that every point of the first class is on the right of every point of the second class, there exists one and only one point at which the line is thus severed.

Following up this idea, Dedekind considered rules by which a separation⁶ or *section* of *all* rational numbers into two classes can be made, these classes (which will be called the L -class and the R -class, or the left class and the right class) being such that they possess the following properties:

- i. At least one member of each class exists.
- ii. Every member of the L -class is less than every member of the R -class.

It is obvious that such a section is made by any rational number x , and x is either the greatest number of the L -class or the least number of the R -class. But sections can be made in which no rational number x plays this part. Thus, since there is no rational number⁷ whose square is 2, it is easy to see that we may form a section in which the R -class consists of the positive rational numbers whose squares exceed 2, and the L -class consists of all other rational numbers. Then this section is such that the R -class has no least member and the L -class has no greatest member, for, if

⁵ The theory, though elaborated in 1858, was not published before the appearance of Dedekind's tract, *Stetigkeit und irrationale Zahlen*, Brunswick, 1872. Other theories are due to Weierstrass [see von Dantscher, *Die Weierstrass'sche Theorie der irrationalen Zahlen* (Leipzig, 1908)] and Cantor, *Math. Ann.* v (1872), pp. 123–130.

⁶ This procedure formed the basis of the treatment of irrational numbers by the Greek mathematicians in the sixth and fifth centuries BC. The advance made by Dedekind consisted in observing that a purely arithmetical theory could be built upon it.

⁷ For if $\frac{p}{q}$ be such a number, this fraction being in its lowest terms, it may be seen that $\frac{2q-p}{p-q}$ is another such number, and $0 < p - q < q$, so that $\frac{p}{q}$ is not in its lowest terms. The contradiction implies that such a rational number does not exist.

x be any positive rational fraction, and $y = \frac{x(x^2+6)}{3x^2+2}$, then $y - x = \frac{2x(2-x^2)}{3x^2+2}$ and $y^2 - 2 = \frac{(x^2-2)^3}{(3x^2+2)^2}$, so x^2 , y^2 and 2 are in order of magnitude, and therefore given any member x of the L -class, we can always find a greater member of the L -class, or given any member x' of the R -class, we can always find a smaller member of the R -class, such numbers being, for instance, y and y' , where y' is the same function of x' as y of x .

In the above, the motivation for defining y as such comes from numerical analysis, specifically algorithms used to find roots of equations.

While the text presents the formula somewhat magically ("out of thin air"), it is actually the formula for Halley's Method applied to finding the square root of 2.

In the context of Dedekind cuts, the authors want to prove that the rational numbers are "dense" right up to the edge of the cut for $\sqrt{2}$.

- *If you have a rational number x such that $x^2 < 2$ (it's in the L -class), you need to prove there is another rational y that is bigger than x but still has $y^2 < 2$. (No greatest member).*
- *If you have a rational number x such that $x^2 > 2$ (it's in the R -class), you need to prove there is another rational y that is smaller than x but still has $y^2 > 2$. (No least member).*

The beauty of using this specific formula is that it preserves the "sidedness" of the approximation while moving closer to the target $\sqrt{2}$.

As shown in the text, the difference between the square of the new value and 2 is:

$$y^2 - 2 = \frac{(x^2 - 2)^3}{(3x^2 + 2)^2}$$

Because the term $(x^2 - 2)$ is cubed:

- *If $x^2 < 2$, then $(x^2 - 2)$ is negative. The cube is negative. So, $y^2 - 2$ is negative ($y^2 < 2$). You stay in the L -class.*
- *If $x^2 > 2$, then $(x^2 - 2)$ is positive. The cube is positive. So, $y^2 - 2$ is positive ($y^2 > 2$). You stay in the R -class.*

Simultaneously, the text shows $y - x = \frac{2x(2-x^2)}{3x^2+2}$. This term ensures you always move towards the cut (increasing if you are below it, decreasing if you are above it).

If a section is made in which the R -class has a least member A_2 , or if the L -class has a greatest member A_1 , the section determines a *rational-real* number, which it is convenient to denote by the same⁸ symbol A_2 or A_1 .

If a section is made, such that the R -class has no least member and the L -class has no greatest member, *the section determines an irrational-real number*⁹.

⁸ This causes no confusion in practice.

⁹ B. A. W. Russell defines the class of real numbers as actually being the class of all L classes; the class of real numbers whose L classes have a greatest member corresponds to the class of rational numbers, and though the rational real number x which corresponds to a rational number x is conceptually distinct from it, no confusion arises from denoting both by the same symbol.

The authors are making a subtle but important distinction here about types of mathematical objects.

- *The Definition: In this theory, a "Real Number" is defined as a Section (a specific way of dividing all rational numbers into two infinite sets).*
- *The Problem: A "Section" (a pair of sets) is not the same thing as a simple fraction like $\frac{1}{2}$. They are different species of objects.*
- *The Solution:*
 - *Rational-Real: If the separation happens exactly at a rational number (e.g., $\frac{1}{2}$), we create a Section that "points" to it. We call this a "rational-real" number and agree to treat it as identical to the original fraction.*
 - *Irrational-Real: If the separation happens where no rational number exists (like at $\sqrt{2}$), the Section defines a new gap-filler.*

Essentially, this definition embeds a copy of the rational numbers into the new, larger system of Real numbers so arithmetic can work consistently for both.

If x, y are real numbers (defined by sections) we say that x is greater than y if the L -class defining x contains at least two¹⁰ members of the R -class defining y .

Let $\alpha, \beta, \dots$ be real numbers and let $A_1, B_1, \dots$ be any members of the corresponding L -classes while $A_2, B_2, \dots$ are any members of the corresponding R -classes. The classes of which $A_1, A_2, \dots$ are respectively members will be denoted by the symbols $(A_1), (A_2), \dots$.

The above notation may appear confusing, and thus requires further explanation. If a real number α is defined by a Dedekind section of the rational numbers into a left class L_α and a right class R_α , the symbols A_1 and A_2 denote arbitrary rational numbers chosen from these classes: $A_1 \in L_\alpha$ and $A_2 \in R_\alpha$. Similarly, if β is another real number determined by a section with classes L_β and R_β , the symbols B_1 and B_2 denote arbitrary rational numbers with $B_1 \in L_\beta$ and $B_2 \in R_\beta$. The subscripts simply label particular rational numbers chosen from the relevant classes; they do not indicate order or magnitude. In modern notation one would usually write $a \in L_\alpha$ and $b \in R_\alpha$ rather than introduce symbols such as A_1 and A_2 .

When the notation (A_1) is used, it denotes the real number (that is, the Dedekind section) determined by the class of rational numbers of which A_1 is a member; the parentheses therefore refer to the section itself rather than to the rational number A_1 .

Then the sum (written $\alpha + \beta$) of two real numbers α and β is defined as the real number (rational or irrational) which is determined by the L -class $(A_1 + B_1)$ and the R -class $(A_2 + B_2)$. It is, of course, necessary to prove that these classes determine a section of the rational numbers.

It is evident that $A_1 + B_1 < A_2 + B_2$ and that at least one member of each of the classes $(A_1 + B_1), (A_2 + B_2)$ exists. It remains to prove that there is, at most, *one* rational number which is greater than every $A_1 + B_1$ and less than every $A_2 + B_2$, suppose, if possible, that there are two, x and y ($y > x$). Let a_1 be a member of (A_1) and let a_2 be a member of (A_2) , and let N be the

¹⁰ If the classes had only one member in common, that member might be the greatest member of the L -class of x and the least member of the R -class of y .

integer next greater than $\frac{2(a_2 - a_1)}{y - x}$. Take the last of the numbers $a_1 + \frac{m}{N}(a_2 - a_1)$, (where $m = 0, 1, \dots, N$), which belongs to (A_1) and the first of them which belongs to (A_2) , let these two numbers be c_1, c_2 .

So, c_1 and c_2 are two consecutive numbers in the list. Thus, there exists some m such that

$$c_1 = a_1 + \frac{m}{N}(a_2 - a_1)$$

$$c_2 = a_1 + \frac{m+1}{N}(a_2 - a_1)$$

Then

$$c_2 - c_1 = \frac{1}{N}(a_2 - a_1) < \frac{1}{2}(y - x)$$

Choose d_1, d_2 in a similar manner from the classes defining β , then

$$c_2 + d_2 - c_1 - d_1 < y - x$$

But $c_2 + d_2 \geq y$, $c_1 + d_1 \leq x$, and therefore $c_2 + d_2 - c_1 - d_1 \geq y - x$, we have therefore arrived at a contradiction by supposing that two rational numbers x, y exist belonging neither to $(A_1 + B_1)$ nor to $(A_2 + B_2)$.

If every rational number belongs either to the class $(A_1 + B_1)$ or to the class $(A_2 + B_2)$, then the classes $(A_1 + B_1), (A_2 + B_2)$ define an irrational number. If one rational number x exists belonging to neither class, then the L -class formed by x and $(A_1 + B_1)$ and the R -class $(A_2 + B_2)$ define the rational-real number x . In either case, the number defined is called the sum $\alpha + \beta$.

The difference $\alpha - \beta$ of two real numbers is defined by the L -class $(A_1 - B_2)$ and the R -class $(A_2 - B_1)$.

The product of two positive real numbers α, β is defined by the R -class $(A_2 B_2)$ and the L -class of all other rational numbers.

The reader will see without difficulty how to define the product of negative real numbers and the quotient of two real numbers, and further, it may be shown that real numbers may be combined in accordance with the associative, distributive and commutative laws.

The aggregate of rational-real and irrational-real numbers is called the aggregate of real numbers; for brevity, rational-real numbers and irrational-real numbers are called rational and irrational numbers respectively.

2.3 Complex Numbers

The construction of complex numbers as ordered pairs of real numbers, motivated by displacements in the plane, follows the presentation used in the original text in this section. While logically sound, this approach is less common in modern introductions to complex numbers. Today it is more usual to introduce complex numbers algebraically by adjoining to the real numbers a quantity i satisfying $i^2 = -1$. From this viewpoint a complex number is written in the form $a + bi$, where a and b are real numbers, and arithmetic is determined by the rule $i^2 = -1$. In other words, the complex numbers arise naturally when one seeks solutions of the equation $x^2 + 1 = 0$, which has no solution within the real numbers.

We have seen that a real number may be visualised as a displacement along a definite straight line. If, however, P and Q are any two points in a plane, the displacement $\overline{PQ}$ needs two real numbers for its specification, for instance, the differences of the coordinates of P and Q referred to fixed rectangular axes.

If the coordinates of P be (ξ, η) and those of Q be $(\xi + x, \eta + y)$, the displacement $\overline{PQ}$ may be described by the symbol $[x, y]$.

We are thus led to consider the association of real numbers in ordered¹¹ pairs. The natural definition of the sum of two displacements $[x, y]$ and $[x', y']$ is the displacement which is the result of the successive applications of the two displacements. It is therefore convenient to define the sum of two number-pairs by the equation

$$[x, y] + [x', y'] = [x + x', y + y']$$

The product of a number-pair and a real number x' is then naturally defined by the equation

$$x' \times [x, y] = [x'x, x'y]$$

We are at liberty to define the product of two number-pairs in any convenient manner, but the only definition which does not give rise to results that are merely trivial is that symbolised by the equation

$$[x, y] \times [x', y'] = [xx' - yy', xy' + x'y]$$

It is then evident that

$$[x, 0] \times [x', y'] = [xx', xy'] = x \times [x', y']$$

and

$$[0, y] \times [x', y'] = [-yy', x'y] = y \times [-y', x']$$

The geometrical interpretation of these results is that the effect of multiplying by the displacement $[x, 0]$ is the same as that of multiplying by the real number x , but the effect of multiplying a displacement by $[0, y]$ is to multiply it by a real number y and turn it through a right angle.

It is convenient to denote the number-pair $[x, y]$ by the compound symbol $x + iy$, and a number-pair is now conveniently called (after Gauss) a complex number; in the fundamental operations of arithmetic the complex number $x + 0i$ may be replaced by the real number x , and defining i to mean $0 + i1$, we have

$$i^2 = [0, 1] \times [0, 1] = [-1, 0]$$

and so i^2 may be replaced by -1 .

The reader will easily convince himself that the definitions of addition and multiplication of number-pairs have been so framed that we may perform the ordinary operations of algebra with complex numbers in exactly the same way as with real numbers, treating the symbol i as a number and replacing the product i^2 by -1 wherever it occurs.

Thus, the reader will verify that, if a, b, c are complex numbers, we have

$$\begin{aligned} a + b &= b + a \\ ab &= ba \end{aligned}$$

¹¹ The order of the two terms distinguishes the ordered number pair $[x, y]$ from the ordered number pair $[y, x]$.

$$(a + b) + c = a + (b + c)$$

$$(ab)c = a(bc)$$

$$a(b + c) = ab + ac$$

and if ab is zero, then either a or b is zero.

It is found that algebraical operations, direct or inverse, when applied to complex numbers, do not suggest numbers of any fresh type; the complex number will therefore for our purposes be taken as the most general type of number.

The introduction of the complex number has led to many important developments in mathematics. Functions which, when real variables only are considered, appear as essentially distinct, are seen to be connected when complex variables are introduced; thus, the circular functions are found to be expressible in terms of exponential functions of a complex argument, by the equations

$$\cos x = \frac{1}{2}(e^{ix} + e^{-ix}), \quad \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$$

At this point, Whittaker and Watson merely state these relations to illustrate the unifying power of complex variables; the identities themselves follow later from Euler's formula $e^{ix} = \cos x + i \sin x$, which is established using the power-series definitions of the exponential and trigonometric functions.

Again, many of the most important theorems of modern analysis are not true if the numbers concerned are restricted to be real; thus, the theorem that every algebraic equation of degree n has n roots is true in general only when regarded as a theorem concerning complex numbers.

Hamilton's quaternions furnish an example of a still further extension of the idea of number. A quaternion

$$w + xi + yj + zk$$

is formed from four real numbers w, x, y, z , and four number-units $1, i, j, k$, in the same way that the ordinary complex number $x + iy$ might be regarded as being formed from two real numbers x, y , and two number-units $1, i$. Quaternions, however, do not obey the commutative law of multiplication.

2.4 The Modulus of a Complex Number

Let $x + iy$ be a complex number, x and y being real numbers. Then the positive square root of $x^2 + y^2$ is called the **modulus** of $x + iy$, and is written

$$|x + iy|$$

Let us consider the complex number which is the sum of two given complex numbers $x + iy$ and $u + iv$. We have

$$(x + iy) + (u + iv) = (x + u) + i(y + v)$$

The modulus of the sum of the two numbers is therefore

$$[(x + u)^2 + (y + v)^2]^{\frac{1}{2}}$$

or

$$[(x^2 + y^2) + (u^2 + v^2) + 2(xu + yv)]^{\frac{1}{2}}$$

But

$$\begin{aligned} [|x + iy| + |u + iv|]^2 &= [(x^2 + y^2)^{1/2} + (u^2 + v^2)^{1/2}]^2 \\ &= (x^2 + y^2) + (u^2 + v^2) + 2(x^2 + y^2)^{1/2}(u^2 + v^2)^{1/2} \\ &= (x^2 + y^2) + (u^2 + v^2) + 2[(xu + yv)^2 + (xv - yu)^2]^{1/2} \end{aligned}$$

And this latter expression is greater than (or at least equal to)

$$(x^2 + y^2) + (u^2 + v^2) + 2(xu + yv)$$

We have therefore

$$|x + iy| + |u + iv| \geq |(x + iy) + (u + iv)|$$

i.e. *the modulus of the sum of two complex numbers cannot be greater than the sum of their moduli*, and it follows by induction that the modulus of the sum of any number of complex numbers cannot be greater than the sum of their moduli.

The inequality obtained above,

$$|(x + iy) + (u + iv)| \leq |x + iy| + |u + iv|$$

*is known as the **triangle inequality** for complex numbers. If complex numbers are interpreted geometrically as points (or vectors) in the plane, the modulus $|x + iy|$ represents the distance from the origin to the point (x, y) . The sum of two complex numbers corresponds to vector addition. The inequality therefore expresses the familiar geometric fact that, in any triangle, the length of one side cannot exceed the sum of the lengths of the other two sides. Thus the modulus behaves like an ordinary distance in the plane. This property is fundamental in analysis and plays a central role in the study of convergence of complex-valued sequences and series.*

Let us consider next the complex number which is the product of two given complex numbers, $x + iy$ and $u + iv$; we have

$$(x + iy)(u + iv) = (xu - yv) + i(xv + yu)$$

and so

$$\begin{aligned} |(x + iy)(u + iv)| &= [(xu - yv)^2 + (xv + yu)^2]^{1/2} \\ &= [(x^2 + y^2)(u^2 + v^2)]^{1/2} \\ &= |x + iy| |u + iv| \end{aligned}$$

The modulus of the product of two complex numbers (and hence, by induction, of any number of complex numbers) is therefore equal to the product of their moduli.

2.5 The Argand diagram

We have seen that complex numbers may be represented in a geometrical diagram by taking rectangular axes Ox, Oy in a plane. Then a point P whose coordinates referred to these axes are x, y may be regarded as representing the complex number $x + iy$. In this way, to every point of the plane there corresponds some one complex number, and conversely, to every possible complex number there

corresponds one, and only one, point of the plane. The complex number $x + iy$ may be denoted by a single letter¹² z . The point P is then called the *representative point* of the number z ; we shall also speak of the number z as being the *affix* of the point P .

If we denote $(x^2 + y^2)^{1/2}$ by r , and choose θ so that

$$r \cos \theta = x, \quad r \sin \theta = y$$

then r and θ are clearly the radius vector and vectorial angle of the point P , referred to the origin O and axis Ox .

The representation of complex numbers thus afforded is often called the **Argand diagram**.¹³

By the definition already given, it is evident that r is the modulus of z . The angle θ is called the argument, or amplitude, or phase, of z . We write

$$\theta = \arg z$$

From geometrical considerations, it appears that (although the modulus of a complex number is unique) the argument is not unique; if θ be a value of the argument, the other values of the argument are comprised in the expression

$$2n\pi + \theta$$

where n is any integer, not zero. The **principal value** of $\arg z$ is that which satisfies the inequality

$$-\pi < \arg z \leq \pi$$

The geometric representation of complex numbers is often called the Argand diagram, after Jean-Robert Argand (1768–1822), who published a clear description of this idea in 1806. Similar representations had been discovered earlier by Caspar Wessel (1797) and later used by Gauss. The notation $\arg z$ for the argument of a complex number is closely related historically to this representation. Many modern books simply interpret “arg” as an abbreviation for “argument,” although the term also echoes the name Argand, whose diagram provides the geometric interpretation of the argument.

If P_1 and P_2 are the representative points corresponding to values z_1 and z_2 respectively of z , then the point which represents the value $z_1 + z_2$ is clearly the terminus of a line drawn from P_1 , equal and parallel to that which joins the origin to P_2 .

In modern language this construction is simply vector addition in the plane. If complex numbers are identified with vectors (x, y) , then the sum $z_1 + z_2$ corresponds to the vector obtained by adding the vectors representing z_1 and z_2 . Geometrically this is the familiar parallelogram rule for vector addition.

¹² It is convenient to call x and y the **real** and **imaginary parts** of z respectively. We frequently write $x = R(z)$, $y = I(z)$.

¹³ It was published by J. R. Argand, *Essai sur une manière de représenter les quantités imaginaires dans les constructions géométriques* (1806); it had however previously been used by Gauss in his Helmstedt dissertation, 1799 (*Werke*, II, pp. 20–23), who had discovered it in Oct. 1797 (*Math. Ann.*, LXVI, p. 18), and Caspar Wessel had discussed it in a memoir presented to the Danish Academy in 1797 and published by that Society in 1798–99. The phrase **complex number** first occurs in 1831, Gauss, *Werke*, II, p. 102.

To find the point which represents the complex number $z_1 z_2$, where z_1 and z_2 are two given complex numbers, we notice that if

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$$

$$z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$$

then, by multiplication,

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

The point which represents the number $z_1 z_2$ has therefore a radius vector measured by the product of the radius vectors of P_1 and P_2 , and a vectorial angle equal to the sum of the vectorial angles of P_1 and P_2 .

The above formula follows from the trigonometric addition identities

$$\cos(\theta_1 + \theta_2) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2$$

$$\sin(\theta_1 + \theta_2) = \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2$$

Substituting these identities into the product

$$(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2)$$

yields

$$\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)$$

2.6 Euler's Formula (new)

In several places, the original text from Whittaker and Watson poses problems that benefit from the use of Euler's formula (which is not introduced until much later). To make these arguments accessible at an earlier stage, we introduce the formula here.

Euler's formula. For any real number θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

From this identity, we can evaluate certain trigonometric sums.

Further, we have $|e^{i\theta}|^2 = e^{i\theta} e^{-i\theta} = e^{i\theta} e^{-i\theta} = 1$, and therefore $|e^{i\theta}| = 1$. The bar over a complex number indicates its complement, i.e., if $z = x + iy$ then $\bar{z} = x - iy$.

As an example of using Euler's formula, consider the finite sum

$$\sum_{n=1}^p e^{in\theta}$$

This is a geometric series with ratio $e^{i\theta}$, and hence

$$\sum_{n=1}^p e^{in\theta} = e^{i\theta} \frac{1 - e^{ip\theta}}{1 - e^{i\theta}} = \frac{e^{i\theta} - e^{i(p+1)\theta}}{1 - e^{i\theta}}$$

To transform this expression, we use the identity

$$e^{ix} - e^{iy} = 2i e^{\frac{i(x+y)}{2}} \sin\left(\frac{x-y}{2}\right)$$

which follows from Euler's formula as shown below. We first note that

$$e^{ix} - e^{iy} = (\cos x + i \sin x) - (\cos y + i \sin y) = (\cos x - \cos y) + i(\sin x - \sin y)$$

Now use the sum-to-product identities:

$$\cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

$$\sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

Substituting,

$$e^{ix} - e^{iy} = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) + 2i \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

Factor out $2 \sin\left(\frac{x-y}{2}\right)$:

$$e^{ix} - e^{iy} = 2 \sin\left(\frac{x-y}{2}\right) \left(-\sin\left(\frac{x+y}{2}\right) + i \cos\left(\frac{x+y}{2}\right) \right)$$

Now observe that

$$ie^{i\theta} = i(\cos \theta + i \sin \theta) = i \cos \theta - \sin \theta$$

Hence

$$-\sin \theta + i \cos \theta = ie^{i\theta}$$

Taking $\theta = \frac{x+y}{2}$, we get

$$-\sin\left(\frac{x+y}{2}\right) + i \cos\left(\frac{x+y}{2}\right) = ie^{\frac{i(x+y)}{2}}$$

Therefore

$$e^{ix} - e^{iy} = 2 \sin\left(\frac{x-y}{2}\right) \cdot ie^{\frac{i(x+y)}{2}} = 2i e^{\frac{i(x+y)}{2}} \sin\left(\frac{x-y}{2}\right)$$

Applying this identity to the numerator and denominator of the summation (well) above, we obtain

$$e^{i\theta} - e^{i(p+1)\theta} = -2i e^{\frac{i(p+2)\theta}{2}} \sin\left(\frac{p\theta}{2}\right)$$

and

$$1 - e^{i\theta} = e^{i0} - e^{i\theta} = -2i e^{\frac{i\theta}{2}} \sin\left(\frac{\theta}{2}\right)$$

Dividing, the factors $-2i$ cancel, and we obtain

$$\sum_{n=1}^p e^{in\theta} = e^{i(p+1)\theta/2} \frac{\sin\left(\frac{p\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

Taking real and imaginary parts of the summation on the left, it follows that

$$\sum_{n=1}^p \cos(n\theta) \text{ and } \sum_{n=1}^p \sin(n\theta)$$

are bounded for all real θ such that $\sin(\theta/2) \neq 0$.

In particular, for $0 < \theta < 2\pi$,

$$\left| \sum_{n=1}^p \sin(n\theta) \right| \leq \frac{1}{\sin\left(\frac{\theta}{2}\right)} = \csc\left(\frac{\theta}{2}\right)$$

and a similar bound holds for the cosine sum.

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Miscellaneous Examples

1. Show that the representative points of the complex numbers

$$1 + 4i, 2 + 7i, 3 + 10i$$

are collinear.

Solution: Consider the complex numbers as points: $(1,4)$, $(2,7)$ and $(3,10)$. The slope of the line is 3. The equation of the line is $y - 4 = 3(x - 1)$ or $y = 3x + 1$. Since a complex number corresponding to a point (x,y) is $z = x + iy$, the points on this line correspond to $x + i(3x + 1)$.

2. Show that a parabola can be drawn to pass through the representative points of the complex numbers

$$2 + i, 4 + 4i, 6 + 9i, 8 + 16i, 10 + 25i$$

Solution: This one is fairly easy to guess, i.e., $y = \frac{x^2}{4}$. Since a complex number corresponding to a point (x,y) is $z = x + iy$, the points on this parabola correspond to $x + i\frac{x^2}{4}$.

3. Determine the n th roots of unity by aid of the Argand diagram, and show that the number of primitive roots (roots the powers of each of which give all the roots) is the number of integers (including unity) less than n and prime to it.

Prove that if $\theta_1, \theta_2, \theta_3, \dots$ be the arguments of the primitive roots,

$$\sum \cos(p\theta) = 0$$

when p is a positive integer less than

$$\frac{n}{abc \cdots k}$$

where $a, b, c, \dots, k$ are the different constituent primes of n ; and that, when

$$p = \frac{n}{abc \cdots k}$$

$$\sum \cos(p\theta) = \frac{(-1)^\mu n}{abc \cdots k}$$

where μ is the number of the constituent primes.

(Math. Trip. 1895 – Cambridge Mathematical Tripos examination.)

This exercise involves properties of the primitive roots of unity. A convenient modern approach uses Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

which allows the roots of unity to be written as $e^{2\pi i k/n}$. The resulting sums may then be evaluated using properties of these complex exponentials. Since Euler's formula and related results are introduced later, we omit a detailed solution here.

3 The Theory of Convergence

If a child comes to believe they are dumb at maths... then it is the teaching method, not the students' ability... that should be questioned." — Esther White

3.1 The definition¹⁴ of the limit of a sequence

Let $z_1, z_2, z_3, \dots$ be an unending sequence of numbers, real or complex. Then, if a number l exists such that, corresponding to every positive¹⁵ number ε , no matter how small, a number n_0 can be found, such that

$$|z_n - l| < \varepsilon$$

for all values of n greater than n_0 , the sequence (z_n) is said to tend to the limit l as n tends to infinity.

Symbolic forms of the statement¹⁶ "the limit of the sequence (z_n) , as n tends to infinity, is l " are

$$\lim_{n \rightarrow \infty} z_n = l, \quad \lim z_n = l, \quad z_n \rightarrow l \text{ as } n \rightarrow \infty.$$

If the sequence be such that, given an arbitrary number N (no matter how large), we can find n_0 such that

$$|z_n| > N$$

for all values of n greater than n_0 , we say that " $|z_n|$ tends to infinity as n tends to infinity", and we write

$$|z_n| \rightarrow \infty$$

In the corresponding case when

$$-x_n > N \text{ when } n > n_0$$

we say that

$$x_n \rightarrow -\infty$$

If a sequence of real numbers does not tend to a limit or to ∞ or to $-\infty$, the sequence is said to *oscillate*.

3.1.1 Definition of the phrase "of the order of"

If (ξ_n) and (z_n) are two sequences such that a number n_0 exists such that

$$\left| \frac{\xi_n}{z_n} \right| < K$$

whenever $n > n_0$, where K is independent of n , we say that ξ_n is "of the order of" z_n , and we write¹⁷

$$\xi_n = O(z_n).$$

¹⁴ A definition equivalent to this was first given by John Wallis in 1655 [Opera, I (1695), p.382].

¹⁵ The number zero is excluded from the class of positive numbers.

¹⁶ The arrow notation is due to *Leathem, Camb. Math. Tract.*, No. 1.

¹⁷ This notation is due to Bachmann, *Zahlentheorie* (1894), p. 401, and Landau, *Primzahlen*, I (1909), p. 61.

Thus

$$\frac{15n + 19}{1 + n^3} = O\left(\frac{1}{n^2}\right)$$

If $\lim \left(\frac{\xi_n}{z_n}\right) = 0$, we write $\xi_n = o(z_n)$.

The symbols $O(\cdot)$ and $o(\cdot)$ are now standard in asymptotic analysis. If $\xi_n = O(z_n)$, this means that the ratio $\left|\frac{\xi_n}{z_n}\right|$ remains bounded for sufficiently large n ; in other words, ξ_n does not grow faster than a constant multiple of z_n . If $\xi_n = o(z_n)$, then $\frac{\xi_n}{z_n} \rightarrow 0$ as $n \rightarrow \infty$; thus ξ_n becomes negligible compared with z_n for large n . These symbols are commonly called “Big-O” and “little-o” notation and are widely used to describe the relative growth of functions or sequences.

3.2 The limit of an increasing sequence

Let (x_n) be a sequence of real numbers such that $x_{n+1} \geq x_n$ for all values of n ; then the sequence tends to a limit or else tends to infinity (and so it does not oscillate).

Let α be any rational-real number; then either

- (1) $x_n \geq \alpha$ for all values of n greater than some number n_0 depending on the value of α , or
- (2) $x_n < \alpha$ for every value of n .

If (2) is not the case for any value of α (no matter how large), then

$$x_n \rightarrow \infty$$

But if values of α exist for which (2) holds, we can divide the rational numbers into two classes, the *L-class* consisting of those rational numbers α for which (1) holds and the *R-class* of those rational numbers α for which (2) holds. This section defines a real number α , rational or irrational.

And if ε be an arbitrary positive number, $\alpha - \frac{1}{2}\varepsilon$ belongs to the L-class which defines α , and so we can find n_1 such that $x_n \geq \alpha - \frac{1}{2}\varepsilon$ whenever $n > n_1$, and $\alpha + \frac{1}{2}\varepsilon$ is a member of the R-class and so $x_n < \alpha + \frac{1}{2}\varepsilon$. Therefore, whenever $n > n_1$,

$$|\alpha - x_n| < \varepsilon$$

Therefore

$$x_n \rightarrow \alpha$$

The construction above may appear confusing at first, since many numbers α may satisfy condition (2). For example, if $x_n = 2 - \frac{1}{n}$, then every number greater than 2 satisfies (2), not just the limit. The idea is not to identify a single such number, but rather to divide the rational numbers into two classes: those which are eventually less than or equal to the sequence and those which are always greater than it. The boundary between these two classes determines a real number, which is the limit of the sequence. In modern terms, this

construction corresponds to defining the limit as the least upper bound (supremum) of the sequence.

This result is now known as the monotone convergence theorem for sequences: every increasing sequence of real numbers either converges to a finite limit or tends to infinity. In modern treatments this is usually proved using the least upper bound (supremum) property of the real numbers; here it is obtained using the construction of real numbers by Dedekind sections.

Corollary. A decreasing sequence tends to a limit or to $-\infty$.

Example 1. If $\lim z_n = l$, $\lim z'_n = l'$, then

$$\lim (z_n + z'_n) = l + l'$$

For, given ε , we can find n and n' such that

$$(i) \text{ when } m > n, |z_m - l| < \frac{1}{2}\varepsilon$$

$$(ii) \text{ when } m > n', |z'_m - l'| < \frac{1}{2}\varepsilon$$

Let n_1 be the greater of n and n' ; then, when $m > n_1$,

$$|(z_m + z'_m) - (l + l')| \leq |z_m - l| + |z'_m - l'| < \varepsilon$$

and this is the condition that

$$\lim (z_m + z'_m) = l + l'$$

Example 2. Prove similarly that

$$\lim (z_n - z'_n) = l - l', \lim (z_n z'_n) = ll'$$

and, if $l' \neq 0$,

$$\lim \left(\frac{z_n}{z'_n} \right) = \frac{l}{l'}$$

Example 3. If $0 < x < 1$, $x^n \rightarrow 0$.

For if $x = (1 + a)^{-1}$, $a > 0$, then

$$0 < x^n = \frac{1}{(1+a)^n} < \frac{1}{1+na}$$

by the binomial theorem for a positive integral index. And it is obvious that, given a positive number ε , we can choose n_0 such that

$$\frac{1}{1+na} < \varepsilon \text{ when } n > n_0$$

and so $x^n \rightarrow 0$.

3.2.1 Limit-points and the Bolzano–Weierstrass¹⁸ theorem

Let (x_n) be a sequence of real numbers. If any number G exists such that, for every positive value of ε , no matter how small, an unlimited number of terms of the sequence can be found such that

$$G - \varepsilon < x_n < G + \varepsilon$$

then G is called a **limit-point**, or **cluster-point**, of the sequence.

Bolzano's theorem is that, if $\lambda < x_n \leq \rho$, where λ, ρ are independent of n , then the sequence (x_n) has at least one limit-point.

To prove the theorem, choose a section in which (i) the R-class consists of all the rational numbers which are such that, if A be any one of them, there are only a limited number of terms x_n satisfying $x_n > A$, and (ii) the L-class is such that there are an unlimited number of terms x_n such that $x_n \geq a$ for all members a of the L-class.

This section defines a real number G , and, if ε be an arbitrary positive number, $G - \frac{1}{2}\varepsilon$ and $G + \frac{1}{2}\varepsilon$ are members of the L and R classes respectively, and so there are an unlimited number of terms of the sequence satisfying

$$G - \varepsilon < G - \frac{1}{2}\varepsilon \leq x_n \leq G + \frac{1}{2}\varepsilon < G + \varepsilon$$

and so G satisfies the condition that it should be a limit-point.

In modern terms, this result is a version of the Bolzano–Weierstrass theorem: every bounded sequence of real numbers has a convergent subsequence, and hence at least one limit-point. A limit-point (or cluster point) of a sequence is a number that is approached arbitrarily closely by infinitely many terms of the sequence, though the sequence itself need not converge. The argument given above constructs such a point using a Dedekind section, whereas modern proofs typically rely on the completeness of the real numbers, expressed through the least upper bound property.

3.2.1.1 Definition of “the greatest of the limits”

The number G obtained in Section 3.2.1 is called *the greatest of the limits* of the sequence (x_n) . The sequence (x_n) cannot have a limit-point greater than G ; for if G' were such a limit-point, and $\varepsilon = \frac{1}{2}(G' - G)$, $G' - \varepsilon$ is a member of the R-class defining G , so that there are only a limited number of terms of the sequence which satisfy $x_n > G' - \varepsilon$. This condition is inconsistent with G' being a limit-point. We write

$$G = \limsup_{n \rightarrow \infty} x_n$$

The *least of the limits*, L , of the sequence (written $\liminf x_n$) is defined to be

$$L = -\limsup_{n \rightarrow \infty} (-x_n)$$

¹⁸ This theorem, frequently ascribed to Weierstrass, was proved by Bolzano, *Abh. d. k. böhmischen Ges. der Wiss.* (1817) [Reprinted in *Klassiker der Exakten Wiss.*, No. 153]. It seems to have been known to Cauchy.

The quantities $\limsup x_n$ and $\liminf x_n$, called the limit superior and limit inferior of the sequence, describe the ultimate upper and lower behavior of the sequence. Roughly speaking, $\limsup x_n$ is the largest value that the sequence approaches infinitely often, while $\liminf x_n$ is the smallest such value. More precisely, $\limsup x_n$ is the greatest limit-point of the sequence and $\liminf x_n$ is the least limit-point.

For example, the sequence $x_n = (-1)^n$ oscillates between 1 and -1 . It has two limit-points, namely 1 and -1 , and hence

$$\limsup x_n = 1, \quad \liminf x_n = -1$$

As another example, if $x_n = 1 + \frac{(-1)^n}{n}$, then the sequence converges to 1, but approaches this value from above and below. In this case both the greatest and least limit-points are equal, and

$$\limsup x_n = \liminf x_n = 1$$

In general, a sequence converges if and only if its limit superior and limit inferior are equal, in which case their common value is the limit.

3.2.2 Cauchy's¹⁹ theorem on the necessary and sufficient condition for the existence of a limit

We shall now show that the necessary and sufficient condition for the existence of a limiting value of a sequence of numbers $z_1, z_2, z_3, \dots$ is that, corresponding to any given positive number ε , however small, it shall be possible to find a number n such that

$$|z_{n+p} - z_n| < \varepsilon$$

for all positive integral values of p . This result is one of the most important and fundamental theorems of analysis. It is sometimes called the **Principle of Convergence**.

First, we have to show that this condition is *necessary*, i.e. that it is satisfied whenever a limit exists. Suppose then that a limit l exists, then (Section 3.1) corresponding to any positive number ε , however small, an integer n can be chosen such that

$$|z_n - l| < \frac{1}{2}\varepsilon, \quad |z_{n+p} - l| < \frac{1}{2}\varepsilon$$

for all positive values of p ; therefore

$$|z_{n+p} - z_n| = |(z_{n+p} - l) - (z_n - l)| \leq |z_{n+p} - l| + |z_n - l| < \varepsilon$$

which shows the necessity of the condition

$$|z_{n+p} - z_n| < \varepsilon$$

and thus establishes the first half of the theorem.

Secondly, we have to prove²⁰ that this condition is *sufficient*, i.e. that if it is satisfied, then a limit exists.

¹⁹ *Analyse Algébrique* (1821), p. 125.

²⁰ This proof is given by Stolz and Gmeiner, *Theoretische Arithmetik*, II (1902), p. 144.

(I) Suppose that the sequence of real numbers (x_n) satisfies Cauchy's condition, that is to say that, corresponding to any positive number ε , an integer n can be chosen such that

$$|x_{n+p} - x_n| < \varepsilon$$

for all positive integral values of p .

Let the value of n , corresponding to the value 1 of ε , be m .

Let λ_1, ρ_1 be the least and greatest of $x_1, x_2, \dots, x_m$; then

$$\lambda_1 - 1 < x_n < \rho_1 + 1$$

for all values of n ; write $\lambda_1 - 1 = \lambda$, $\rho_1 + 1 = \rho$.

Then, for all values of n , $\lambda < x_n < \rho$. Therefore, by the theorem of Section 3.2.1, the sequence (x_n) has at least one limit-point G .

The above argument shows that a Cauchy sequence is bounded. Taking $\varepsilon = 1$, there exists an index m such that all terms beyond the m th differ from one another by less than 1, and hence lie within 1 of x_m . The first m terms form a finite set and therefore have a least and greatest element. By enlarging this finite interval slightly, one obtains bounds λ and ρ such that $\lambda < x_n < \rho$ for all n .

Further, there cannot be more than one limit-point, for if there were two, G and H ($H < G$), take $\varepsilon < \frac{1}{4}(G - H)$. Then, by hypothesis, a number n exists such that

$$|x_{n+p} - x_n| < \varepsilon$$

for every positive value of p . But since G and H are limit-points, positive numbers q and r exist such that

$$|G - x_{n+q}| < \varepsilon, \quad |H - x_{n+r}| < \varepsilon$$

Then

$$|G - x_{n+q}| + |x_{n+q} - x_n| + |x_n - x_{n+r}| + |x_{n+r} - H| < 4\varepsilon$$

But, by Section 2.4, the sum on the left is greater than or equal to $|G - H|$. Therefore $G - H < 4\varepsilon$, which is contrary to hypothesis, so there is only one limit-point. Hence there are only a finite number of terms of the sequence outside the interval $(G - \delta, G + \delta)$, where δ is an arbitrary positive number, for, if there were an unlimited number of such terms, these would have a limit-point which would be a limit-point of the given sequence and which would not coincide with G , and therefore G is the limit of (x_n) .

(II) Now let the sequence (z_n) of real or complex numbers satisfy Cauchy's condition, and let $z_n = x_n + iy_n$, where x_n and y_n are real, then for all values of n and p

$$|x_{n+p} - x_n| \leq |z_{n+p} - z_n|, \quad |y_{n+p} - y_n| \leq |z_{n+p} - z_n|$$

Therefore, the sequences of real numbers (x_n) and (y_n) satisfy Cauchy's condition, and so, by (I), the limits of (x_n) and (y_n) exist. Therefore, by Section 3.2 Example 1, the limit of (z_n) exists. The result is therefore established.

3.3 Convergence of an infinite series

Let $u_1, u_2, u_3, \dots, u_n, \dots$ be a sequence of numbers, real or complex. Let the sum

$$u_1 + u_2 + \dots + u_n$$

be denoted by S_n .

Then, if S_n tends to a limit S as n tends to infinity, the infinite series

$$u_1 + u_2 + u_3 + u_4 + \dots$$

is said to be **convergent**, or to converge to the sum S . In other cases, the infinite series is said to be **divergent**. When the series converges, the expression $S - S_n$, which is the sum of the series

$$u_{n+1} + u_{n+2} + u_{n+3} + \dots$$

is called the *remainder after n terms*, and is frequently denoted by the symbol R_n .

The sum

$$u_{n+1} + u_{n+2} + \dots + u_{n+p}$$

will be denoted by $S_{n,p}$.

It follows at once, by combining the above definition with the results of the last paragraph, that the necessary and sufficient condition for the convergence of an infinite series is that, given an arbitrary positive number ε , we can find n such that

$$|S_{n,p}| = |S_{n+p} - S_n| < \varepsilon \text{ for every positive value of } p$$

Equation 1. Necessary and sufficient condition for the convergence of an infinite series

Since $u_{n+1} = S_{n,1}$, it follows as a particular case that

$$\lim u_{n+1} = 0$$

in other words, the n th term of a convergent series must tend to zero as n tends to infinity. But this last condition, though necessary, is not sufficient in itself to ensure the convergence of the series, as appears from a study of the *harmonic* series

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$

In this series,

$$S_{n,n} = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n}$$

The expression on the right is decreased by replacing each term by $(2n)^{-1}$ in place of each term, and so

$$S_{n,n} > \frac{1}{2}$$

Therefore

$$S_{2^{n+1}} = 1 + S_{1,1} + S_{2,2} + S_{4,4} + S_{8,8} + \dots + S_{2^n, 2^n} > \frac{1}{2}(n+3) \rightarrow \infty$$

so the series is divergent; this result was noticed by Leibniz in 1673.

The partial sums may be written by grouping the terms in blocks of increasing length:

$$S_{2^{n+1}} = 1 + (u_2) + (u_3 + u_4) + (u_5 + \cdots + u_8) + \cdots + (u_{2^n+1} + \cdots + u_{2^{n+1}})$$

In general, the k th block consists of the terms from $2^k + 1$ to 2^{k+1} , and is equal to $S_{2^k, 2^k}$. From the preceding result,

$$S_{2^k, 2^k} > \frac{1}{2}$$

for every k . Hence

$$S_{2^{n+1}} > 1 + \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2}$$

where there are $n + 1$ blocks, each exceeding $\frac{1}{2}$. Therefore

$$S_{2^{n+1}} > \frac{1}{2}(n + 3) \rightarrow \infty$$

and the series diverges.

There are two general classes of problems which we are called upon to investigate in connection with the convergence of series.

(i) We may arrive at a series by some formal process, e.g. that of solving a linear differential equation by a series, and then to justify the process it will usually have to be proved that the series thus formally obtained is convergent. Simple conditions for establishing convergence in such circumstances are obtained in Sections 3.3.1-3.6.1.

(ii) Given an expression S , it may be possible to obtain a development

$$S = \sum_{m=1}^n u_m + R_n,$$

valid for all values of n , and, from the definition of a limit, it follows that, if we can prove that $R_n \rightarrow 0$, then the series

$$\sum_{m=1}^{\infty} u_m$$

converges and its sum is S .

Infinite series were used²¹ by Lord Brouncker in *Phil. Trans.* II (1668), pp. 615–649, and the term *convergent* was introduced by James Gregory, Professor of Mathematics at Edinburgh, in the same year; the term *divergent* was introduced by N. Bernoulli in 1713. Infinite series were used systematically by Newton in 1669, *De analysi per aequationes numero terminorum infinitas*, and he investigated the convergence of hypergeometric series in 1704. But the great mathematicians of the eighteenth century used infinite series freely without, for the most part, examining their convergence. Thus, Euler gave the sum of the series

$$\cdots + \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + z^3 + \cdots \quad (\text{a})$$

²¹ See also the note to Section 3.7.

as zero, on the ground that

$$z + z^2 + z^3 + \cdots = \frac{z}{1 - z} \quad (\text{b})$$

and

$$1 + \frac{1}{z} + \frac{1}{z^2} + \cdots = \frac{z}{z - 1} \quad (\text{c})$$

The error, of course, arises from the fact that the series (b) converges only when $|z| < 1$, and the series (c) converges only when $|z| > 1$, so the series (a) never converges.

For the history of researches on convergence, see Pringsheim and Molk, *Encyclopédie des Sci. Math.*, I (1) and Reiff, *Geschichte der unendlichen Reihen* (Tübingen, 1889).

3.3.1 Abel's inequality²² and Dirichlet's Test for Convergence

Let $f_n \geq f_{n+1} > 0$ for all integer values of n . Then

$$\left| \sum_{n=1}^m a_n f_n \right| \leq A f_1$$

where A is the greatest of the sums

$$|a_1|, |a_1 + a_2|, |a_1 + a_2 + a_3|, \dots, |a_1 + a_2 + \cdots + a_m|$$

Let

$$s_n = \sum_{k=1}^n a_k$$

Then, by summation by parts,

$$\begin{aligned} \sum_{n=1}^m a_n f_n &= s_1 f_1 + \sum_{n=2}^m (s_n - s_{n-1}) f_n \\ &= \sum_{n=1}^{m-1} s_n (f_n - f_{n+1}) + s_m f_m \end{aligned}$$

Since the sequence (f_n) is decreasing, the differences

$$f_n - f_{n+1} \geq 0$$

Let

$$A = \max_{1 \leq p \leq m} |s_p|$$

Then, for each n ,

$$|s_n| (f_n - f_{n+1}) \leq A(f_n - f_{n+1}), \quad |s_m| f_m \leq A f_m$$

²² *Journal für Math.* 8 (1826), pp. 311–339. A particular case of the theorem of Section 3.3.1, Corollary (i), also appears in that memoir.

Summing, we obtain

$$\left| \sum_{n=1}^m a_n f_n \right| \leq A \left(\sum_{n=1}^{m-1} (f_n - f_{n+1}) + f_m \right) = A f_1$$

Corollary. If a_n and w_n are any real or complex numbers, then

$$\left| \sum_{n=1}^m a_n w_n \right| \leq A \left(\sum_{n=1}^{m-1} |w_{n+1} - w_n| + |w_m| \right)$$

where

$$A = \max_{1 \leq p \leq m} \left| \sum_{n=1}^p a_n \right|$$

This corollary appears in the work of Hardy and is closely related to Abel's inequality.

Dirichlet's²³ test for convergence

Let $\left| \sum_{n=1}^p a_n \right| < K$, where K is independent of p . Then, if $f_n \geq f_{n+1} > 0$ and $\lim_{n \rightarrow \infty} f_n = 0$,

$$\sum_{n=1}^{\infty} a_n f_n \text{ converges}$$

For, since $\lim_{n \rightarrow \infty} f_n = 0$ ²⁴, given an arbitrary positive number ε , we can find m such that

$$f_{m+1} < \frac{\varepsilon}{2K}$$

Then

$$\left| \sum_{n=m+1}^{m+q} a_n \right| \leq \left| \sum_{n=1}^{m+q} a_n \right| + \left| \sum_{n=1}^m a_n \right| < 2K$$

for all positive values of q , so that, by Abel's inequality, we have, for all positive values of p ,

$$\left| \sum_{n=m+1}^{m+p} a_n f_n \right| \leq A f_{m+1}$$

where $A < 2K$

²³ Journal für Math. (2), vii (1862), pp. 253–255. Before the publication of the 2nd edition of Jordan's *Cours d'Analyse* (1893), Dirichlet's test and Abel's test were frequently jointly described as the Dirichlet–Abel test, see e.g. Pringsheim, Math. Ann. xxv (1885), p. 423

²⁴ In these circumstances, we say $f_n \rightarrow 0$ steadily.

Therefore,

$$\left| \sum_{n=m+1}^{m+p} a_n f_n \right| < 2K f_{m+1} < \varepsilon$$

and so, by Equation 1,

$$\sum_{n=1}^{\infty} a_n f_n \text{ converges}$$

Note. (On cancellation in infinite series)

It might be thought that inserting terms whose sum is zero does not affect the convergence of a series. This is not the case.

For example, start with the convergent series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ and form a new series by inserting cancelling pairs:

$$1 + \frac{1}{2^2} + \cdots + \frac{1}{10^2} + 10 + \frac{1}{11^2} + \cdots + \frac{1}{100^2} - 10 + 100 + \frac{1}{101^2} + \cdots + \frac{1}{1000^2} - 100 + 1000 \cdots$$

Although each inserted pair has sum zero, the new series does not converge. Indeed, after the term $+10$ the partial sum is increased by 10, after the term $+100$ it is increased by 100, and so on. Hence the partial sums are not bounded, and therefore the series cannot converge.

This shows that the convergence of an infinite series depends on the behavior of its partial sums, and not merely on the formal appearance of cancellation among its terms.

Example. Show that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges.

Solution. If we write $a_n = (-1)^n$, $f_n = \frac{1}{n}$ then $\sum_{n=1}^p a_n = \sum_{n=1}^p (-1)^n$ has bounded partial sums, since the partial sums take only the values 0 and -1 , and are therefore bounded. Also, the sequence (f_n) is positive, decreasing, and satisfies $\lim_{n \rightarrow \infty} f_n = 0$. Therefore, by Dirichlet's

test, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges.

Exercise. Show that the series $\sum_{n=1}^{\infty} \frac{\sin n}{n}$ converges.

Corollary (i). Abel's test for convergence.

If $\sum_{n=1}^{\infty} a_n$ converges and the sequence (u_n) is monotonic (i.e. $u_n \geq u_{n+1}$ always or else $u_n \leq u_{n+1}$ always) and $|u_n| < \kappa$, where κ is independent of n , then

$$\sum_{n=1}^{\infty} a_n u_n \text{ converges}$$

For, by statements at the beginning of Section 3.2, u_n tends to a limit u , let $|u - u_n| = f_n$. Then $f_n \rightarrow 0$ steadily, and therefore $\sum_{n=1}^{\infty} a_n f_n$ converges.

But, if (u_n) is an increasing sequence, $f_n = u - u_n$, and so $\sum_{n=1}^{\infty} (u - u_n) a_n$ converges,

Therefore, since $\sum_{n=1}^{\infty} u a_n$ converges, $\sum_{n=1}^{\infty} u_n a_n$ also converges.

If (u_n) is a decreasing sequence, $f_n = u_n - u$, and a similar proof holds.

Example. Show that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n+1}$ converges.

Solution. Write $\frac{(-1)^n}{n+1} = \frac{(-1)^n}{n} \cdot \frac{n}{n+1}$

Thus, the given series may be written in the form $\sum_{n=1}^{\infty} a_n u_n$ where $a_n = \frac{(-1)^n}{n}$, $u_n = \frac{n}{n+1}$.

Now the series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges, and the sequence (u_n) is monotonic and bounded, since $u_n = \frac{n}{n+1} < 1$ and u_n increases to 1. Therefore, by Abel's test, the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n+1} \text{ converges.}$$

Corollary (ii). Taking $a_n = (-1)^{n-1}$ in Dirichlet's test, it follows that, if $f_n \geq f_{n+1}$ and $\lim f_n = 0$,

$$f_1 - f_2 + f_3 - f_4 + \cdots$$

converges.

Exercise. Show that if $0 < \theta < 2\pi$,

$$\left| \sum_{n=1}^p \sin n\theta \right| < \csc\left(\frac{\theta}{2}\right)$$

and deduce that, if $f_n \rightarrow 0$ steadily,

$$\sum_{n=1}^{\infty} f_n \sin n\theta$$

converges for all real values of θ , and that

$$\sum_{n=1}^{\infty} f_n \cos n\theta$$

converges if θ is not an even multiple of π .

Solution.

By Section 2.6, for $0 < \theta < 2\pi$, we have

$$\left| \sum_{n=1}^p \sin(n\theta) \right| \leq \frac{1}{\sin\left(\frac{\theta}{2}\right)} = \csc \frac{\theta}{2}$$

Thus, the partial sums of $\sum \sin(n\theta)$ are bounded.

Now let $f_n \rightarrow 0$ steadily. Then (f_n) is positive, decreasing, and tends to 0. Therefore, by Dirichlet's test,

$$\sum_{n=1}^{\infty} f_n \sin(n\theta)$$

converges for all real values of θ . If θ is an integral multiple of 2π , every term is zero, so the conclusion is trivial.

Again, by Section 2.6, taking real parts gives

$$\sum_{n=1}^p \cos(n\theta) = \frac{\sin(p\theta/2) \cos((p+1)\theta/2)}{\sin(\theta/2)}$$

Hence the partial sums of $\sum \cos(n\theta)$ are bounded whenever $\sin\left(\frac{\theta}{2}\right) \neq 0$, that is, whenever θ is not an even multiple of π . Therefore, by Dirichlet's test,

$$\sum_{n=1}^{\infty} f_n \cos(n\theta)$$

converges if θ is not an even multiple of π .

Exercise. Show that, if $f_n \rightarrow 0$ steadily,

$$\sum_{n=1}^{\infty} (-1)^n f_n \cos(n\theta)$$

converges if θ is real and not an odd multiple of π , and

$$\sum_{n=1}^{\infty} (-1)^n f_n \sin(n\theta)$$

converges for all real values of θ .

[Write $\pi + \theta$ for θ in the previous exercise.]

3.3.2 Absolute and conditional convergence

In order that a series $\sum_{n=1}^{\infty} u_n$ of real or complex terms may converge, it is sufficient (but not necessary) that the series of moduli $\sum_{n=1}^{\infty} |u_n|$ should converge.

For, if

$$\sigma_{n,p} = |u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}|$$

and if

$$\sum_{n=1}^{\infty} |u_n|$$

converges, we can find n , corresponding to a given number ε , such that $\sigma_{n,p} < \varepsilon$

for all values of p . By the triangle inequality,

$$|S_{n,p}| = |u_{n+1} + \cdots + u_{n+p}| \leq \sigma_{n,p}$$

and so $\sum_{n=1}^{\infty} u_n$ converges.

The condition is not necessary, for, writing $f_n = \frac{1}{n}$ in Section 3.3.1, Corollary (ii), we see that

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

converges, though (Section 3.3) the series of moduli

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is known to diverge.

In this case, therefore, the divergence of the series of moduli does not entail the divergence of the series itself.

Series, which are such that the series formed by the moduli of their terms are convergent, possess special properties of great importance, and are called **absolutely convergent series**. Series which, though convergent, are not absolutely convergent (i.e. the series themselves converge, but the series of moduli diverge) are said to be **conditionally convergent**.

3.3.3 The Geometric Series and the Riemann Zeta Function

The convergence of a particular series is in most cases investigated, not by the direct consideration of the sum $S_{n,p}$, but (as will appear from the following articles) by a comparison of the given series with some other series which is known to be convergent or divergent. We shall now investigate the convergence of two of the series which are most frequently used as standards for comparison.

(I) The geometric series

The geometric series is defined to be the series

$$1 + z + z^2 + z^3 + z^4 + \cdots$$

Consider the series of moduli

$$1 + |z| + |z|^2 + |z|^3 + \dots$$

for this series

$$S_{n,p} = |z|^{n+1} + |z|^{n+2} + \dots + |z|^{n+p} = |z|^{n+1} \frac{1 - |z|^p}{1 - |z|}$$

Hence, if $|z| < 1$, then

$$S_{n,p} < \frac{|z|^{n+1}}{1 - |z|}$$

for all values of p , and, by Section 3.2, Example 3, given any positive number ε , we can find n such that

$$\frac{|z|^{n+1}}{1 - |z|} < \varepsilon$$

Thus, given ε , we can find n such that, for all values of p , $S_{n,p} < \varepsilon$. Hence, by Section 3.2, the series

$$1 + |z| + |z|^2 + \dots$$

is convergent so long as $|z| < 1$, and therefore the geometric series is absolutely convergent if $|z| < 1$.

When $|z| \geq 1$, the terms of the geometric series do not tend to zero as n tends to infinity, and the series is therefore divergent.

(II) The series

$$\frac{1}{1^s} + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \frac{1}{5^s} + \dots$$

Consider now the series

$$S_n = \sum_{m=1}^n \frac{1}{m^s} \text{ which defines the **Riemann Zeta function**}$$

where s is greater than 1.

In more advanced treatments, s is allowed to be complex, and the series converges when $R(s) > 1$.

We have

$$\frac{1}{2^s} + \frac{1}{3^s} < \frac{2}{2^s} = \frac{1}{2^{s-1}}$$

$$\frac{1}{4^s} + \frac{1}{5^s} + \frac{1}{6^s} + \frac{1}{7^s} < \frac{4}{4^s} = \frac{1}{4^{s-1}}$$

and so on.

Thus, the sum of $2^p - 1$ terms is bounded (as shown below).

$$\frac{1}{1^s} + \frac{1}{2^{s-1}} + \frac{1}{4^{s-1}} + \frac{1}{8^{s-1}} + \dots + \frac{1}{2^{(p-1)(s-1)}} < \frac{1}{1 - 2^{1-s}}$$

and so, the sum of any number of terms is less than $(1-2^{1-s})^{-1}$. Therefore, the increasing sequence $\sum_{m=1}^n \frac{1}{m^s}$ cannot tend to infinity; therefore, by Section 3.2, the series $\sum_{n=1}^{\infty} \frac{1}{n^s}$ is convergent if $s > 1$, and since its terms are all real and positive, they are equal to their own moduli, and so the series of moduli of the terms is convergent, that is, the convergence is absolute.

If $s = 1$, the series becomes

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

which we have already shown to be divergent, and when $s < 1$, it is *a fortiori* (i.e., with even stronger reason) divergent, since the effect of diminishing s is to increase the terms of the series. The series $\sum_{n=1}^{\infty} \frac{1}{n^s}$ is therefore divergent if $s \leq 1$.

3.3.4 The Comparison Theorem

We shall now show that a series

$$u_1 + u_2 + u_3 + \dots$$

is absolutely convergent, provided that

$$|u_n| \leq C|v_n| \quad (\text{more commonly stated as } \left| \frac{u_n}{v_n} \right| \leq C, \text{ provided } v_n \neq 0)$$

where C is some number independent of n , and v_n is the n th term of another series which is known to be absolutely convergent.

For, under these conditions, we have

$$|u_{n+1}| + |u_{n+2}| + \dots + |u_{n+p}| < C(|v_{n+1}| + |v_{n+2}| + \dots + |v_{n+p}|)$$

where n and p are any integers. But since the series $\sum v_n$ is absolutely convergent, the series $\sum |v_n|$ is convergent, and so, given ε , we can find n such that

$$|v_{n+1}| + |v_{n+2}| + \dots + |v_{n+p}| < \frac{\varepsilon}{C}$$

for all values of p . It follows therefore that we can find n such that

$$|u_{n+1}| + |u_{n+2}| + \dots + |u_{n+p}| < \varepsilon$$

for all values of p , i.e. the series $\sum |u_n|$ is convergent. The series $\sum u_n$ is therefore absolutely convergent.

Corollary. A series is absolutely convergent if the ratio of its n th term to the n th term of a series which is known to be absolutely convergent is less than some number independent of n .

Example 1. Show that the series

$$\cos z + \frac{1}{2^2} \cos 2z + \frac{1}{3^2} \cos 3z + \frac{1}{4^2} \cos 4z + \cdots$$

is absolutely convergent for all real values of z .

When z is real, we have $|\cos nz| \leq 1$, and therefore

$$\left| \frac{\cos nz}{n^2} \right| \leq \frac{1}{n^2}$$

The moduli of the terms of the given series are therefore less than, or at most equal to, the corresponding terms of the series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

which by Section 3.3 is absolutely convergent. The given series is therefore absolutely convergent.

Example 2. Show that the series

$$\frac{1}{1^2(z - z_1)} + \frac{1}{2^2(z - z_2)} + \frac{1}{3^2(z - z_3)} + \frac{1}{4^2(z - z_4)} + \cdots$$

where

$$z_n = e^{in}, \quad n = 1, 2, 3, \dots$$

is convergent for all values of z which are not on the circle $|z| = 1$.

Solution. Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^2(z - z_n)}$, $z_n = e^{in}$

Since $|z_n| = 1$ for every n , it is enough to show that $|z - z_n|$ is bounded below by a positive constant independent of n . There are two cases.

Case 1. $|z| < 1$

By the triangle inequality,

$$|z_n| = |(z_n - z) + z| \leq |z_n - z| + |z|$$

Since $|z_n| = 1$, this gives

$$1 \leq |z_n - z| + |z|$$

and therefore

$$|z - z_n| \geq 1 - |z| > 0$$

Since $|z| < 1$, the quantity $1 - |z|$ is positive.

Case 2. $|z| > 1$

Again, by the triangle inequality,

$$|z| = |(z - z_n) + z_n| \leq |z - z_n| + |z_n|$$

and since $|z_n| = 1$,

$$|z| \leq |z - z_n| + 1$$

So, $|z - z_n| \geq |z| - 1 > 0$.

Since $|z| > 1$, the quantity $|z| - 1$ is positive.

Thus, in either case, there is a positive constant δ , independent of n , such that

$$|z - z_n| \geq \delta$$

Hence

$$\left| \frac{1}{n^2(z - z_n)} \right| \leq \frac{1}{\delta} \cdot \frac{1}{n^2}$$

Now $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. Therefore, by the comparison theorem, the series $\sum_{n=1}^{\infty} \frac{1}{n^2(z - z_n)}$ is absolutely convergent, and therefore convergent, for all values of z which are not on the circle $|z| = 1$.

The geometric representation of complex numbers is helpful in discussing a question of this kind (referring to Example 2 above). Let values of the complex number z be represented on a plane, then the numbers $z_1, z_2, z_3, \dots$ will give a sequence of points which lie on the circumference of the circle whose centre is the origin and whose radius is unity, and it can be shown that every point on the circle is a limit point (Section 3.2.1) of the points z_n .

For these special values z_n of z , the given series does not exist (referring to the summation in Example 2 above), since the denominator of the n th term vanishes when $z = z_n$. For simplicity we do not discuss the series for any point z situated on the circumference of the circle of radius unity.

Suppose now that $|z| \neq 1$. Then for all values of n , $|z - z_n| \geq ||z| - |z_n|| > c^{-1}$, for some value of c , so the moduli of the terms of the given series are less than the corresponding terms of the series

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

which is known to be absolutely convergent. The given series is therefore absolutely convergent for all values of z , except those which are on the circle $|z| = 1$.

It is interesting to notice that the area in the z -plane over which the series converges is divided into two parts, between which there is no intercommunication, by the circle $|z| = 1$.

Example 3. Show that the series

$$2 \sin\left(\frac{z}{3}\right) + 4 \sin\left(\frac{z}{9}\right) + 8 \sin\left(\frac{z}{27}\right) + \dots + 2^n \sin\left(\frac{z}{3^n}\right) + \dots$$

converges absolutely for all values of z .

Since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$,

$$\lim_{n \rightarrow \infty} 3^n \sin\left(\frac{z}{3^n}\right) = \lim_{n \rightarrow \infty} z \frac{\sin\left(\frac{z}{3^n}\right)}{\left(\frac{z}{3^n}\right)} = z$$

Since $3^n \sin(z/3^n) \rightarrow z$, the sequence is bounded; hence there exists a constant k , independent of n , such that $\left|3^n \sin\left(\frac{z}{3^n}\right)\right| < k$ and therefore $\left|2^n \sin\left(\frac{z}{3^n}\right)\right| \leq k \left(\frac{2}{3}\right)^n$. Since the geometric series $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ converges, the given series converges absolutely.

3.3.5 Cauchy's test for absolute convergence²⁵

If $\lim_{n \rightarrow \infty} \sup |u_n|^{\frac{1}{n}} < 1$, $\sum_{n=1}^{\infty} u_n$ converges absolutely.

The use of $\lim \sup$ is necessary because the limit of $|u_n|^{\frac{1}{n}}$ need not exist. For example, if $|u_n|^{\frac{1}{n}} = \frac{1}{4}$ for n even and $\frac{1}{2}$ for n odd, then the limit does not exist but $\lim \sup |u_n|^{\frac{1}{n}} = \frac{1}{2}$.

For we can find m such that, when $n \geq m$, $|u_n|^{\frac{1}{n}} \leq \rho < 1$, where ρ is independent of n . Then, when $n > m$, $|u_n| < \rho^n$, and since $\sum_{n=m+1}^{\infty} \rho^n$ converges, it follows from Section 3.3.4 that $\sum_{n=m+1}^{\infty} u_n$ (and therefore $\sum_{n=1}^{\infty} u_n$) converges absolutely.

Note. If $\lim_{n \rightarrow \infty} \sup |u_n|^{\frac{1}{n}} > 1$, u_n does not tend to zero, and, by Section 3.3, $\sum_{n=1}^{\infty} u_n$ does not converge.

3.3.6 D'Alembert's²⁶ Ratio Test for Absolute Convergence

We shall now show that a series $u_1 + u_2 + u_3 + \dots$ is absolutely convergent, provided that for all values of n greater than some fixed value r , the ratio $\frac{u_{n+1}}{u_n}$ is less than ρ , where ρ is a positive number independent of n and less than unity.

For the terms of the series

$$|u_{r+1}| + |u_{r+2}| + |u_{r+3}| + \dots$$

are respectively less than the corresponding terms of the series

$$|u_{r+1}|(1 + \rho + \rho^2 + \rho^3 + \dots)$$

which is absolutely convergent when $\rho < 1$, therefore $\sum_{n=r+1}^{\infty} u_n$ (and hence the given series) is absolutely convergent.

A particular case of this theorem is that if

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = l < 1,$$

the series is absolutely convergent.

²⁵ *Analyse Algébrique*, pp. 132–135

²⁶ *Opuscles*, t. v (1768), pp. 171–182

For, by the definition of a limit, we can find r such that

$$\left| \frac{u_{n+1}}{u_n} - l \right| < \frac{1}{2}(1 - l), \text{ when } n > r$$

and then

$$\left| \frac{u_{n+1}}{u_n} \right| < \frac{1}{2}(1 + l) < 1, \text{ when } n > r$$

Note. If $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| > 1$, u_n does not tend to zero, and, by Section 3.3, $\sum_{n=1}^{\infty} u_n$ does not converge.

Example 1. If $|c| < 1$, show that the series $\sum_{n=1}^{\infty} c^{n^2} e^{nz}$ converges absolutely for all values of z .

[For $\frac{u_{n+1}}{u_n} = c^{(n+1)^2 - n^2} e^z = c^{2n+1} e^z \rightarrow 0$, as $n \rightarrow \infty$, if $|c| < 1$.]

Example 2. Show that the series

$$z + \frac{(a-b)}{2!} z^2 + \frac{(a-b)(a-2b)}{3!} z^3 + \frac{(a-b)(a-2b)(a-3b)}{4!} z^4 + \dots$$

converges absolutely if $|z| < |b|^{-1}$

[For $\frac{u_{n+1}}{u_n} = \frac{a-nb}{n+1} z \rightarrow -bz$ as $n \rightarrow \infty$ so the condition for absolute convergence is $|bz| < 1$, i.e. $|z| < |b|^{-1}$.]

Example 3. Show that the series

$$\sum_{n=1}^{\infty} \frac{nz^{n-1}}{z^n - (1 + n^{-1})^n}$$

converges absolutely if $|z| < 1$.

The following solution is provided in lieu of the solution from Whittaker and Watson.

We consider the modulus of the general term:

$$\left| \frac{nz^{n-1}}{z^n - (1 + n^{-1})^n} \right| = \frac{n|z|^{n-1}}{|z^n - (1 + n^{-1})^n|}$$

We first obtain a lower bound for the denominator. By the triangle inequality,

$$|z^n - (1 + n^{-1})^n| \geq (1 + n^{-1})^n - |z|^n, \text{ since } (1 + n^{-1})^n > |z|^n$$

Now $|z| < 1$, so $|z|^n \leq 1$ and $(1 + n^{-1})^n \geq 2$ for all $n \geq 1$. Hence

$$|z^n - (1 + n^{-1})^n| \geq 2 - 1 = 1$$

Therefore

$$\left| \frac{nz^{n-1}}{z^n - (1 + n^{-1})^n} \right| \leq n|z|^{n-1}$$

Now consider the series $\sum_{n=1}^{\infty} n|z|^{n-1}$

Let $r = |z| < 1$. Then $\sum_{n=1}^{\infty} nr^{n-1}$ is a standard series and is convergent (it is, in fact, the derivative of the geometric series).

Hence the given series is dominated term-by-term by a convergent series, and therefore it converges absolutely.

3.3.7 A General Theorem for the Case When the Ratio Tends to Unity

It is obvious that if, for all values of n greater than some fixed value r , $|u_{n+1}|$ is greater than $|u_n|$, then the terms of the series do not tend to zero as $n \rightarrow \infty$, and the series is therefore divergent. On the other hand, if $\left| \frac{u_{n+1}}{u_n} \right|$ is less than some number which is itself less than unity and independent of n (when $n > r$), we have shown in Section 3.3.6 that the series is absolutely convergent. The critical case is that in which, as n increases, $\left| \frac{u_{n+1}}{u_n} \right|$ tends to the value unity. In this case, a further investigation is necessary.

We shall now show that²⁷ a series $u_1 + u_2 + u_3 + \dots$ in which $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = 1$ will be absolutely convergent if a positive number c exists such that

$$\limsup n \left(\left| \frac{u_{n+1}}{u_n} \right| - 1 \right) = -1 - c$$

For, compare the series $\sum |u_n|$ with the series $\sum v_n$, where $v_n = An^{-1-\frac{c}{2}}$, and A is a constant. The series $\sum v_n$ is convergent, since it is a p -series with exponent $1 + \frac{c}{2} > 1$ (see Section 3.3.3)

We have

$$\frac{v_{n+1}}{v_n} = \left(\frac{n}{n+1} \right)^{1+\frac{c}{2}} = \left(1 + \frac{1}{n} \right)^{-(1+\frac{c}{2})} = 1 - \frac{1+\frac{c}{2}}{n} + O\left(\frac{1}{n^2}\right)$$

by the generalized binomial theorem [2].

As $n \rightarrow \infty$, $n \left(\frac{v_{n+1}}{v_n} - 1 \right) \rightarrow -1 - \frac{c}{2}$.

²⁷ This is the second (D'Alembert's theorem given in Section 3.3.6 being the first) of a hierarchy of theorems due to De Morgan. See Chrystal, *Algebra*, Ch. xxvi for an historical account of these theorems.

[Alternate completion of proof] From the above discussion, we can find m such that, for all $n > m$,

$$\left| \frac{u_{n+1}}{u_n} \right| \leq \frac{v_{n+1}}{v_n}$$

Since $v_n > 0$ for all n , the ratio $\frac{v_{n+1}}{v_n}$ is positive, and no modulus is required on the right-hand side of the above inequality.

We now show that, by a suitable choice of the constant A , we may ensure that $|u_n| < v_n$ for all positive integers n .

First choose A so large that $|u_m| < v_m$. This is possible since $v_m = Am^{-1-\frac{c}{2}}$, and so increasing A increases v_m .

Now, for $n \geq m$, we prove by induction that $|u_n| < v_n$. Suppose this holds for some $n \geq m$. Then

$$|u_{n+1}| = |u_n| \cdot \left| \frac{u_{n+1}}{u_n} \right| \leq |u_n| \cdot \frac{v_{n+1}}{v_n} < v_n \cdot \frac{v_{n+1}}{v_n} = v_{n+1}$$

Thus $|u_{n+1}| < v_{n+1}$, and the result follows for all $n \geq m$.

For the finitely many terms $n < m$, we may increase A if necessary, so that $|u_n| < v_n$ also holds for these values of n .

Hence, for all n , $|u_n| < v_n$.

Since $\sum v_n$ is convergent, it follows by comparison that $\sum |u_n|$ is convergent, and therefore $\sum u_n$ is absolutely convergent.

Corollary. If

$$\left| \frac{u_{n+1}}{u_n} \right| = 1 + \frac{A_1}{n} + O\left(\frac{1}{n^2}\right),$$

where A_1 is independent of n , then the series is absolutely convergent if $A_1 < -1$.

Example. Investigate the convergence of

$$\sum_{n=1}^{\infty} n^r \exp\left(-k \sum_{m=1}^n \frac{1}{m}\right)$$

when $r > k$ and when $r < k$.

Solution: Let

$$u_n = n^r \exp\left(-k \sum_{m=1}^n \frac{1}{m}\right)$$

Then, after some calculations, we have

$$\frac{u_{n+1}}{u_n} = \left(1 + \frac{1}{n}\right)^r \exp\left(-\frac{k}{n+1}\right)$$

For large n ,

$$\left(1 + \frac{1}{n}\right)^r = 1 + \frac{r}{n} + O\left(\frac{1}{n^2}\right)$$

and

$$\exp\left(-\frac{k}{n+1}\right) = 1 - \frac{k}{n+1} + O\left(\frac{1}{n^2}\right) = 1 - \frac{k}{n} + O\left(\frac{1}{n^2}\right)$$

The last equality above is true since

$$\begin{aligned} \frac{1}{n+1} &= \frac{1}{n} \cdot \frac{1}{1+\frac{1}{n}}, & \frac{1}{1+\frac{1}{n}} &= 1 - \frac{1}{n} + O\left(\frac{1}{n^2}\right) \\ \Rightarrow \frac{1}{n+1} &= \frac{1}{n} \left(1 - \frac{1}{n} + O\left(\frac{1}{n^2}\right)\right) = \frac{1}{n} - \frac{1}{n^2} + O\left(\frac{1}{n^3}\right) = \frac{1}{n} + O\left(\frac{1}{n^2}\right) \end{aligned}$$

So, we have

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{u_{n+1}}{u_n} = \left(1 + \frac{r}{n} + O\left(\frac{1}{n^2}\right)\right) \left(1 - \frac{k}{n} + O\left(\frac{1}{n^2}\right)\right) = 1 + \frac{r-k}{n} + O\left(\frac{1}{n^2}\right)$$

Thus, using the notation from the above corollary, we have $A_1 = r - k$. So, the series is absolutely convergent if $A_1 < -1$ or when $k - r > 1$. On the other hand, if $r - k > -1$, the series is divergent.

In summary,

- if $r > k$, then $r - k > 0$, and the series is divergent
- if $r < k$, the result depends on how much smaller r is than k :
 - if $k - r > 1$, the series converges absolutely
 - if $k - r < 1$, the series diverges

The borderline case is $k - r = 1$ for which the present corollary does not decide the question.

3.3.8 Convergence of the hypergeometric series

The theorems which have been given may be illustrated by a discussion of the convergence of the hypergeometric series

$$1 + \frac{ab}{1 \cdot c}z + \frac{a(a+1)b(b+1)}{1 \cdot 2 \cdot c(c+1)}z^2 + \frac{a(a+1)(a+2)b(b+1)(b+2)}{1 \cdot 2 \cdot 3 \cdot c(c+1)(c+2)}z^3 + \dots$$

which is generally denoted by $F(a, b, c, z)$

If c is a negative integer, all the terms after the $(1-c)$ th have zero denominators, and if either a or b is a negative integer the series will terminate at the $(1-a)$ th or $(1-b)$ th term as the case may be. We shall suppose these cases set aside, so that a , b , and c are assumed not to be negative integers

In this series

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{(a+n-1)(b+n-1)}{n(c+n-1)}z\right| \rightarrow |z| \text{ as } n \rightarrow \infty$$

We see therefore, by Section 3.3.6, that the series is absolutely convergent when $|z| < 1$, and divergent when $|z| > 1$.

When $|z| = 1$, we have

$$\begin{aligned} \left| \frac{u_{n+1}}{u_n} \right| &= \left| 1 + \frac{a-1}{n} \right| \cdot \left| 1 + \frac{b-1}{n} \right| \cdot \left| 1 - \frac{c-1}{n} \right| + O\left(\frac{1}{n^2}\right) \\ &= \left| 1 + \frac{a+b-c-1}{n} + O\left(\frac{1}{n^2}\right) \right| \end{aligned}$$

Let a, b, c be complex numbers, and let them be given in terms of their real and imaginary parts by the equations

$$a = a' + ia'', b = b' + ib'', c = c' + ic''$$

Then we have

$$\begin{aligned} \left| \frac{u_{n+1}}{u_n} \right| &= \left| 1 + \frac{a' + b' - c' - 1}{n} + i \frac{a'' + b'' - c''}{n} + O\left(\frac{1}{n^2}\right) \right| \\ &= \left\{ \left(1 + \frac{a' + b' - c' - 1}{n} \right)^2 + \left(\frac{a'' + b'' - c''}{n} \right)^2 + O\left(\frac{1}{n^2}\right) \right\}^{\frac{1}{2}} \\ &= \left\{ 1 + \frac{2(a' + b' - c' - 1)}{n} + O\left(\frac{1}{n^2}\right) \right\}^{\frac{1}{2}} = 1 + \frac{a' + b' - c' - 1}{n} + O\left(\frac{1}{n^2}\right) \end{aligned}$$

The last equality follows from the expansion $(1 + \alpha)^{\frac{1}{2}} = 1 + \frac{\alpha}{2} + O(\alpha^2)$ for small α . In our case, $\alpha = \frac{2(a' + b' - c' - 1)}{n}$.

By Section 3.3.7, Corollary, a condition for absolute convergence is $a' + b' - c' < 0$.

Hence when $|z|=1$, a sufficient condition for the absolute convergence of the hypergeometric series is that the real part of $a + b - c$ shall be negative.

3.4 Effect of Changing the Order of the Terms in a Series (Rewritten)

In a finite sum, the order of the terms may be altered without affecting the result. In an infinite series, however, this is not generally the case. The following example illustrates this fact.

The alternating harmonic series

Consider the series $S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$

Group the terms in pairs: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \dots$

If we define $S_n = \sum_{k=1}^n \left(\frac{1}{2k-1} - \frac{1}{2k} \right)$, then S_n is the sum of the first $2n$ terms of the series.

If we let $\sigma_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$, then $S_n = \sigma_{2n} - \sigma_n$.

A rearrangement of the same terms

Next, consider the rearranged series $\Sigma = 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots$

This series is formed from the same terms as S , but in a different order. The pattern is:

- two positive terms (odd denominators),
- followed by one negative term (even denominator).

We group the terms as $\left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k}\right)$ and then define $\Sigma_n = \sum_{k=1}^n \left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k}\right)$.

Expressing Σ_n in terms of harmonic sums

The positive terms in Σ_n are $1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{4n-1}$.

The negative terms are $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}$.

Thus, $\Sigma_n = \left(1 + \frac{1}{3} + \cdots + \frac{1}{4n-1}\right) - \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}\right)$

The goal is to rewrite this in terms of $\sigma_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$.

Breaking σ_{4n} into its odd and even terms, we have

$$\sigma_{4n} = \left(1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{4n-1}\right) + \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{4n}\right)$$

So, the sum of the terms in Σ_n with odd denominators is given by

$$\sigma_{4n} - \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{4n}\right) = \sigma_{4n} - \frac{1}{2}\sigma_{2n}$$

Next, we note that

$$\frac{1}{2}\sigma_n = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}$$

Thus, we have

$$\Sigma_n = \left(\sigma_{4n} - \frac{1}{2}\sigma_{2n}\right) - \frac{1}{2}\sigma_n = \sigma_{4n} - \frac{1}{2}\sigma_{2n} - \frac{1}{2}\sigma_n$$

Rearranging,

$$\Sigma_n = (\sigma_{4n} - \sigma_{2n}) + \frac{1}{2}(\sigma_{2n} - \sigma_n)$$

But from earlier,

$$\sigma_{2n} - \sigma_n = S_n \text{ and } \sigma_{4n} - \sigma_{2n} = S_{2n}$$

Therefore,

$$\Sigma_n = S_{2n} + \frac{1}{2}S_n$$

Passing to the limit: Since S_n converges to S , any subsequence (such as S_{2n}) converges to the same limit S . Hence,

$$\Sigma = S + \frac{1}{2}S = \frac{3}{2}S$$

Conclusion: The series Σ is formed from the same terms as S , but in a different order, and yet $\Sigma \neq S$. Thus, changing the order of the terms of an infinite series may alter its sum.

This phenomenon occurs because the series S is convergent but not absolutely convergent.

Exercise. If in the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

the order of the terms be altered, so that the ratio of the number of positive terms to the number of negative terms in the first n terms is ultimately a^2 , show that the sum of the series will become $\ln(2a)$. (Manning)

One thing should be made explicit here (but not stated in the original text), i.e.,

We assume that the positive terms retain their original order

$$1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$$

and the negative terms retain their original order

$$-\frac{1}{2}, -\frac{1}{4}, -\frac{1}{6}, \dots$$

but that the two lists are interlaced in a different way.

Without that assumption, the original statement of the problem is ambiguous.

Solution

Let T_n denote the sum of the first n terms of the rearranged series. Suppose that among these first n terms there are p_n positive terms and q_n negative terms. Then

$$p_n + q_n = n, \quad \lim_{n \rightarrow \infty} \frac{p_n}{q_n} = a^2$$

Under the interpretation stated above, the first p_n positive terms are $1, \frac{1}{3}, \frac{1}{5}, \dots, \frac{1}{2p_n-1}$.

and the first q_n negative terms are $-\frac{1}{2}, -\frac{1}{4}, -\frac{1}{6}, \dots, -\frac{1}{2q_n}$.

Hence,

$$T_n = \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2p_n - 1}\right) - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2q_n}\right).$$

Next, let

$$\sigma_m = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m}$$

Then

$$1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2p_n - 1} = \sigma_{2p_n} - \frac{1}{2}\sigma_{p_n}$$

because subtracting the even terms from σ_{2p_n} leaves the odd terms. Also,

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2q_n} = \frac{1}{2}\sigma_{q_n}$$

Therefore

$$T_n = \sigma_{2p_n} - \frac{1}{2}\sigma_{p_n} - \frac{1}{2}\sigma_{q_n}$$

Using the asymptotic formula [3],

$$\sigma_m = \ln m + \gamma + o(1) \text{ where } \gamma \approx 0.577 \text{ is the Euler-Mascheroni constant.}$$

we obtain

$$T_n = \ln(2p_n) - \frac{1}{2}\ln p_n - \frac{1}{2}\ln q_n + o(1)$$

[**Note:** We say that a function $f(n)$ is $o(1)$ as $n \rightarrow \infty$ if $\lim_{n \rightarrow \infty} f(n) = 0$.]

Thus

$$T_n = \ln 2 + \frac{1}{2}\ln\left(\frac{p_n}{q_n}\right) + o(1)$$

Since $\frac{p_n}{q_n} \rightarrow a^2$, it follows that

$$T_n \rightarrow \ln 2 + \frac{1}{2}\ln a^2 = \ln 2 + \ln a = \ln(2a)$$

Hence the sum of the rearranged series is $\ln(2a)$.

3.4.1 The fundamental property of absolutely convergent series

We shall show that the sum of an absolutely convergent series is not affected by changing the order in which the terms occur.

Let $S = u_1 + u_2 + u_3 + \dots$ be an absolutely convergent series, and let S' be a series formed by the same terms in a different order.

Let ε be an arbitrary positive number, and let n be chosen so that

$$|u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \frac{1}{2}\varepsilon$$

for all values of p .

Since S' is a rearrangement of S , each of the terms $u_1, \dots, u_n$ appears somewhere in S' ; hence there exists m such that all of these terms occur among the first m terms of S' .

Suppose that in order to obtain the first n terms of S we have to take m terms of S' ; then if $k > m$, $S'_k = S_n + \text{terms of } S \text{ with suffixes greater than } n$.

That is, once all of $u_1, \dots, u_n$ have appeared in S' , any further terms in the partial sum S'_k must come from terms u_j with $j > n$.

So, $S'_k - S = S_n - S + \text{terms of } S \text{ with suffixes greater than } n$.

Now the modulus of the sum of any number of terms of S with suffixes greater than n does not exceed the sum of their moduli, and therefore is less than $\frac{1}{2}\varepsilon$.

Therefore

$$|S'_k - S| < |S_n - S| + \frac{1}{2}\varepsilon$$

But

$$|S_n - S| \leq \lim_{p \rightarrow \infty} \{|u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}|\} \leq \frac{1}{2}\varepsilon.$$

Therefore, given ε , we can find m such that

$$|S'_k - S| < \varepsilon$$

when $k > m$; therefore $S'_k \rightarrow S$, which is the required result.

If a series of real terms converges, but not absolutely, and if S_p be the sum of the first p positive terms, and σ_n the sum of the first n negative terms, then $S_p \rightarrow \infty$, $\sigma_n \rightarrow -\infty$, and $\lim (S_p + \sigma_n)$ does not exist unless we are given some relation between p and n . It has, in fact, been shewn by Riemann that it is possible, by choosing a suitable relation, to make $\lim (S_p + \sigma_n)$ equal to any given real number. *This phenomenon is a consequence of the Riemann rearrangement theorem: a conditionally convergent series may be rearranged to converge to any prescribed real value, or even to diverge.*[4]

Table 1. Tests for Convergence and Absolute Convergence

Test	Applies to	Condition	Conclusion	Absolute vs conditional
Term Test (Divergence)	Series $\sum a_n$	a_n does not tend to 0	Diverges	Necessary condition only
Comparison Test	Series $\sum a_n$	$0 \leq a_n \leq b_n$, and $\sum b_n$ converges	$\sum a_n$ converges	Absolute (nonnegative terms)
Limit Comparison	Series $\sum a_n$, $\sum b_n$	$\lim (a_n/b_n) = c > 0$	Same behavior	Absolute (with moduli)
Ratio Test	Series $\sum a_n$	$\lim a_{n+1}/a_n = L$	$L < 1$: converges; $L > 1$: diverges	Absolute if $L < 1$
Root Test	Series $\sum a_n$	$\limsup a_n ^{1/n} = L$	$L < 1$: converges; $L > 1$: diverges	Absolute if $L < 1$
Integral Test	Series $\sum a_n$	Compare with $\int f(x) dx$	Same behavior	Absolute (positive terms)
Alternating Series	Series $\sum (-1)_n a_n$	$a_n \downarrow 0$	Converges	Usually conditional
Absolute Convergence	Series $\sum a_n$	$\sum a_n $ converges	$\sum a_n$ converges	Absolute $\Rightarrow$ convergence
Dirichlet's Test	$\sum a_n b_n$	Partial sums of a_n bounded, $b_n \rightarrow 0$	Converges	Conditional
Abel's Test	$\sum a_n b_n$	$\sum a_n$ converges, b_n monotone bounded	Converges	Conditional

3.5 Double series²⁸

Let $u_{m,n}$ be a number determinate for all positive integral values of m and n ; consider the array

$$\begin{array}{cccc}
 u_{1,1} & u_{1,2} & u_{1,3} & \cdots \\
 u_{2,1} & u_{2,2} & u_{2,3} & \cdots \\
 u_{3,1} & u_{3,2} & u_{3,3} & \cdots \\
 \vdots & \vdots & \vdots &
 \end{array}$$

Let the sum of the terms inside the rectangle formed by the first m rows of the first n columns of this array of terms be denoted by $S_{m,n}$.

²⁸ A complete theory of double series, on which this account is based, is given by Pringsheim, *Münchener Sitzungsberichte*, XXVI (1897), pp. 101–152. See further memoirs by that writer, *Math. Ann.* LIII (1900), pp. 289–321 and by London, *ibid.* pp. 322–370, and also Bromwich, *Infinite Series*, which, in addition to an account of Pringsheim's theory, contains many developments of the subject. Other important theorems are given by Bromwich, *Proc. London Math. Soc.* (2), I (1904), pp. 176–201.

If a number S exists such that, given any arbitrary positive number ε , it is possible to find integers m and n such that

$$|S_{\mu,\nu} - S| < \varepsilon$$

whenever both $\mu > m$ and $\nu > n$, we say²⁹ that the double series of which the general element is $u_{m,n}$ converges to the sum S , and we write

$$\lim_{\mu \rightarrow \infty, \nu \rightarrow \infty} S_{\mu,\nu} = S$$

If the double series of which the general element is $|u_{m,n}|$ is convergent, we say that the given double series is **absolutely convergent**.

Since $u_{m,n} = (S_{m,n} - S_{m,n-1}) - (S_{m-1,n} - S_{m-1,n-1})$, it is easily seen that, if the double series is convergent, then

$$\lim_{\mu \rightarrow \infty, \nu \rightarrow \infty} u_{\mu,\nu} = 0$$

Stolz' necessary and sufficient³⁰ condition for convergence

A condition for convergence which is obviously necessary (see Section 3.2.2) is that, given ε , we can find m and n such that

$$|S_{\mu+\rho,\nu+\sigma} - S_{\mu,\nu}| < \varepsilon$$

whenever $\mu > m$ and $\nu > n$, and ρ, σ may take any of the values $0, 1, 2, \dots$

The condition is also sufficient; for, suppose it satisfied, then, when $\mu > m + n$,

$$|S_{\mu+\rho,\mu+\rho} - S_{\mu,\mu}| < \varepsilon$$

Therefore, by Section 3.2.2, $S_{\mu,\mu}$ has a limit S , and then making ρ and σ tend to infinity in such a way that $\mu + \rho = \nu + \sigma$, we see that

$$|S - S_{\mu,\nu}| \leq \varepsilon$$

whenever $\mu > m$ and $\nu > n$; that is to say, the double series converges.

The idea of the above argument is to restrict attention to the diagonal sequence $S_{\mu,\mu}$, which reduces the problem to a single sequence; the general case then follows from the same inequality.

Corollary. An absolutely convergent double series is convergent.

²⁹ This definition is practically due to Cauchy, *Analyse Algebrique*, p. 540

³⁰ This condition, stated by Stolz, *Math. Ann.* XXIV (1884), pp. 157–171, appears to have been first proved by Pringsheim.

For if the double series converges absolutely, and if $t_{m,n}$ be the sum of m rows of n columns of the series of moduli, then, given ε , we can find μ such that when $p > m > \mu$ and $q > n > \mu$, $t_{p,q} - t_{m,n} < \varepsilon$. But $|S_{p,q} - S_{m,n}| \leq t_{p,q} - t_{m,n}$ (This follows from the triangle inequality, since the modulus of a sum is not greater than the sum of the moduli of its terms.) So, $|S_{p,q} - S_{m,n}| < \varepsilon$ when $p > m > \mu$, $q > n > \mu$, and this condition is sufficient for convergence.

Comparison theorem for double series

Let $a_{m,n}$ and $b_{m,n}$ be double sequences, and suppose that $|a_{m,n}| \leq b_{m,n}$ for all relevant values of m and n , where $b_{m,n} \geq 0$.

If the double series $\sum_{m,n} b_{m,n}$ converges, then the double series $\sum_{m,n} a_{m,n}$ converges absolutely, and hence converges.

Indeed, since $|a_{m,n}| \leq b_{m,n}$, the rectangular partial sums of the series of moduli $\sum_{m,n} a_{m,n}$ are bounded above by the corresponding rectangular partial sums of $\sum_{m,n} b_{m,n}$.

Thus, the series of moduli converges whenever the comparison series converges. Therefore, $\sum_{m,n} a_{m,n}$ is absolutely convergent, and by the preceding corollary it is convergent.

This is the double-series version of the ordinary comparison test for single series. It is especially useful when the terms of a complicated double series can be bounded by the terms of a simpler positive double series whose convergence is already known.

3.5.1 Methods of Summing Double Series

Let us suppose that $\sum_{v=1}^{\infty} u_{\mu,v}$ converges to the sum S_{μ} . Then $\sum_{\mu=1}^{\infty} S_{\mu}$ is called the sum by rows of the double series; that is to say, the sum by rows is

$$\sum_{\mu=1}^{\infty} \left(\sum_{v=1}^{\infty} u_{\mu,v} \right)$$

In other words, we first fix μ and sum along each row (letting $v \rightarrow \infty$), thereby obtaining a sequence (S_{μ}) , and then sum this sequence.

Similarly, the sum by columns is defined as

$$\sum_{v=1}^{\infty} \left(\sum_{\mu=1}^{\infty} u_{\mu,v} \right)$$

Here we instead fix v , sum down each column (letting $\mu \rightarrow \infty$), and then sum the resulting sequence.

That these two sums are not necessarily the same is shown by the example

$$S_{\mu,v} = \frac{\mu - v}{\mu + v},$$

in which the sum by rows is -1 , the sum by columns is $+1$, and S does not exist.

The quantity $S_{\mu,v} = \frac{\mu-v}{\mu+v}$ is best viewed as the rectangular partial sum, not the general term of the double series. So, to determine the actual row sums, we use

$$(\text{sum of the } \mu\text{-th row up to column } v) = S_{\mu,v} - S_{\mu-1,v}$$

Then we let $v \rightarrow \infty$.

For the first row, $S_{1,v} = \frac{1-v}{1+v}$. Hence the sum of the first row is

$$\lim_{v \rightarrow \infty} S_{1,v} = \lim_{v \rightarrow \infty} \frac{1-v}{1+v} = -1.$$

For the second row, we subtract the first rectangular sum from the second:

$$\begin{aligned} S_{2,v} - S_{1,v} &= \frac{2-v}{2+v} - \frac{1-v}{1+v} \\ &= \frac{(2-v)(1+v) - (1-v)(2+v)}{(2+v)(1+v)} = \frac{2v}{(v+2)(v+1)} \rightarrow 0 \text{ as } v \rightarrow \infty \end{aligned}$$

In general, for $\mu \geq 2$,

$$S_{\mu,v} - S_{\mu-1,v} = \frac{\mu-v}{\mu+v} - \frac{\mu-1-v}{\mu-1+v}$$

A similar calculation gives

$$S_{\mu,v} - S_{\mu-1,v} = \frac{2v}{(\mu+v)(\mu+v-1)}$$

and therefore

$$\lim_{v \rightarrow \infty} (S_{\mu,v} - S_{\mu-1,v}) = 0$$

So, the sum of the first row is -1 and the sum of each other row is 0. Thus, the sum by rows is -1.

The column calculations are analogous, i.e., the sum of the first column is 1 and the sum of each other column is 0. Thus, the sum by columns is 1.

Pringsheim's Theorem

The name "Pringsheim's theorem" is used here in the classical sense, referring to results on the convergence of double series; it is not to be confused with other theorems bearing Pringsheim's name in the theory of power series.

If S exists and the sums by rows and columns exist, then each of these sums is equal to S .

For since S exists, then (by definition) we can find m such that

$$|S_{\mu,v} - S| < \varepsilon \text{ if } \mu > m, v > m$$

For fixed $\mu > m$, all sufficiently late terms $S_{\mu, \nu}$ lie within ε of S ; since $S_{\mu, \nu}$ has a limit as $\nu \rightarrow \infty$, that limit must also lie within distance ε of S .

And therefore, since $\lim_{\nu \rightarrow \infty} S_{\mu, \nu}$ exists,

$$\left| \left(\lim_{\nu \rightarrow \infty} S_{\mu, \nu} \right) - S \right| \leq \varepsilon$$

that is to say,

$$|S_{\mu} - S| \leq \varepsilon \text{ when } \mu > m$$

and so (Section 3.2.2) the sum by rows converges to S .

In like manner the sum by columns converges to S .

The hypothesis that S exists is essential; the preceding example shows that the sums by rows and columns may exist and yet be unequal if the double limit S does not exist.

3.5.2 Absolutely convergent double series

The following is an alternative explanation from what is in Whittaker and Watson.

We prove the analogue of Section 3.4.1 for double series: if a double series is absolutely convergent, then every rearrangement of its terms, regarded as an ordinary (single-index) series formed from the terms $u_{\mu, \nu}$ themselves (i.e., not the series formed from the moduli $|u_{\mu, \nu}|$), in which every term occurs exactly once, converges to the same limit.

Since the series is absolutely convergent, the double series whose general term is $u_{\mu, \nu}$ converges, and therefore has a sum S . Let $\sigma_{\mu, \nu}$ denote the sum of the rectangle of μ rows and ν columns of the series whose general term is $|u_{\mu, \nu}|$, and let the sum of this series of absolute values be σ . Then, given ε , we can find integers m and n such that

$$\sigma - \sigma_{\mu, \nu} < \varepsilon$$

whenever both $\mu > m$ and $\nu > n$.

Let t_N be the sum of the first N terms in a rearrangement of the terms $u_{\mu, \nu}$.

Given $\varepsilon > 0$, we can find integers m and n such that the sum of the absolute values of all terms lying outside the rectangle $p \leq m, q \leq n$ is less than $\frac{1}{2}\varepsilon$; that is,

$$\sigma - \sigma_{m, n} < \frac{1}{2}\varepsilon$$

Next, let $M \geq \max(m, n)$, and choose N sufficiently large so that the first N terms of the rearranged series include all terms $u_{p, q}$ with $p \leq M + 1$ and $q \leq M + 1$, i.e., the first N terms of the rearranged series include all the terms in the square $S_{M+1, M+1}$. So, the difference $t_N - S_{M+1, M+1}$ consists of a finite collection of terms lying outside the square but contained among the first N terms of the rearranged

series. Therefore, its modulus is bounded by the sum of the moduli of these terms, and hence by the sum of the moduli of all terms outside the square, which is less than $\frac{1}{2}\varepsilon$, i.e.,

$$|t_N - S_{M+1,M+1}| \leq \sigma - \sigma_{M+1,M+1} \leq \sigma - \sigma_{m,n} < \frac{1}{2}\varepsilon$$

The difference $S - S_{M+1,M+1}$ represents the contribution of all terms lying outside the rectangle $p \leq M+1$ and $q \leq M+1$; that is, all terms for which $p > M+1$ or $q > M+1$. Its modulus is therefore bounded by the sum of the absolute values of these terms, namely $\sigma - \sigma_{M+1,M+1} \leq \sigma - \sigma_{m,n} < \frac{1}{2}\varepsilon$.

It follows from the triangle inequality that

$$|S - t_N| \leq |S - S_{M+1,M+1}| + |S_{M+1,M+1} - t_N| < \varepsilon.$$

Since ε is arbitrary, it follows that $t_N \rightarrow S$.

The preceding results illustrate three complementary aspects of convergence for double series. Stolz' condition provides a Cauchy-type criterion ensuring convergence of the rectangular partial sums $S_{m,n}$. Pringsheim's theorem shows that, when the double limit exists, any sums formed by first summing along rows or columns must agree with it. Finally, absolute convergence guarantees not only the existence of the sum S , but also that the terms may be rearranged arbitrarily into a single-index series without altering this sum. Thus, absolute convergence restores for double series the same invariance under rearrangement that holds for ordinary series.

Example 1

Prove that in an absolutely convergent double series, $\sum_{v=1}^{\infty} u_{m,n}$ exists, and hence that the sums by rows and columns respectively converge to S .

[Let the sum of the first μ rows and ν columns of the series of moduli be $t_{\mu,\nu}$, and let t be the sum of the series of moduli.

$$|b_1| + \cdots + |b_\mu| \leq \sum_{r=1}^{\mu} \sum_{v=1}^{\infty} |u_{r,v}|.$$

Then $\sum_{v=1}^{\infty} |u_{\mu,v}| < t$, and so $\sum_{v=1}^{\infty} u_{\mu,v}$ converges; let its sum be b_μ .

Thus, for each fixed μ , the sum of the μ -th row exists, and the sequence (b_μ) represents the sums of the rows of the double series.

Thus,

$$|b_1| + |b_2| + \cdots + |b_\mu| \leq \sum_{r=1}^{\mu} \sum_{v=1}^{\infty} |u_{r,v}| \leq \lim_{\nu \rightarrow \infty} t_{\mu,\nu} \leq t$$

and so, $\sum_{\mu=1}^{\infty} b_{\mu}$ converges absolutely.

For each v , the quantity $t_{\mu,v}$ is the sum of the absolute values of the first μ rows and v columns, and hence bounds the sum of the absolute values of the first μ row sums. Passing to the limit as $v \rightarrow \infty$ gives the inequality above.

Therefore, the sum by rows of the double series exists, and similarly the sum by columns exists, and the required result then follows from Pringsheim's theorem.]

Example 2

Show from first principles that if the terms of an absolutely convergent double series be arranged in the order

$$u_{1,1} + (u_{2,1} + u_{1,2}) + (u_{3,1} + u_{2,2} + u_{1,3}) + (u_{4,1} + u_{3,2} + u_{2,3} + u_{1,4}) + \dots$$

this series converges to S .

Solution

Let S denote the sum of the absolutely convergent double series, and let σ denote the sum of the corresponding series of moduli.

Since the double series is absolutely convergent, given $\varepsilon > 0$, we can find an integer m such that the sum of the absolute values of all terms lying outside the square $p \leq m, q \leq m$ is less than $\frac{1}{2}\varepsilon$; that is,

$$\sigma - \sigma_{m,m} < \frac{1}{2}\varepsilon$$

Now consider the series formed by summing along successive diagonals:

$$u_{1,1}, u_{2,1} + u_{1,2}, u_{3,1} + u_{2,2} + u_{1,3}, \dots$$

After the first $2m - 1$ diagonals have been taken, every term $u_{p,q}$ with $p \leq m$ and $q \leq m$ has appeared. Indeed, if $p \leq m$ and $q \leq m$, then

$$p + q - 1 \leq 2m - 1,$$

so $u_{p,q}$ occurs in one of the first $2m - 1$ diagonals.

Let T_{2m-1} denote the sum of the terms in these first $2m - 1$ diagonals. Then T_{2m-1} contains all terms in the square $p \leq m, q \leq m$, together with some additional terms lying outside that square. Hence $T_{2m-1} - S_{m,m}$ is a finite sum of terms lying outside the square, and therefore

$$|T_{2m-1} - S_{m,m}| \leq \sigma - \sigma_{m,m} < \frac{1}{2}\varepsilon$$

Also, $S - S_{m,m}$ represents the contribution of all terms lying outside the square $p \leq m, q \leq m$, and so

$$|S - S_{m,m}| \leq \sigma - \sigma_{m,m} < \frac{1}{2}\varepsilon$$

Therefore, by the triangle inequality,

$$|S - T_{2m-1}| \leq |S - S_{m,m}| + |S_{m,m} - T_{2m-1}| < \varepsilon$$

Since ε is arbitrary, it follows that the series formed by summing along successive diagonals converges to S .

3.5.3 Cauchy's Theorem³¹ on the Multiplication of Absolutely Convergent Series

We shall now show that if two series

$$S = u_1 + u_2 + u_3 + \cdots$$

and

$$T = v_1 + v_2 + v_3 + \cdots$$

are absolutely convergent, then the series

$$P = u_1 v_1 + u_2 v_1 + u_1 v_2 + u_2 v_2 + \cdots$$

formed by the products of their terms, written in any order, is absolutely convergent, and has for sum ST .

Let $S_n = u_1 + u_2 + \cdots + u_n$ and $T_n = v_1 + v_2 + \cdots + v_n$.

Then $ST = \lim S_n \cdot \lim T_n = \lim (S_n T_n)$, by Example 2 of Section 3.2.

Now

$$S_n T_n = \begin{array}{c} u_1 v_1 + u_2 v_1 + \cdots + u_n v_1 \\ + u_1 v_2 + u_2 v_2 + \cdots + u_n v_2 \\ \vdots \\ + u_1 v_n + u_2 v_n + \cdots + u_n v_n \end{array}$$

But this double series is absolutely convergent, for if these terms are replaced by their moduli, the result is $\sigma_n \tau_n$, where

$$\sigma_n = |u_1| + |u_2| + \cdots + |u_n|, \quad \tau_n = |v_1| + |v_2| + \cdots + |v_n|$$

and $\sigma_n \tau_n$ is known to have a limit. Therefore, by Section 3.5.2, if the elements of the double series, of which the general term is $u_m v_n$, be taken in any order, their sum converges to ST .

Example

Show that the series obtained by multiplying the two series

$$1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots, \quad 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots$$

and rearranging according to powers of z , converges so long as the representative point of z lies in the ring-shaped region (*annulus*) bounded by the circles $|z| = 1$ and $|z| = 2$.

Solution

If the double series of products is absolutely convergent, then rearrangement is valid.

So, we check absolute convergence of:

³¹ *Analyse Algébrique*, Note VII

$$\sum_{m,n} \left| \frac{z^m}{2^m} \cdot \frac{1}{z^n} \right| = \sum_{m,n} \frac{|z|^m}{2^m} \cdot \frac{1}{|z|^n} = \left(\sum_{m=0}^{\infty} \left(\frac{|z|}{2} \right)^m \right) \left(\sum_{n=0}^{\infty} \left(\frac{1}{|z|} \right)^n \right)$$

The first series in the product converges if $\frac{|z|}{2} < 1 \Rightarrow |z| < 2$

The second series in the product converges if $\frac{1}{|z|} < 1 \Rightarrow |z| > 1$

Both series converge if and only if $1 < |z| < 2$. So, the double series is absolutely convergent in this region, and therefore, the rearranged series (grouped by powers of z) converges to the same sum.

3.6 Power-Series³²

A series of the type

$$a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots$$

in which the coefficients $a_0, a_1, a_2, a_3, \dots$ are independent of z , is called a series proceeding according to ascending powers of z , or briefly a **power series**.

We shall now show that if a power-series converges for any value z_0 of z , it will be absolutely convergent for all values of z whose representative points are within a circle which passes through z_0 and has its centre at the origin.

For, if z be such a point, we have $|z| \leq |z_0|$. Now, since $\sum_{n=0}^{\infty} a_n z_0^n$ converges, $a_n z_0^n$ must tend to zero as $n \rightarrow \infty$, and so we can find M (independent of n) such that $|a_n z_0^n| < M$. Thus $|a_n z^n| < M \left| \frac{z}{z_0} \right|^n$.

Therefore, every term in the series $\sum_{n=0}^{\infty} a_n z^n$ is less than the corresponding term in the convergent geometric series $\sum_{n=0}^{\infty} M \left| \frac{z}{z_0} \right|^n$, the series is therefore convergent, and so the power-series is absolutely convergent, as the series of moduli of its terms is a convergent series; the result stated is therefore established.

Let $\lim |a_n|^{-1/n} = r$; then, from Section 3.3.5, $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely when $|z| < r$; if $|z| > r$, $a_n z^n$ does not tend to zero and so, $\sum_{n=0}^{\infty} a_n z^n$ diverges (Section 3.3).

From the preceding result, if the series converges for some value z_0 , then it converges absolutely for all z with $|z| \leq |z_0|$. Thus, the set of values of z for which the series converges is a region consisting of all points within some circle centered at the origin (possibly including or excluding the boundary).

*The circle $|z| = r$, which bounds this region, is called the **circle of convergence** of the series, and r is called the **radius of convergence**.*

In practice there is usually a simpler way of finding r , derived from d'Alembert's test (Section 3.3.6), $r = \lim \left(\frac{a_n}{a_{n+1}} \right)$, if this limit exists.

³² The results of this section are due to Cauchy, *Analyse Algébrique*, Ch. IX

A power-series may converge for all values of the variable, as happens, for instance, in the case of the series³³

$$z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

which represents the function $\sin z$; in this case the series converges over the whole z -plane.

On the other hand, the radius of convergence of a power-series may be zero, thus in the case of the series

$$1 + 1!z + 2!z^2 + 3!z^3 + 4!z^4 + \dots$$

We have $\left| \frac{u_{n+1}}{u_n} \right| = n|z|$, which, for all values of n greater than some fixed value N (depending on z), is greater than unity when z has any value different from zero. The series converges therefore only at the point $z = 0$, and the radius of its circle of convergence vanishes.

A power-series may or may not converge for points which are actually on the periphery of the circle, thus the series

$$1 + \frac{z}{1^s} + \frac{z^2}{2^s} + \frac{z^3}{3^s} + \frac{z^4}{4^s} + \dots,$$

whose radius of convergence is unity, converges or diverges at the point $z = 1$, according as s is greater or not greater than unity, as was seen in Section 3.3.3.

Corollary. If (a_n) be a sequence of positive terms such that $\lim_{n \rightarrow \infty} \left(\frac{a_{n+1}}{a_n} \right)$ exists, this limit is equal to $\lim a_n^{1/n}$.

If $\frac{a_{n+1}}{a_n} \rightarrow L$, then

$$\log a_n = \sum_{k=1}^{n-1} \log \left(\frac{a_{k+1}}{a_k} \right) + \log a_1$$

and dividing by n shows that $\frac{1}{n} \log a_n \rightarrow \log L$. Hence $a_n^{1/n} \rightarrow L$.

3.6.1 Convergence of series derived from a power-series

Let $a_0 + a_1z + a_2z^2 + a_3z^3 + a_4z^4 + \dots$ be a power-series, and consider the series

$$a_1 + 2a_2z + 3a_3z^2 + 4a_4z^3 + \dots$$

which is obtained by differentiating the power-series term by term. We shall now show that the derived series has the same circle of convergence as the original series.

For let z be a point within the circle of convergence of the power-series, and choose a positive number r_1 , intermediate in value between $|z|$ and r , the radius of convergence. Then, since the series

³³ The series for e^z , $\sin z$, $\cos z$ and the fundamental properties of these functions and of $\log z$ will be assumed throughout.

$\sum_{n=0}^{\infty} a_n r_1^n$ converges absolutely, its terms must tend to zero as $n \rightarrow \infty$, and it must therefore be possible to find a positive number M , independent of n , such that $|a_n| < M r_1^{-n}$ for all values of n .

Then the terms of the series $\sum_{n=1}^{\infty} n |a_n| |z|^{n-1}$ are less than the corresponding terms of the series

$$\frac{M}{r_1} \sum_{n=1}^{\infty} n \left(\frac{|z|}{r_1} \right)^{n-1}$$

But this series converges, by Section 3.3.6, since $|z| < r_1$. Therefore, by Section 3.3.4, the

series $\sum_{n=1}^{\infty} n |a_n| |z|^{n-1}$ converges, that is, the series $\sum_{n=1}^{\infty} n a_n z^{n-1}$ converges absolutely for all points z situated within the circle of convergence of the original series $\sum_{n=0}^{\infty} a_n z^n$. When $|z| > r$, $a_n z^n$ does not tend to zero, and *a fortiori* (even more so) $n a_n z^{n-1}$ does not tend to zero, and so the two series have the same circle of convergence.

Corollary. The series $\sum_{n=0}^{\infty} \frac{a_n z^{n+1}}{n+1}$, obtained by integrating the original power-series term by term, has the same circle of convergence as $\sum_{n=0}^{\infty} a_n z^n$.

Table 2. Convergence Tests for Power Series

Test	Applies To	Condition	Conclusion	Absolute vs Conditional
Radius (Ratio)	Power series $\sum a_n z^n$	$\lim a_{(n+1)}/a_n = L$	$R = 1/L$	Absolute for $ z < R$
Radius (Root)	Power series $\sum a_n z^n$	$\lim \sup a_n ^{(1/n)} = L$	$R = 1/L$	Absolute for $ z < R$
Inside Circle	Power series $\sum a_n z^n$	$ z < R$	Converges	Absolutely convergent
Outside Circle	Power series $\sum a_n z^n$	$ z > R$	Diverges	—
Boundary	Power series $\sum a_n z^n$	$ z = R$	Inconclusive	Must test separately
Differentiation	Power series $\sum a_n z^n$	$ z < R$	Same radius of convergence	Absolute inside circle
Integration	Power series $\sum a_n z^n$	$ z < R$	Same radius of convergence	Absolute inside circle

3.7 Infinite Products

We next consider a class of limits, known as *infinite products*.

Let $1 + a_1, 1 + a_2, 1 + a_3, \dots$ be a sequence such that none of its members vanish. If, as $n \rightarrow \infty$, the product

$$(1 + a_1)(1 + a_2)(1 + a_3) \cdots (1 + a_n)$$

(which we denote by Π_n) tends to a definite limit other than zero, this limit is called the value of the infinite product

$$\Pi = (1 + a_1)(1 + a_2)(1 + a_3) \cdots$$

and the product is said to be convergent. It is almost obvious that a necessary condition for convergence is that $\lim_{n \rightarrow \infty} a_n = 0$, since $\lim \Pi_{n-1} = \lim \Pi_n \neq 0$.

Since $\Pi_n = \Pi_{n-1}(1 + a_n)$, we have $1 + a_n = \frac{\Pi_n}{\Pi_{n-1}}$. If the product converges to a nonzero limit, both Π_n and Π_{n-1} tend to the same limit, and therefore $1 + a_n \rightarrow 1$, so that $a_n \rightarrow 0$.

The limit of the product is written $\prod_{n=1}^{\infty} (1 + a_n)$.

Now

$$\prod_{n=1}^m (1 + a_n) = \exp \left(\sum_{n=1}^m \log (1 + a_n) \right)$$

and³⁴

$$\exp \left(\lim_{m \rightarrow \infty} u_m \right) = \lim_{m \rightarrow \infty} \exp u_m$$

if the former limit exists; hence a sufficient condition that the product should converge is that

$$\sum_{n=1}^{\infty} \log (1 + a_n)$$

should converge when the logarithms have their principal values. If this series of logarithms converges absolutely, the convergence of the product is said to be absolute.

The condition for absolute convergence is given by the following theorem:

In order that the infinite product

$$(1 + a_1)(1 + a_2)(1 + a_3) \cdots$$

may be absolutely convergent, it is necessary and sufficient that the series

$$a_1 + a_2 + a_3 + \cdots$$

should be absolutely convergent.

³⁴ The convergence of the product in which $a_n = (-1)^n/n$ was investigated by Wallis as early as 1655.

By definition, the product $\prod_{n=1}^{\infty} (1 + a_n)$ is absolutely convergent if and only if the series $\sum_{n=1}^{\infty} \log (1 + a_n)$ is absolutely convergent. To relate this condition to the series $\sum a_n$, we observe that, for small values of a_n ,

$$\log (1 + a_n) = a_n - \frac{a_n^2}{2} + \frac{a_n^3}{3} - \dots$$

so that $\log (1 + a_n) = a_n + O(a_n^2)$ as $a_n \rightarrow 0$. Hence, for all sufficiently large n , the quantities $|\log (1 + a_n)|$ and $|a_n|$ are comparable; more precisely, there exist positive constants C_1 and C_2 such that

$$C_1 |a_n| \leq |\log (1 + a_n)| \leq C_2 |a_n|$$

for all large n . It follows that the series $\sum |\log (1 + a_n)|$ converges if and only if the series $\sum |a_n|$ converges. Thus, the infinite product is absolutely convergent if and only if the series $\sum a_n$ is absolutely convergent.

This establishes the result.³⁵

If, in a product, a finite number of factors vanish, and if, when these are suppressed, the resulting product converges, the original product is said to converge to zero. But such a product as

$$\prod_{n=2}^{\infty} (1 - n^{-1})$$

is said to diverge to zero.

Corollary. Since, if $S_n \rightarrow l$, $\exp (S_n) \rightarrow \exp l$, it follows from Section 3.4.1 that the factors of an absolutely convergent product can be deranged without affecting the value of the product.

Example 1. Show that if $\prod_{n=1}^{\infty} (1 + a_n)$ converges, so does $\sum_{n=1}^{\infty} \log (1 + a_n)$, if the logarithms have their principal values.

Let $P_n = \prod_{k=1}^n (1 + a_k)$, and suppose that $P_n \rightarrow P \neq 0$. Since none of the factors $1 + a_n$ vanish, the principal values of $\log (1 + a_n)$ are defined, and we may write $S_n = \sum_{k=1}^n \log (1 + a_k)$.
Then

$$\exp(S_n) = \prod_{k=1}^n (1 + a_k) = P_n$$

³⁵ A discussion of the convergence of infinite products, in which the results are obtained without making use of the logarithmic function, is given by Pringsheim, *Math. Ann.* xxxiii (1889), pp. 119–154, and also by Bromwich, *Infinite Series*, Ch. vi.

Since $P_n \rightarrow P \neq 0$, the values of P_n are, for sufficiently large n , confined to a region not containing the origin, in which the principal logarithm is continuous. Hence $S_n = \log P_n \rightarrow \log P$, and therefore, the series $\sum_{n=1}^{\infty} \log(1 + a_n)$ converges.

Example 2. Show that the infinite product

$$\frac{\sin z}{z} \cdot \frac{\sin\left(\frac{z}{2}\right)}{z/2} \cdot \frac{\sin\left(\frac{z}{3}\right)}{z/3} \cdot \frac{\sin\left(\frac{z}{4}\right)}{z/4} \dots$$

is absolutely convergent for all values of z .

[For $\left(\frac{\sin(z/n)}{z/n}\right)$ can be written in the form $1 - \frac{\lambda_n}{n^2}$ where $|\lambda_n| < k$ and k is independent of n .

From the expansion $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots$, we obtain, on putting $x = z/n$,

$$\frac{\sin(z/n)}{z/n} = 1 - \frac{1}{n^2} \left(\frac{z^2}{6} - \frac{z^4}{120n^2} + \dots \right) = 1 - \frac{\lambda_n}{n^2}$$

where λ_n remains bounded as $n \rightarrow \infty$.

Hence, the series $\sum_{n=1}^{\infty} \frac{\lambda_n}{n^2}$ is absolutely convergent, as is seen on comparing it with $\sum_{n=1}^{\infty} \frac{1}{n^2}$. The infinite product is therefore absolutely convergent.]

3.7.1 Some examples of infinite products

Consider the infinite product

$$\left(1 - \frac{z^2}{\pi^2}\right) \left(1 - \frac{z^2}{2^2\pi^2}\right) \left(1 - \frac{z^2}{3^2\pi^2}\right) \dots$$

which can be shown to represent the function $z^{-1} \sin z$. [5]

In order to find whether it is absolutely convergent, we must consider the series

$$\sum_{n=1}^{\infty} \frac{z^2}{n^2\pi^2} = \frac{z^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

This series is absolutely convergent, and so the product is absolutely convergent for all values of z .

Now let the product be written in the form

$$\left(1 - \frac{z}{\pi}\right) \left(1 + \frac{z}{\pi}\right) \left(1 - \frac{z}{2\pi}\right) \left(1 + \frac{z}{2\pi}\right) \dots$$

The absolute convergence of this product depends on that of the series

$$-\frac{z}{\pi} + \frac{z}{\pi} - \frac{z}{2\pi} + \frac{z}{2\pi} - \dots$$

But this series is only conditionally convergent, since its series of moduli

$$\frac{|z|}{\pi} + \frac{|z|}{\pi} + \frac{|z|}{2\pi} + \frac{|z|}{2\pi} + \dots$$

is divergent. In this form therefore the infinite product is not absolutely convergent, and so, if the order of the factors $\left(1 \pm \frac{z}{n\pi}\right)$ is deranged, there is a risk of altering the value of the product.

Writing $1 - \frac{z^2}{n^2\pi^2} = \left(1 - \frac{z}{n\pi}\right)\left(1 + \frac{z}{n\pi}\right)$, we may express the product either as a product of quadratic factors or as a product of linear factors. In the first form, the terms differ from unity by quantities of order $1/n^2$, and the product is absolutely convergent. In the second form, the individual factors differ from unity by quantities of order $1/n$, and the corresponding series is only conditionally convergent. Hence the product is not absolutely convergent in this form, and a rearrangement of the factors may alter its value.

Lastly, let the same product be written in the form

$$\left\{\left(1 - \frac{z}{\pi}\right)e^{\frac{z}{\pi}}\right\}\left\{\left(1 + \frac{z}{\pi}\right)e^{-\frac{z}{\pi}}\right\}\left\{\left(1 - \frac{z}{2\pi}\right)e^{\frac{z}{2\pi}}\right\}\left\{\left(1 + \frac{z}{2\pi}\right)e^{-\frac{z}{2\pi}}\right\}\dots,$$

in which each of the expressions

$$\left(1 \pm \frac{z}{n\pi}\right)e^{\mp \frac{z}{n\pi}}$$

is counted as a single factor of the infinite product. The absolute convergence of this product depends on that of the series of which the $(2m-1)$ th and $(2m)$ th terms are

$$\left(1 \pm \frac{z}{m\pi}\right)e^{\mp \frac{z}{m\pi}} - 1$$

But it is easy to verify that

$$\left(1 \pm \frac{z}{m\pi}\right)e^{\mp \frac{z}{m\pi}} = 1 + O\left(\frac{1}{m^2}\right)$$

and so, the absolute convergence of the series in question follows by comparison with the series

$$1 + \frac{1}{2^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{4^2} + \dots$$

The infinite product in this last form is therefore again absolutely convergent, the adjunction of the factors $e^{\mp \frac{z}{n\pi}}$ having changed the convergence from conditional to absolute. This result is a particular case of the first part of the factor theorem of Weierstrass.[5]

Example 1. Prove that

$$\prod_{n=1}^{\infty} \left\{\left(1 - \frac{z}{c+n}\right)e^{\frac{z}{n}}\right\}$$

is absolutely convergent for all values of z , if c is a constant other than a negative integer.

For the infinite product converges absolutely with the series

$$\sum_{n=1}^{\infty} \left\{ \left(1 - \frac{z}{c+n}\right) e^{\frac{z}{n}} - 1 \right\}$$

Solution: Let

$$u_n = \left(1 - \frac{z}{c+n}\right) e^{z/n} - 1$$

To prove absolute convergence of the product, it is enough to show that the series $\sum_{n=1}^{\infty} |u_n|$ converges.

We begin by expanding the exponential factor in powers of $\frac{1}{n}$:

$$e^{z/n} = 1 + \frac{z}{n} + \frac{z^2}{2n^2} + O(n^{-3})$$

Substituting this into the expression for u_n , we obtain

$$\left(1 - \frac{z}{c+n}\right) e^{z/n} = \left(1 - \frac{z}{c+n}\right) \left(1 + \frac{z}{n} + \frac{z^2}{2n^2} + O(n^{-3})\right)$$

On multiplying out, this gives

$$\left(1 - \frac{z}{c+n}\right) e^{z/n} = 1 + \frac{z}{n} - \frac{z}{c+n} + O(n^{-2}),$$

and therefore

$$u_n = \frac{z}{n} - \frac{z}{c+n} + O(n^{-2})$$

Now the two leading terms almost cancel. Indeed,

$$\frac{z}{n} - \frac{z}{c+n} = z \left(\frac{1}{n} - \frac{1}{c+n} \right) = \frac{zc}{n(c+n)}$$

Thus

$$u_n = \frac{zc}{n(c+n)} + O(n^{-2})$$

The point is that this remaining term is itself of order n^{-2} . Since c is fixed, we have $|c+n| \geq n - |c|$, and so, for all sufficiently large n , $|c+n| \geq \frac{1}{2}n$. Hence

$$\left| \frac{zc}{n(c+n)} \right| \leq \frac{2|z||c|}{n^2}$$

It follows that $u_n = O\left(\frac{1}{n^2}\right)$.

Therefore, the series $\sum_{n=1}^{\infty} |u_n|$ converges by comparison with the convergent series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

Accordingly, the infinite product $\prod_{n=1}^{\infty} \left\{ \left(1 - \frac{z}{c+n}\right) e^{z/n} \right\}$ converges absolutely.

Example 2. Show that $\prod_{n=2}^{\infty} \left\{ 1 - \left(1 - \frac{1}{n} \right)^{-n} z^{-n} \right\}$ converges for all points z situated outside a circle whose centre is the origin and radius unity.

For the infinite product is absolutely convergent provided that the series $\sum_{n=2}^{\infty} \left(1 - \frac{1}{n} \right)^{-n} z^{-n}$ is absolutely convergent. But $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n} \right)^{-n} = e$, and so, the limit of the ratio of the $(n+1)$ th term of the series to the n th term is $\frac{1}{z}$; there is therefore absolute convergence when $\left| \frac{1}{z} \right| < 1$, i.e., when $|z| > 1$.

Example 3. Show that

$$\frac{1 \cdot 2 \cdot 3 \cdots (n-1)}{(z+1)(z+2) \cdots (z+n-1)} n^z = \prod_{k=1}^{n-1} \frac{k}{z+k} \cdot n^z$$

tends to a finite limit as $n \rightarrow \infty$, unless z is a negative integer.

Solution. Let

$$P_n = \frac{1 \cdot 2 \cdot 3 \cdots (n-1)}{(z+1)(z+2) \cdots (z+n-1)} n^z$$

We wish to show that P_n tends to a finite limit as $n \rightarrow \infty$, unless z is a negative integer.

To do this, we consider the ratio of successive terms. After some cancellation, we have

$$\frac{P_{n+1}}{P_n} = \frac{n}{z+n} \left(\frac{n+1}{n} \right)^z$$

Since $\frac{n}{z+n} = \left(1 + \frac{z}{n} \right)^{-1}$ and $\left(\frac{n+1}{n} \right)^z = \left(1 + \frac{1}{n} \right)^z$,

$$\frac{P_{n+1}}{P_n} = \left(1 + \frac{1}{n} \right)^z \left(1 + \frac{z}{n} \right)^{-1}$$

Expanding in powers of $\frac{1}{n}$, we have (by the general binomial expansion [2])

$$\left(1 + \frac{1}{n} \right)^z = 1 + \frac{z}{n} + \frac{z(z-1)}{2n^2} + O\left(\frac{1}{n^3}\right)$$

and

$$\left(1 + \frac{z}{n} \right)^{-1} = 1 - \frac{z}{n} + \frac{z^2}{n^2} + O\left(\frac{1}{n^3}\right)$$

Multiplying these two expansions, the terms of order $1/n$ cancel, and we obtain

$$\frac{P_{n+1}}{P_n} = 1 + \frac{z(z-1)}{2n^2} + O\left(\frac{1}{n^3}\right)$$

Thus, the successive ratios are of the form $1 + O\left(\frac{1}{n^2}\right)$. It follows that the infinite product formed from these ratios is absolutely convergent, since the corresponding series $\sum_{n=1}^{\infty} \left\{ \frac{z(z-1)}{2n^2} + O\left(\frac{1}{n^3}\right) \right\}$ is absolutely convergent by comparison with $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Hence P_n tends to a finite limit as $n \rightarrow \infty$. (Note that z is fixed, and so, the coefficient $\frac{z(z-1)}{2}$ is constant and hence the terms are $O\left(\frac{1}{n^2}\right)$.)

If z is a negative integer, then one of the factors in the denominator vanishes, and the expression is therefore not defined.

Example 4. Prove that

$$z \left(1 - \frac{z}{\pi}\right) \left(1 - \frac{z}{2\pi}\right) \left(1 + \frac{z}{\pi}\right) \left(1 - \frac{z}{3\pi}\right) \left(1 - \frac{z}{4\pi}\right) \left(1 + \frac{z}{2\pi}\right) \cdots = e^{-\frac{z}{\pi} \log 2} \sin z$$

Solution. Let

$$P_k = z \left(1 - \frac{z}{\pi}\right) \left(1 - \frac{z}{2\pi}\right) \left(1 + \frac{z}{\pi}\right) \cdots \left(1 - \frac{z}{(2k-1)\pi}\right) \left(1 - \frac{z}{2k\pi}\right) \left(1 + \frac{z}{k\pi}\right)$$

We examine the limit of P_k as $k \rightarrow \infty$.

For each negative factor, write

$$1 - \frac{z}{m\pi} = \left(1 - \frac{z}{m\pi}\right) e^{z/(m\pi)} \cdot e^{-z/(m\pi)}$$

and for each positive factor, write

$$1 + \frac{z}{m\pi} = \left(1 + \frac{z}{m\pi}\right) e^{-z/(m\pi)} \cdot e^{z/(m\pi)}$$

Substituting these into P_k , we obtain $P_k = E_k Q_k$ where

$$Q_k = z \prod_{m=1}^k \left(1 - \frac{z}{m\pi}\right) e^{z/(m\pi)} \left(1 + \frac{z}{m\pi}\right) e^{-z/(m\pi)}$$

and

$$E_k = \exp \left(-\frac{z}{\pi} - \frac{z}{2\pi} + \frac{z}{\pi} - \frac{z}{3\pi} - \frac{z}{4\pi} + \frac{z}{2\pi} - \cdots - \frac{z}{(2k-1)\pi} - \frac{z}{2k\pi} + \frac{z}{k\pi} \right)$$

Now each factor in Q_k differs from 1 by terms of order $1/m^2$, since

$$\left(1 \pm \frac{z}{m\pi}\right) e^{\mp z/(m\pi)} = 1 + O\left(\frac{1}{m^2}\right)$$

Hence, the corresponding infinite product is absolutely convergent. Therefore, its factors may be rearranged without altering its value, and so

$$Q = z \prod_{m=1}^{\infty} \left(1 - \frac{z}{m\pi}\right) e^{z/(m\pi)} \left(1 + \frac{z}{m\pi}\right) e^{-z/(m\pi)}$$

is well defined.

Combining the two factors for each m , we get $\left(1 - \frac{z}{m\pi}\right)\left(1 + \frac{z}{m\pi}\right) = 1 - \frac{z^2}{m^2\pi^2}$ and the exponential factors cancel. Thus $Q = z \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2\pi^2}\right)$. Using the standard product for $\sin z$, this is equal to $Q = \sin z$. [6]

To simplify E_k , we collect together the terms corresponding to each m . This gives

$$E_k = \exp\left(\frac{z}{\pi} \sum_{m=1}^k \left(\frac{1}{m} - \frac{1}{2m-1} - \frac{1}{2m}\right)\right)$$

But $\frac{1}{m} - \frac{1}{2m-1} - \frac{1}{2m} = -\left(\frac{1}{2m-1} - \frac{1}{2m}\right)$, and hence $E_k = \exp\left(-\frac{z}{\pi} \sum_{m=1}^k \left(\frac{1}{2m-1} - \frac{1}{2m}\right)\right)$.

Since $\sum_{m=1}^{\infty} \left(\frac{1}{2m-1} - \frac{1}{2m}\right) = \log 2$, it follows that $\lim_{k \rightarrow \infty} E_k = e^{-(z/\pi) \log 2}$.

Hence $\lim_{k \rightarrow \infty} P_k = e^{-(z/\pi) \log 2} \sin z$, which proves the result.

Table 3. Infinite Product Convergence

Test	Applies To	Condition	Conclusion	Absolute vs Conditional
Product convergence	Product $\prod (1 + a_n)$	$\sum \log(1 + a_n)$ converges	Product converges	Depends on log-series
Absolute convergence	Product $\prod (1 + a_n)$	$\sum \log(1 + a_n) $ converges	Absolutely convergent	Absolute
Equivalent condition	Product $\prod (1 + a_n)$	$\sum a_n $ converges	Absolutely convergent	Absolute
Necessary condition	Product $\prod (1 + a_n)$	$a_n \rightarrow 0$	Required	—
Rearrangement	Product $\prod (1 + a_n)$	Absolute convergence	Unchanged	Order independent
Conditional case	Product $\prod (1 + a_n)$	Conditional only	Order matters	Value may change

3.8 Infinite Determinants

Infinite series and infinite products are not by any means the only known cases of limiting processes which can lead to intelligible results. The researches of G. W. Hill in the Lunar Theory³⁶ brought into notice the possibilities of **infinite determinants**.

³⁶ Reprinted in *Acta Mathematica*, viii (1886), pp. 1–36. Infinite determinants had previously occurred in the researches of Furstenau on the algebraic equation of the n th degree, *Darstellung der reellen Wurzeln algebraischer Gleichungen durch Determinanten* (Marburg, 1860). Special types of infinite determinants (known as *continuants*) occur in the theory of infinite continued fractions; see Sylvester, *Math. Papers*, i, p. 504 and iii, p. 249.

The actual investigation of the convergence is due not to Hill but to Poincaré (*Bull. de la Soc. Math. de France*, xiv (1886), p. 87). We shall follow the exposition given by H. von Koch, *Acta Math.*, xvi (1892), p. 217.

Let A_{ik} be defined for all integer values (positive and negative) of i, k , and denote by

$$D_m = [A_{ik}], \quad i, k \in \{-m, \dots, m\}$$

the determinant formed of the numbers A_{ik} , $i, k \in \{-m, \dots, m\}$; then if, as $m \rightarrow \infty$, the expression D_m tends to a determinate limit D , we shall say that the infinite determinant

$$[A_{ik}], \quad i, k \in \mathbb{Z}$$

is *convergent* and has the value D . If the limit D does not exist, the determinant in question will be said to be *divergent*.

Here $D_m = [A_{ik}]$ denotes the ordinary determinant of the finite square matrix (A_{ik}) with indices $i, k \in \{-m, \dots, m\}$, as defined in the usual way (for example, by the expansion over permutations). Thus, an infinite determinant is defined as the limit of such finite determinants as the range of indices increases without bound.

The elements A_{ii} (where i takes all values) are said to form the *principal diagonal* of the determinant D ; the elements A_{ik} (where i is fixed and k takes all values) are said to form the *row* i ; and the elements A_{ik} (where k is fixed and i takes all values) are said to form the *column* k . Any element A_{ik} is called a *diagonal* or a *non-diagonal* element, according as $i = k$ or $i \neq k$. The element $A_{0,0}$ is called the *origin* of the determinant.

3.8.1 Convergence of an infinite determinant

We shall now show that an infinite determinant converges, provided the product of the diagonal elements converges absolutely, and the sum of the non-diagonal elements converges absolutely.

For let the diagonal elements of an infinite determinant D be denoted by $1 + a_{ii}$, and let the non-diagonal elements be denoted by a_{ik} ($i \neq k$), so that the determinant is

$$\begin{vmatrix} \dots & \dots & \dots & \dots & \dots \\ \dots & 1 + a_{-1,-1} & a_{-1,0} & a_{-1,1} & \dots \\ \dots & a_{0,-1} & 1 + a_{0,0} & a_{0,1} & \dots \\ \dots & a_{1,-1} & a_{1,0} & 1 + a_{1,1} & \dots \\ \dots & \dots & \dots & \dots & \dots \end{vmatrix}$$

Then, since the series $\sum_{i,k \in \mathbb{Z}} |a_{ik}|$ is convergent, the product $\bar{P} = \prod_{i \in \mathbb{Z}} (1 + \sum_{k \in \mathbb{Z}} |a_{ik}|)$ is convergent.

Rationale: If we write $b_i = \sum_{k \in \mathbb{Z}} |a_{i,k}|$, then $\sum_{i \in \mathbb{Z}} b_i = \sum_{i,k \in \mathbb{Z}} |a_{i,k}|$ is convergent. Hence, by the theorem on absolutely convergent infinite products from Section 3.7, the product $\bar{P} = \prod_{i \in \mathbb{Z}} (1 + b_i)$ is convergent.

Next, form the products

$$P_m = \prod_{i=-m}^m \left(1 + \sum_{k=-m}^m a_{ik} \right), \quad \bar{P}_m = \prod_{i=-m}^m \left(1 + \sum_{k=-m}^m |a_{ik}| \right)$$

To clarify the relation between P_m and D_m , consider the expansion of

$$P_m = \prod_{i=-m}^m \left(1 + \sum_{k=-m}^m a_{i,k} \right)$$

Each term in this expansion is obtained by selecting, for each index i , either the term 1 or one of the elements $a_{i,k}$, and forming the corresponding product.

The determinant D_m , on the other hand, is the sum of those products in which exactly one element is selected from each row and each column, that is, products of the form

$$\prod_{i=-m}^m a_{i,\sigma(i)}$$

where σ is a permutation of the index set $\{-m, \dots, m\}$, together with the appropriate sign.

Thus, D_m may be obtained from the expansion of P_m by discarding all terms which do not correspond to permutations (that is, replacing them by zero), and assigning to the remaining terms their proper signs.

If we now consider the product

$$\overline{P}_m = \prod_{i=-m}^m \left(1 + \sum_{k=-m}^m |a_{i,k}| \right)$$

each term in the expansion of D_m corresponds to a term in the expansion of $\overline{P}_m$ of the same form but with all factors replaced by their absolute values. It follows that each such term has modulus not exceeding the corresponding term in $\overline{P}_m$.

Now $D_{m+p} - D_m$ represents the sum of those terms in the expansion of D_{m+p} which involve at least one element $a_{i,k}$ with $|i| > m$ or $|k| > m$.

Indeed, if all such elements are replaced by zero, the determinant D_{m+p} reduces to D_m ; hence, the remaining terms are precisely those which vanish under this truncation.

To each of these terms there corresponds a term in the expansion of $\overline{P}_{m+p} - \overline{P}_m$ of the same form but with all factors replaced by their absolute values. It follows that each such term has modulus not exceeding the corresponding term in $\overline{P}_{m+p} - \overline{P}_m$. Hence

$$|D_{m+p} - D_m| < |\overline{P}_{m+p} - \overline{P}_m|$$

Since the product $\bar{P} = \prod_{i \in \mathbb{Z}} (1 + \sum_{k \in \mathbb{Z}} |a_{i,k}|)$ is convergent, the sequence $(\bar{P}_m)$ tends to a finite limit as $m \rightarrow \infty$. It follows that $\overline{P}_{m+p} - \bar{P}_m \rightarrow 0$ as $m \rightarrow \infty$ uniformly in p . Hence,

$$D_{m+p} - D_m \rightarrow 0 \text{ as } m \rightarrow \infty$$

Therefore, the sequence (D_m) is Cauchy (Section 3.2.2) and converges to a finite limit. This establishes the proposition.

3.8.2 The rearrangement theorem for convergent infinite determinants

We shall now show that a determinant, of the convergent form already considered, remains convergent when the elements of any row are replaced by any set of elements whose moduli are all less than some fixed positive number.

Replace, for example, the elements $A_{0,k}, k \in \mathbb{Z}$ of the row through the origin by elements $\mu_k, k \in \mathbb{Z}$ which satisfy the inequality $|\mu_k| < \mu$ where μ is a positive number, and let the new values of D_m and D be denoted by D'_m and D' .

Moreover, denote by $\bar{P}'_m$ and $\bar{P}'$ the products obtained by suppressing in $\bar{P}_m$ and $\bar{P}$ the factor corresponding to the index $i = 0$. We see that no terms of D'_m can have a greater modulus than the corresponding term in the expansion of $\bar{P}'_m$, and consequently, reasoning as in the last article, we have

$$|D'_{m+p} - D'_m| < |\bar{P}'_{m+p} - \bar{P}'_m|$$

which is sufficient to establish the result stated.

Example

Show that the necessary and sufficient condition for the absolute convergence of the infinite determinant

$$\lim_{m \rightarrow \infty} \begin{vmatrix} 1 & \alpha_1 & 0 & 0 & \dots & 0 \\ \beta_1 & 1 & \alpha_2 & 0 & \dots & 0 \\ 0 & \beta_2 & 1 & \alpha_3 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \beta_m & 1 \end{vmatrix}$$

is that the series $\alpha_1\beta_1 + \alpha_2\beta_2 + \alpha_3\beta_3 + \dots$ shall be absolutely convergent.

Let D_m denote the finite determinant of order $m + 1$. Since all entries outside the principal diagonal and the two adjacent diagonals are zero, a nonzero term in the expansion of D_m can arise only by selecting diagonal entries 1, or else by selecting an entry α_i in row i together with the entry β_i in row $i + 1$. Thus, every non-diagonal contribution occurs in the form $\alpha_i\beta_i$, and two such factors $\alpha_i\beta_i$ and $\alpha_j\beta_j$ can occur together only when $|i - j| > 1$.

It follows that the expansion of D_m is of the form

$$D_m = 1 - \sum_{i=1}^m \alpha_i \beta_i + \sum_{\substack{1 \leq i < j \leq m \\ |i-j| > 1}} \alpha_i \beta_i \alpha_j \beta_j - \dots$$

where the later terms are products of factors $\alpha_i\beta_i$ with no two indices consecutive.

Hence, the absolute convergence of the determinant is equivalent to the convergence of the series formed by the moduli of these terms. If $\sum_{i=1}^{\infty} |a_i \beta_i|$ converges, then the product $\prod_{i=1}^{\infty} (1 + |a_i \beta_i|)$ also converges, and its expansion contains all products of the quantities $|a_i \beta_i|$, including those with no two consecutive indices. Therefore, the series of moduli corresponding to the determinant converges, and so the determinant is absolutely convergent.

Conversely, if the determinant is absolutely convergent, then in particular the sum of the moduli of its one-factor terms $|a_1 \beta_1| + |a_2 \beta_2| + |a_3 \beta_3| + \dots$ must converge. Thus $\sum_{i=1}^{\infty} a_i \beta_i$ must be absolutely convergent.

Therefore, the necessary and sufficient condition for the absolute convergence of the infinite determinant is that the series $a_1 \beta_1 + a_2 \beta_2 + a_3 \beta_3 + \dots$ be absolutely convergent.

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Miscellaneous Examples

1. Evaluate $\lim_{n \rightarrow \infty} (e^{-na} n^b)$, $\lim_{n \rightarrow \infty} (n^{-a} \log n)$ when $a > 0, b > 0$.

Solution:

First, we write $e^{-na} n^b = \frac{n^b}{e^{an}}$. The denominator grows exponentially, while the numerator grows only polynomially, so the quotient tends to 0.

A direct proof uses l'Hospital's rule repeatedly:

$$\lim_{n \rightarrow \infty} \frac{n^b}{e^{an}} = \lim_{n \rightarrow \infty} \frac{b n^{b-1}}{a e^{an}} = \dots = \lim_{n \rightarrow \infty} \frac{C n^{b-m}}{a^m e^{an}}$$

and after enough differentiations the numerator becomes constant (or zero), while the denominator remains exponential. Hence the limit is 0.

For the second problem, we write $n^{-a} \log n = \frac{\log n}{n^a}$. Since $\log n$ grows more slowly than any positive power of n , this tends to 0.

Using l'Hospital's rule:

$$\lim_{n \rightarrow \infty} \frac{\log n}{n^a} = \lim_{n \rightarrow \infty} \frac{1/n}{a n^{a-1}} = \lim_{n \rightarrow \infty} \frac{1}{a n^a} = 0$$

Thus, $\lim_{n \rightarrow \infty} n^{-a} \log n = 0$.

2. Investigate the convergence of $\sum_{n=1}^{\infty} \left\{ 1 - n \log \left(\frac{2n+1}{2n-1} \right) \right\}$. (Trinity, 1904)

Solution:

Let $u_n = 1 - n \log \left(\frac{2n+1}{2n-1} \right)$. We must determine whether the series $\sum_{n=1}^{\infty} u_n$ converges.

We first simplify the logarithm. Write

$$\frac{2n+1}{2n-1} = \frac{1 + \frac{1}{2n}}{1 - \frac{1}{2n}}$$

so that

$$\log \left(\frac{2n+1}{2n-1} \right) = \log \left(1 + \frac{1}{2n} \right) - \log \left(1 - \frac{1}{2n} \right)$$

Now use the standard expansions

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, \quad \log(1-x) = -(x + \frac{x^2}{2} + \frac{x^3}{3} + \dots)$$

which are valid for $|x| < 1$. Subtracting, we obtain

$$\log\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots\right)$$

Substituting $x = \frac{1}{2n}$, this gives

$$\log\left(\frac{2n+1}{2n-1}\right) = 2\left(\frac{1}{2n} + \frac{1}{3(2n)^3} + \frac{1}{5(2n)^5} + \dots\right)$$

Hence

$$n \log\left(\frac{2n+1}{2n-1}\right) = 1 + \frac{1}{12n^2} + \frac{1}{80n^4} + \dots$$

and therefore

$$u_n = 1 - \left(1 + \frac{1}{12n^2} + \frac{1}{80n^4} + \dots\right) = -\frac{1}{12n^2} + O\left(\frac{1}{n^4}\right)$$

Thus $u_n \sim -\frac{1}{12n^2}$. Since the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, it follows by comparison that

$\sum_{n=1}^{\infty} u_n$ converges absolutely.

3. Investigate the convergence of $\sum_{n=1}^{\infty} \left\{ \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} \cdot \frac{4n+3}{2n+2} \right\}^2$. (Peterhouse, 1908)

Hint: Rewrite the product $\frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot (2n)}$ in terms of factorials, and compare the general term

with a power of $n^{-1/2}$; then use a comparison test.

4. Find the range of values of z for which the series

$$2\sin^2 z - 4\sin^4 z + 8\sin^6 z - \dots + (-1)^n 2^n \sin^{2n} z + \dots$$

is convergent.

Solution: Write the series as

$$2\sin^2 z - 4\sin^4 z + 8\sin^6 z - \dots = \sum_{n=1}^{\infty} (-1)^{n-1} 2^n \sin^{2n} z$$

This is a geometric series with ratio $-2\sin^2 z$. Hence it converges when $|2\sin^2 z| < 1$.

Next, examine separately the boundary cases $2\sin^2 z = \pm 1$.

You should find that:

- if $2\sin^2 z = 1$, the series becomes $1 - 1 + 1 - 1 + \dots$ and does not converge;
- if $2\sin^2 z = -1$, the terms are all equal to 1, and the series also diverges.

Thus, the condition for convergence is $|\sin z| < \frac{1}{\sqrt{2}}$.

5. Show that the series $\frac{1}{z} - \frac{1}{z+1} + \frac{1}{z+2} - \frac{1}{z+3} + \dots$

is conditionally convergent (except for certain exceptional values of z), but that the series

$$\left(\frac{1}{z} + \frac{1}{z+1} + \dots + \frac{1}{z+p-1}\right) - \left(\frac{1}{z+p} + \frac{1}{z+p+1} + \dots + \frac{1}{z+2p+q-1}\right) + \frac{1}{z+2p+q} + \dots$$

in which each group of p positive terms is followed by $p+q$ negative terms, is divergent. (Simon)

Solution:

The exceptional values are $z = 0, -1, -2, \dots$, for then one of the denominators vanishes.

Assume therefore that z is not a non-positive integer, noting the z can be a complex number.

First, consider the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{z+n}$. Let $a_n = (-1)^n$, $f_n = \frac{1}{z+n}$. Then the partial sums of a_n are bounded, while $f_n \rightarrow 0$ as $n \rightarrow \infty$, and f_n is ultimately monotone. Hence, by Dirichlet's test, the series converges.

It is not absolutely convergent, since $\left|\frac{1}{z+n}\right| \sim \frac{1}{n}$ ($n \rightarrow \infty$), and therefore, $\sum_{n=0}^{\infty} \left|\frac{1}{z+n}\right|$ has the same behavior as the harmonic series and diverges. Thus, the series is conditionally convergent.

Next, consider the original series in which each group of p positive terms is followed by $p+q$ negative terms. Since $\frac{1}{z+n} = \frac{1}{n} + O\left(\frac{1}{n^2}\right)$, the convergence or divergence is the same as that of the corresponding real series $\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{p} - \frac{1}{p+1} - \dots - \frac{1}{2p+q} + \frac{1}{2p+q+1} + \dots$.

Let $L = 2p+q$, and group the terms into blocks of length L . The r th block sum is

$$B_r = \sum_{j=1}^p \frac{1}{rL+j} - \sum_{j=p+1}^{2p+q} \frac{1}{rL+j}, \quad r = 0, 1, 2, \dots$$

For large r , we have

$$\frac{1}{rL+j} = \frac{1}{rL} \cdot \frac{1}{1+\frac{j}{rL}} = \frac{1}{rL} + O\left(\frac{1}{r^2}\right)$$

since j is bounded while $r \rightarrow \infty$. This estimate holds uniformly for $1 \leq j \leq p+q$. So,

$$\sum_{j=1}^p \frac{1}{rL+j} = \frac{p}{rL} + O\left(\frac{1}{r^2}\right), \quad \sum_{j=p+1}^{2p+q} \frac{1}{rL+j} = \frac{p+q}{rL} + O\left(\frac{1}{r^2}\right)$$

Hence $B_r = -\frac{q}{r^L} + O\left(\frac{1}{r^2}\right)$, and so $\sum B_r$ diverges, since the harmonic series diverges. Therefore, the grouped series diverges.

Conclusion: For $z \neq 0, -1, -2, \dots$, the series

$$\frac{1}{z} - \frac{1}{z+1} + \frac{1}{z+2} - \frac{1}{z+3} + \dots$$

is conditionally convergent, whereas the series formed by taking p positive terms followed by $p + q$ negative terms is divergent.

6. Show that $1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \frac{1}{7} - \dots = \frac{1}{2} \log 2$. (Trinity, 1908)

Solution: We begin with the alternating harmonic series, whose sum we know

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots = \log 2$$

Now separate its positive and negative terms. The positive terms are $1 + \frac{1}{3} + \frac{1}{5} + \dots$, and the negative terms are $-\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots\right)$.

The given series may be written in the form

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) - \left(\frac{1}{6} - \frac{1}{8}\right) + \dots$$

but it is more convenient to group it as

$$\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \dots$$

Observe that

$$\frac{1}{2m-1} - \frac{1}{2(2m-1)} = \frac{1}{2(2m-1)}$$

So, the terms

$$1 - \frac{1}{2}, \frac{1}{3} - \frac{1}{6}, \frac{1}{5} - \frac{1}{10}, \dots$$

sum to

$$\frac{1}{2} \left(1 + \frac{1}{3} + \frac{1}{5} + \dots\right)$$

Thus, the whole series becomes

$$\frac{1}{2} \left(1 + \frac{1}{3} + \frac{1}{5} + \dots\right) - \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \dots\right)$$

But

$$\frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \dots = \frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots\right)$$

Hence, the given series is

$$\frac{1}{2} \left[\left(1 + \frac{1}{3} + \frac{1}{5} + \cdots \right) - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \cdots \right) \right]$$

The expression in brackets is exactly the alternating harmonic series, and therefore equals $\log 2$.

So, the sum of the given series is $\frac{1}{2} \log 2$.

7. Show that the series $\frac{1}{1^\alpha} + \frac{1}{2^\beta} + \frac{1}{3^\alpha} + \frac{1}{4^\beta} + \cdots$ is convergent (for $1 < \alpha < \beta$), although

$$\lim_{n \rightarrow \infty} \frac{u_{2n+1}}{u_{2n}} = \infty. \text{ (Cesàro)}$$

Hint: Separate the series into two subsequences (odd and even indices) and compare each with a p -series; note that the ratio test fails because successive terms come from different subsequences.

8. Show that the series $\alpha + \beta^2 + \alpha^3 + \beta^4 + \cdots$ is convergent (for $0 < \alpha < \beta < 1$), although

$$\lim_{n \rightarrow \infty} \frac{u_{2n}}{u_{2n-1}} = \infty. \text{ (Cesàro)}$$

9. Show that the series

$$\sum_{n=1}^{\infty} \frac{nz^{n-1}[(1+n^{-1})^n - 1]}{(z^n - 1)(z^n - (1+n^{-1})^n)}$$

converges absolutely for all values of z , except the values

$$z = \left(1 + \frac{\alpha}{m} \right) e^{\frac{2k\pi i}{m}}, \quad \alpha = 0, 1, \quad k = 0, 1, \dots, m-1, \quad m = 1, 2, 3, \dots$$

Hint: Use the limit $(1 + 1/n)^n \rightarrow e$ and expand the numerator to first order; then compare the general term with a rational function of n to determine convergence.

10. Show that, when $s > 1$,

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \frac{1}{s-1} + \sum_{n=1}^{\infty} \left[\frac{1}{n^s} + \frac{1}{s-1} \left(\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} \right) \right]$$

and that the series on the right converges when $0 < s < 1$.

(de la Vallée Poussin, *Mem de l'Acad de Belgique*, LIII (1896), no 6)

Solution: Let $u_n = \frac{1}{n^s} + \frac{1}{s-1} \left(\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} \right)$.

We first simplify u_n asymptotically.

Expansion of the difference term:

For $n \neq 0$, we have the following identity

$$(n+1)^{-(s-1)} = \left(n \left(1 + \frac{1}{n}\right)\right)^{-(s-1)} = n^{-(s-1)} \left(1 + \frac{1}{n}\right)^{-(s-1)}$$

Using the generalized binomial expansion,

$$\left(1 + \frac{1}{n}\right)^{-(s-1)} = 1 - \frac{s-1}{n} + \frac{s(s-1)}{2n^2} + O\left(\frac{1}{n^3}\right)$$

Hence

$$\frac{1}{(n+1)^{s-1}} = \frac{1}{n^{s-1}} - \frac{s-1}{n^s} + \frac{s(s-1)}{2n^{s+1}} + O\left(\frac{1}{n^{s+2}}\right)$$

Therefore

$$\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} = -\frac{s-1}{n^s} + \frac{s(s-1)}{2n^{s+1}} + O\left(\frac{1}{n^{s+2}}\right)$$

Dividing by $s-1$, we get

$$\frac{1}{s-1} \left(\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} \right) = -\frac{1}{n^s} + \frac{s}{2n^{s+1}} + O\left(\frac{1}{n^{s+2}}\right)$$

So,

$$u_n = \frac{1}{n^s} - \frac{1}{n^s} + \frac{s}{2n^{s+1}} + O\left(\frac{1}{n^{s+2}}\right) = \frac{s}{2n^{s+1}} + O\left(\frac{1}{n^{s+2}}\right)$$

Thus, $u_n = O\left(\frac{1}{n^{s+1}}\right)$. Since $s+1 > 1$ whenever $s > 0$, it follows that $\sum_{n=1}^{\infty} u_n$ converges for every $s > 0$, $s \neq 1$. In particular, the series on the right converges when $0 < s < 1$.

Returning to the summation, suppose $s > 1$ and consider the partial sum

$$\sum_{n=1}^N u_n = \sum_{n=1}^N \frac{1}{n^s} + \frac{1}{s-1} \sum_{n=1}^N \left(\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} \right)$$

The second sum is telescopic:

$$\sum_{n=1}^N \left(\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} \right) = \frac{1}{(N+1)^{s-1}} - 1$$

Hence,

$$\sum_{n=1}^N u_n = \sum_{n=1}^N \frac{1}{n^s} + \frac{1}{s-1} \left(\frac{1}{(N+1)^{s-1}} - 1 \right)$$

Rearranging,

$$\sum_{n=1}^N \frac{1}{n^s} = \frac{1}{s-1} + \sum_{n=1}^N u_n - \frac{1}{s-1} \cdot \frac{1}{(N+1)^{s-1}}$$

Now $s > 1$, so

$$\frac{1}{(N+1)^{s-1}} \rightarrow 0 \text{ as } N \rightarrow \infty$$

Passing to the limit gives

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \frac{1}{s-1} + \sum_{n=1}^{\infty} \left[\frac{1}{n^s} + \frac{1}{s-1} \left(\frac{1}{(n+1)^{s-1}} - \frac{1}{n^{s-1}} \right) \right].$$

This proves the required identity.

11. In the series whose general term is $u_n = q^{n-v} \cdot x^{\frac{1}{2}v(v+1)}$, $0 < q < 1 < x$ (where $v(n)$ = number of digits in n), show: $\lim_{n \rightarrow \infty} u_n^{1/n} = q$, and that the series is convergent, yet $\lim_{n \rightarrow \infty} \sup \frac{u_{n+1}}{u_n} = \infty$.

Hint: Let $v(n)$ be the number of digits of n , so $v(n) \sim \log_{10} n$; use this to evaluate $u_n^{1/n}$, and note that $v(n)$ increases only at powers of 10 to analyze the ratio $\frac{u_{n+1}}{u_n}$.

12. Show that the series $q_1 + q_1^2 + q_2^3 + q_1^4 + q_2^5 + q_3^6 + q_1^7 + \dots$ where $q_n = q^{1+\frac{4}{n}}$, $0 < q < 1$ is convergent, although ratio test fails at triangular numbers. (Cesàro)

Hint: Observe that the indices at which the pattern changes are the triangular numbers; group the series into blocks between successive triangular numbers and compare the resulting sums with a geometric series.

13. Show that the series

$$\sum_{n=0}^{\infty} \frac{e^{2n\pi i x}}{(w+n)^s}$$

where w is real, and where $(w+n)^s$ is understood to mean $e^{s \log(w+n)}$, the logarithm being taken in its arithmetic sense, is convergent for all values of s , when $I(x)$ is positive, and is convergent for values of s whose real part is positive, when x is real and not an integer.

14. If $u_n > 0$, show that if $\sum u_n$ converges, then $\lim_{n \rightarrow \infty} \inf n u_n = 0$, and that, if in addition $u_n \geq u_{n+1}$, then $\lim_{n \rightarrow \infty} n u_n = 0$.

Solution: Assume throughout that $u_n > 0$ and $\sum_{n=1}^{\infty} u_n$ converges.

We first show that $\lim_{n \rightarrow \infty} \inf nu_n = 0$.

Suppose on the contrary that there exists a constant $c > 0$ such that $nu_n \geq c$ for all sufficiently large n . Then $u_n \geq \frac{c}{n}$ for all sufficiently large n , and so the series $\sum u_n$ would dominate a constant multiple of the harmonic series $\sum \frac{1}{n}$, which diverges. This contradicts the convergence of $\sum u_n$.

Hence, it is impossible that nu_n should remain bounded away from zero for all large n , and therefore, $\lim_{n \rightarrow \infty} \inf nu_n = 0$.

Next, we show that if $u_n \geq u_{n+1}$, then $nu_n \rightarrow 0$ as $n \rightarrow \infty$.

To that end, we note the standard block estimate:

$$\frac{n}{2} u_n \leq u_{[n/2]} + u_{[n/2]+1} + \cdots + u_n$$

The right-hand side is part of a tail of the convergent series, and so tends to 0. Hence $\frac{n}{2} u_n \rightarrow 0$, and therefore, $nu_n \rightarrow 0$. Thus, when (u_n) is decreasing, $\lim_{n \rightarrow \infty} nu_n = 0$.

[If the original statement of the problem had stipulated $\lim_{n \rightarrow \infty} \sup nu_n = 0$, then the result would be false. We can illustrate this with a counterexample.

Define

$$u_n = \begin{cases} \frac{1}{2^k}, & n = 2^k \\ \frac{1}{n^2}, & \text{otherwise} \end{cases}$$

Then $\sum_{n=1}^{\infty} u_n$ still converges, because $\sum_{k=1}^{\infty} \frac{1}{2^k} < \infty$ and $\sum 1/n^2$ converges. However, at the

powers of 2, $nu_n = 2^k \cdot \frac{1}{2^k} = 1$. Hence, $\lim_{n \rightarrow \infty} \sup nu_n = 1$, and so, certainly $\lim_{n \rightarrow \infty} \sup nu_n \neq 0$ in this example.]

15. Given $a_{m,n} = \frac{m-n}{2^{m+n}} \cdot \frac{(m+n-1)!}{m!n!}$, $a_{m,0} = 2^{-m}$, $a_{0,n} = -2^{-n}$, $a_{0,0} = 0$, $m, n > 0$

show that $\sum_{m,n} a_{m,n} = -1$, $\sum_{m,n} |a_{m,n}| = 1$. (Trinity, 1904)

16. By converting the series $1 + \frac{8q}{1-q} + \frac{16q^2}{1+q^2} + \frac{24q^3}{1-q^3} \cdots$, $|q| < 1$ into a double series, show it equals

$$1 + \frac{8q}{(1-q)^2} + \frac{8q^2}{(1+q^2)^2} + \frac{8q^3}{(1-q^3)^2} \cdots \text{ (Jacobi)}$$

Hint: Expand each term $\frac{q^n}{1 \pm q^n}$ as a geometric series and rearrange the result as a double series; then interchange the order of summation.

17. Assuming $\sin z = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2 \pi^2}\right)$, show that if $m, n \rightarrow \infty$ in such a way that $\lim \left(\frac{m}{n}\right) = k$, where k is finite, then

$$\lim \prod_{r=-n}^m \left(1 + \frac{z}{r\pi}\right) = k^{\frac{z}{\pi}} \cdot \frac{\sin z}{z}$$

noting that the factor for which $r = 0$ is omitted. (*Math Trip*, 1904)

18. If $u_0 = u_1 = u_2 = 0$, and if, when $n > 1$, $u_{2n-1} = -\frac{1}{\sqrt{n}}$, $u_{2n} = \frac{1}{\sqrt{n}} + \frac{1}{n} + \frac{1}{n\sqrt{n}}$

then $\prod_{n=0}^{\infty} (1 + u_n)$ converges, though $\sum_{n=0}^{\infty} u_n$ and $\sum_{n=0}^{\infty} u_n^2$ are divergent. (*Math. Trip.*, 1906)

19. Prove that $\prod_{n=1}^{\infty} \left\{ \left(1 - \frac{z}{n}\right)^{n^k} \exp \left(\sum_{m=1}^{k+1} \frac{n^{k-m} z^m}{m} \right) \right\}$ where k is any positive integer, converges absolutely for all values of z .

Solution: Let

$$u_n = \left(1 - \frac{z}{n}\right)^{n^k} \exp \left(\sum_{m=1}^{k+1} \frac{n^{k-m} z^m}{m} \right)$$

We shall show that $u_n = 1 + O\left(\frac{1}{n^2}\right)$, for large n . It will then follow that the product converges absolutely.

Since z is fixed, we have $\left|\frac{z}{n}\right| < 1$ for all sufficiently large n . For such n , we may use the

expansion $\log(1 - w) = -\sum_{m=1}^{\infty} \frac{w^m}{m}$, $|w| < 1$, with $w = \frac{z}{n}$. Thus

$$\log \left(1 - \frac{z}{n}\right) = -\sum_{m=1}^{\infty} \frac{z^m}{mn^m}$$

Multiplying by n^k , we obtain

$$n^k \log \left(1 - \frac{z}{n}\right) = -\sum_{m=1}^{\infty} \frac{n^{k-m} z^m}{m}$$

Hence

$$\log u_n = n^k \log \left(1 - \frac{z}{n}\right) + \sum_{m=1}^{k+1} \frac{n^{k-m} z^m}{m}$$

The first $k + 1$ terms cancel, and so

$$\log u_n = - \sum_{m=k+2}^{\infty} \frac{n^{k-m} z^m}{m}$$

The first term on the right is $-\frac{z^{k+2}}{(k+2)n^2}$ and all remaining terms are of still smaller order.

Therefore, $\log u_n = O\left(\frac{1}{n^2}\right)$. It follows that $u_n = \exp(\log u_n) = 1 + O\left(\frac{1}{n^2}\right)$. Thus, $u_n - 1 =$

$O\left(\frac{1}{n^2}\right)$, and therefore, $\sum_{n=1}^{\infty} |u_n - 1|$ converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Hence, by the

theorem on absolutely convergent infinite products, the product $\prod_{n=1}^{\infty} u_n$ converges absolutely.

20. If $\sum_{n=1}^{\infty} a_n$ be a conditionally convergent series of real terms, then $\prod_{n=1}^{\infty} (1 + a_n)$ converges (but not absolutely) or diverges to zero according as $\sum_{n=1}^{\infty} a_n^2$ converges or diverges. (Cauchy)

21. Let $\sum_{n=1}^{\infty} \theta_n$ be an absolutely convergent series. Show that the infinite determinant

$$\Delta(c) = \begin{vmatrix} \ddots & \dots & \dots & \dots & \dots \\ \frac{(c-4)^2 - \theta_0}{4^2 - \theta_0} & \frac{-\theta_1}{4^2 - \theta_0} & \frac{-\theta_2}{4^2 - \theta_0} & \frac{-\theta_3}{4^2 - \theta_0} & \frac{-\theta_4}{4^2 - \theta_0} \\ \frac{-\theta_1}{2^2 - \theta_0} & \frac{(c-2)^2 - \theta_0}{2^2 - \theta_0} & \frac{-\theta_1}{2^2 - \theta_0} & \frac{-\theta_2}{2^2 - \theta_0} & \frac{-\theta_3}{2^2 - \theta_0} \\ \frac{-\theta_2}{0^2 - \theta_0} & \frac{-\theta_1}{0^2 - \theta_0} & \frac{c^2 - \theta_0}{0^2 - \theta_0} & \frac{-\theta_1}{0^2 - \theta_0} & \frac{-\theta_2}{0^2 - \theta_0} \\ \frac{-\theta_3}{2^2 - \theta_0} & \frac{-\theta_2}{2^2 - \theta_0} & \frac{-\theta_1}{2^2 - \theta_0} & \frac{(c+2)^2 - \theta_0}{2^2 - \theta_0} & \frac{-\theta_1}{2^2 - \theta_0} \\ \frac{-\theta_4}{4^2 - \theta_0} & \frac{-\theta_3}{4^2 - \theta_0} & \frac{-\theta_2}{4^2 - \theta_0} & \frac{-\theta_1}{4^2 - \theta_0} & \frac{(c+4)^2 - \theta_0}{4^2 - \theta_0} \\ \dots & \dots & \dots & \dots & \ddots \end{vmatrix}$$

converges, and show that the equation $\Delta(c) = 0$ is equivalent to the equation

$$\sin^2 \left(\frac{1}{2} \pi c \right) = \Delta(0) \sin^2 \left(\frac{1}{2} \pi \theta_0^{1/2} \right)$$

(Hill)

4 Continuous Functions and Uniform Convergence

In mathematics, you don't understand things. You just get used to them. — John von Neumann

4.1 The dependence of one complex number on another

The problems with which Analysis is mainly occupied relate to the *dependence* of one complex number on another. If z and ζ are two complex numbers, so connected that, if z is given any one of a certain set of values, corresponding values of ζ can be determined, e.g. if ζ is the square of z , or if $\zeta = 1$ when z is real and $\zeta = 0$ for all other values of z , then ζ is said to be a **function** of z .

This dependence must not be confused with the most important case of it, which will be explained later under the title of *analytic functionality*.

If ζ is a real function of a real variable z , then the relation between ζ and z , which may be written

$$\zeta = f(z)$$

can be visualized by a curve in a plane, namely the locus of a point whose coordinates referred to rectangular axes in the plane are (z, ζ) . No such simple and convenient geometrical method can be found for visualizing an equation

$$\zeta = f(z)$$

considered as defining the dependence of one complex number $\zeta = \xi + i\eta$ on another complex number $z = x + iy$. A representation strictly analogous to the one already given for real variables would require four-dimensional space, since the number of variables ξ, η, x, y is now four.

One suggestion (made by Lie and Weierstrass) is to use a doubly-manifold system of lines in the quadruply-manifold totality of lines in three-dimensional space.

“Doubly-manifold” and “quadruply-manifold” are 19th-century usage for what we would now call a 2-parameter and 4-parameter family of functions, respectively. The most geometrically powerful approach, the theory of Riemann surfaces, belongs to the study of analytic functions and is developed in later chapters of Whittaker & Watson.

Another suggestion is to represent ξ and η separately by means of surfaces

$$\xi = \xi(x, y), \quad \eta = \eta(x, y)$$

Example: $f(z) = z^2$

Take $z = x + iy$, so $f(z) = z^2 = (x + iy)^2$. Expanding:

$$f(z) = x^2 - y^2 + 2ixy$$

The two surfaces are therefore:

$$\xi(x, y) = x^2 - y^2, \quad \eta(x, y) = 2xy$$

Both are hyperbolic paraboloids (saddle surfaces). What the graphs make clear is that the two surfaces are actually identical in shape — $\eta = 2xy$ is simply $\xi = x^2 - y^2$ rotated by 45° about the z -axis. This is not a coincidence: for any analytic function, ξ and η are not independent but are linked by the Cauchy–Riemann equations

$$\frac{\partial \xi}{\partial x} = \frac{\partial \eta}{\partial y}, \frac{\partial \xi}{\partial y} = -\frac{\partial \eta}{\partial x}$$

which is exactly what the authors mean when they warn that this visualization method — though geometrically clear — requires two separate surfaces and obscures the deep relationship between the two components.

$$f(z) = z^2 = \zeta, \quad \zeta = \xi + i\eta, \quad z = x + iy$$

$$\xi(x, y) = x^2 - y^2$$

$$\eta(x, y) = 2xy$$

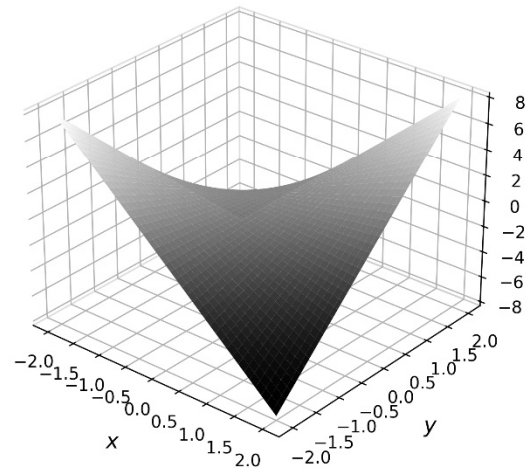
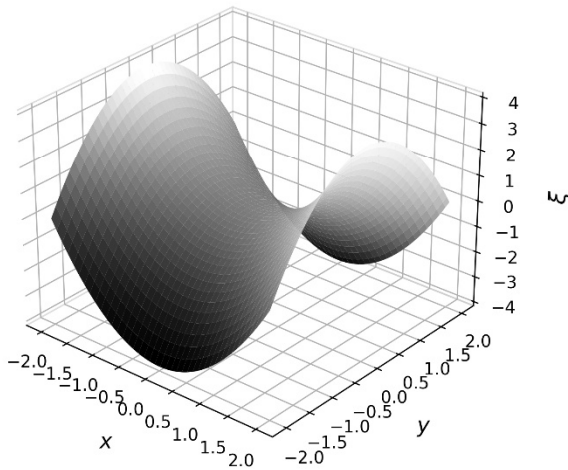


Figure 1. Hyperbolic paraboloids

A third suggestion, due to Heffter,³⁷ is to write

$$\zeta = re^{i\theta}$$

then draw the surface $r = r(x, y)$ — which may be called the *modular-surface* of the function — and on it to express the values of θ by surface-markings. It might be possible to modify this suggestion in various ways by representing θ by curves drawn on the surface $r = r(x, y)$.

4.2 Continuity of functions of real variables

The reader will have a general idea (derived from the graphical representation of functions of a real variable) as to what is meant by continuity.

We now have to give a precise definition which shall embody this vague idea.

Let $f(x)$ be a function of x defined when $a \leq x \leq b$.

Let x_1 be such that $a \leq x_1 \leq b$. If there exists a number l such that, corresponding to an arbitrary positive number ϵ , we can find a positive number η such that

$$|f(x) - l| < \epsilon$$

whenever $|x - x_1| < \eta$, $x \neq x_1$, and $a \leq x \leq b$, then l is called the **limit** of $f(x)$ as $x \rightarrow x_1$.

³⁷ *Zeitschrift für Math. und Phys.* **xlix** (1899), p. 235.

It may happen that we can find a number l_+ (even when l does not exist) such that $|f(x) - l_+| < \epsilon$ when $x_1 < x < x_1 + \eta$. We call l_+ the limit of $f(x)$ when x approaches x_1 from the right and denote it by $f(x_1 + 0)$; in a similar manner we define $f(x_1 - 0)$ if it exists.

If $f(x_1 + 0)$, $f(x_1)$, $f(x_1 - 0)$ all exist and are equal, we say that $f(x)$ is *continuous* at x_1 , so that if $f(x)$ is **continuous** at x_1 , then, given ϵ , we can find η such that

$$|f(x) - f(x_1)| < \epsilon$$

whenever $|x - x_1| < \eta$ and $a \leq x \leq b$.

If l_+ and l_- exist but are unequal, $f(x)$ is said to have an *ordinary discontinuity*³⁸ at x_1 , and if $l_+ = l_- \neq f(x_1)$, $f(x)$ is said to have a *removable discontinuity* at x_1 .

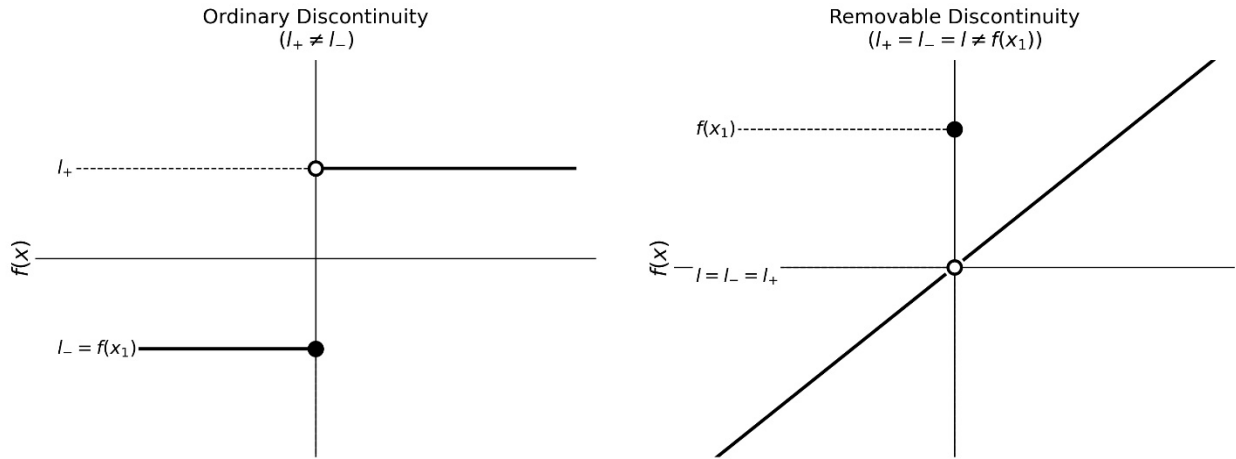


Figure 2. Discontinuity examples

If $f(x)$ is a complex function of a real variable, and if $f(x) = g(x) + ih(x)$ where $g(x)$ and $h(x)$ are real, the continuity of $f(x)$ at x_1 implies the continuity of $g(x)$ and of $h(x)$ at x_1 . For when $|f(x) - f(x_1)| < \epsilon$, then $|g(x) - g(x_1)| < \epsilon$ and $|h(x) - h(x_1)| < \epsilon$, and the result stated is obvious.

In the above, it is asserted without proof that $|f(x) - f(x_1)| < \epsilon$ implies both $|g(x) - g(x_1)| < \epsilon$ and $|h(x) - h(x_1)| < \epsilon$. This follows from the fact that for any complex number $w = u + iv$, both $|u| \leq |w|$ and $|v| \leq |w|$.

Example. From examples 1 and 2 (in Section 3.2) deduce that if $f(x)$ and $\phi(x)$ are continuous at x_1 , so are $f(x) \pm \phi(x)$, $f(x) \times \phi(x)$ and, if $\phi(x_1) \neq 0$, $f(x)/\phi(x)$.

The popular idea of continuity, so far as it relates to a real variable $f(x)$ depending on another real variable x , is somewhat different from that just considered, and may perhaps best be expressed by the statement: "The function $f(x)$ is said to depend continuously on x if, as x passes through the set of all values intermediate between any two adjacent values x_1 and x_2 , $f(x)$ passes through the set of all values intermediate between the corresponding values $f(x_1)$ and $f(x_2)$."

³⁸ If a function is said to have ordinary discontinuities at certain points of an interval it is implied that it is continuous at all other points of the interval.

The question thus arises, how far this popular definition is equivalent to the precise definition given above.

Cauchy showed that if a real function $f(x)$, of a real variable x , satisfies the precise definition, then it also satisfies what we have called the popular definition; this result will be proved in Section 4.6.3. But the converse is not true, as was shown by Darboux. This fact may be illustrated by the following example.³⁹

Between $x = -1$ and $x = +1$ (except at $x = 0$), let $f(x) = \sin \frac{\pi}{2x}$, and let $f(0) = 0$.

The graph of $\sin \frac{\pi}{2x}$ is shown in Figure 3. The graph oscillates more violently as one approaches 0 from the left and right.

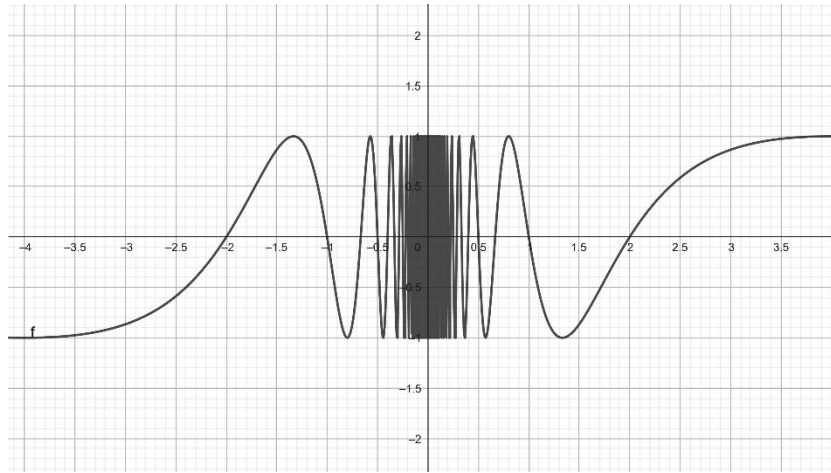


Figure 3. Violently oscillating sine function

It can then be proved that $f(x)$ depends continuously on x near $x = 0$, in the sense of the popular definition, but is not continuous at $x = 0$ in the sense of the precise definition.

We show the epsilon-delta condition fails. Take $\varepsilon = \frac{1}{2}$. We must show that for every $\eta > 0$, however small, there exists a point x with $|x| < \eta$ such that $|f(x) - f(0)| \geq \varepsilon$.

Given any $\eta > 0$, choose a positive integer n large enough that

$$x_n = \frac{1}{2n+1} < \eta$$

(this is always possible since $x_n \rightarrow 0$). Then

$$f(x_n) = \sin \frac{\pi(2n+1)}{2} = \sin \left(n\pi + \frac{\pi}{2} \right) = (-1)^n$$

so $|f(x_n) - f(0)| = 1 > \frac{1}{2} = \varepsilon$. The condition fails for this ε , so f is not continuous at 0 in the precise sense.

³⁹ Due to Mansion, *Mathesis*, (2) xix (1899), pp. 129–131.

The function $f(x) = \sin\left(\frac{\pi}{2x}\right)$ with $f(0) = 0$ provides an instructive example of the distinction between limit superior/inferior and ordinary limits. Although the limit $\lim_{x \rightarrow 0} f(x)$ does not exist, the limit superior and limit inferior do exist:

$$\limsup_{x \rightarrow 0} \sin\left(\frac{\pi}{2x}\right) = 1, \quad \liminf_{x \rightarrow 0} \sin\left(\frac{\pi}{2x}\right) = -1$$

The oscillating behavior near $x = 0$ ensures that in every punctured neighborhood of the origin, the function achieves values arbitrarily close to both 1 and -1 . This is why the epsilon-delta definition of continuity fails: for $\varepsilon = \frac{1}{2}$, no matter how small we choose $\delta > 0$, we can find points x with $|x| < \delta$ where $|f(x) - f(0)| = |f(x)| \geq 1 > \varepsilon$.

The key insight is that limit superior and limit inferior always exist (as extended real numbers) for any real-valued function. When they are equal, the ordinary limit exists; when they differ, as in this example, the ordinary limit fails to exist. This makes $\limsup$ and $\liminf$ valuable tools for analyzing the behavior of oscillating functions.

Example. If $f(x)$ is an increasing function on an open interval (a, b) , then the one-sided limits $f(x - 0)$ and $f(x + 0)$ exist at every interior point x .

Proof sketch: At any interior point c , consider the values of f to the left of c . Since f is increasing, these values form a bounded set (bounded above by $f(c)$), so their supremum exists. This supremum is precisely $f(c - 0)$, the left-hand limit. Similarly, the values of f to the right of c have an infimum, which equals $f(c + 0)$, the right-hand limit.

Note: At the endpoints a and b , only one-sided limits may exist. At the left endpoint a , only $f(a + 0)$ is defined; at the right endpoint b , only $f(b - 0)$ is defined.

4.2.1 Simple Curves and Continua

Let x and y be two real functions of a real variable t which are continuous for every value of t such that $a \leq t \leq b$. We denote the dependence of x and y on t by writing

$$x = x(t), \quad y = y(t) \quad (a \leq t \leq b)$$

The functions $x(t)$, $y(t)$ are supposed to be such that they do not assume the same pair of values for any two different values of t in the range $a < t < b$, i.e., *the curve does not cross (intersect) itself*.

Then the set of points with coordinates (x, y) corresponding to these values of t is called a **simple curve**. If

$$x(a) = x(b), \quad y(a) = y(b)$$

the simple curve is said to be **closed**.

Example. The circle $x^2 + y^2 = 1$ is a simple closed curve, for we may write⁴⁰

$$x = \cos t, \quad y = \sin t \quad (0 \leq t \leq 2\pi)$$

⁴⁰ For a proof that the sine and cosine are continuous functions, see the *Appendix*, §A.41 of Whittaker & Watson.

A **two-dimensional continuum** is a set of points in a plane possessing the following two properties.

1. If (x, y) are the Cartesian coordinates of any point of it, a *positive* number δ (depending on x and y) can be found such that every point whose distance from (x, y) is less than δ belongs to the set.
2. Any two points of the set can be joined by a simple curve consisting entirely of points of the set.

Example. The points for which $x^2 + y^2 < 1$ form a continuum. For if P is any point inside the unit circle such that $OP = r < 1$, we may take $\delta = 1 - r$, and any two points inside the circle may be joined by a straight line lying wholly inside the circle.

The following two theorems⁴¹ will be assumed in this work; simple cases of them appear obvious from geometrical intuitions and, generally, theorems of a similar nature will be taken for granted, as formal proofs are usually extremely long and difficult.

Theorem I. (Jordan Closed Curve Theorem) A simple closed curve divides the plane into two continua (the interior and the exterior).

Theorem II. If P is a point on the curve and Q is a point not on the curve, the angle between QP and Ox increases by $\pm 2\pi$ or by zero, as P describes the curve, according as Q is an interior point or an exterior point. If the increase is $+2\pi$, P is said to describe the curve counterclockwise.

On Winding Numbers: *The winding number is a topological measure that counts how many times a closed curve winds around a given point. Specifically, as point P traces along the curve, the angle that the vector from Q to P makes (measured from a fixed direction, say the positive x -axis) changes continuously. For a point Q in the interior of a simple closed curve, this angle increases (or decreases) by exactly 2π as P completes one full circuit around the curve. For a point Q in the exterior, the net angular change is zero. For simple closed curves, the only possible winding numbers are $+1$, -1 , and 0 ; interior points cannot have winding numbers with absolute value greater than 1 . This difference in winding number—whether it equals $\pm 2\pi$ or 0 —provides a criterion for determining whether a point lies inside or outside the curve, independent of the curve's shape.*

A continuum formed by the interior of a simple curve is sometimes called an *open two-dimensional region*, or briefly an *open region*, and the curve is called its *boundary*; such a continuum with its boundary is then called a *closed two-dimensional region*, or briefly a *closed region* or *domain*.

A simple curve is sometimes called a *closed one-dimensional region*; a simple curve with its end-points omitted is then called an *open one-dimensional region*.

4.2.2 Continuous functions of complex variables

Let $f(z)$ be a function of z defined at all points of a closed region (one- or two-dimensional) in the Argand diagram, and let z_1 be a point of the region.

⁴¹ Formal proofs will be found in Watson's *Complex Integration and Cauchy's Theorem* (Cambridge Math. Tracts, No. 15).

Then $f(z)$ is said to be continuous at z_1 , if given any positive number ε , we can find a corresponding positive number η such that

$$|f(z) - f(z_1)| < \varepsilon$$

whenever $|z - z_1| < \eta$ and z is a point of the region.

4.3 Series of variable terms — Uniformity of convergence

Consider the series

$$x^2 + \frac{x^2}{1+x^2} + \frac{x^2}{(1+x^2)^2} + \cdots + \frac{x^2}{(1+x^2)^n} + \cdots$$

This series converges absolutely (Section 3.3.3) for all real values of x .

If $S_n(x)$ be the sum of n terms, then

$$S_n(x) = 1 + x^2 - \frac{1}{(1+x^2)^{n-1}}$$

and so

$$\lim_{n \rightarrow \infty} S_n(x) = 1 + x^2, \quad (x \neq 0)$$

but $S_n(0) = 0$, and therefore

$$\lim_{n \rightarrow \infty} S_n(0) = 0$$

Consequently, although the series is an absolutely convergent series of continuous functions of x , the sum is a discontinuous function of x . We naturally enquire the reason of this rather remarkable phenomenon, which was investigated in 1841–1848 by Stokes⁴², Seidel⁴³ and Weierstrass⁴⁴, who showed that it cannot occur except in connection with another phenomenon, that of non-uniform convergence, which will now be explained.

Let the functions $u_1(z), u_2(z), \dots$ be defined at all points of a closed region of the Argand diagram. Let

$$S_n(z) = u_1(z) + u_2(z) + \cdots + u_n(z)$$

The condition that the series $\sum_{n=1}^{\infty} u_n(z)$ should converge for any particular value of z is that, given ε , a number n should exist such that

$$|S_{n+p}(z) - S_n(z)| < \varepsilon$$

for all positive values of p , the value of n of course depending on ε .

Let n have the smallest integer value for which the condition is satisfied. This integer will, in general, depend on the particular value of z which has been selected for consideration. We denote this dependence by writing $n(z)$ in place of n . Now it may happen that we can find a number N , independent of z , such that $n(z) < N$ for all values of z in the region under consideration.

⁴² *Camb Phil Trans VIII (1847), pp 533–583 [Collected Papers, I pp 296–313]*

⁴³ *Münchener Abhandlungen, v (1848), p 381*

⁴⁴ *Ges Math Werke, I pp. 67, 75*

If this number N exists, the series is said to **converge uniformly** throughout the region.

If no such number N exists, the convergence is said to be **non-uniform**.⁴⁵

Uniformity of convergence is thus a property depending on a whole set of values of z , whereas previously we have considered the convergence of a series for various particular values of z , the convergence for each value being considered without reference to the other values.

The distinction here is between pointwise convergence and uniform convergence. For a fixed value of z , Cauchy's convergence condition allows the required integer n to depend on that particular value of z . Uniform convergence is stronger: after ε is chosen, there must be one integer N that works simultaneously for every z in the region. In modern notation, this is the uniform Cauchy condition for the series of functions $\sum u_n(z)$.

We define the phrase "uniformity of convergence near a point z_1 " to mean that there is a definite positive number δ such that the series converges uniformly in the domain common to the circle

$$|z - z_1| \leq \delta$$

and the region in which the series converges.

Example. *Uniform convergence of a geometric series of functions*

Consider the series of functions

$$z + z^2 + z^3 + \dots$$

on the closed disk

$$|z| \leq r$$

where $0 < r < 1$. For any z in this disk, we have

$$|z^n| \leq r^n$$

The numerical series

$$r + r^2 + r^3 + \dots$$

is a convergent geometric series. Hence the tail

$$r^{N+1} + r^{N+2} + \dots + r^{N+p}$$

can be made smaller than any prescribed positive number ε , by choosing N large enough. Importantly, this choice of N depends only on ε and r , not on the particular value of z in the disk. More explicitly,

$$|z^{N+1} + z^{N+2} + \dots + z^{N+p}| \leq |z|^{N+1} + |z|^{N+2} + \dots + |z|^{N+p}$$

and therefore,

$$|z^{N+1} + z^{N+2} + \dots + z^{N+p}| \leq r^{N+1} + r^{N+2} + \dots + r^{N+p}$$

⁴⁵ The reader who is unacquainted with the concept of uniformity of convergence will find it made much clearer by consulting Bromwich, *Infinite Series*, Ch. VII, where an illuminating account of Osgood's graphical investigation is given.

Since $r^{N+1} + r^{N+2} + \dots + r^{N+p} < \frac{r^{N+1}}{1-r}$, we may choose N so large that $\frac{r^{N+1}}{1-r} < \varepsilon$.

It follows that

$$|z^{N+1} + z^{N+2} + \dots + z^{N+p}| < \varepsilon$$

for all positive integers p and for all z satisfying $|z| \leq r$. Therefore, the series

$$z + z^2 + z^3 + \dots$$

converges uniformly on every closed disk $|z| \leq r$, where $0 < r < 1$.

This example illustrates the essential point in uniform convergence: once ε is chosen, the same integer N works for every z in the region under consideration. The required N may depend on the region, here through the number r , but it does not depend on the individual value of z .

4.3.1 On the condition for uniformity of convergence

If $R_{n,p}(z) = u_{n+1}(z) + u_{n+2}(z) + \dots + u_{n+p}(z)$, we have seen that the necessary and sufficient condition that

$$\sum_{n=1}^{\infty} u_n(z)$$

should converge uniformly in a region is that, given any positive number ε , it should be possible to choose N independent of z (but depending on ε) such that

$$|R_{N,p}(z)| < \varepsilon$$

for all positive integral values of p .⁴⁶

The expression $R_{n,p}(z)$ is the sum of a finite block of terms in the tail of the series, starting just after the n th term and ending at the $(n+p)$ th term. Since

$$R_{n,p}(z) = S_{n+p}(z) - S_n(z),$$

this condition is exactly Cauchy's criterion applied to the partial sums, but with the important additional requirement that the integer N be chosen independently of z . Thus, the same cutoff point in the series must control all tails throughout the entire region.

If the condition is satisfied, then by Section 3.2.2, the sequence of partial sums $S_n(z)$ tends to a limit, say $S(z)$, for each value of z under consideration. Since ε is independent of p , we may let $p \rightarrow \infty$ in the inequality $|R_{N,p}(z)| < \varepsilon$ and obtain $\left| \lim_{p \rightarrow \infty} R_{N,p}(z) \right| \leq \varepsilon$.

Now suppose that $n > N$. The finite remainder beginning after the N -th term may be split into two parts: first the finite block from the $(N+1)$ -st term through the n -th term, and then the remaining tail after the n -th term. Thus,

$$\begin{aligned} R_{N,p}(z) &= u_{N+1}(z) + u_{N+2}(z) + \dots + u_{N+p}(z) \\ &= R_{N,n-N}(z) + (u_{n+1}(z) + u_{n+2}(z) + \dots + u_{N+p}(z)) \end{aligned}$$

⁴⁶ This section shows that it is indifferent whether uniformity of convergence is defined by means of the partial remainder $R_{n,p}(z)$ or by $R_n(z)$. Writers differ in the definition taken as fundamental.

Since $n > N$, the finite remainder $R_{N,p}(z)$ contains two parts: the finite block from $N + 1$ through n , and the remaining terms from $n + 1$ onward. The latter part is the beginning of the tail $S(z) - S_n(z)$. Hence, after subtracting the finite block $R_{N,n-N}(z)$ and then letting $p \rightarrow \infty$, we obtain

$$S(z) - S_n(z) = \left\{ \lim_{p \rightarrow \infty} R_{N,p}(z) \right\} - R_{N,n-N}(z).$$

Since both $\left| \lim_{p \rightarrow \infty} R_{N,p}(z) \right|$ and $|R_{N,n-N}(z)|$ are less than ε , it follows from the triangle inequality that

$$|S(z) - S_n(z)| < 2\varepsilon$$

Thus, writing $\frac{1}{2}\varepsilon$ in place of ε , a necessary condition for uniformity of convergence is that

$$|S(z) - S_n(z)| < \varepsilon$$

whenever $n > N$, where N is independent of z .

Conversely, this condition is also sufficient. For if

$$|S(z) - S_n(z)| < \varepsilon$$

whenever $n > N$, with N independent of z , then the difference between two sufficiently far-out partial sums is small. In particular,

$$R_{n,p}(z) = S_{n+p}(z) - S_n(z)$$

and hence

$$|R_{n,p}(z)| \leq |S_{n+p}(z) - S(z)| + |S(z) - S_n(z)|$$

Each term on the right is less than $\frac{\varepsilon}{2}$, so

$$|R_{n,p}(z)| < \varepsilon$$

This is precisely the Cauchy condition for uniform convergence. Therefore, it is equivalent to define uniform convergence either by requiring the remainders $R_{n,p}(z)$ to become uniformly small, or by requiring the difference between the partial sums $S_n(z)$ and the limiting function $S(z)$ to become uniformly small.

Example 1. Show that, if x be real, the sum of the series

$$\frac{x}{1(x+1)} + \frac{x}{(x+1)(2x+1)} + \cdots + \frac{x}{\{(n-1)x+1\}\{nx+1\}} + \cdots$$

is discontinuous at $x = 0$ and the series is non-uniformly convergent near $x = 0$.

To find the sum of the first n terms, we use a telescoping process. We add the first two terms to get

$$\frac{x}{1(x+1)} + \frac{x}{(x+1)(2x+1)} = \frac{x}{x+1} \left(1 + \frac{1}{2x+1} \right) = \frac{x}{x+1} \cdot \frac{2x+2}{2x+1} = \frac{2x}{2x+1}$$

Next, add the above result to the third term to get

$$\frac{2x}{2x+1} + \frac{x}{(2x+1)(3x+1)} = \frac{x}{2x+1} \left(2 + \frac{1}{(3x+1)} \right) = \frac{x}{2x+1} \cdot \frac{6x+3}{3x+1} = \frac{3x}{(3x+1)}$$

Continuing in the manner we get that the sum of the first n terms is

$$\frac{nx}{nx+1} = 1 - \frac{1}{nx+1}$$

and so, when $x = 0$ the sum is 0, while when $x \neq 0$, the sum is 1.

The value of $R_n(x) = S(x) - S_n(x)$ is $\frac{1}{nx+1}$ if $x \neq 0$. Thus when x is small, say x is one-hundred-

millionth, the remainder after a million terms is $\frac{1}{\frac{1}{100}+1}$ or $1 - \frac{1}{101}$. So, the first million terms of the series

do not contribute one per cent of the sum. And in general, to make $\frac{1}{nx+1} < \varepsilon$ it is necessary to take $n >$

$\frac{1}{x} \left(\frac{1}{\varepsilon} - 1 \right)$. Corresponding to a given ε , no number N exists, independent of x , such that $n < N$ for all values of x in any interval including $x = 0$, for by taking x sufficiently small we can make n greater than any number N which is independent of x . There is therefore non-uniform convergence near $x = 0$.

Example 2. Discuss the series

$$\sum_{n=1}^{\infty} \frac{x\{n(n+1)x^2 - 1\}}{\{1 + n^2x^2\}\{1 + (n+1)^2x^2\}}$$

in which x is real.

The n th term can be written

$$\frac{nx}{1 + n^2x^2} - \frac{(n+1)x}{1 + (n+1)^2x^2}$$

Writing the sum of the first n using the above, we get

$$S_n(x) = \frac{x}{1 + x^2} - \frac{(n+1)x}{1 + (n+1)^2x^2}$$

Taking the limit as n approaches infinity, we have that

$$S(x) = \frac{x}{1 + x^2}$$

Thus,

$$R_n(x) = S(x) - S_n(x) = \frac{(n+1)x}{1 + (n+1)^2x^2}$$

*This remainder tends to zero for each fixed x . However, the convergence is **not uniform near** $x = 0$. To see why, observe that if we take $x = \frac{1}{n+1}$, then*

$$R_n(x) = \frac{(n+1) \cdot \frac{1}{n+1}}{1 + (n+1)^2 \left(\frac{1}{n+1} \right)^2} = \frac{1}{2}$$

So, no matter how large n is, we can choose x close to 0 so that the remainder is still equal to $\frac{1}{2}$. Therefore, there is no single integer N , independent of x , which makes

$$|R_n(x)| < \varepsilon$$

for all $n > N$ and all x sufficiently near 0, for example when $\varepsilon < \frac{1}{2}$.

Thus, the series converges pointwise to $S(x) = \frac{x}{1+x^2}$, but the convergence is not uniform near $x = 0$.

This example is important because it shows that failure of uniform convergence does not always reveal itself through a discontinuous limiting function. In Example 1, the limiting function was discontinuous at $x = 0$, which made non-uniform convergence easy to suspect. In the present example, however, the limiting function

$$S(x) = \frac{x}{1+x^2}$$

is continuous at $x = 0$. The failure of uniform convergence is instead detected by the behavior of the remainder

$$R_n(x) = \frac{(n+1)x}{1+(n+1)^2x^2}$$

which can remain as large as $\frac{1}{2}$ when x is chosen to be approximately $\frac{1}{n+1}$. This illustrates that uniform convergence is a stronger condition than pointwise convergence, and cannot be checked merely by looking at the continuity of the limiting function!

4.3.2 Connection of Discontinuity with Non-Uniform Convergence

We shall now show that if a series of continuous functions of z is uniformly convergent for all values of z in a given closed domain, the sum is a continuous function of z at all points of the domain.

For let the series be

$$f(z) = u_1(z) + u_2(z) + \cdots + u_n(z) + \cdots = S_n(z) + R_n(z),$$

where $R_n(z)$ is the remainder after n terms.

Since the series is uniformly convergent, given any positive number ε , we can find a corresponding integer n , independent of z , such that $|R_n(z)| < \frac{1}{3}\varepsilon$ for all values of z within the domain.

Now n and ε being thus fixed, we can, on account of the continuity of $S_n(z)$, find a positive number η such that $|S_n(z) - S_n(z')| < \frac{1}{3}\varepsilon$ whenever $|z - z'| < \eta$.

We then have

$$\begin{aligned} |f(z) - f(z')| &= |\{S_n(z) + R_n(z)\} - \{S_n(z') + R_n(z')\}| \\ &= |\{S_n(z) - S_n(z')\} + \{R_n(z) - R_n(z')\}| \end{aligned}$$

and hence, by the triangle inequality,

$$|f(z) - f(z')| \leq |S_n(z) - S_n(z')| + |R_n(z)| + |R_n(z')|$$

Since each term on the right is less than $\frac{1}{3}\varepsilon$, it follows that

$$|f(z) - f(z')| < \varepsilon,$$

which is the condition for continuity at z .

The proof uses the triangle inequality. Since both remainders are less than $\varepsilon/3$, and since the finite partial sum $S_n(z)$ is continuous, the total change in $f(z)$ is bounded by the sum of three quantities, each less than $\varepsilon/3$. The important point is that uniform convergence allows the same integer n to be chosen for all z in the domain.

Example 1. Show that near $x = 0$ the series

$$u_1(x) + u_2(x) + u_3(x) + \cdots,$$

where

$$u_1(x) = x, \quad u_n(x) = x^{1/(2n-1)} - x^{1/(2n-3)}, \quad n \geq 2,$$

and real values of x are concerned, is discontinuous and non-uniformly convergent.

In this example it is convenient to take a slightly different form of the test. We shall show that, given an arbitrarily small number ε , it is possible to choose values of x , as small as we please, depending on n , in such a way that $|R_n(x)|$ is not less than ε for any value of n , no matter how large. The reader will easily see that the existence of such values of x is inconsistent with the condition for uniformity of convergence.

The value of $S_n(x)$ is

$$S_n(x) = x^{1/(2n-1)}$$

As n tends to infinity, $S_n(x)$ tends to 1, 0, or -1 , according as x is positive, zero, or negative. The series is therefore absolutely convergent for all values of x , and has a discontinuity at $x = 0$.

The telescoping is as follows:

$$S_n(x) = x + (x^{1/3} - x) + (x^{1/5} - x^{1/3}) + \cdots + (x^{1/(2n-1)} - x^{1/(2n-3)})$$

and so, all intermediate terms cancel, leaving

$$S_n(x) = x^{1/(2n-1)}$$

The limiting function in this example requires a little care. Since $S_n(x) = x^{1/(2n-1)}$, the exponent tends to 0 as $n \rightarrow \infty$. For $x > 0$, this gives

$$\lim_{n \rightarrow \infty} x^{1/(2n-1)} = x^0 = 1$$

For $x = 0$, however, each term of the sequence is already equal to 0, since $0^{1/(2n-1)} = 0$.

Hence, $\lim_{n \rightarrow \infty} S_n(0) = 0$.

The case $x < 0$ is more subtle. Since $2n - 1$ is always odd, the expression $x^{1/(2n-1)}$ is interpreted as the real odd root of x . Thus, for example, $(-8)^{1/3} = -2$.

If $x < 0$, write $x = -a$, where $a > 0$. Then $x^{1/(2n-1)} = (-a)^{1/(2n-1)} = -a^{1/(2n-1)}$.

Since $\lim_{n \rightarrow \infty} a^{1/(2n-1)} = 1$, it follows that $\lim_{n \rightarrow \infty} x^{1/(2n-1)} = -1$.

Therefore, the limiting function is

$$S(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

This function has a jump discontinuity at $x = 0$. In particular, as x approaches 0 from the right, $S(x)$ approaches 1, while as x approaches 0 from the left, $S(x)$ approaches -1 . This discontinuity shows that the convergence cannot be uniform in any interval containing 0, because a uniformly convergent series of continuous functions has a continuous sum.

Example 2. Show that near $z = 0$ the series

$$\sum_{n=1}^{\infty} \frac{-2z(1+z)^{n-1}}{\{1 + (1+z)^{n-1}\}\{1 + (1+z)^n\}}$$

is non-uniformly convergent and its sum is discontinuous.

The n -th term can be written

$$\frac{1 - (1+z)^n}{1 + (1+z)^n} - \frac{1 - (1+z)^{n-1}}{1 + (1+z)^{n-1}},$$

and so, the sum of the first n terms is

$$S_n(z) = \frac{1 - (1+z)^n}{1 + (1+z)^n}.$$

Thus, considering real values of z greater than -1 , it is seen that the sum to infinity is 1, 0, or -1 , according as z is negative, zero, or positive. There is thus a discontinuity at $z = 0$.

This discontinuity is explained by the fact that the series is non-uniformly convergent near $z = 0$.

Indeed, when z is positive, the remainder after n terms is $R_n(z) = -1 - S_n(z) = \frac{-2}{1 + (1+z)^n}$. Hence,

$$|R_n(z)| = \frac{2}{1 + (1+z)^n}$$

Now take $z = \frac{1}{n}$. Then $z \rightarrow 0$ as $n \rightarrow \infty$, so these values of z lie arbitrarily close to 0. But for this choice of z ,

$$\left| R_n\left(\frac{1}{n}\right) \right| = \frac{2}{1 + \left(1 + \frac{1}{n}\right)^n}$$

Using the fact that $\left(1 + \frac{1}{n}\right)^n < e$, we have $\left| R_n\left(\frac{1}{n}\right) \right| > \frac{2}{1+e}$ where e is Euler's number.

Thus, however large n may be, one can choose a positive value of z , namely $z = \frac{1}{n}$, arbitrarily close to 0, for which the remainder is still greater than $\frac{2}{1+e}$. This number is not arbitrarily small.

Therefore, no single integer N , independent of z , can make the remainder small for all values of z near 0. The series is therefore non-uniformly convergent near $z = 0$.

The key point is that the value $z = 1/n$ approaches 0 as n increases. Thus, no matter how far out in the series one goes, there are still values of z arbitrarily close to 0 for which the remainder is not small. This is precisely what prevents the convergence from being uniform near $z = 0$.

4.3.3 The Distinction Between Absolute and Uniform Convergence

[The following is an extensive rewrite of §3 33 of Whittaker and Watson.]

Uniform convergence and absolute convergence are different concepts. Neither one automatically implies the other in every setting.

Absolute convergence concerns the convergence of the series obtained by taking the absolute values of the individual terms. Uniform convergence concerns whether the remainders of the series can be made small by choosing an index N that works throughout an entire region, independently of the particular point in that region.

Thus, absolute convergence is a pointwise condition unless additional estimates are supplied, while uniform convergence is a condition on a whole set of values of the variable.

For example, consider the series

$$\sum_{n=1}^{\infty} \frac{z^2}{(1+z^2)^n}$$

If z is real, this is a geometric series with ratio $\frac{1}{1+z^2}$. For every real $z \neq 0$, $\left| \frac{1}{1+z^2} \right| < 1$, and therefore the series converges absolutely. At $z = 0$, every term is zero, so the series also converges. Hence, for real z , the series converges absolutely for every value of z .

However, this absolute convergence is not uniform near $z = 0$. The sum of the first n terms is

$$S_n(z) = z^2 + \frac{z^2}{1+z^2} + \frac{z^2}{(1+z^2)^2} + \cdots + \frac{z^2}{(1+z^2)^{n-1}}$$

Using the formula for a finite geometric series, this becomes

$$S_n(z) = 1 + z^2 - \frac{1}{(1+z^2)^{n-1}}$$

For fixed real $z \neq 0$, $\lim_{n \rightarrow \infty} S_n(z) = 1 + z^2$. But at $z = 0$, every term of the original series is zero, and hence $S_n(0) = 0$ for every n . Therefore, $S(0) = 0$. Thus, the limiting function is

$$S(z) = \begin{cases} 1 + z^2, & z \neq 0, \\ 0, & z = 0. \end{cases}$$

This function is discontinuous at $z = 0$. Since a uniformly convergent series of continuous functions has a continuous sum, the convergence cannot be uniform in any interval containing 0.

There is an important subtlety here. The authors write the variable as z , which usually indicates a complex variable. However, the statement that the series

$$\sum_{n=1}^{\infty} \frac{z^2}{(1+z^2)^n}$$

converges absolutely for all z is correct only when z is being treated as real. For complex z , the ratio test would require $\left| \frac{1}{1+z^2} \right| < 1$, or equivalently $|1+z^2| > 1$. This condition is not satisfied throughout any full complex neighborhood of 0. For example, if $z = i/2$, then $1+z^2 = 1 - \frac{1}{4} = \frac{3}{4}$, and so $\left| \frac{1}{1+z^2} \right| = \frac{4}{3} > 1$.

Thus, in this example, the author's statement should be understood as a real-variable illustration of the distinction between absolute convergence and uniform convergence.

Conversely, a series may be uniformly convergent without being absolutely convergent. Consider

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{z^2 + n}$$

where z is real. The series of moduli is

$$\sum_{n=1}^{\infty} \frac{1}{z^2 + n}$$

For each real z , this behaves like the harmonic series for large n , and therefore diverges. Hence the original series is not absolutely convergent.

Nevertheless, the original series is uniformly convergent for all real z . The terms $\frac{1}{z^2+n}$ are positive, decrease as n increases, and satisfy $\frac{1}{z^2+n} \leq \frac{1}{n}$.

Since the series is alternating with decreasing terms, the alternating-series estimate gives

$$|R_n(z)| \leq \frac{1}{z^2 + n + 1} \leq \frac{1}{n + 1}$$

[**Note:** Here we use the standard remainder estimate for alternating series. If

$$a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$$

and $a_n \rightarrow 0$, then the remainder after n terms in the alternating series

$$a_1 - a_2 + a_3 - a_4 + \cdots$$

has absolute value not exceeding the first omitted term:

$$|R_n| \leq a_{n+1}$$

This follows because the successive partial sums alternately overestimate and underestimate the limiting sum, while the size of the oscillation is bounded by the next term.]

Given any positive number ε , choose N so large that $\frac{1}{N+1} < \varepsilon$. Then, whenever $n > N$, $|R_n(z)| < \varepsilon$ for every real value of z . The same N works for all z , so the convergence is uniform.

This example shows that uniform convergence does not require absolute convergence.

The distinction may be summarized as follows. Absolutely convergent series behave, in many respects, like finite sums: their terms may often be rearranged or grouped safely. Uniformly convergent series behave, in many respects, like finite sums of functions: if the individual terms are continuous, then the sum is continuous, and under suitable hypotheses the series may be integrated term by term. The two ideas are related, but they measure different kinds of control over an infinite series.

4.3.4 A Condition, Due to Weierstrass, for Uniform Convergence

A sufficient, though not necessary, condition for the uniform convergence of a series may be stated as follows.

If, for all values of z within a domain, the absolute values (*i.e., complex moduli*) of the terms of a series

$$S = u_1(z) + u_2(z) + u_3(z) + \cdots$$

are respectively less than the corresponding terms in a convergent series of positive terms

$$T = M_1 + M_2 + M_3 + \cdots,$$

where M_n is independent of z , then the series S is uniformly convergent in this region.⁴⁷

In modern terminology, this is the Weierstrass M-test.

This follows from the fact that, since the series T is convergent, it is always possible to choose n so that the remainder after the first n terms of T is less than any assigned positive number ϵ . Therefore, the absolute value of the remainder after the first n terms of S is also less than ϵ . Since the value of n thus found is independent of z , it follows, by Section 4.3.1, that the series S is uniformly convergent. Moreover, by Section 3.3.4, the series S also converges absolutely.

Example. The series

$$\cos z + \frac{1}{2^2} \cos^2 z + \frac{1}{3^2} \cos^3 z + \cdots$$

is uniformly convergent for all real values of z , because the absolute values of its terms are not greater than the corresponding terms of the convergent series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots,$$

whose terms are positive constants.

4.3.4.1 Uniformity of Convergence of Infinite Products

A convergent product

$$\prod_{n=1}^{\infty} (1 + u_n(z))$$

is said to converge uniformly in a domain of values of z if, given ϵ , we can find m , independent of z , such that

⁴⁷ Weierstrass, *Abhandlungen aus der Functionenlehre*, p. 70. The test given by this condition is usually described, for example by Osgood, *Annals of Mathematics*, VII (1889), p. 130, as the *M*-test.

$$\left| \prod_{n=1}^{m+p} (1 + u_n(z)) - \prod_{n=1}^m (1 + u_n(z)) \right| < \epsilon$$

for all positive integral values of p .⁴⁸

The only condition for uniformity of convergence of infinite products which will be used in this work is the following: the product converges uniformly if

$$|u_n(z)| < M_n,$$

where M_n is independent of z , and the series

$$\sum_{n=1}^{\infty} M_n$$

converges.

To prove the validity of this condition, observe that

$$\prod_{n=1}^{\infty} (1 + M_n)$$

converges by Section 3.7. Therefore, we can choose m such that

$$\prod_{n=1}^{m+p} (1 + M_n) - \prod_{n=1}^m (1 + M_n) < \epsilon$$

Then, putting

$$A(z) = \left| \prod_{n=1}^{m+p} (1 + u_n(z)) - \prod_{n=1}^m (1 + u_n(z)) \right|$$

Split the longer product into two pieces as follows

$$\prod_{n=1}^{m+p} (1 + u_n(z)) = \left(\prod_{n=1}^m (1 + u_n(z)) \right) \left(\prod_{n=m+1}^{m+p} (1 + u_n(z)) \right)$$

and substitute into the formula for $A(z)$ to get

$$A(z) = \left| \prod_{n=1}^m (1 + u_n(z)) \left[\prod_{n=m+1}^{m+p} (1 + u_n(z)) - 1 \right] \right|$$

Hence,

$$A(z) \leq \prod_{n=1}^m (1 + M_n) \left[\prod_{n=m+1}^{m+p} (1 + M_n) - 1 \right]$$

⁴⁸ The definition is, effectively, that given by Osgood, *Funktionentheorie*, p. 462. The condition here given for uniformity of convergence is also established in that work.

The index n in each product is only a dummy index, the second product may equivalently be written with a different index, say j , and the product from 1 to m multiplied by the product from $m + 1$ to $m + p$ combines into the product from 1 to $m + p$, and so,

$$A(z) \leq \prod_{n=1}^{m+p} (1 + M_n) - \prod_{n=1}^m (1 + M_n) < \epsilon$$

The choice of m is independent of z , and hence the infinite product converges uniformly.

The proof is the product analogue of the Weierstrass M-test for series. The ordinary M-test controls the tails of a series by comparing them with the tails of a convergent numerical series. Here the same idea is applied to products: the factors $1 + u_n(z)$ are controlled by the numerical factors $1 + M_n$, and the convergence of the numerical product provides a bound that is independent of z .

4.3.5 Hardy's Tests for Uniform Convergence⁴⁹

[The following is an extensive rewrite of §3 35 of Whittaker and Watson.]

The reader will see, from Section 3.3.1, that the following is a uniform version of Dirichlet's test. Suppose that, in a given domain,

$$\left| \sum_{n=1}^p a_n(z) \right| < k$$

where $a_n(z)$ is real, and where k is finite and independent of both p and z . In other words, the partial sums of the series $\sum_{n=1}^{\infty} a_n(z)$ are bounded by the same constant k throughout the whole domain.

Further, suppose also that $f_n(z) \geq f_{n+1}(z)$ and that $f_n(z) \rightarrow 0$ uniformly as $n \rightarrow \infty$. The first condition says that, for each fixed z , the sequence $f_n(z)$ decreases as n increases. The second condition says something stronger than ordinary pointwise convergence: given any positive number ϵ , we can choose N , independent of z , such that $f_n(z) < \epsilon$ whenever $n > N$, for all z in the domain.

Under these assumptions, the series $\sum_{n=1}^{\infty} a_n(z) f_n(z)$ converges uniformly in the domain.

The idea is that the bounded partial sums of $a_n(z)$ provide cancellation, while the factors $f_n(z)$ decrease uniformly to zero. Thus, the tails of the product series can be made uniformly small, not merely small for each individual value of z . This is the same mechanism as Dirichlet's test for ordinary convergence, but with the additional requirement that the bounds used in the argument do not depend on z .

Also, if $k \geq u_n(z) \geq u_{n+1}(z) \geq 0$, where k is independent of z , and $\sum_{n=1}^{\infty} a_n(z)$ converges uniformly, then $\sum_{n=1}^{\infty} a_n(z) u_n(z)$ converges uniformly.

⁴⁹ G. H. Hardy, "Theorems concerning infinite series," *Proceedings of the London Mathematical Society*, Series 2, Vol. 4 (1907), pp. 247-265. W&W note that these results, which are generalizations of Abel's theorem, though well known, do not appear to have been published before 1907. From their resemblance to the tests of Dirichlet and Abel for convergence, Bromwich proposes to call them Dirichlet's and Abel's tests, respectively.

To prove the latter result, let $\varepsilon > 0$ be given. Since $\sum_{n=1}^{\infty} a_n(z)$ converges uniformly, we can choose an integer N , independent of z , such that for every $m \geq N$, all sums

$$a_{m+1}(z), \quad a_{m+1}(z) + a_{m+2}(z), \quad \dots, \quad a_{m+1}(z) + a_{m+2}(z) + \dots + a_{m+p}(z)$$

are numerically less than $\frac{\varepsilon}{k}$. That is, $|\sum_{n=m+1}^{m+p} a_n(z)| < \frac{\varepsilon}{k}$ for all $m \geq N$, all positive integers p , and all z in the domain.

By the corollary to Abel's inequality, we have

$$\left| \sum_{n=m+1}^{m+p} a_n(z) u_n(z) \right| < \frac{\varepsilon}{k} u_{m+1}(z).$$

To see how the corollary is being applied, regard the tail beginning at $m+1$ as a new finite sum. That is, set $b_r(z) = a_{m+r}(z)$, $w_r(z) = u_{m+r}(z)$, $r = 1, 2, \dots, p$. Then

$$\sum_{n=m+1}^{m+p} a_n(z) u_n(z) = \sum_{r=1}^p b_r(z) w_r(z)$$

Since the original series $\sum_{n=1}^{\infty} a_n(z)$ converges uniformly, we can choose m sufficiently large, independently of z , so that every partial sum of this tail satisfies

$$\left| \sum_{r=1}^q b_r(z) \right| = \left| \sum_{n=m+1}^{m+q} a_n(z) \right| < \frac{\varepsilon}{k}, \quad q = 1, 2, \dots, p$$

Thus, in Abel's corollary, we may take $A = \frac{\varepsilon}{k}$.

Also, since $k \geq u_n(z) \geq u_{n+1}(z) \geq 0$, the sequence $u_n(z)$ is decreasing and nonnegative. Hence

$$\sum_{r=1}^{p-1} |w_{r+1}(z) - w_r(z)| + |w_p(z)| = \sum_{r=1}^{p-1} (w_r(z) - w_{r+1}(z)) + w_p(z)$$

This sum telescopes to $w_1(z) = u_{m+1}(z)$. Therefore, Abel's corollary gives

$$\left| \sum_{n=m+1}^{m+p} a_n(z) u_n(z) \right| < \frac{\varepsilon}{k} u_{m+1}(z)$$

Since $0 \leq u_{m+1}(z) \leq k$, we have $\frac{\varepsilon}{k} u_{m+1}(z) \leq \varepsilon$. Therefore,

$$\left| \sum_{n=m+1}^{m+p} a_n(z) u_n(z) \right| < \varepsilon$$

Since the required choice of m is independent of z , the series $\sum_{n=1}^{\infty} a_n(z) u_n(z)$ converges uniformly.

Example 1. Show that, if $\delta > 0$, the series

$$\sum_{n=1}^{\infty} \frac{\cos n\theta}{n}, \quad \sum_{n=1}^{\infty} \frac{\sin n\theta}{n}$$

converge uniformly in the range $\delta < \theta < 2\pi - \delta$.

Obtain the corresponding result for the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n \cos n\theta}{n}, \quad \sum_{n=1}^{\infty} \frac{(-1)^n \sin n\theta}{n},$$

by writing $\theta + \pi$ for θ .

Solution: We use Hardy's uniform version of Dirichlet's test. The relevant point is that the partial sums of $\sum_{n=1}^{\infty} e^{in\theta}$ are uniformly bounded when θ stays away between 0 and 2π . Using the formula for the sum of a finite geometric series, we have

$$\sum_{n=1}^p e^{in\theta} = e^{i\theta} \frac{1 - e^{ip\theta}}{1 - e^{i\theta}}$$

Taking moduli,

$$\left| \sum_{n=1}^p e^{in\theta} \right| = |e^{i\theta}| \frac{|1 - e^{ip\theta}|}{|1 - e^{i\theta}|}$$

Since $|e^{i\theta}| = 1$ and $|1 - e^{ip\theta}| \leq 2$, we obtain

$$\left| \sum_{n=1}^p e^{in\theta} \right| \leq \frac{2}{|1 - e^{i\theta}|} = \frac{2}{2 \left| \sin \frac{\theta}{2} \right|}$$

In the range $0 < \delta < \theta < 2\pi - \delta$, $\left| \sin \frac{\theta}{2} \right| \geq \sin \frac{\delta}{2}$ and therefore,

$$\left| \sum_{n=1}^p e^{in\theta} \right| \leq \frac{1}{\sin (\delta/2)}$$

This bound is independent of both p and θ . Hence the partial sums of $\sum_{n=1}^{\infty} e^{in\theta}$ are uniformly bounded in the given range. Taking real and imaginary parts, it follows that the partial sums of $\sum_{n=1}^{\infty} \cos n\theta$ and $\sum_{n=1}^{\infty} \sin n\theta$ are also uniformly bounded in the same range.

Next, set $f_n(\theta) = \frac{1}{n}$. Then $f_n(\theta) \geq f_{n+1}(\theta)$ and $f_n(\theta) \rightarrow 0$ uniformly, since $\frac{1}{n}$ does not depend on θ . Therefore, by Hardy's uniform Dirichlet test, $\sum_{n=1}^{\infty} \frac{\cos n\theta}{n}$ and $\sum_{n=1}^{\infty} \frac{\sin n\theta}{n}$ converge uniformly for $\delta < \theta < 2\pi - \delta$. This proves the desired result.

For the corresponding alternating series, $\sum_{n=1}^{\infty} \frac{(-1)^n \cos n\theta}{n}$ and $\sum_{n=1}^{\infty} \frac{(-1)^n \sin n\theta}{n}$ we use the identities

$$(-1)^n \cos n\theta = \cos n(\theta + \pi), \quad (-1)^n \sin n\theta = \sin n(\theta + \pi)$$

For the alternating series, replacing θ by $\theta + \pi$ shifts the exceptional point. The original argument fails when θ is near 0 or 2π , because then $e^{i\theta}$ is near 1. After replacing θ by $\theta + \pi$, the corresponding failure occurs when $e^{i(\theta+\pi)}$ is near 1, that is, when θ is near π . Hence the alternating series converge uniformly on intervals that remain a positive distance away from $\theta = \pi$.

Example 2. If, when $a \leq x \leq b$, $|\omega_n(x)| < k_1$ for all positive integers n , and

$$\sum_{n=1}^{\infty} |\omega_{n+1}(x) - \omega_n(x)| < k_2,$$

where k_1 and k_2 are independent of n and x , and if $\sum_{n=1}^{\infty} a_n$ is a convergent series independent of x , then $\sum_{n=1}^{\infty} a_n \omega_n(x)$ converges uniformly when $a \leq x \leq b$. (Hardy)

Solution: To prove this, let $\varepsilon > 0$ be given. Since $\sum_{n=1}^{\infty} a_n$ converges, it satisfies Cauchy's condition. Therefore, we can choose m sufficiently large so that, for every positive integer p ,

$$\left| \sum_{n=m+1}^{m+p} a_n \right| < \varepsilon$$

This choice of m is independent of x , since the series $\sum_{n=1}^{\infty} a_n$ does not involve x .

Now consider the corresponding tail of the series $\sum_{n=1}^{\infty} a_n \omega_n(x)$. We want to estimate

$$\sum_{n=m+1}^{m+p} a_n \omega_n(x)$$

Apply the corollary to Abel's inequality to the finite sum beginning at $m + 1$. To match the notation of Abel's inequality, relabel the tail by setting

$$b_r = a_{m+r}, \quad w_r(x) = \omega_{m+r}(x), \quad r = 1, 2, \dots, p.$$

Then

$$\sum_{n=m+1}^{m+p} a_n \omega_n(x) = \sum_{r=1}^p b_r w_r(x)$$

The partial sums of the b_r 's satisfy

$$\left| \sum_{r=1}^q b_r \right| = \left| \sum_{n=m+1}^{m+q} a_n \right| < \varepsilon$$

for every $q = 1, 2, \dots, p$.

Thus, in Abel's inequality, letting $A = \varepsilon$ yields the following

$$\left| \sum_{r=1}^p b_r w_r(x) \right| \leq \varepsilon \left(\sum_{r=1}^{p-1} |w_{r+1}(x) - w_r(x)| + |w_p(x)| \right)$$

Returning to the original notation, this becomes

$$\left| \sum_{n=m+1}^{m+p} a_n \omega_n(x) \right| \leq \varepsilon \left(\sum_{n=m+1}^{m+p-1} |\omega_{n+1}(x) - \omega_n(x)| + |\omega_{m+p}(x)| \right)$$

By hypothesis, $|\omega_{m+p}(x)| < k_1$, and

$$\sum_{n=m+1}^{m+p-1} |\omega_{n+1}(x) - \omega_n(x)| \leq \sum_{n=1}^{\infty} |\omega_{n+1}(x) - \omega_n(x)| < k_2$$

Therefore,

$$\left| \sum_{n=m+1}^{m+p} a_n \omega_n(x) \right| < \varepsilon(k_1 + k_2).$$

Since k_1 and k_2 are fixed constants independent of x , this shows that the tails of the series

$\sum_{n=1}^{\infty} a_n \omega_n(x)$ can be made uniformly small. More formally, if we begin by choosing m so that

$$\left| \sum_{n=m+1}^{m+p} a_n \right| < \frac{\varepsilon}{k_1 + k_2}$$

then the above argument gives

$$\left| \sum_{n=m+1}^{m+p} a_n \omega_n(x) \right| < \varepsilon$$

The integer m is independent of x . Therefore, by the Cauchy criterion for uniform convergence, $\sum_{n=1}^{\infty} a_n \omega_n(x)$ converges uniformly on $a \leq x \leq b$.

The idea behind this example is that the factors $\omega_n(x)$ are allowed to vary with x , but not too wildly. The condition $|\omega_n(x)| < k_1$ keeps the individual factors bounded, while

$$\sum_{n=1}^{\infty} |\omega_{n+1}(x) - \omega_n(x)| < k_2$$

controls the total variation of the sequence $\omega_n(x)$. Abel's inequality then shows that multiplying a convergent numerical series by such a controlled sequence of functions preserves uniform convergence.

4.4 Discussion of a Particular Double Series

[The following is a significant rewrite of the passage from the authors.]

Let ω_1 and ω_2 be complex constants whose ratio ω_2/ω_1 is not real, and let $\alpha > 0$.

The series

$$\sum_{m,n} \frac{1}{(z + 2m\omega_1 + 2n\omega_2)^\alpha}$$

in which the summation extends over all positive, negative, and zero integral values of m and n , is of great importance in the theory of elliptic functions. At each of the points

$$z = -2m\omega_1 - 2n\omega_2$$

the series does not exist. It can be shown that the series converges absolutely for all other values of z if α is greater than 2, and that the convergence is uniform for those values of z such that

$$|z + 2m\omega_1 + 2n\omega_2| > \delta$$

for all integral values of m and n , where δ is an arbitrary positive number.⁵⁰

⁵⁰ The reader will easily define uniformity of convergence of double series; see Section 3.5.

Now, if m and n are not both zero, and if

$$|z + 2m\omega_1 + 2n\omega_2| \geq \delta > 0$$

for all integral values of m and n , then we can find a positive number C , depending on δ but not on z , such that

$$\left| \frac{1}{(z + 2m\omega_1 + 2n\omega_2)^\alpha} \right| < C \left| \frac{1}{(2m\omega_1 + 2n\omega_2)^\alpha} \right|$$

The constant C exists because z is assumed to stay at least a distance δ away from every point of the lattice at which the denominator vanishes. Thus, the shifted denominator $z + 2m\omega_1 + 2n\omega_2$ cannot become arbitrarily small. The constant C may depend on δ , since allowing z to come closer to a lattice point forces a larger comparison constant, but C is independent of z , m , and n within the region under consideration.

Consequently, by Section 4.3.4, the given series is absolutely and uniformly convergent in the domain considered if

$$\sum_{\substack{m,n \in \mathbb{Z} \\ (m,n) \neq (0,0)}} \frac{1}{|m\omega_1 + n\omega_2|^\alpha}$$

converges.

To discuss the convergence of the above series, let

$$\omega_1 = \alpha_1 + i\beta_1, \quad \omega_2 = \alpha_2 + i\beta_2$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are real. Since ω_2/ω_1 is not real,

$$\alpha_1\beta_2 - \alpha_2\beta_1 \neq 0$$

With the given substitutions, the series becomes

$$\sum_{\substack{m,n \in \mathbb{Z} \\ (m,n) \neq (0,0)}} \frac{1}{\{(\alpha_1 m + \alpha_2 n)^2 + (\beta_1 m + \beta_2 n)^2\}^{\alpha/2}}$$

We compare this with the simpler double series

$$S = \sum_{\substack{m,n \in \mathbb{Z} \\ (m,n) \neq (0,0)}} \frac{1}{(m^2 + n^2)^{\alpha/2}}$$

Before applying the comparison theorem, we must show that the denominator in the transformed series is bounded below by a positive constant multiple of $m^2 + n^2$. This is where the assumption that ω_2/ω_1 is not real is used. Put $\mu = \frac{n}{m}$, when $m \neq 0$. (The case $m = 0$ is handled separately and causes no difficulty, since then the quotient reduces to a positive constant involving ω_2 .) Consider the quotient of the corresponding denominator terms:

$$\frac{(\alpha_1 m + \alpha_2 n)^2 + (\beta_1 m + \beta_2 n)^2}{m^2 + n^2} = \frac{(\alpha_1 + \alpha_2 \mu)^2 + (\beta_1 + \beta_2 \mu)^2}{1 + \mu^2}$$

This expression is a continuous function of the real variable μ . Moreover, it cannot vanish. If it did vanish, then both $\alpha_1 + \alpha_2\mu = 0$ and $\beta_1 + \beta_2\mu = 0$ would hold for the same real value of μ . This would imply that $\alpha_1\beta_2 - \alpha_2\beta_1 = 0$, contrary to the assumption that ω_2/ω_1 is not real.

As $|\mu| \rightarrow \infty$, this quotient tends to $\alpha_2^2 + \beta_2^2 = |\omega_2|^2 > 0$. Hence the quotient has a positive lower bound on the extended real line, say $K > 0$. Therefore,

$$(\alpha_1 m + \alpha_2 n)^2 + (\beta_1 m + \beta_2 n)^2 \geq K(m^2 + n^2)$$

Raising both sides to the power $\alpha/2$, we obtain

$$\{(\alpha_1 m + \alpha_2 n)^2 + (\beta_1 m + \beta_2 n)^2\}^{\alpha/2} \geq K^{\alpha/2}(m^2 + n^2)^{\alpha/2}$$

Consequently,

$$\frac{1}{\{(\alpha_1 m + \alpha_2 n)^2 + (\beta_1 m + \beta_2 n)^2\}^{\alpha/2}} \leq \frac{1}{K^{\alpha/2}} \frac{1}{(m^2 + n^2)^{\alpha/2}}$$

Thus, the terms of the original double series are bounded above by a constant multiple of the terms of S . By the comparison theorem for double series (see Section 3.5), it is therefore enough to prove that S converges.

To prove that S converges, consider the sum over the first quadrant. By symmetry, it is enough to estimate

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{1}{(m^2 + n^2)^{\alpha/2}}$$

where the term $m = n = 0$ is omitted.

Separate this sum into three parts:

- the terms for which $m = n$;
- the terms for which $m > n$;
- the terms for which $m < n$.

For the diagonal terms $m = n$, we have $m^2 + n^2 = 2m^2$, and hence

$$\sum_{m=1}^{\infty} \frac{1}{(2m^2)^{\alpha/2}} = \frac{1}{2^{\alpha/2}} \sum_{m=1}^{\infty} \frac{1}{m^{\alpha}}$$

This series converges when $\alpha > 1$.

Next, consider the terms for which $m > n$. Then $0 \leq n \leq m - 1$, and $m^2 + n^2 \geq m^2$. Therefore,

$$\frac{1}{(m^2 + n^2)^{\alpha/2}} \leq \frac{1}{m^{\alpha}}$$

For each fixed m , there are m possible values of n , namely $n = 0, 1, \dots, m - 1$. Thus,

$$\sum_{n=0}^{m-1} \frac{1}{(m^2 + n^2)^{\alpha/2}} \leq \frac{m}{m^\alpha} = \frac{1}{m^{\alpha-1}}$$

Hence

$$\sum_{m=1}^{\infty} \sum_{n=0}^{m-1} \frac{1}{(m^2 + n^2)^{\alpha/2}} \leq \sum_{m=1}^{\infty} \frac{1}{m^{\alpha-1}}$$

This converges when $\alpha - 1 > 1$, that is, when $\alpha > 2$.

The case $m < n$ is handled in the same way.

Hence, the comparison series S converges when $\alpha > 2$. Therefore, the original double series is absolutely convergent. Moreover, by the Weierstrass condition in Section 4.3.4, the convergence is uniform for the specified range of values of z .

Example. Prove that the series

$$\sum \frac{1}{(m_1^2 + m_2^2 + \dots + m_r^2)^\mu}$$

in which the summation extends over all positive, negative, and zero integer values of $m_1, m_2, \dots, m_r$, except the set of simultaneous zero values, is absolutely convergent if

$$\mu > \frac{r}{2}$$

(Eisenstein, Journal für Mathematik, xxxv)

Solution: Since every term is positive, absolute convergence is the same as ordinary convergence.

We group the lattice (integer) points into shells according to the size of the largest absolute value among their coordinates. This converts the multiple series into a single series indexed by N . Each nonzero integer r -tuple belongs to exactly one shell, namely the shell for which

$$N = \max \{|m_1|, |m_2|, \dots, |m_r|\}$$

For a fixed positive integer N , consider all points satisfying this condition. For any such point, at least one coordinate has absolute value N . Hence,

$$m_1^2 + m_2^2 + \dots + m_r^2 \geq N^2$$

and therefore

$$\frac{1}{(m_1^2 + m_2^2 + \dots + m_r^2)^\mu} \leq \frac{1}{N^{2\mu}}$$

Next, we estimate how many integer points have $\max \{|m_1|, |m_2|, \dots, |m_r|\} = N$.

All such points lie in the hypercube $-N \leq m_j \leq N, j = 1, 2, \dots, r$.

The number of integer points **in** this cube is $(2N+1)^r$.

The points for which $\max |m_j| = N$ lie on the boundary shell of the r -dimensional cube $[-N, N]^r$.

The number of such lattice points is $(2N+1)^r - (2N-1)^r$ since this is the number of lattice points in the cube of radius N minus the number in the smaller cube of radius $N-1$.

This difference grows like a constant multiple of N^{r-1} . Hence there is a constant C , independent of N , such that $(2N+1)^r - (2N-1)^r \leq CN^{r-1}$.

Therefore, the total contribution from all terms with maximum coordinate equal to N is at most

$$CN^{r-1} \cdot \frac{1}{N^{2\mu}} = \frac{C}{N^{2\mu-r+1}}$$

Thus, the given multiple series is dominated by the single series

$$C \sum_{N=1}^{\infty} \frac{1}{N^{2\mu-r+1}}.$$

This is a p -series, and it converges when $2\mu - r + 1 > 1$. This condition is equivalent to $2\mu > r$, or $\mu > \frac{r}{2}$. Therefore, the original series is absolutely convergent whenever $\mu > \frac{r}{2}$. This proves the result.

4.5 The Concept of Uniformity

There are processes other than summing a series in which the idea of uniformity is important.

Let ε be an arbitrary positive number, and let $f(z, \zeta)$ be a function of the two variables z and ζ . Suppose that, for each point z of a closed region, the inequality

$$|f(z, \zeta)| < \varepsilon$$

is satisfied when ζ is chosen from a certain set of values, which we shall denote by ζ_z . The particular set of allowable values may, of course, depend on the particular value of z under consideration (thus the subscript z on ζ).

If a single set of values, denoted by ζ_0 , can be found such that every member of ζ_0 belongs to all the sets ζ_z , then the function $f(z, \zeta)$ is said to satisfy the inequality uniformly for all points z of the region.

In other words, the same choice of allowable values of ζ works simultaneously for every z in the region, rather than having to be chosen separately for each z .

Similarly, if a function $\varphi(z)$ possesses some property for every positive value of ε by virtue of the inequality

$$|f(z, \zeta)| < \varepsilon$$

then $\varphi(z)$ is said to possess that property uniformly if the same set of values of ζ can be chosen independently of z .

In addition to uniform convergence of series and products, we shall also have to consider uniform convergence of integrals and uniform continuity. For example, a series is uniformly convergent when

$$R_n(z) < \varepsilon$$

for all sufficiently large n , where the required integer n depends on ε but is independent of z .

Further, a function $f(z)$ is continuous in a closed region if, given ε , we can find a positive number η_z such that $|f(z + \zeta_z) - f(z)| < \varepsilon$ whenever $0 < |\zeta_z| < \eta_z$ and $z + \zeta$ is a point of the region.

The function will be uniformly continuous if we can find a positive number η , independent of z , such that $\eta < \eta_z$ and $|f(z + \zeta) - f(z)| < \varepsilon$ whenever $0 < |\zeta| < \eta$ and $z + \zeta$ is a point of the region. In this case, the common set ζ_0 consists of all ζ whose moduli are less than η .

We shall find later that continuity on a closed region involves uniformity of continuity. This is in marked contrast with convergence, since pointwise convergence does not, by itself, imply uniform convergence.

The essential distinction is whether the auxiliary quantity can be chosen once and for all. Ordinary continuity at a point allows η to depend on the point z . Uniform continuity requires one η that works throughout the whole region. Likewise, ordinary convergence of a series allows the number of terms needed to depend on z , while uniform convergence requires a single threshold that works for all z in the region.

4.6 The Modified Heine-Borel Theorem

The following theorem is of great importance in connection with properties of uniformity. We give a proof for a one-dimensional closed region.⁵¹

Theorem (modified Heine-Borel theorem). Given a straight line segment CD , and a rule by which, corresponding to each point P of CD , we can determine a closed interval $I(P)$ of CD , with P an interior point of $I(P)$,^{52 53} then the line segment CD can be divided into a finite number of closed intervals $J_1, J_2, \dots, J_k$ such that each interval J_r contains at least one point P_r , not an endpoint of J_r , with the property that no point of J_r lies outside the interval $I(P_r)$ associated (by means of the given law) with that point P_r .⁵⁴

The theorem is depicted in Figure 4. In the figure, each subinterval J_r is chosen so that it lies inside one of the intervals $I(P_r)$ associated with a point P_r in that subinterval.

Equivalently, each J_r is contained in the corresponding interval $I(P_r)$: $J_r \subseteq I(P_r)$.

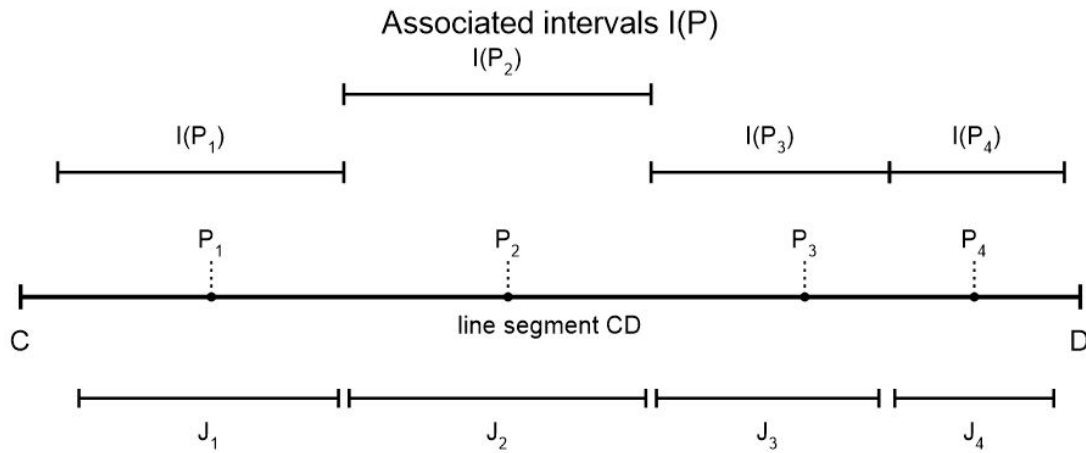
⁵¹ A formal proof of the theorem for a two-dimensional region will be found in Watson's *Complex Integration and Cauchy's Theorem* (Cambridge Mathematical Tracts, No. 15).

⁵² Examples of such rules associating intervals with points will be found in W&W §§3.61 and 5.13.

⁵³ Except when P is at C or D , in which case it is an endpoint.

⁵⁴ This statement of the Heine-Borel theorem, which is sometimes called the Borel-Lebesgue theorem, is due to Baker, *Proceedings of the London Mathematical Society*, Series 2, Vol. 1 (1904), p. 24. Hobson, *The Theory of Functions of a Real Variable* (1907), p. 87, points out that the theorem is practically given in Goursat's proof of Cauchy's theorem, *Transactions of the American Mathematical Society*, Vol. 1 (1900), p. 14. The ordinary form of the Heine-Borel theorem will be found in the treatise cited.

The intervals J_r form a finite subdivision of CD . Thus, adjacent intervals may share an endpoint, but their interiors are disjoint. Each J_r contains at least one point P_r , not an endpoint of J_r , such that $J_r \subseteq I(P_r)$.



Note: each J_r lies inside its associated $I(P_r)$.

Figure 4. Suitable intervals in the modified Heine-Borel theorem

A closed interval of the nature just described will be called a *suitable interval*, and will be said to satisfy condition (A).

If CD satisfies condition (A), what is required is already proved. If not, bisect CD . If either or both of the intervals into which CD is divided is not suitable, bisect that interval or those intervals.⁵⁵

The theorem says that a local choice made around each point of a closed interval can be replaced by finitely many choices that cover the whole interval in a controlled way. This is the compactness idea behind the Heine-Borel theorem: local information, if available around every point of a closed and bounded interval, can be reduced to a finite collection of intervals.

Proof: Thus, the process of bisecting intervals which are not suitable either will terminate or it will not. If it does terminate, the theorem is proved, for CD will have been divided into suitable intervals.

Suppose that the process does not terminate, and let an interval which can be divided into suitable intervals by the process of bisection just described be said to satisfy condition (B).

Then, by hypothesis, CD does not satisfy condition (B). Therefore, at least one of the bisected portions of CD does not satisfy condition (B). Take that one which does not. If neither satisfies condition (B), take the left-hand one. Bisect it and select that bisected part which does not satisfy condition (B).

This process of bisection and selection gives an unending sequence of intervals $s_0, s_1, s_2, \dots$ such that:

1. the length of s_n is $2^{-n}CD$;
2. no point of s_{n+1} is outside s_n ;
3. the interval s_n does not satisfy condition (A)

⁵⁵ A suitable interval is not to be bisected, since one of the parts into which it is divided might not be suitable.

Let the distances of the endpoints of s_n from C be x_n and y_n . Then

$$x_n \leq x_{n+1} < y_{n+1} \leq y_n.$$

Therefore, by Section 3.2, x_n and y_n have limits, and by condition (1) above, these limits are the same. Say this common limit is ξ .

Let Q be the point whose distance from C is ξ .

Since x_n and y_n denote distances from C , their common limit ξ is also a distance from C , measured along the segment CD . It therefore determines a unique point Q between C and D , namely the point satisfying $CQ = \xi$. Since $x_n \leq \xi \leq y_n$ for every n , this point Q lies in every interval s_n .

But, by hypothesis, there is a number δ_Q such that every point of CD whose distance from Q is less than δ_Q is a point of the associated interval $I(Q)$.

This follows because Q , like every point of CD , has an associated interval $I(Q)$, and by the hypothesis of the theorem Q is an interior point of that interval. Therefore, there is some positive distance on both sides of Q that remains inside $I(Q)$. Choosing δ_Q smaller than those distances ensures that every point of CD within distance δ_Q of Q lies in $I(Q)$. Notice that Q need not be the midpoint of $I(Q)$; it only needs to be an interior point.

Choose n so large that

$$2^{-n}CD < \delta_Q.$$

Then Q is either an internal point or an endpoint of s_n , and the distance of every point of s_n from Q is less than δ_Q . Therefore, the interval s_n satisfies condition (A), which is contrary to condition (3) above.

The hypothesis that the process of bisecting intervals does not terminate therefore involves a contradiction. Hence the process does terminate, and the theorem is proved. ■

*In modern terminology, the theorem just proved is closely related to the compactness of a closed interval. A set is called **bounded** if it fits inside some finite interval, and **closed** if it contains all of its limiting points. A set is called **compact** if every collection of open intervals that covers it contains a finite subcollection that still covers it. The Heine–Borel theorem says that, in the real line, a set is compact exactly when it is both closed and bounded. Thus, the closed interval $[a, b]$ is compact, and this compactness is the reason one can pass from many local choices of intervals to a finite collection that works for the whole interval.*

The modified version of the Heine–Borel theorem presented here is a partitioned form of the general Heine–Borel theorem. It does not merely assert the existence of finitely many associated intervals covering CD . It also produces a finite subdivision of CD into closed intervals J_r , whose interiors are disjoint, and whose adjacent intervals meet only at a common endpoint, with each J_r contained in one associated interval $I(P_r)$. This extra structure is useful in later arguments, especially when one wants to move across the interval one subinterval at a time.

In the two-dimensional form of the theorem⁵⁶, the interval CD is replaced by a closed two-dimensional region, the interval $I(P)$ by a circle⁵⁷ with center P , and the interval J_r by a square with sides parallel to the axes.

⁵⁶ The reader will see that a proof may be constructed on similar lines by drawing a square circumscribing the region and carrying out a process of dividing squares into four equal squares.

⁵⁷ Or the portion of the circle which lies inside the region.

4.6.1 Uniformity of Continuity

From the theorem just proved, it follows without difficulty that if a function $f(x)$ of a real variable x is continuous when $a \leq x \leq b$, then $f(x)$ is uniformly continuous⁵⁸ throughout the range $a \leq x \leq b$.

*The important point is that the interval is both **closed and bounded**. Continuity at each point gives a local δ that may depend on the point, but the Heine–Borel theorem applies to a closed and bounded interval and allows these local intervals to be reduced to finitely many intervals covering $[a, b]$. From finitely many intervals one can choose a single positive lower bound, which then gives a δ that is independent of the point x . This is what turns ordinary continuity into uniform continuity on $[a, b]$.*

Proof: For let ε be an arbitrary positive number. Then, by virtue of the continuity of $f(x)$, corresponding to any value of x , we can find a positive number δ_x , depending on x , such that

$$|f(x') - f(x)| < \frac{1}{4}\varepsilon$$

for all values of x' such that

$$|x' - x| < \delta_x$$

Here the associated interval $I(x)$ in the Heine–Borel theorem is the interval around x supplied by the continuity of f at x , chosen so that $|f(x') - f(x)| < \frac{1}{4}\varepsilon$ whenever $x' \in I(x)$. Thus, the theorem gives finitely many closed subintervals of $[a, b]$, each lying inside one such associated interval.

Then by Section 4.6, we can divide the range (a, b) into a **finite** number of closed intervals with the property that, in each interval, there is a number x_1 such that

$$|f(x') - f(x_1)| < \frac{1}{4}\varepsilon$$

whenever x' lies in the interval in which x_1 lies.

Let δ_0 be the length of the smallest of these intervals, and let ξ, ξ' be any two numbers in the closed range (a, b) such that

$$|\xi - \xi'| < \delta_0$$

Then ξ and ξ' lie in the same interval or in adjacent intervals.

If they lie in adjacent intervals, let ξ_0 be the common endpoint. Then we can find numbers x_1, x_2 , one in each interval, such that

$$|f(\xi) - f(x_1)| < \frac{1}{4}\varepsilon, \quad |f(\xi_0) - f(x_1)| < \frac{1}{4}\varepsilon$$

and

$$|f(\xi') - f(x_2)| < \frac{1}{4}\varepsilon, \quad |f(\xi_0) - f(x_2)| < \frac{1}{4}\varepsilon$$

Thus,

$$|f(\xi) - f(\xi')|$$

⁵⁸ This result is due to Heine; see *Journal für Math.*, lxxi (1870), p. 361, and lxxiv (1872), p. 188.

$$= |f(\xi) - f(x_1)| - |f(\xi_0) - f(x_1)| - |f(\xi') - f(x_2)| + |f(\xi_0) - f(x_2)| < \varepsilon$$

If ξ and ξ' lie in the same interval, the argument is even simpler. Using the point x_1 associated with that interval, we obtain

$$|f(\xi) - f(\xi')| \leq |f(\xi) - f(x_1)| + |f(x_1) - f(\xi')| < \frac{1}{4}\varepsilon + \frac{1}{4}\varepsilon = \frac{1}{2}\varepsilon < \varepsilon$$

In either case, we have shown that, for any number ξ in the range,

$$|f(\xi) - f(\xi + \zeta)| < \varepsilon$$

whenever $\xi + \zeta$ is in the range and $-\delta_0 < \zeta < \delta_0$ where δ_0 is independent of ξ . The uniformity of the continuity is therefore established. ■

Corollary (i). From the two-dimensional form of the theorem of Section 4.6, we can prove that a function of a complex variable, continuous at all points of a closed region of the Argand diagram, is uniformly continuous throughout that region.

Corollary (ii). A function $f(x)$ which is continuous throughout the range

$$a \leq x \leq b$$

is bounded in the range. That is to say, we can find a number K , independent of x , such that

$$|f(x)| < K$$

for all points x in the range.

Proof: Let n be the number of parts into which the range is divided. Let

$$a, \xi_1, \xi_2, \dots, \xi_{n-1}, b$$

be their endpoints. Then, if x is any point of the r -th interval, we can find numbers

$$x_1, x_2, \dots, x_r$$

such that $|f(a) - f(x_1)| < \frac{1}{4}\varepsilon$, $|f(x_1) - f(\xi_1)| < \frac{1}{4}\varepsilon$, $|f(\xi_1) - f(x_2)| < \frac{1}{4}\varepsilon$, $|f(x_2) - f(\xi_2)| < \frac{1}{4}\varepsilon$, and so on, up to $|f(\xi_{r-1}) - f(x)| < \frac{1}{4}\varepsilon$.

The point is that we can move from a to x through the subdivision one interval at a time. Each step changes the value of f by less than $\frac{1}{4}\varepsilon$. Since x lies in the r -th interval, at most $2r$ such small changes are needed to get from a to x . Therefore, by the triangle inequality, the total change from $f(a)$ to $f(x)$ is less than $2r \cdot \frac{1}{4}\varepsilon = \frac{1}{2}r\varepsilon$.

Since $r \leq n$, where n is the total number of intervals in the subdivision, $|f(a) - f(x)| < \frac{1}{2}n\varepsilon$.

Finally, $|f(x)| = |f(a) + (f(x) - f(a))| \leq |f(a)| + |f(x) - f(a)| < |f(a)| + \frac{1}{2}n\varepsilon$.

This gives a bound independent of x , namely $K = |f(a)| + \frac{1}{2}n\varepsilon$.

So, the function is bounded on the interval. ■

The corresponding theorem for functions of complex variables is left to the reader.

4.6.2 Continuous Functions Attain Their Bounds

A real function, of a real variable, continuous in a closed interval, attains its upper bound.

Proof: Let $f(x)$ be a real continuous function of x for $a \leq x \leq b$. By Corollary (ii) in the previous section, $f(x)$ is bounded on this interval. Hence, the set of values taken by $f(x)$ has an upper bound. Let α denote the least upper bound of these values. Thus $f(x) \leq \alpha$ for all $x \in [a, b]$, while for every positive number δ , there exists some $x \in [a, b]$ such that $f(x) > \alpha - \delta$.

Then α is called the **upper bound** of $f(x)$, and the theorem states that a number x' in the range can be found such that $f(x') = \alpha$. For, no matter how small δ may be, we can find values of x for which $|f(x) - \alpha|^{-1} > \delta^{-1}$. Therefore, $|f(x) - \alpha|^{-1}$ is not bounded in the range. Hence, by Section 4.6.1, Corollary (ii), it is not continuous at some point or points of the range. But since $|f(x) - \alpha|$ is continuous at all points of the range, its reciprocal is continuous at all points of the range, except those points at which $f(x) = \alpha$. Therefore, $f(x) = \alpha$ at some point of the range. The theorem is therefore proved. ■

Corollary (i). The lower bound of a continuous function may be defined in a similar manner, and a continuous function attains its lower bound.

Corollary (ii). If $f(z)$ is a function of a complex variable continuous in a closed region, then $|f(z)|$ attains its upper bound.

4.6.3 Intermediate Value Theorem

A real function of a real variable, continuous in a closed interval, attains all values between its upper and lower bounds.

Proof: [The following completely replaces the proof given by the authors.]

Let $f(x)$ be a continuous function on the closed interval $a \leq x \leq b$.

Let M and m be the upper and lower bounds of $f(x)$ on this interval. By the preceding section, $f(x)$ attains both bounds. Thus, there are points $\bar{x}$ and $\underline{x}$ in $[a, b]$ such that $f(\bar{x}) = M$, $f(\underline{x}) = m$.

Let μ be any number between m and M . We shall show that there is some point c between $\underline{x}$ and $\bar{x}$ such that $f(c) = \mu$.

Consider the function $g(x) = f(x) - \mu$. Since $f(x)$ is continuous, $g(x)$ is also continuous. Moreover,

$$g(\underline{x}) = f(\underline{x}) - \mu = m - \mu < 0$$

while

$$g(\bar{x}) = f(\bar{x}) - \mu = M - \mu > 0.$$

Thus $g(x)$ has opposite signs at the two points $\underline{x}$ and $\bar{x}$.

If $\underline{x} < \bar{x}$, apply the intermediate sign-change argument on the interval $[\underline{x}, \bar{x}]$.

If $\bar{x} < \underline{x}$, apply the same argument on $[\bar{x}, \underline{x}]$.

In either case, continuity prevents $g(x)$ from passing from a negative value to a positive value without taking the value zero somewhere in between. Hence, there exists a point c between $\underline{x}$ and $\bar{x}$ such that $g(c) = 0$. Therefore, $f(c) - \mu = 0$, and so, $f(c) = \mu$. Since μ was an arbitrary number satisfying $m < \mu < M$, it follows that $f(x)$ attains every value between its lower and upper bounds. ■

This is the usual intermediate value theorem in the special case where the interval endpoints are chosen to be points at which the function attains its lower and upper bounds. Whittaker and Watson's proof approaches the result through uniform continuity and bounds, but the essential idea is the same: a continuous real-valued function cannot move from below μ to above μ without equaling μ at some intermediate point.

4.6.4 The Fluctuation of a Function of a Real Variable⁵⁹

Let $f(x)$ be a real bounded function, defined when $a \leq x \leq b$. Let $a \leq x_1 \leq x_2 \leq \dots \leq x_n \leq b$. Then

$$|f(a) - f(x_1)| + |f(x_1) - f(x_2)| + \dots + |f(x_n) - f(b)|$$

is called the fluctuation of $f(x)$ in the range (a, b) for the set of subdivisions $x_1, x_2, \dots, x_n$.

If the fluctuation has an upper bound F_a^b , independent of n , for all choices of $x_1, x_2, \dots, x_n$, then $f(x)$ is said to have limited total fluctuation in the range (a, b) . F_a^b is called the total fluctuation in the range.

Example 1. If $f(x)$ be monotonic⁶⁰ in the range (a, b) , its total fluctuation in the range is

$$|f(a) - f(b)|$$

Example 2. A function with limited total fluctuation can be expressed as the difference of two positive increasing monotonic functions.

These functions may be taken to be $\frac{1}{2}\{F_a^x + f(x)\}$ and $\frac{1}{2}\{F_a^x - f(x)\}$.

Example 3. If $f(x)$ have limited total fluctuation in the range (a, b) , then the limits $f(x \pm 0)$ exist at all points in the interior of the range. [See the examples in Section 4.2.]

Example 4. If $f(x), g(x)$ have limited total fluctuation in the range (a, b) , so has $f(x)g(x)$.

For $|f(x')g(x') - f(x)g(x)| \leq |f(x')| |g(x') - g(x)| + |g(x)| |f(x') - f(x)|$

and so, the total fluctuation of $f(x)g(x)$ cannot exceed $gF_a^b + fG_a^b$ where f, g are the upper bounds of $|f(x')|, |g(x')|$.

In modern terminology, a function with limited total fluctuation is usually called a function of bounded variation. Examples 2–4 record some standard properties of such functions: they can be decomposed into the difference of two increasing functions, they have one-sided limits at interior points, and they are closed under multiplication.

⁵⁹ The terminology of this section is partly that of Hobson, *The Theory of Functions of a Real Variable* (1907), and partly that of Young, *The Theory of Sets of Points* (1906).

⁶⁰ The function is monotonic if $\frac{f(x) - f(x')}{x - x'}$ is one-signed or zero for all pairs of different values of x and x' .

4.7 Uniformity of Convergence of Power Series

Let the power series $a_0 + a_1 z + \cdots + a_n z^n + \cdots$ converge absolutely when $z = z_0$.

Then, if $|z| \leq |z_0|$, $|a_n z^n| \leq |a_n z_0^n|$.

But since $\sum_{n=0}^{\infty} |a_n z_0^n|$ converges, it follows, by Section 4.3.4, that $\sum_{n=0}^{\infty} a_n z^n$ converges uniformly with regard to the variable z when $|z| \leq |z_0|$.

Hence, by Section 4.3.2, a power series is a continuous function of the variable throughout the closed region formed by the interior and boundary of any circle concentric with the circle of convergence and of smaller radius (Section 3.6).

4.7.1 Abel's Theorem⁶¹ on Continuity up to the Circle of Convergence

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series, whose radius of convergence is unity, and let it be such that $\sum_{n=0}^{\infty} a_n$ converges, and let $0 \leq x < 1$; then Abel's theorem asserts that

$$\lim_{x \rightarrow 1} \left(\sum_{n=0}^{\infty} a_n x^n \right) = \sum_{n=0}^{\infty} a_n$$

For, with the notation of Section 4.3.5, the function x^n satisfies the conditions laid on $u_n(x)$, when $0 \leq x \leq 1$, consequently $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges uniformly throughout the range $0 \leq x \leq 1$, it is therefore, by Section 4.3.2, a continuous function of x throughout the range, and so

$\lim_{x \rightarrow 1-0} f(x) = f(1)$, which is the theorem stated.

4.7.2 Abel's Theorem⁶² on Multiplication of Series

This is a modification of the theorem of Section 3.5.3 for absolutely convergent series.

Let $c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0$

Then the convergence of $\sum_{n=0}^{\infty} a_n$, $\sum_{n=0}^{\infty} b_n$, and $\sum_{n=0}^{\infty} c_n$ is a sufficient condition that

$$\left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} c_n$$

For, let

$$A(x) = \sum_{n=0}^{\infty} a_n x^n, \quad B(x) = \sum_{n=0}^{\infty} b_n x^n, \quad C(x) = \sum_{n=0}^{\infty} c_n x^n$$

⁶¹ *Journal für Math.* 1 (1826), pp. 311–339, theorem IV. Abel's proof employs directly the arguments by which the theorems of Sections 4.3.2 and 4.3.5 are proved. In the case when $\sum a_n$ converges, the theorem is obvious from Section 4.7.

⁶² *Journal für Math.* 1 (1826), pp. 311–339, Theorem VI. This is Abel's original proof. In some textbooks a more elaborate proof, by the use of Cesàro's sums, is given.

Then the series for $A(x)$, $B(x)$, $C(x)$ are absolutely convergent when $|x| < 1$, (Section 3.6), and consequently, by Section 3.5.3, $A(x)B(x) = C(x)$ when $0 < x < 1$; therefore, by Example 2 in Section 3.2,

$$\left\{ \lim_{x \rightarrow 1-0} A(x) \right\} \left\{ \lim_{x \rightarrow 1-0} B(x) \right\} = \left\{ \lim_{x \rightarrow 1-0} C(x) \right\}$$

provided that these three limits exist; but, by Section 4.7.1, these three limits are

$\sum_{n=0}^{\infty} a_n$, $\sum_{n=0}^{\infty} b_n$, $\sum_{n=0}^{\infty} c_n$ and the theorem is proved.

4.7.3 Power Series Which Vanish Identically

If a convergent power series vanishes for all values of z such that $|z| < r_1$, where $r_1 > 0$, then all the coefficients in the power series vanish.

For, if not, let a_m be the first coefficient which does not vanish.

Then $a_m + a_{m+1}z + a_{m+2}z^2 + \dots$ vanishes for all values of z , zero excepted, and converges absolutely when $|z| \leq r < r_1$.

Since the tail after a_m is absolutely convergent for $|z| \leq r$, there is a constant $M \geq 0$ such that

$$|a_{m+1} + a_{m+2}z + a_{m+3}z^2 + \dots| \leq M$$

for $|z| \leq r$. Hence,

$$|a_{m+1}z + a_{m+2}z^2 + a_{m+3}z^3 + \dots| \leq |z| M$$

Choosing $\delta \leq r$ so that $\delta M \leq \frac{1}{2}|a_m|$, the tail satisfies

$$|a_{m+1}z + a_{m+2}z^2 + \dots| \leq \frac{1}{2}|a_m|$$

whenever $|z| \leq \delta$. Therefore, it cannot cancel a_m , and the expression

$$a_m + a_{m+1}z + a_{m+2}z^2 + \dots$$

cannot vanish for $0 < |z| \leq \delta$. This contradicts the hypothesis. Therefore, all coefficients vanish.

Corollary 1. We may “equate corresponding coefficients” in two power series whose sums are equal throughout the region $|z| < \delta$, where $\delta > 0$.

Corollary 2. We may also equate coefficients in two power series which are proved equal only when z is real.

Table 4. Key Results from Section 4: Continuous Functions and Uniform Convergence

Section	Result	Main Point
4.2	Continuity of a real function	A function is continuous at a point if small changes in the input produce small changes in the value of the function.
4.2.2	Continuity of functions of a complex variable	Continuity extends naturally to functions of a complex variable by using the modulus as the measure of distance.
4.3	Uniform convergence	A series of functions converges uniformly when the same index works throughout the whole region, not separately at each point.
4.3.2	Continuity and uniform convergence	A uniformly convergent series of continuous functions has a continuous sum. Non-uniform convergence can produce discontinuities.
4.3.4	Weierstrass condition	If the terms of a series are bounded in modulus by the terms of a convergent numerical series, then the series converges uniformly.
4.5	Uniformity	A property holds uniformly when the auxiliary choice needed to establish it can be made independently of the point under consideration.
4.6	Modified Heine–Borel theorem	A local choice of intervals around each point of a closed interval can be reduced to finitely many suitable intervals. This is the compactness principle used in later proofs.
4.6.1	Uniform continuity on a closed interval	A function continuous on a closed interval is uniformly continuous there. As a corollary, such a function is bounded.
4.6.2– 4.6.3	Extreme value and intermediate value theorems	A continuous real function on a closed interval attains its upper and lower bounds and takes every value between them.
4.7	Power series	A power series is uniformly convergent inside any closed circle lying strictly within its circle of convergence. Hence it defines a continuous function there.
4.7.1	Abel’s theorem	Under suitable convergence at the endpoint, a power series is continuous as the variable approaches the boundary point from within the interval.
4.7.3	Identity theorem for power series	If a convergent power series vanishes throughout a neighborhood of zero, then all of its coefficients vanish.

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Miscellaneous Examples

1. Show that the series $\sum_{n=1}^{\infty} \frac{z^{n-1}}{(1-z^n)(1-z^{n+1})}$ is equal to $\frac{1}{(1-z)^2}$ when $|z| < 1$, and is equal to $\frac{1}{z(1-z)^2}$ when $|z| > 1$. Is this fact connected with the theory of uniform convergence?

Solution: The useful observation is that

$$\frac{z^{n-1}}{(1-z^n)(1-z^{n+1})} = \frac{1}{z(1-z)} \left\{ \frac{1}{1-z^n} - \frac{1}{1-z^{n+1}} \right\}$$

This is a telescoping sum, and so, the partial sum from $n = 1$ to N is

$$\sum_{n=1}^N \frac{z^{n-1}}{(1-z^n)(1-z^{n+1})} = \frac{1}{z(1-z)} \left\{ \frac{1}{1-z} - \frac{1}{1-z^{N+1}} \right\}$$

If $|z| < 1$, then $z^{N+1} \rightarrow 0$, and so $\frac{1}{1-z^{N+1}} \rightarrow 1$. Hence,

$$\sum_{n=1}^{\infty} \frac{z^{n-1}}{(1-z^n)(1-z^{n+1})} = \frac{1}{z(1-z)} \left\{ \frac{1}{1-z} - 1 \right\} = \frac{1}{(1-z)^2}$$

If $|z| > 1$, then $|z|^{N+1} \rightarrow \infty$, and so $\frac{1}{1-z^{N+1}} \rightarrow 0$. Hence,

$$\sum_{n=1}^{\infty} \frac{z^{n-1}}{(1-z^n)(1-z^{n+1})} = \frac{1}{z(1-z)} \cdot \frac{1}{1-z} = \frac{1}{z(1-z)^2}$$

Thus, the series has the stated sums.

This example is connected with uniform convergence because the limiting process depends on whether $|z| < 1$ or $|z| > 1$. The convergence is uniform on closed regions lying strictly inside the unit circle, and also on closed regions lying strictly outside it, but not in any region which approaches the boundary $|z| = 1$, where the terms have singularities.

2. Show that the series $2 \sin \frac{1}{3z} + 4 \sin \frac{1}{9z} + \cdots + 2^n \sin \frac{1}{3^n z} + \cdots$ converges absolutely for all values of z , $z = 0$ excepted, but does not converge uniformly near $z = 0$.

Solution: For fixed $z \neq 0$, the argument $\frac{1}{3^n z}$ tends to zero as $n \rightarrow \infty$. We may use the estimate

$|\sin w| \leq 2|w|$ for small w . Hence, for all sufficiently large n , $\frac{1}{3^n z}$ is small and we have

$$\left| 2^n \sin \frac{1}{3^n z} \right| \leq 2^n \cdot 2 \left| \frac{1}{3^n z} \right| = \frac{2}{|z|} \left(\frac{2}{3} \right)^n$$

The series $\sum_{n=1}^{\infty} \frac{2}{|z|} \left(\frac{2}{3}\right)^n$ is a convergent geometric series. Therefore, the given series converges absolutely for each fixed $z \neq 0$.

However, the convergence is not uniform near $z = 0$. For uniform convergence, the individual terms would have to tend to zero uniformly. But this fails. Indeed, for each n , choose $z_n = \frac{2}{\pi 3^n}$. Then $z_n \rightarrow 0$, and $\frac{1}{3^n z_n} = \frac{\pi}{2}$. Therefore, $2^n \sin\left(\frac{1}{3^n z_n}\right) = 2^n \sin \frac{\pi}{2} = 2^n$. Thus, the n -th term can be made arbitrarily large at points z_n arbitrarily close to zero. Hence, the terms do not tend to zero uniformly near $z = 0$, and so the series cannot converge uniformly near $z = 0$.

3. If $u_n(x) = -2(n-1)xe^{-(n-1)^2x^2} + 2nxe^{-n^2x^2}$, show that $\sum_{n=1}^{\infty} u_n(x)$ does not converge uniformly near $x = 0$. (Math. Trip., 1907)

Solution: Define $v_n(x) = 2nxe^{-n^2x^2}$.

Then $u_n(x) = v_n(x) - v_{n-1}(x)$ since $v_{n-1}(x) = 2(n-1)xe^{-(n-1)^2x^2}$.

Therefore, the partial sums telescope:

$$\sum_{n=1}^N u_n(x) = v_N(x) - v_0(x)$$

But $v_0(x) = 0$, and so,

$$S_N(x) = \sum_{n=1}^N u_n(x) = 2Nxe^{-N^2x^2}$$

For each fixed x , this expression tends to zero as $N \rightarrow \infty$. Thus, the series converges pointwise to zero.

However, the convergence is not uniform near $x = 0$. To see this, choose $x_N = \frac{1}{N\sqrt{2}}$.

Then $x_N \rightarrow 0$ as $N \rightarrow \infty$, but

$$2Nx_N e^{-N^2x_N^2} = 2N \cdot \frac{1}{N\sqrt{2}} \cdot e^{-1/2} = \sqrt{2}e^{-1/2}$$

This is a positive constant independent of N . Hence, the partial sums $S_N(x)$ do not tend to their limiting value 0 uniformly near $x = 0$. Therefore, $\sum_{n=1}^{\infty} u_n(x)$ does not converge uniformly near $x = 0$.

4. Show that the series $\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \dots$ is convergent, but that its square, formed by Abel's rule, $\frac{1}{1} - \frac{2}{\sqrt{2}} + \left(\frac{2}{\sqrt{3}} + \frac{1}{2}\right) - \left(\frac{2}{\sqrt{4}} + \frac{2}{\sqrt{6}}\right) - \dots$ is divergent.

Solution: Consider the alternating series $\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \dots$

This series converges by the alternating series test, since the terms $1/\sqrt{n}$ decrease monotonically to zero.

Now form the square by Abel's rule. Let $a_n = \frac{(-1)^{n+1}}{\sqrt{n}}$.

The coefficient of the n -th term in the product is

$$c_n = a_1 a_n + a_2 a_{n-1} + \dots + a_n a_1$$

Thus

$$c_n = (-1)^{n+1} \left\{ \frac{1}{\sqrt{1 \cdot n}} + \frac{1}{\sqrt{2(n-1)}} + \dots + \frac{1}{\sqrt{n \cdot 1}} \right\}$$

So, the signs alternate, but the magnitude of c_n is

$$|c_n| = \sum_{k=1}^n \frac{1}{\sqrt{k(n+1-k)}}$$

We now show that these terms do not tend to zero. By the arithmetic-geometric mean inequality [7],

$$\sqrt{k(n+1-k)} \leq \frac{k + (n+1-k)}{2} = \frac{n+1}{2}$$

Therefore

$$\frac{1}{\sqrt{k(n+1-k)}} \geq \frac{2}{n+1}$$

Hence,

$$|c_n| = \sum_{k=1}^n \frac{1}{\sqrt{k(n+1-k)}} \geq \sum_{k=1}^n \frac{2}{n+1} = \frac{2n}{n+1} \rightarrow 2, n \rightarrow \infty$$

Thus, the terms c_n of the product series do not tend to zero. Therefore, the product series is divergent.

This shows that although the original alternating series is convergent, its square, formed by Abel's rule, is divergent.

5. If the convergent series $s = \frac{1}{1^r} - \frac{1}{2^r} + \frac{1}{3^r} - \frac{1}{4^r} + \dots$ ($r > 0$) be multiplied by itself, the terms of the product being arranged as in Abel's result, show that the resulting series diverges if $r \leq \frac{1}{2}$, but converges to the sum s^2 if $r > \frac{1}{2}$. (Cauchy and Cajori)

Hint: Use Abel's multiplication rule. The coefficient of the n -th term in the product is essentially

$$(-1)^{n+1} \sum_{k=1}^n \frac{1}{k^r (n+1-k)^r}$$

Estimate this inner sum. For large n , it behaves like a constant multiple of n^{1-2r} .

Thus, the product terms fail to tend to zero if $r \leq \frac{1}{2}$. If $r > \frac{1}{2}$, Abel's theorem on multiplication of series applies, since the resulting product series converges.

6. If the two conditionally convergent series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^r}$ and $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s}$, where r and s lie between 0 and 1, be multiplied together, and the product arranged as in Abel's result, show that the necessary and sufficient condition for the convergence of the resulting series is $r + s > 1$. (Cajori)

Hint: As in Exercise 5, write the coefficient of the product series as

$$(-1)^{n+1} \sum_{k=1}^n \frac{1}{k^r (n+1-k)^s}$$

Estimate this convolution sum. Its size is comparable to n^{1-r-s} .

Therefore, the terms tend to zero only when $r + s > 1$.

This condition is also sufficient, using the corresponding convergence estimate for the product series.

7. Show that if the series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ be multiplied by itself any number of times, the terms of the product being arranged as in Abel's result, the resulting series converges. (Cajori)

Hint: The given series is

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$$

It is the power series for $\arctan x$ evaluated at $x = 1$. Consider first the product of the corresponding power series for $\arctan x$ when $0 \leq x < 1$, where multiplication is justified inside the interval of convergence. Then use Abel's theorem to pass to the limit as $x \rightarrow 1 - 0$ (in modern notation, this would be written as $x \rightarrow 1^-$).

8. Show that the q th power of the series $a_1 \sin \theta + a_2 \sin 2\theta + \dots + a_n \sin n\theta + \dots$ is convergent whenever $q(1 - r) < 1$, r being the greatest number satisfying the relation $a_n < n^{-r}$ for all values of n .

Hint: Use Dirichlet's test for trigonometric series. The partial sums of $\sum \sin n\theta$ are bounded when θ is not a multiple of 2π . The condition $a_n \leq n^{-r}$ gives control over the decay of the coefficients. When the series is raised to the q -th power, the relevant product coefficients are

controlled by repeated convolution estimates. The condition $q(1-r) < 1$ ensures that these product coefficients still decrease rapidly enough for convergence.

9. Show that if θ is not equal to 0 or a multiple of 2π , and if $u_0, u_1, u_2, \dots$ be a sequence such that $u_n \rightarrow 0$ steadily, then the series $\sum u_n \cos(n\theta + \alpha)$ is convergent.

Show also that, if the limit of u_n is not zero, but u_n is still monotonic, the sum of the series is oscillatory if $\frac{\theta}{\pi}$ is rational, but that, if $\frac{\theta}{\pi}$ is irrational, the sum may have any value between certain bounds whose difference is $a \operatorname{cosec} \frac{\theta}{2}$, where $a = \lim_{n \rightarrow \infty} u_n$. (Math. Trip., 1896)

Hint: For the first part, apply Dirichlet's test to $\sum u_n \cos(n\theta + \alpha)$.

The sequence u_n decreases steadily to 0, while the partial sums of $\sum \cos(n\theta + \alpha)$ are bounded when θ is not 0 or a multiple of 2π .

For the second part, if $u_n \rightarrow a \neq 0$, separate $u_n = a + (u_n - a)$. The series involving $u_n - a$ is handled by Dirichlet's test. The remaining part behaves like the partial sums of $a \sum \cos(n\theta + \alpha)$, whose behavior depends on whether $\frac{\theta}{\pi}$ is rational or irrational.

5 The Theory of Riemann Integration

Understand. Don't memorize. Learn principles, not formulas. —Richard Feynman

5.1 The Concept of Integration

The reader is doubtless familiar with the idea of integration as the operation inverse to that of differentiation, and he or she is equally well aware that the integral, in this sense, of a given elementary function is not always expressible in terms of elementary functions. In order therefore to give a definition of the integral of a function which shall be always available, even though it is not practicable to obtain a function of which the given function is the differential coefficient, we have recourse to the result that the integral⁶³ of $f(x)$ between the limits a and b is the area bounded by the curve $y = f(x)$, the axis of x , and the ordinates $x = a$, $x = b$. We proceed to frame a formal definition of integration with this idea as the starting-point.

5.1.1 Upper and Lower Integrals⁶⁴

Let $f(x)$ be a bounded function of x in the range (a, b) . Divide the interval at the points

$$x_1, x_2, \dots, x_{n-1}$$

where

$$a < x_1 \leq x_2 \leq \dots \leq x_{n-1} < b$$

Let U, L be the bounds of $f(x)$ in the range (a, b) , and let U_r, L_r be the bounds of $f(x)$ in the range (x_{r-1}, x_r) , where $x_0 = a$, $x_n = b$.

Consider the sums⁶⁵

$$\begin{aligned} S_n &= U_1(x_1 - a) + U_2(x_2 - x_1) + \dots + U_n(b - x_{n-1}), \\ s_n &= L_1(x_1 - a) + L_2(x_2 - x_1) + \dots + L_n(b - x_{n-1}) \end{aligned}$$

Then

$$U(b - a) \geq S_n \geq s_n \geq L(b - a)$$

For a given n , S_n and s_n are bounded functions of $x_1, x_2, \dots, x_{n-1}$. Let their lower and upper bounds⁶⁶ respectively be $\underline{S}_n, \bar{s}_n$, so that $\underline{S}_n, \bar{s}_n$ depend only on n and on the form of $f(x)$, and not on the particular way of dividing the interval into n parts. In Figure 5, note that each U_i rectangle extends

⁶³ Defined as the elementary function whose differential coefficient is $f(x)$.

⁶⁴ The following procedure for establishing existence theorems concerning integrals is based on that given by Goursat, *Cours d'Analyse*, I, Ch. IV. The concepts of upper and lower integrals are due to Darboux, *Ann. de l'École norm. sup.* (2) IV (1875), p. 64.

⁶⁵ The reader will find a figure of great assistance in following the argument of this section. S_n and s_n represent the sums of the areas of a number of rectangles which are respectively greater and less than the area bounded by $y = f(x)$, $x = a$, $x = b$, and $y = 0$, if this area be assumed to exist.

⁶⁶ The bounds of a function of n variables are defined in just the same manner as the bounds of a function of a single variable (Section 4.6.2).

down to the x -axis; it is not merely the portion above the corresponding L_i rectangle. Thus, the upper and lower rectangles overlap, and their respective areas are $U_i(x_i - x_{i-1})$ and $L_i(x_i - x_{i-1})$.

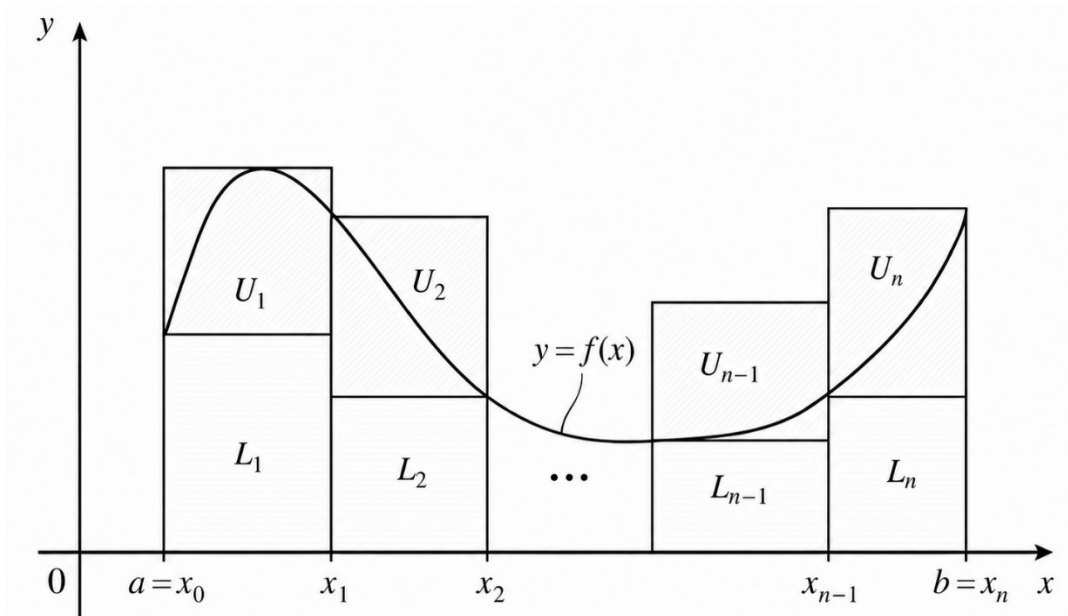


Figure 5. Upper and lower sums

Let the lower and upper bounds of these functions of n be S, s . Then $S_n \geq S, s_n \leq s$.

We proceed to show that s is at most equal to S , i.e. $S \geq s$.

Let the intervals $(a, x_1), (x_1, x_2), \dots$ be divided into smaller intervals by new points of subdivision, and let

$$a, y_1, y_2, \dots, y_{k-1}, y_k (= x_1), y_{k+1}, \dots, y_{l-1}, y_l (= x_2), y_{l+1}, \dots, y_{m-1}, b$$

be the end-points of the smaller intervals, let U'_r, L'_r be the bounds of $f(x)$ in the interval (y_{r-1}, y_r) .

Let

$$T_m = \sum_{r=1}^m (y_r - y_{r-1}) U'_r, \quad t_m = \sum_{r=1}^m (y_r - y_{r-1}) L'_r$$

Since $U'_1, U'_2, \dots, U'_k$ do not exceed U_1 , it follows without difficulty that

$$S_n \geq T_m \geq t_m \geq s_n$$

Now consider the subdivision of (a, b) into intervals by the points $x_1, x_2, \dots, x_{n-1}$ and the subdivision by a different set of points $x'_1, x'_2, \dots, x'_{n'-1}$. Let $S'_{n'}, s'_{n'}$ be the sums for the second kind of subdivision which correspond to the sums S_n, s_n for the first kind of subdivision. Take all the points $x_1, \dots, x_{n-1}, x'_1, \dots, x'_{n'-1}$ as the points $y_1, y_2, \dots, y_m$. Then $S_n \geq T_m \geq t_m \geq s_n$ and $S'_{n'} \geq T_m \geq t_m \geq s'_{n'}$. Hence every expression of the type S_n exceeds, or at least equals, every expression of the type $s'_{n'}$, and therefore S cannot be less than s .

[For if $S < s$ and $s - S = 2\eta$, we could find an S_n and an s'_n , such that $S_n - S < \eta$, $s - s'_n < \eta$ and so $s'_n > S_n$, which is impossible.]

The bound S is called the upper integral of $f(x)$, and we denote it by $\bar{I}(f; a, b)$. The bound s is called the lower integral of $f(x)$, and we denote it by $\underline{I}(f; a, b)$.

If $S = s$, their common value is called the **integral** of $f(x)$ taken between the limits⁶⁷ of integration a and b .

The integral is written $\int_a^b f(x) dx$

We define $\int_b^a f(x) dx$, when $a < b$, to mean $-\int_a^b f(x) dx$

Example 1.

$$\int_a^b \{f(x) + \phi(x)\} dx = \int_a^b f(x) dx + \int_a^b \phi(x) dx$$

Example 2. By means of example 1, define the integral of a continuous complex function of a real variable.

5.1.2 Riemann's Condition of Integrability⁶⁸

A function is said to be “integrable in the sense of Riemann” if, with the notation of Section 5.1.1, S_n and s_n have a common limit, called the **Riemann integral** of the function, when the number of intervals (x_{r-1}, x_r) tends to infinity in such a way that the length of the longest of them tends to zero.

The necessary and sufficient condition that a bounded function should be integrable is that $S_n - s_n$ should tend to zero when the number of intervals (x_{r-1}, x_r) tends to infinity in such a way that the length of the longest tends to zero.

The condition is obviously necessary, for if S_n and s_n have a common limit, $S_n - s_n \rightarrow 0$ as $n \rightarrow \infty$.

And it is sufficient, for, since $S_n \geq S \geq s \geq s_n$, it follows that if $\lim (S_n - s_n) = 0$, then $\lim S_n = \lim s_n = S = s$

Note. A continuous function $f(x)$ is “integrable.” For, given ε , we can find δ such that

$$|f(x') - f(x'')| < \frac{\varepsilon}{b-a}$$

whenever $|x' - x''| < \delta$.

Take all the intervals (x_{s-1}, x_s) less than δ , and then $U_s - L_s < \frac{\varepsilon}{b-a}$.

⁶⁷ “Extreme values” would be a more appropriate term, but “limits” has the sanction of custom. “Terminal” has been suggested by Lamb, *Infinitesimal Calculus* (1897), p. 207.

⁶⁸ Riemann (*Ges. Math. Werke*, p. 239) bases his definition of an integral on the limit of the sum occurring in §4.1.3, but it is then difficult to prove the uniqueness of the limit. A more general definition of integration, which is of very great importance in the modern theory of Functions of Real Variables, has been given by Lebesgue, *Annali di Mat.* (3) VII (1902), pp. 231–359. See also his *Leçons sur l'intégration* (Paris, 1904).

Next, compute the difference between the upper and lower sums:

$$S_n - s_n = \sum_{s=1}^n (U_s - L_s)(x_s - x_{s-1}) < \sum_{s=1}^n \frac{\epsilon}{b-a} (x_s - x_{s-1}) = \frac{\epsilon}{b-a} \sum_{s=1}^n (x_s - x_{s-1})$$

Since the subinterval lengths add up to the whole interval length, i.e., $\sum_{s=1}^n (x_s - x_{s-1}) = b - a$, we have that $S_n - s_n < \frac{\epsilon}{b-a} (b-a) = \epsilon$.

Therefore $S_n - s_n \rightarrow 0$ under the circumstances specified in the condition of integrability.

Corollary. If S_n and s_n have the same limit S for one mode of subdivision of (a, b) into intervals of the specified kind, the limits of S_n and of s_n for any other such mode of subdivision are both S .

Example 0. Not every bounded function is Riemann integrable. For example, define f on $[a, b]$ such that $f(x) = 1$ when x is rational and $f(x) = 0$ when x is irrational. Every interval contains both rational and irrational numbers, so every lower sum is 0, while every upper sum is $b - a$. Hence $S_n - s_n$ does not tend to zero, and the function is not Riemann integrable. In the Lebesgue theory, however, this function is integrable and has integral 0, since the rational numbers have measure zero. This example is revisited in the Appendix on the Lebesgue integral.

Example 1. The product of two integrable functions is an integrable function.

Example 2. A bounded function f on $[a, b]$ which is continuous except at a finite number of ordinary discontinuities is integrable.

Suppose first that $f(x)$ is continuous on $[a, b]$ except for one ordinary discontinuity at c . Choose a small interval I containing c , say $I = (c - \eta, c + \eta)$, where I has total length δ_1 .

Outside the interval I , f is continuous on a closed set consisting of one or two closed intervals. Hence, f is uniformly continuous outside I . Therefore, given $\epsilon > 0$, we can choose $\delta > 0$ such that

$$|f(x') - f(x'')| < \frac{\epsilon}{b-a}$$

whenever x' and x'' lie outside I and $|x' - x''| < \delta$.

Now choose a subdivision of $[a, b]$ whose largest subinterval has length less than δ , and include the endpoints of I among the subdivision points. On every subinterval outside I , the oscillation of f is less than $\frac{\epsilon}{b-a}$. Thus, the contribution of all such subintervals to $S_n - s_n$ is at most

$$\frac{\epsilon}{b-a} (b-a-\delta_1) < \epsilon$$

Further, since f is bounded throughout $[a, b]$ which includes the interval I , there exists a constant K such that $U_I - L_I \leq K$. Hence, the contribution of I to $S_n - s_n$ is at most $K\delta_1$. Therefore, $S_n - s_n < \epsilon + K\delta_1$. Since δ_1 may be chosen arbitrarily small, the term $K\delta_1$ can be made as small as desired. It follows that $S_n - s_n \rightarrow 0$ as the mesh of the subdivision tends to zero. Therefore, f is Riemann integrable.

If f has finitely many ordinary discontinuities, enclose each one in a small interval, with the total length of all these exceptional intervals less than δ_1 . The same argument applies, i.e., outside

these exceptional intervals, f is uniformly continuous; inside them, the total contribution to $S_n - s_n$ is bounded by a constant times their total length. Hence $S_n - s_n \rightarrow 0$, and f is integrable.

The finite-discontinuity assumption can be weakened. In modern terminology, a bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has measure zero. Therefore, a bounded function with only countably many discontinuities is also Riemann integrable. Proving this more general result, however, requires ideas closer to measure theory and is better viewed in connection with the Lebesgue integral discussed in the Appendix.

Example 3. A function with limited total fluctuation and a finite number of ordinary discontinuities is integrable. (See Section 4.6.4, Example 2.)

5.1.3 A General Theorem on Integration

Let $f(x)$ be integrable, and let ϵ be any positive number. Then it is possible to choose δ so that

$$\left| \sum_{p=1}^n (x_p - x_{p-1})f(x'_{p-1}) - \int_a^b f(x) dx \right| < \epsilon,$$

provided that

$$x_p - x_{p-1} \leq \delta, \quad x_{p-1} \leq x'_{p-1} \leq x_p$$

Here $x_0 = a$ and $x_n = b$, and x'_{p-1} denotes an arbitrary point of the subinterval $[x_{p-1}, x_p]$. Thus, the displayed sum is a general Riemann sum.

To prove the theorem we observe that, given ϵ , we can choose the length of the longest interval δ so small that $S_n - s_n < \epsilon$. Also, $S_n \geq \sum_{p=1}^n (x_p - x_{p-1})f(x'_{p-1}) \geq s_n$ and $S_n \geq \int_a^b f(x) dx \geq s_n$. Therefore, $\left| \sum_{p=1}^n (x_p - x_{p-1})f(x'_{p-1}) - \int_a^b f(x) dx \right| \leq S_n - s_n < \epsilon$.

Example. Evaluate the following definite integral directly from the theorem of this section:

$$\int_0^X \frac{dx}{(1-x^2)^{1/2}}, \quad 0 < X < 1$$

Let $\alpha = \arcsin X$, so that $X = \sin \alpha$. [This looks like a leap but recall that $\frac{d}{dx} \arcsin x$ is known to be $\frac{1}{\sqrt{1-x^2}}$ for $|x| < 1$.]

Divide the interval $[0, \alpha]$ into p equal parts: $\delta = \frac{\alpha}{p}$.

Now define a corresponding subdivision of $[0, X]$ by $x_s = \sin(s\delta)$, $s = 0, 1, 2, \dots, p$ which implies that $x_0 = 0$, $x_p = \sin(p\delta) = \sin \alpha = X$.

For the interval $[x_{s-1}, x_s]$, choose the sample point $x'_s = \sin\left(s - \frac{1}{2}\right)\delta$ where $x'_s \in [x_{s-1}, x_s]$. This point lies in $[x_{s-1}, x_s]$, since $x_{s-1} = \sin((s-1)\delta)$, $x_s = \sin(s\delta)$ and $(s-1)\delta < \left(s - \frac{1}{2}\right)\delta < s\delta$. Since $\sin x$ is increasing on $0, \frac{\pi}{2}]$, it follows that $x_{s-1} < x'_s < x_s$.

We have that $\sqrt{1 - x_s'^2} = \sqrt{1 - \sin^2\left(s - \frac{1}{2}\right)\delta} = \cos\left(s - \frac{1}{2}\right)\delta$, since all the angles involved lie between 0 and $\pi/2$, where cosine is nonnegative.

Also, $x_s - x_{s-1} = \sin(s\delta) - \sin((s-1)\delta)$.

Using $\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$, we get $x_s - x_{s-1} = 2 \cos\left(s - \frac{1}{2}\right)\delta \sin \frac{\delta}{2}$.

Therefore, the s -th term in the Riemann sum is

$$\frac{x_s - x_{s-1}}{\sqrt{1 - x_s'^2}} = \frac{2 \cos\left(s - \frac{1}{2}\right)\delta \sin \frac{\delta}{2}}{\cos\left(s - \frac{1}{2}\right)\delta} = 2 \sin \frac{\delta}{2}$$

Thus, every term in the Riemann sum is the same, so

$$\sum_{s=1}^p \frac{x_s - x_{s-1}}{\sqrt{1 - x_s'^2}} = 2p \sin \frac{\delta}{2}$$

Since $\delta = \frac{\alpha}{p}$, this becomes

$$2p \sin \frac{\delta}{2} = p\delta \cdot \frac{\sin(\delta/2)}{\delta/2} = \alpha \cdot \frac{\sin(\delta/2)}{\delta/2}$$

As $p \rightarrow \infty$, we have $\delta \rightarrow 0$, and therefore

$$\frac{\sin(\delta/2)}{\delta/2} \rightarrow 1$$

Hence, the Riemann sums tend to $\alpha = \arcsin X$.

By the general theorem on integration, these Riemann sums converge to the integral. Therefore

$$\int_0^X \frac{dx}{\sqrt{1 - x^2}} = \arcsin X$$

The examples that follow use the general theorem on integration to evaluate limits of Riemann sums and definite integrals. However, some of the choices made in these examples are motivated by prior knowledge from differential calculus, such as the antiderivatives of $\cos x$ and $1/\sqrt{1-x^2}$. Thus, these examples should be read as illustrations of the theorem rather than as a systematic development of integration from first principles. A more modern treatment would first establish the basic algebraic properties of the Riemann integral and then prove the fundamental theorem of calculus, which connects definite integrals with antiderivatives.

Example 1. Show that

$$\lim_{n \rightarrow \infty} \frac{1 + \cos \frac{x}{n} + \cos \frac{2x}{n} + \cdots + \cos \frac{(n-1)x}{n}}{n} = \frac{\sin x}{x}$$

Hint: Interpret the expression as a Riemann sum for $\int_0^1 \cos(tx) dt$, using the subdivision $t_r = r/n$. Then evaluate the integral by the substitution $u = tx$, giving $\int_0^1 \cos(tx) dt = \frac{\sin x}{x}$. For $x = 0$, the expression is identically 1, consistent with the limiting value.

Example 2. If $f(x)$ has ordinary discontinuities at the points $a_1, a_2, \dots, a_k$, then

$$\int_a^b f(x) dx = \lim \left\{ \int_a^{a_1 - \delta_1} + \int_{a_1 + \epsilon_1}^{a_2 - \delta_2} + \cdots + \int_{a_k + \epsilon_k}^b f(x) dx \right\}$$

where the limit is taken by making $\delta_1, \delta_2, \dots, \delta_k, \epsilon_1, \epsilon_2, \dots, \epsilon_k$ tend to $+0$ independently.

Hint: Surround each discontinuity a_i by a small, excluded interval $(a_i - \delta_i, a_i + \epsilon_i)$. Since f is bounded, the total contribution from the excluded intervals tends to zero as their total length tends to zero.

Example 3. If $f(x)$ is integrable when $a_1 \leq x \leq b_1$, and if, when $a_1 \leq a < b \leq b_1$, we write

$$\int_a^b f(x) dx = \phi(a, b)$$

and if $f(b+0)$ exists, then

$$\lim_{\delta \rightarrow +0} \frac{\phi(a, b + \delta) - \phi(a, b)}{\delta} = f(b+0)$$

Deduce that, if $f(x)$ is continuous at a and b ,

$$\frac{d}{da} \int_a^b f(x) dx = -f(a), \quad \frac{d}{db} \int_a^b f(x) dx = f(b)$$

Hint: Observe that

$$\phi(a, b + \delta) - \phi(a, b) = \int_b^{b+\delta} f(x) dx.$$

Divide by δ . If $f(b+0)$ exists, the average value of f over $[b, b + \delta]$ tends to $f(b+0)$.

Example 4. Prove by differentiation that, if $\phi(x)$ is a continuous function of x and $\frac{dx}{dt}$ a continuous function of t , then

$$\int_{x_0}^{x_1} \phi(x) dx = \int_{t_0}^{t_1} \phi(x) \frac{dx}{dt} dt$$

Hint: Let $F(x) = \int_{x_0}^x \phi(u) du$. Then the left side is $F(x_1) - F(x_0)$. Use the chain rule: $\frac{d}{dt} F(x(t)) = \phi(x(t)) \frac{dx}{dt}$.

Example 5. If $f'(x)$ and $\phi'(x)$ are continuous when $a \leq x \leq b$, show from Example 3 that

$$\int_a^b f'(x) \phi(x) dx + \int_a^b \phi'(x) f(x) dx = f(b) \phi(b) - f(a) \phi(a)$$

Hint: Apply Example 3 to the product $f(x) \phi(x)$. Since $\frac{d}{dx} \{f(x) \phi(x)\} = f'(x) \phi(x) + f(x) \phi'(x)$, integrating over $[a, b]$ gives the result.

Example 6. If $f(x)$ is integrable in the range (a, c) , and $a \leq b \leq c$, show that $\int_a^b f(x) dx$ is a continuous function of b .

Hint: Write $\int_a^{b+h} f(x) dx - \int_a^b f(x) dx = \int_b^{b+h} f(x) dx$. Since f is bounded, the absolute value of this difference is at most $M|h|$, where M bounds $|f|$. Hence it tends to zero as $h \rightarrow 0$.

5.1.4 Mean Value Theorems

The two following general theorems are frequently useful.

(I) Let U and L be the upper and lower bounds of the integrable function $f(x)$ in the range (a, b) .

Then from the definition of an integral it is obvious that

$$\int_a^b \{U - f(x)\} dx, \quad \int_a^b \{f(x) - L\} dx$$

are not negative, and so

$$U(b - a) \geq \int_a^b f(x) dx \geq L(b - a)$$

This is known as the **First Mean Value Theorem**.

If $f(x)$ is continuous, we can find a number ξ such that $a < \xi < b$, and such that $f(\xi)$ has any given value lying between U and L (Section 4.6.3). Therefore, we can find ξ such that

$$\int_a^b f(x) dx = (b - a) f(\xi)$$

Figure 6 illustrates the first mean value theorem for integrals for the function $f(x) = x + \sin 2x$, $0 \leq x \leq 5$. In this case, $\int_0^5 f(x) dx \approx 13.4195$. Since $b - a = 5$, the average value of f on $[0, 5]$ is

approximately $\frac{1}{5} \int_0^5 f(x) dx \approx 2.6839$. The theorem asserts that there is some $\xi \in [0, 5]$ such that $f(\xi) = 2.6839$, or equivalently, $5f(\xi) \approx 13.4195$. In the figure, one such value is $\xi \approx 2.98741$.

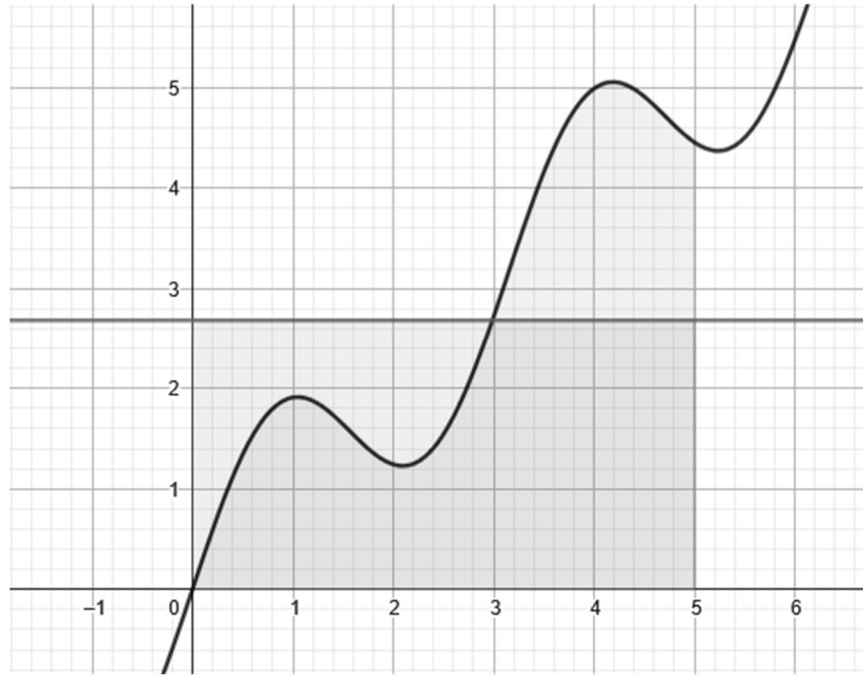


Figure 6. Example of first mean value theorem of integrals

If $F(x)$ has a continuous differential coefficient $F'(x)$ in the range (a, b) , we have, on writing $F'(\xi)$ for $f(\xi)$,

$$F(b) - F(a) = (b - a)F'(\xi)$$

for some value of ξ such that $a < \xi < b$.

Example. If $f(x)$ is continuous and $\phi(x) \geq 0$, show that ξ can be found such that

$$\int_a^b f(x)\phi(x) dx = f(\xi) \int_a^b \phi(x) dx$$

(II) Let $f(x)$ and $\phi(x)$ be integrable in the range (a, b) , and let $\phi(x)$ be a positive decreasing function of x . Then Bonnet's⁶⁹ form of the **Second Mean Value Theorem** is that a number ξ exists such that $a \leq \xi \leq b$, and

$$\int_a^b f(x)\phi(x) dx = \phi(a) \int_a^\xi f(x) dx$$

The proof by the authors (which is omitted here) is based on a summation-by-parts argument applied to Riemann sums. The main idea is that $\phi(x)$, being positive and decreasing, gives the largest weight to the values of $f(x)$ near a and smaller weights to values farther to the right. Bonnet's theorem says that this

⁶⁹ *Journal de Math.* XIV (1849), p. 249. The proof given is a modified form of an investigation due to Hölder, *Gött. Nach.* (1889), pp. 38–47.

weighted integral can be reproduced by using the largest weight $\phi(a)$, but stopping the integral of f at some suitable point ξ .

Let $F(t) = \int_a^t f(x) dx, a \leq t \leq b$.

Since f is integrable, $F(t)$ is continuous on $[a, b]$. Hence $F(t)$ attains an upper bound M and a lower bound m , and takes every value between them.

The decreasing function $\phi(x)$ acts as a system of positive weights. Since its largest value is $\phi(a)$, the weighted integral $\int_a^b f(x)\phi(x) dx$ must lie between $\phi(a)m$ and $\phi(a)M$.

This is the essential content of the summation-by-parts estimate in the proof by the author. Therefore, there is some value between m and M , say $F(\xi)$, such that

$$\int_a^b f(x)\phi(x) dx = \phi(a)F(\xi)$$

Substituting the definition of $F(\xi)$, we obtain

$$\int_a^b f(x)\phi(x) dx = \phi(a) \int_a^\xi f(x) dx$$

This proves Bonnet's form of the Second Mean Value Theorem.

The key point is that the decreasing factor $\phi(x)$ does not create values outside the range already produced by the accumulated integrals $\int_a^t f(x) dx$. It merely forms a weighted average, in a suitable sense, of these accumulated values. Since $F(t)$ is continuous, some point ξ realizes the required accumulated value.

Example. By writing $|\phi(x) - \phi(b)|$ in place of $\phi(x)$ in Bonnet's form of the mean value theorem, show that if $\phi(x)$ is a monotonic function, then a number ξ exists such that $a \leq \xi \leq b$, and

$$\int_a^b f(x)\phi(x) dx = \phi(a) \int_a^\xi f(x) dx + \phi(b) \int_\xi^b f(x) dx.$$

(Du Bois Reymond)

The formula says that, for a monotonic weight $\phi(x)$, the effect of multiplying $f(x)$ by the continuously changing weight $\phi(x)$ can be represented by a simpler step weight: use $\phi(a)$ on the part of the interval from a to ξ , and use $\phi(b)$ on the part from ξ to b . The point ξ is chosen so that this two-level approximation gives exactly the same integral as the original monotonic weight.

5.2 Differentiation of Integrals Containing a Parameter

The equation⁷⁰

$$\frac{d}{d\alpha} \int_a^b f(x, \alpha) dx = \int_a^b \frac{\partial f}{\partial \alpha} dx$$

is true if $f(x, \alpha)$ possesses a Riemann integral with respect to x , and if $f_\alpha \equiv \frac{\partial f}{\partial \alpha}$ is a continuous function of both the variables x and α .

The above theorem states a familiar and important principle: under suitable continuity assumptions, differentiation with respect to a parameter may be passed under the integral sign. In modern language, this is a form of Leibniz's rule. The technical hypotheses ensure that the difference quotients $\frac{f(x, \alpha+h) - f(x, \alpha)}{h}$ approach $\frac{\partial f}{\partial \alpha}$ uniformly enough in x that their integrals converge to the integral of the limit.

Counterexample to the Differentiation Under the Integral Sign Theorem

Define the function $f(x, y)$ as follows:

$$f(x, y) = \begin{cases} \frac{x^3 y}{(x^2 + y^2)^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Consider the integral $F(x) = \int_0^1 f(x, y) dy$.

It can be shown that:

$$F(x) = \frac{x}{2(1 + x^2)} \text{ for all } x.$$

In particular, F is differentiable at $x = 0$ and $F'(0) = \frac{1}{2}$.

The partial derivative with respect to x is:

$$\frac{\partial f}{\partial x}(x, y) = \begin{cases} \frac{x^2 y(3y^2 - x^2)}{(x^2 + y^2)^3} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases}$$

Evaluating the integral of the partial at $x = 0$:

$$\int_0^1 \frac{\partial f}{\partial x}(0, y) dy = 0.$$

In conclusion, the left side of the equation (derivative of the integral) equals $\frac{1}{2}$, while the right side (integral of the partial derivative) equals 0. Thus, the two sides are not equal.

⁷⁰ This formula was given by Leibniz, without specifying the restrictions laid on $f(x, \alpha)$.

Footnote from original text : $\phi(x, y)$ is defined to be a continuous function of both variables if, given ϵ , we can find δ such that $|\phi(x', y') - \phi(x, y)| < \epsilon$ whenever $\{(x' - x)^2 + (y' - y)^2\}^{1/2} < \delta$.

It can be shown by Section 4.6 that if $\phi(x, y)$ is a continuous function of both variables at all points of a closed region in a Cartesian diagram, it is uniformly continuous throughout the region. The proof is almost identical with that of Section 4.6.1. It should be noticed that, if $\phi(x, y)$ is a continuous function of each variable, it is not necessarily a continuous function of both; as an example, take

$$\phi(x, y) = \frac{(x + y)^2}{x^2 + y^2}, \phi(0, 0) = 1$$

This is a continuous function of x and of y at $(0, 0)$, but not of both x and y .

Proof: We have that

$$\frac{d}{d\alpha} \int_a^b f(x, \alpha) dx = \lim_{h \rightarrow 0} \int_a^b \frac{f(x, \alpha + h) - f(x, \alpha)}{h} dx$$

if this limit exists. But, by the first mean value theorem, since f_α is a continuous function of α , the second integrand is $f_\alpha(x, \alpha + \theta h)$, where $0 \leq \theta \leq 1$.

But, for any given ϵ , a number δ , independent of x , exists, since the continuity of f_α is uniform⁷¹ with respect to the variable x , such that $|f_\alpha(x, \alpha') - f_\alpha(x, \alpha)| < \frac{\epsilon}{b-a}$ whenever $|\alpha' - \alpha| < \delta$.

Taking $|h| < \delta$, we see that $|\theta h| < \delta$, and so whenever $|h| < \delta$,

$$\left| \int_a^b \frac{f(x, \alpha + h) - f(x, \alpha)}{h} dx - \int_a^b f_\alpha(x, \alpha) dx \right| \leq \int_a^b |f_\alpha(x, \alpha + \theta h) - f_\alpha(x, \alpha)| dx < \epsilon$$

Therefore, by the definition of a limit of a function (Section 4.2), $\lim_{h \rightarrow 0} \int_a^b \frac{f(x, \alpha + h) - f(x, \alpha)}{h} dx$ exists and is equal to $\int_a^b f_\alpha dx$. ■

Example 1. If a, b be not constants but functions of α with continuous differential coefficients, show that

$$\frac{d}{d\alpha} \int_a^b f(x, \alpha) dx = f(b, \alpha) \frac{db}{d\alpha} - f(a, \alpha) \frac{da}{d\alpha} + \int_a^b \frac{\partial f}{\partial \alpha} dx$$

Example 2. If $f(x, \alpha)$ is a continuous function of both variables, $\int_a^b f(x, \alpha) dx$ is a continuous function of α .

⁷¹ It is obvious that it would have been sufficient to assume that f_α had a Riemann integral and was a continuous function of α , the continuity being uniform with respect to x , instead of assuming that f_α was a continuous function of both variables. This is actually done by Hobson, *Functions of a Real Variable*, p. 599.

5.3 Double Integrals and Repeated Integrals

Let $f(x, y)$ be a function which is continuous with regard to both of the variables x and y , when $a \leq x \leq b, \alpha \leq y \leq \beta$. By Section 5.2, Example 2, it is clear that

$$\int_a^b \left\{ \int_\alpha^\beta f(x, y) dy \right\} dx, \quad \int_\alpha^\beta \left\{ \int_a^b f(x, y) dx \right\} dy$$

both exist. These are called **repeated integrals**.

Here Section 5.2, Example 2 is used only to show that the repeated integrals exist. It does not by itself justify changing the order of integration. The equality of the two repeated integrals with each other, and with the double integral, is a separate result: for continuous functions on a rectangle, this is the elementary rectangular case of Fubini's theorem.[8]

Also, as in Section 4.6.2, $f(x, y)$, being a continuous function of both variables, attains its upper and lower bounds.

Consider the range of values of x and y to be the points inside and on a rectangle in a Cartesian diagram; divide it into $n\nu$ rectangles by lines parallel to the axes.

Let $U_{m,\mu}, L_{m,\mu}$ be the upper and lower bounds of $f(x, y)$ in one of the smaller rectangles whose area is, say, $A_{m,\mu}$, and let

$$\sum_{m=1}^n \sum_{\mu=1}^\nu U_{m,\mu} A_{m,\mu} = S_{n,\nu}, \quad \sum_{m=1}^n \sum_{\mu=1}^\nu L_{m,\mu} A_{m,\mu} = s_{n,\nu}.$$

Then $S_{n,\nu} > s_{n,\nu}$, and, as in Section 5.1.1, we can find numbers $\underline{S}_{n,\nu}, \bar{s}_{n,\nu}$, which are the lower and upper bounds of $S_{n,\nu}, s_{n,\nu}$ respectively, the values of $\underline{S}_{n,\nu}, \bar{s}_{n,\nu}$ depending only on the number of the rectangles and not on their shapes, and $\underline{S}_{n,\nu} \geq \bar{s}_{n,\nu}$.

We then find the lower and upper bounds S and s , respectively, of $\underline{S}_{n,\nu}, \bar{s}_{n,\nu}$, as functions of n and ν , and $\underline{S}_{n,\nu} \geq S \geq s \geq \bar{s}_{n,\nu}$, as in Section 5.1.1.

Also, from the uniformity of the continuity of $f(x, y)$, given ϵ , we can find δ such that $U_{m,\mu} - L_{m,\mu} < \epsilon$, for all values of m and μ , whenever the sides of all the small rectangles are less than the number δ , which depends only on the form of the function $f(x, y)$ and on ϵ .

And then $S_{n,\nu} - s_{n,\nu} < \epsilon(b - a)(\beta - \alpha)$, and so $S - s < \epsilon(b - a)(\beta - \alpha)$.

But S and s are independent of ϵ , and so $S = s$.

The common value of S and s is called the **double integral** of $f(x, y)$, and is written

$$\int_a^b \int_\alpha^\beta f(x, y) (dx dy).$$

It is easy to show that the repeated integrals and the double integral are all equal when $f(x, y)$ is a continuous function of both variables.

For let Γ_m, Λ_m be the upper and lower bounds of $\int_\alpha^\beta f(x, y) dy$ as x varies between x_{m-1} and x_m .

Then

$$\sum_{m=1}^n \Gamma_m (x_m - x_{m-1}) \geq \int_a^b \left\{ \int_{\alpha}^{\beta} f(x, y) dy \right\} dx \geq \sum_{m=1}^n \Lambda_m (x_m - x_{m-1})$$

But⁷²

$$\sum_{\mu=1}^v U_{m,\mu} (y_{\mu} - y_{\mu-1}) \geq \Gamma_m \geq \Lambda_m \geq \sum_{\mu=1}^v L_{m,\mu} (y_{\mu} - y_{\mu-1})$$

Multiplying these last inequalities by $x_m - x_{m-1}$, using the preceding inequalities and summing, we get

$$\sum_{m=1}^n \sum_{\mu=1}^v U_{m,\mu} A_{m,\mu} \geq \int_a^b \left\{ \int_{\alpha}^{\beta} f(x, y) dy \right\} dx \geq \sum_{m=1}^n \sum_{\mu=1}^v L_{m,\mu} A_{m,\mu},$$

and so, proceeding to the limit,

$$S \geq \int_a^b \left\{ \int_{\alpha}^{\beta} f(x, y) dy \right\} dx \geq s.$$

But

$$S = s = \int_a^b \int_{\alpha}^{\beta} f(x, y) (dx dy),$$

and so, one of the repeated integrals is equal to the double integral. Similarly, the other repeated integral is equal to the double integral.

Corollary. If $f(x, y)$ be a continuous function of both variables,

$$\int_0^1 dx \left\{ \int_0^{1-x} f(x, y) dy \right\} = \int_0^1 dy \left\{ \int_0^{1-y} f(x, y) dx \right\}.$$

Although double integrals over rectangles can often be evaluated as repeated integrals, the double integral is the fundamental object: it represents accumulation over a two-dimensional region. The repeated integrals are computational tools obtained by slicing the region in one coordinate direction and then integrating the resulting one-dimensional integrals. The equality of the two repeated integrals is therefore not merely a notational convenience; it justifies computing the same two-dimensional quantity by either order of integration.

The equality of the two repeated integrals depends on the continuity, or at least sufficient integrability, of the function on the rectangle. If the function has a singularity, the order of integration may matter.

⁷² The upper bound of $f(x, y)$ in the rectangle $A_{m,\mu}$ is not less than the upper bound of $f(x, y)$ on that portion of the line $x = \xi$ which lies in the rectangle.

For example, consider the unit square $[0,1] \times [0,1]$ and the function

$$f(x,y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

The issue is not the value of the function at $(0,0)$, which by itself would not affect an integral. The issue is that the function is unbounded near $(0,0)$, so it falls outside the continuity and boundedness assumptions under which the equality of repeated integrals was proved.

The repeated integrals below are understood as improper repeated integrals. For example, when integrating first with respect to y , the inner integral is evaluated for $x > 0$ and then integrated with respect to x over $[0,1]$; the exceptional value at $x = 0$ does not affect the outer integral.

Integrating First with Respect to y , Then x :

$$\int_0^1 \left(\int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2} dy \right) dx = \int_0^1 \frac{1}{1 + x^2} dx = [\arctan x]_0^1 = \frac{\pi}{4}$$

Integrating First with Respect to x , Then y :

$$\int_0^1 \left(\int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2} dx \right) dy = -\frac{\pi}{4}$$

The calculations involve a nice antiderivative using substitution or recognizing derivatives of $\frac{y}{x^2 + y^2}$ or $\frac{x}{x^2 + y^2}$.

Why This Happens: Near the origin, f becomes arbitrarily large in magnitude and changes sign depending on the direction of approach. This causes the positive and negative contributions to cancel differently depending on the order of integration.

In modern terms, the problem is that f is not Lebesgue integrable on $[0,1] \times [0,1]$, since

$$\iint_{[0,1]^2} |f(x,y)| dA = \infty$$

Equivalently, the singularity at $(0,0)$ is too strong for the positive and negative contributions to be controlled absolutely. Therefore, Fubini's theorem does not apply, and the two repeated improper integrals need not agree.

5.3.1 Examples of Double and Repeated Integrals

In this section (not in the original text from Whittaker and Watson), we present two problems where one example is naturally formulated as a double integral and the other as a repeated integral.

Problem 1: Find the mass of a thin metal plate (lamina) in the shape of the unit disk $D = \{(x,y): x^2 + y^2 \leq 1\}$ if the density at each point is given by

$$\rho(x,y) = x^2 + y^2$$

Why this is naturally a double integral:

- We are accumulating a 2D quantity (mass) directly over a 2D region.
- The integral doesn't suggest any particular order — we're just summing up density $\times$ tiny area elements dA across the whole plate at once.

Formulation:

$$m = \iint_D \rho(x, y) dA = \iint_{x^2+y^2 \leq 1} (x^2 + y^2) dA$$

This is the clean, conceptual way to write it. (In practice, we would switch to polar coordinates to evaluate it, turning it into a repeated integral in r and θ .)

Result (for completeness): $m = \frac{\pi}{2}$.

This type of problem (volume under a surface, total charge, total population, average value, flux through a surface, etc.) is almost always best introduced as a double integral.

Problem 2: A company's profit rate depends on two variables: advertising expenditure x (in thousands of dollars) and time t (in months). The instantaneous profit rate is given by

$$P(x, t) = 50xe^{-0.1t} - 2x^2$$

for $0 \leq x \leq 10$ and $0 \leq t \leq 12$.

For each fixed advertising level x , first find the total profit over the 12-month period. Then accumulate these totals over all advertising levels $0 \leq x \leq 10$.

Why this is naturally a repeated integral:

- We first integrate profit *rate* with respect to time (for a fixed advertising level) to get total profit for that x .
- Then we integrate *across* different possible advertising levels x (perhaps weighted by some distribution, or if we're averaging, etc.).
- The sequential nature (accumulate over time, then over choices of x) makes the iterated form feel more natural.

Formulation:

$$\text{Accumulated Total Profit} = \int_0^{10} \left(\int_0^{12} P(x, t) dt \right) dx = \int_0^{10} \left(\int_0^{12} (50xe^{-0.1t} - 2x^2) dt \right) dx$$

Here the inner integral has a clear physical meaning: "total profit over 12 months for a fixed advertising budget x ."

Table 5. Summary of the Contrast (Double versus Repeated Integral)

Situation	Natural Formulation	Reason
Mass/volume/charge over a region	Double integral $\iint_D f \, dA$	Direct 2D accumulation
Accumulate one variable, then another (time then parameter, inner then outer)	Repeated integral $\int (\int f \, dx) dy$	Sequential process

In practice, we often evaluate double integrals as repeated integrals when the hypotheses allowing the change of order are satisfied. Conceptually, however, the starting point may be different: a double integral describes accumulation over a region, while a repeated integral emphasizes a sequential accumulation process.

5.4 Infinite Integrals

If $\lim_{b \rightarrow \infty} \left(\int_a^b f(x) \, dx \right)$ exists, we denote it by $\int_a^\infty f(x) \, dx$, and the limit in question is called an **infinite integral**.⁷³

Examples

$$(1) \int_a^\infty \frac{dx}{x^2} = \lim_{b \rightarrow \infty} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{1}{a}$$

$$(2) \int_0^\infty \frac{x \, dx}{(x^2 + a^2)^2} = \lim_{b \rightarrow \infty} \left(-\frac{1}{2(b^2 + a^2)} + \frac{1}{2a^2} \right) = \frac{1}{2a^2}$$

(3) By integrating by parts, show that $\int_0^\infty t^n e^{-t} \, dt = n!$ (Euler)

Using integration by parts repeatedly, we get the antiderivative, i.e.,

$$\int t^n e^{-t} \, dt = -e^{-t} \sum_{k=0}^n \frac{n!}{(n-k)!} t^{n-k} + C$$

Similarly, we define $\int_{-\infty}^b f(x) \, dx$ to mean $\lim_{a \rightarrow -\infty} \int_a^b f(x) \, dx$, if this limit exists.

Further, $\int_{-\infty}^\infty f(x) \, dx$ is defined as $\int_{-\infty}^a f(x) \, dx + \int_a^\infty f(x) \, dx$. In this definition, the choice of a is a matter of indifference.

The definition of $\int_{-\infty}^\infty f(x) \, dx$ requires the two improper integrals on the left and right to exist separately. This avoids misleading cancellations between divergent tails. For example, $f(x) = x/(1+x^2)$ has symmetric integrals $\int_{-R}^R f(x) \, dx = 0$, but $\int_0^\infty f(x) \, dx = +\infty$ and

⁷³ This phrase, due to Hardy, *Proc. London Math. Soc.* XXXIV (1902), p. 16, suggests the analogy between an infinite integral and an infinite series.

$\int_{-\infty}^0 f(x) dx = -\infty$. Thus, the ordinary improper integral does not exist, even though its Cauchy principal value is 0.

5.4.1 Infinite Integrals of Continuous Functions: Conditions for Convergence

A necessary and sufficient condition for the convergence of $\int_a^\infty f(x) dx$ is that, corresponding to any positive number ϵ , a positive number X should exist such that $\int_{x'}^{x''} f(x) dx < \epsilon$ whenever $x'' \geq x' \geq X$.

The condition is obviously necessary. To prove that it is sufficient, suppose it is satisfied; then, if $n \geq X - a$, and n is a positive integer, and $S_n = \int_a^{a+n} f(x) dx$, we have $|S_{n+p} - S_n| < \epsilon$. Hence, by Section 3.2.2, S_n tends to a limit, S , and then, if $\xi > a + n$,

$$\left| S - \int_a^\xi f(x) dx \right| < \left| S - \int_a^{a+n} f(x) dx \right| + \left| \int_{a+n}^\xi f(x) dx \right| < 2\epsilon,$$

and so,

$$\lim_{\xi \rightarrow \infty} \int_a^\xi f(x) dx = S,$$

so that the condition is sufficient.

5.4.2 Uniformity of Convergence of an Infinite Integral

The integral $\int_a^\infty f(x, \alpha) dx$ is said to converge uniformly with regard to α in a given domain of values of α if, corresponding to an arbitrary positive number ϵ , there exists a number X , independent of α , such that

$$\left| \int_{x'}^\infty f(x, \alpha) dx \right| < \epsilon$$

for all values of α in the domain and all values of $x' \geq X$.

The reader will see without difficulty, on comparing Sections 3.2.2 and 4.3.1 with Section 5.4.1, that a necessary and sufficient condition that $\int_a^\infty f(x, \alpha) dx$ should converge uniformly in a given domain is that, corresponding to any positive number ϵ , there exists a number X , independent of α , such that

$$\left| \int_{x'}^{x''} f(x, \alpha) dx \right| < \epsilon$$

for all values of α in the domain whenever $x'' \geq x' \geq X$.

5.4.3 Tests for the convergence of an infinite integral

There are conditions for the convergence of an infinite integral analogous to those given in Section 3 for the convergence of an infinite series.

The following tests are of special importance.

(I) Absolutely convergent integrals. It may be shown that $\int_a^\infty f(x) dx$ certainly converges if $\int_a^\infty |f(x)| dx$ does so, and the former integral is then said to be **absolutely convergent**. The proof is similar to that of Section 3.3.2.

Example. *The comparison test.* If $|f(x)| \leq g(x)$ and $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges absolutely.

Note. It was observed by Dirichlet⁷⁴ that it is *not necessary* for the convergence of $\int_a^\infty f(x) dx$ that $f(x) \rightarrow 0$ as $x \rightarrow \infty$. The reader may see this by considering the function

$$f(x) = 0, \quad n \leq x \leq (n+1) - (n+1)^{-2}$$

$$f(x) = (n+1)^4(n+1-x)\{x - (n+1) + (n+1)^{-2}\}, \quad (n+1) - (n+1)^{-2} \leq x \leq n+1$$

where n takes all integral values.

For $\int_0^\xi f(x) dx$ increases with ξ and $\int_n^{n+1} f(x) dx = \frac{1}{6}(n+1)^{-2}$, whence it follows without difficulty that $\int_a^\infty f(x) dx$ converges. But when $x = (n+1) - \frac{1}{2}(n+1)^{-2}$, $f(x) = 1/4$, and so $f(x)$ does *not* tend to zero.

For a series $\sum a_n$ to converge, it is necessary that $a_n \rightarrow 0$. You might expect the same to be true for $\int_a^\infty f(x) dx$, i.e., that convergence forces $f(x) \rightarrow 0$, but Dirichlet showed this is wrong.

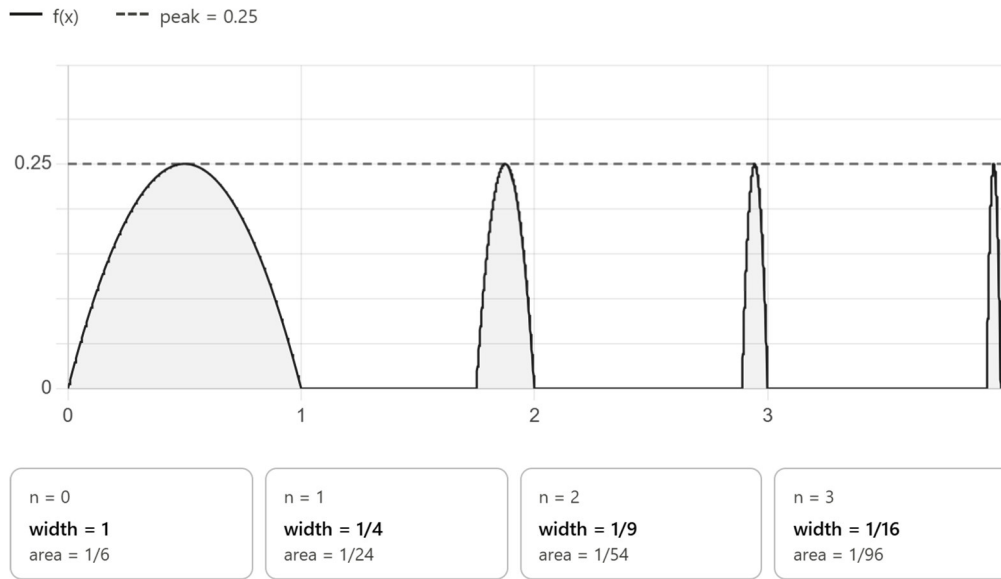
The specific function in the note is constructed to make this concrete. On each unit interval $[n, n+1]$, it is zero almost everywhere, except for a narrow spike crammed near the right endpoint $x = n+1$. As n increases, each spike gets narrower but taller, shaped so that:

- *The area under each spike is $\frac{1}{6}(n+1)^{-2}$, which goes to zero fast enough that the areas sum to a finite total — so the integral converges.*
- *The peak of each spike is always $\frac{1}{4}$, regardless of n . So, no matter how large x gets, f periodically hits $\frac{1}{4}$ again, and therefore cannot tend to zero.*

The intuitive takeaway: the function can "hide" its mass in ever-narrower spikes that become negligible in area but not in height. This is a genuinely different behavior from series, and the note flags it as a trap of which to be aware.

The following figure shows $f(x)$ for the first few values of n .

⁷⁴ Dirichlet's example was $f(x) = \sin x^2$. *Journal für Math.* XVII (1837), p. 60.



The $n = 0$ spike fills the entire unit interval — it's just the parabola $x(1 - x)$. From $n = 1$ onward the spikes compress rapidly toward integer values, but each one still touches $\frac{1}{4}$. The total area under all four spikes is $\frac{1}{6} + \frac{1}{24} + \frac{1}{54} + \frac{1}{96} + \cdots = \frac{1}{6} \sum (n + 1)^{-2}$, which converges — but the function keeps returning to $\frac{1}{4}$ indefinitely, so it cannot tend to zero.

(II) The Maclaurin–Cauchy test.⁷⁵ If $f(x) > 0$ and $f(x) \rightarrow 0$ steadily, $\int_1^\infty f(x) dx$ and $\sum_{n=1}^\infty f(n)$ converge or diverge together.

For

$$f(m) \geq \int_m^{m+1} f(x) dx \geq f(m+1)$$

and so

$$\sum_{m=1}^n f(m) \geq \int_1^{n+1} f(x) dx \geq \sum_{m=2}^{n+1} f(m)$$

The first inequality shows that, if the series converges, the increasing sequence $\int_1^{n+1} f(x) dx$ converges (Section 3.2) when $n \rightarrow \infty$ through integral values, and hence it follows without difficulty that $\int_1^y f(x) dx$ converges when $y \rightarrow \infty$; also, if the integral diverges, so does the series.

The second shows that if the series diverges so does the integral, and if the integral converges so does the series (Section 3.2).

⁷⁵ Maclaurin (*Fluxions*, I, pp. 289, 290) makes a verbal statement practically equivalent to this result. Cauchy's result is given in his *Oeuvres* (2), VII, p. 269.

(III) Bertrand's test.⁷⁶ If $f(x) = O(x^{\lambda-1})$, $\int_a^\infty f(x) dx$ converges when $\lambda < 0$, and if

$$f(x) = O(x^{-1}(\log x)^{\lambda-1})$$

$\int_a^\infty f(x) dx$ converges when $\lambda < 0$.

These results are particular cases of the comparison test given in (I).

(IV) Chartier's test⁷⁷ **for integrals involving periodic functions.**

If $f(x) \rightarrow 0$ steadily as $x \rightarrow \infty$ and if $|\int_a^x \phi(x) dx|$ is bounded as $x \rightarrow \infty$, then $\int_a^\infty f(x) \phi(x) dx$ is convergent.

For if the upper bound of $|\int_a^x \phi(x) dx|$ be A , we can choose X such that $f(x) < \frac{\epsilon}{2A}$ when $x \geq X$, and then by the second mean value theorem, when $x'' \geq x' \geq X$, we have

$$\left| \int_{x'}^{x''} f(x) \phi(x) dx \right| = \left| f(x') \int_{x'}^{\xi} \phi(x) dx \right| = f(x') \left| \int_a^{\xi} \phi(x) dx - \int_a^{x'} \phi(x) dx \right| \leq 2Af(x') < \epsilon$$

which is the condition for convergence.

Example 1. $\int_0^\infty \frac{\sin x}{x} dx$ converges.

Take $f(x) = \frac{1}{x}$ and $\phi(x) = \sin x$. The integrand has a removable singularity at $x = 0$ (since $\frac{\sin x}{x} \rightarrow 1$), so the integral is proper there; Chartier's test is applied from some $a > 0$.

- $f(x) = \frac{1}{x} \rightarrow 0$ steadily
- $|\int_a^x \sin t dt| = |\cos x - \cos a| \leq 2$, which is bounded

Example 2. $\int_0^\infty x^{-1} \sin(x^3 - ax) dx$ converges.

Take $f(x) = \frac{1}{x}$ and $\phi(x) = \sin(x^3 - ax)$. Again, $f(x) \rightarrow 0$ steadily. For the boundedness condition, note that the phase $x^3 - ax$ has derivative $3x^2 - a$, which is positive and growing for

$x > \sqrt{\frac{a}{3}}$. Integrating by parts from some $A > \sqrt{\frac{a}{3}}$, we have

$$\int_A^x \sin(t^3 - at) dt = \left[\frac{\cos(t^3 - at)}{3t^2 - a} \right]_A^x - \int_A^x \frac{6t}{(3t^2 - a)^2} \cos(t^3 - at) dt$$

⁷⁶ *Journal de Math.* VII (1842), pp. 38, 39.

⁷⁷ *Journal de Math.* * XVIII (1853), pp 201–212. It is remarkable that this test for *conditionally* convergent integrals should have been given some years before formal definitions of absolutely convergent integrals.

The first term on the right is bounded by $\frac{2}{3A^2 - a}$, and the remaining term converges absolutely since $\int_A^\infty \frac{6t}{(3t^2 - a)^2} dt < \infty$. So, the boundedness condition holds.

5.4.3.1 Tests for uniformity of convergence of an infinite integral⁷⁸

(I) De la Vallée Poussin's test⁷⁹ The reader will easily see by using the reasoning of Section 4.3.4 that $\int_a^\infty f(x, \alpha) dx$ converges uniformly with regard to α in a domain of values of α if $|f(x, \alpha)| < \mu(x)$, where $\mu(x)$ is independent of α and $\int_a^\infty \mu(x) dx$ converges. [For, choosing X so that $\int_{x'}^{x''} \mu(x) dx < \epsilon$ when $x'' \geq x' \geq X$, we have $\left| \int_{x'}^{x''} f(x, \alpha) dx \right| < \epsilon$, and the choice of X is independent of α .]

Example. $\int_0^\infty x^{\alpha-1} e^{-x} dx$ converges uniformly in any interval $\alpha \in (A, B)$ such that $1 \leq A \leq B$.

The example asserts that the integral converges uniformly as long as α stays within a bounded interval $[A, B]$ with $A \geq 1$. Since $\alpha \leq B$, we have $x^{\alpha-1} \leq x^{B-1}$ for all $\alpha \in [A, B]$, so $|f(x, \alpha)| \leq x^{B-1} e^{-x} = \mu(x)$, which is independent of α and integrable over $[0, \infty)$ (the condition $A \geq 1$ ensures there is no singularity at $x = 0$). De la Vallée Poussin's test then applies directly.

(II) The method of change of variable.

This may be illustrated by an example.

Consider $\int_0^\infty \frac{\sin ax}{x} dx$ where a is real.

We have $\int_{x'}^{x''} \frac{\sin ax}{x} dx = \int_{ax'}^{ax''} \frac{\sin y}{y} dy$.

Since $\int_0^\infty \frac{\sin y}{y} dy$ converges, we can find Y such that $\left| \int_{y'}^{y''} \frac{\sin y}{y} dy \right| < \epsilon$ when $y'' \geq y' \geq Y$.

So $\left| \int_{x'}^{x''} \frac{\sin ax}{x} dx \right| < \epsilon$ whenever $|ax'| \geq Y$; if $|a| \geq \delta > 0$, we therefore get $\left| \int_{x'}^{x''} \frac{\sin ax}{x} dx \right| < \epsilon$

when $x'' \geq x' \geq X = Y/\delta$, and this choice of X is independent of a . So, the convergence is uniform when $a \geq \delta > 0$ and when $a \leq -\delta < 0$.

⁷⁸ The results of this section and of the next section are due to de la Vallée Poussin, *Ann. de la Soc. Scientifique de Bruxelles*, XVI (1892), pp 150–180.

⁷⁹ This name is due to Osgood.

Example. $\int_1^\infty \left\{ \int_0^a \sin(\beta^2 x^3) d\beta \right\} dx$ is uniformly convergent in any range of real values of a . (de la Vallée Poussin.)

Write $z = \beta^2 x^3$, so that $dz = 2\beta x^3 d\beta$. Since $\beta = z^{1/2} x^{-3/2}$, this gives $d\beta = \frac{dz}{2z^{1/2} x^{3/2}}$, and the limits of integration become $z = 0$ and $z = a^2 x^3$. The inner integral transforms to

$$\int_0^a \sin(\beta^2 x^3) d\beta = \frac{1}{2x^{3/2}} \int_0^{a^2 x^3} z^{-\frac{1}{2}} \sin z dz$$

Since $\int_0^\infty z^{-\frac{1}{2}} \sin z dz$ converges, the partial integral $\int_0^t z^{-\frac{1}{2}} \sin z dz$ is bounded uniformly in $t \geq 0$; let M denote this bound. Then

$$\left| \int_0^a \sin(\beta^2 x^3) d\beta \right| \leq \frac{M}{2x^{3/2}} = \mu(x)$$

where $\mu(x)$ is independent of a . Since $\int_1^\infty x^{-3/2} dx$ converges, de la Vallée Poussin's test applies, and the integral is uniformly convergent for all real values of a .

(III) The method of integration by parts

If $\int f(x, \alpha) dx = \phi(x, \alpha) + \int \chi(x, \alpha) dx$, and if $\phi(x, \alpha) \rightarrow 0$ uniformly as $x \rightarrow \infty$ and $\int_a^\infty \chi(x, \alpha) dx$ converges uniformly with regard to α , then obviously, $\int_a^\infty f(x, \alpha) dx$ converges uniformly with regard to α .

(IV) The method of decomposition

Example.

$$\int_0^\infty \frac{\cos x \sin \alpha x}{x} dx = \frac{1}{2} \int_0^\infty \frac{\sin(\alpha + 1)x}{x} dx + \frac{1}{2} \int_0^\infty \frac{\sin(\alpha - 1)x}{x} dx$$

both of the latter integrals converge uniformly in any closed domain of real values of α from which the points $\alpha = \pm 1$ are excluded.

The product-to-sum identity: $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$

Setting $A = x$ and $B = \alpha x$, we have $\sin(1 + \alpha)x - \sin(1 - \alpha)x = 2 \cos x \sin \alpha x$

So, $\cos x \sin \alpha x = \frac{1}{2} \sin(1 + \alpha)x - \frac{1}{2} \sin(1 - \alpha)x = \frac{1}{2} \sin(\alpha + 1)x + \frac{1}{2} \sin(\alpha - 1)x$

5.4.4 Theorems Concerning Uniformly Convergent Infinite Integrals

In this section, the word “domain” means the set of allowable values of the parameter α . The essential point is that the estimates in the definition of uniform convergence must hold for all α in this set. When S is a closed bounded interval, such as $[A, B]$, the required uniformity is often easier to verify. For more general sets, especially unbounded intervals or intervals with excluded endpoints, the necessary uniform estimates must be checked explicitly.

(I) Let $\int_a^\infty f(x, \alpha) dx$ converge uniformly when α lies in a domain S . If $f(x, \alpha)$ is a continuous function of both variables when $x \geq a$ and α lies in S , then $\int_a^\infty f(x, \alpha) dx$ is a continuous function⁸⁰ of α .

For, given ϵ , we can find X , independent of α , such that $\left| \int_\xi^\infty f(x, \alpha) dx \right| < \epsilon$ whenever $\xi \geq X$.

Also, we can find δ , independent of x and α , such that $|f(x, \alpha) - f(x, \alpha')| < \frac{\epsilon}{X-a}$ whenever $|\alpha - \alpha'| < \delta$. That is to say, given ϵ , we can find δ , independent of α , such that

$$\begin{aligned} & \left| \int_a^\infty f(x, \alpha') dx - \int_a^\infty f(x, \alpha) dx \right| \\ & \leq \left| \int_a^X \{f(x, \alpha) - f(x, \alpha')\} dx \right| + \left| \int_X^\infty f(x, \alpha') dx \right| + \left| \int_X^\infty f(x, \alpha) dx \right| < 3\epsilon \end{aligned}$$

whenever $|\alpha' - \alpha| < \delta$, and this is the condition for continuity.

(II) If $f(x, \alpha)$ satisfies the same conditions as in (I), and if α lies in S when $A \leq \alpha \leq B$, then

$$\int_A^B \left\{ \int_a^\infty f(x, \alpha) dx \right\} d\alpha = \int_a^\infty \left\{ \int_A^B f(x, \alpha) d\alpha \right\} dx$$

For, by Section 5.3,

$$\int_A^B \left\{ \int_a^\xi f(x, \alpha) dx \right\} d\alpha = \int_a^\xi \left\{ \int_A^B f(x, \alpha) d\alpha \right\} dx$$

Therefore

$$\begin{aligned} & \left| \int_A^B \left\{ \int_a^\infty f(x, \alpha) dx \right\} d\alpha - \int_a^\xi \left\{ \int_A^B f(x, \alpha) d\alpha \right\} dx \right| \\ & = \left| \int_A^B \left\{ \int_\xi^\infty f(x, \alpha) dx \right\} d\alpha \right| < \int_A^B \epsilon d\alpha < \epsilon(B - A) \end{aligned}$$

for all sufficiently large values of ξ .

⁸⁰ This result is due to Stokes. His statement is that the integral is a continuous function of α if it does not “converge infinitely slowly.”

But, from Sections 3.2 and 5.4.1, this is the condition that $\lim_{\xi \rightarrow \infty} \int_a^\xi \left\{ \int_A^B f(x, \alpha) d\alpha \right\} dx$

should exist, and be equal to $\int_A^B \left\{ \int_a^\infty f(x, \alpha) dx \right\} d\alpha$.

Corollary. The equation

$$\frac{d}{d\alpha} \int_a^\infty \phi(x, \alpha) dx = \int_a^\infty \frac{\partial \phi}{\partial \alpha} dx$$

is true if the integral on the right converges uniformly and the integrand is a continuous function of both variables, when $x \geq a$ and α lies in a domain S , and if the integral on the left is convergent.

Let A be a point of S , and let $\frac{\partial \phi}{\partial \alpha} = f(x, \alpha)$, so that, by Section 5.1.3 Example 3,

$$\int_A^\alpha f(x, \alpha) d\alpha = \phi(x, \alpha) - \phi(x, A)$$

Then $\int_a^\infty \left\{ \int_A^\alpha f(x, \alpha) d\alpha \right\} dx$ converges, that is, $\int_a^\infty \{ \phi(x, \alpha) - \phi(x, A) \} dx$ converges, and therefore, since $\int_a^\infty \phi(x, \alpha) dx$ converges, so does $\int_a^\infty \phi(x, A) dx$.

Then

$$\begin{aligned} \frac{d}{d\alpha} \left[\int_a^\infty \phi(x, \alpha) dx \right] &= \frac{d}{d\alpha} \left[\int_a^\infty \{ \phi(x, \alpha) - \phi(x, A) \} dx \right] \\ &= \frac{d}{d\alpha} \left[\int_a^\infty \left\{ \int_A^\alpha f(x, \alpha) d\alpha \right\} dx \right] \\ &= \frac{d}{d\alpha} \int_A^\alpha \left\{ \int_a^\infty f(x, \alpha) dx \right\} d\alpha \\ &= \int_a^\infty f(x, \alpha) dx = \int_a^\infty \frac{\partial \phi}{\partial \alpha} dx \end{aligned}$$

which is the required result, the change of the order of the integrations having been justified above, and the differentiation of $\int_A^\alpha f(x, \alpha) d\alpha$ with regard to α is justified by Section 5.4.4 (I) and Section 5.1.3 Example 3.

Counterexample to (I): pointwise convergence is not enough

Let $S = (-1, 1)$, and define

$$f(x, \alpha) = |\alpha| e^{-|\alpha|x}, x \geq 0, \alpha \in S$$

For each fixed α , the improper integral $\int_0^\infty f(x, \alpha) dx$ converges.

Indeed, if $\alpha \neq 0$, then $\int_0^\infty |\alpha| e^{-|\alpha|x} dx = 1$, while if $\alpha = 0$, then $f(x, 0) = 0$ for all x , and hence, $\int_0^\infty f(x, 0) dx = 0$.

Thus, the function defined by the improper integral is

$$F(\alpha) = \int_0^\infty f(x, \alpha) dx = \begin{cases} 1, & \alpha \neq 0 \\ 0, & \alpha = 0 \end{cases}$$

This function is not continuous at $\alpha = 0$, even though $f(x, \alpha)$ itself is continuous in both variables.

The reason this does not contradict the theorem of part (I) is that the convergence is not uniform in α . For $\alpha \neq 0$,

$$\int_X^\infty |\alpha| e^{-|\alpha|x} dx = e^{-|\alpha|X}$$

No matter how large X is chosen, we can choose $|\alpha|$ so small that $e^{-|\alpha|X}$ is still close to 1. Therefore, the tail cannot be made small uniformly for all $\alpha \in (-1, 1)$.

So, this example satisfies ordinary pointwise convergence for each fixed α , but the resulting integral is not a continuous function of α . This shows why the theorem of part (I) requires uniform convergence.

5.4.4.1 Pointwise Convergence Is Not Enough

[This section does not appear in the original text from Whittaker and Watson.]

The preceding result shows that uniform convergence is a strong enough condition to justify passing a limiting process through an integral. The following elementary example, on a finite interval, illustrates why pointwise convergence alone is not sufficient.

Define a sequence of functions on $[0, 1]$ as follows:

$$f_n(x) = \begin{cases} n, & 0 < x < 1/n \\ 0, & \text{otherwise} \end{cases}$$

Each function f_n is Riemann integrable on $[0, 1]$. For each fixed x in $[0, 1]$, we have

$$\lim_{n \rightarrow \infty} f_n(x) = 0$$

Thus, the pointwise limit function is the zero function. However, for every n ,

$$\int_0^1 f_n(x) dx = 1$$

Indeed, f_n is equal to n on an interval of length $\frac{1}{n}$, and is zero elsewhere. Hence the area under the graph is $n \cdot \frac{1}{n} = 1$. Therefore,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = 1$$

while

$$\int_0^1 \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx = \int_0^1 0 dx = 0$$

Thus,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \neq \int_0^1 \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx$$

This example shows that pointwise convergence alone does not justify passing a limit through an integral. The functions f_n converge pointwise to 0, but their integrals do not converge to the integral of the limiting function.

5.5 Improper Integrals and Principal Values

If $|f(x)| \rightarrow \infty$ as $x \rightarrow a + 0$, then $\lim_{\delta \rightarrow +0} \int_{a+\delta}^b f(x) dx$ may exist, and is written simply $\int_a^b f(x) dx$.

This limit is called an improper integral. For example,

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{a \rightarrow 0+} \int_a^1 \frac{1}{\sqrt{x}} dx = \lim_{a \rightarrow 0+} [2\sqrt{x}]_a^1 = \lim_{a \rightarrow 0+} (2\sqrt{1} - 2\sqrt{a}) = 2 - 0 = 2$$

If $|f(x)| \rightarrow \infty$ as $x \rightarrow c$, where $a < c < b$, then $\lim_{\delta \rightarrow +0} \int_a^{c-\delta} f(x) dx + \lim_{\delta' \rightarrow +0} \int_{c+\delta'}^b f(x) dx$

may exist. This is also written $\int_a^b f(x) dx$ and is also called an improper integral.

A simple example is $f(x) = \frac{1}{\sqrt{|x-c|}}$, $a < c < b$.

Then $|f(x)| \rightarrow \infty$ as $x \rightarrow c$, but the improper integral exists because the singularity is not too strong.

For a concrete case, take $a = 0$, $b = 1$, and $c = \frac{1}{2}$. Define $f(x) = \frac{1}{\sqrt{|x-\frac{1}{2}|}}$

$$\text{We then have } \int_0^1 f(x) dx = \lim_{\delta \rightarrow 0+} \int_0^{\frac{1}{2}-\delta} \frac{dx}{\sqrt{\frac{1}{2}-x}} + \lim_{\delta' \rightarrow 0+} \int_{\frac{1}{2}+\delta'}^1 \frac{dx}{\sqrt{x-\frac{1}{2}}} = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

So, this is an improper integral with an infinite discontinuity at an interior point, but the integral exists and is finite. The reason is that a singularity behaving like $\frac{1}{\sqrt{|x-c|}}$ is integrable near c , while a singularity like $\frac{1}{|x-c|}$ would not be integrable.

It might, however, happen that neither of these limits exists when $\delta, \delta' \rightarrow 0$ independently, but that

$$\lim_{\delta \rightarrow +0} \left\{ \int_a^{c-\delta} f(x) dx + \int_{c+\delta}^b f(x) dx \right\}$$

exists. This is called Cauchy's principal value of $\int_a^b f(x) dx$ and is written, for brevity, $P \int_a^b f(x) dx$.

Results similar to those of Sections 5.4-5.4.4 may be obtained for improper integrals. But all that is required in practice is:

- i. the idea of absolute convergence;
- ii. the analogue of Bertrand's test for convergence;
- iii. the analogue of de la Vallée Poussin's test for uniformity of convergence.

The construction of these tests is left to the reader, as is also the consideration of integrals in which the integrand has an infinite limit at more than one point of the range of integration.⁸¹

Examples.

(1) $\int_0^\pi x^{-1/2} \cos x \, dx$ is an improper integral.

(2) $\int_0^1 x^{\lambda-1} (1-x)^{\mu-1} \, dx$ is an improper integral if $0 < \lambda < 1$, $0 < \mu < 1$. It does not converge for negative values of λ and μ .

(3) $P \int_0^2 \frac{x^{a-1}}{1-x} \, dx$ is the principal value of an improper integral when $0 < a < 1$.

5.5.1 The Inversion of the Order of Integration of a Certain Repeated Integral

This section is included primarily as a technical illustration of the difficulties that can arise when changing the order of integration in the presence of singularities. The details of the estimates are not essential for a first reading and may be skipped without loss of continuity. The main point is that, for improper integrals, changing the order of integration is not automatic; it must be justified by showing that the singularities are sufficiently well controlled.

General conditions for the legitimacy of inverting the order of integration, when the integrand is not continuous, are difficult to obtain.

The following is a good example of the difficulties to be overcome in inverting the order of integration in a repeated improper integral.

Let $f(x, y)$ be a continuous function of both variables, and let

$$0 < \lambda \leq 1, \quad 0 < \mu \leq 1, \quad 0 < \nu \leq 1$$

Then

$$\begin{aligned} & \int_0^1 dx \left\{ \int_0^{1-x} x^{\lambda-1} y^{\mu-1} (1-x-y)^{\nu-1} f(x, y) \, dy \right\} \\ &= \int_0^1 dy \left\{ \int_0^{1-y} x^{\lambda-1} y^{\mu-1} (1-x-y)^{\nu-1} f(x, y) \, dx \right\} \end{aligned}$$

This integral, which was first employed by Dirichlet, is of importance in the theory of integral equations. The investigation given here is due to W. A. Hurwitz.⁸²

Let $\phi(x, y) = x^{\lambda-1} y^{\mu-1} (1-x-y)^{\nu-1} f(x, y)$, and let M be an upper bound of $|f(x, y)|$.

⁸¹ For a detailed discussion of improper integrals, the reader is referred either to Hobson's or to Pierpont's *Functions of a Real Variable*. The connection between infinite integrals and improper integrals is exhibited by Bromwich, *Infinite Series*, §164.

⁸² *Annals of Mathematics*, IX (1908), p. 183.

The possible singularities of $\phi(x, y)$ occur on the sides of the triangle $x = 0$, $y = 0$, $x + y = 1$.

To avoid these singularities, let δ be a positive number less than $\frac{1}{3}$, and consider the smaller triangle bounded by $x = \delta$, $y = \delta$, $x + y = 1 - \delta$.

On and inside this smaller triangle, $\phi(x, y)$ is continuous. Therefore, by the result on double integrals in Section 5.3,

$$\int_{\delta}^{1-2\delta} dx \left\{ \int_{\delta}^{1-x-\delta} \phi(x, y) dy \right\} = \int_{\delta}^{1-2\delta} dy \left\{ \int_{\delta}^{1-y-\delta} \phi(x, y) dx \right\}$$

The question is whether this equality remains true as $\delta \rightarrow 0$. The omitted parts of the full triangle consist of thin boundary regions near $x = 0$, $y = 0$, $x + y = 1$.

In the original text, Whittaker and Watson estimate the integrals over these boundary regions and show that their contributions tend to zero as $\delta \rightarrow 0$. The key point is that the singular factors are still integrable because $\lambda > 0$, $\mu > 0$, $\nu > 0$.

For example, near $y = 0$, one obtains an estimate of the form

$$\left| \int_{\delta}^{1-2\delta} \left\{ \int_0^{\delta} \phi(x, y) dy \right\} dx \right| \leq C\delta^{\mu},$$

where C is independent of δ . Since $\mu > 0$, this tends to zero as $\delta \rightarrow 0$. Similar estimates apply near $x = 0$ and near $x + y = 1$.

Hence the equality on the smaller triangle passes to the limit, and we obtain

$$\int_0^1 dx \left\{ \int_0^{1-x} \phi(x, y) dy \right\} = \int_0^1 dy \left\{ \int_0^{1-y} \phi(x, y) dx \right\}$$

Substituting back for $\phi(x, y)$, this proves the stated theorem.

Corollary. Writing $\xi = a + (b - a)x$, $\eta = b - (b - a)y$, we see that, if $\phi(\xi, \eta)$ is continuous, then

$$\begin{aligned} & \int_a^b d\xi \left\{ \int_{\xi}^b (\xi - a)^{\lambda-1} (b - \eta)^{\mu-1} (\eta - \xi)^{\nu-1} \phi(\xi, \eta) d\eta \right\} \\ &= \int_a^b d\eta \left\{ \int_a^{\eta} (\xi - a)^{\lambda-1} (b - \eta)^{\mu-1} (\eta - \xi)^{\nu-1} \phi(\xi, \eta) d\xi \right\} \end{aligned}$$

This is called Dirichlet's formula.

[Note. What are now called infinite and improper integrals were defined by Cauchy in *Leçons sur le calcul infinitésimal* in 1823, though the idea of infinite integrals seems to date from Maclaurin (1742). The test for convergence was employed by Chartier (1853). Stokes (1847) distinguished between “essentially,” or absolutely, convergent integrals and non-essentially convergent integrals, though he did not give a formal definition. Such a definition was given by Dirichlet in 1854 and 1858. In the early part of the nineteenth century improper integrals received more attention than infinite integrals, probably because it was not fully realized that an infinite integral is really the limit of an integral.]

5.6 Complex Integration

Integration with respect to a real variable x may be regarded as integration along a particular path, namely part of the real axis, in the Argand diagram. Let $f(z) = P + iQ$ be a function of a complex variable z which is continuous along a simple curve AB in the Argand diagram.⁸³

Let the equations of the curve be $x = x(t), y = y(t), a \leq t \leq b$.

Let $x(a) + iy(a) = z_0$ and $x(b) + iy(b) = z_1$.

Then, if $x(t)$ and $y(t)$ have continuous derivatives,⁸⁴ we define

$$\int_{z_0}^{z_1} f(z) dz$$

taken along the simple curve AB to mean

$$\int_a^b (P + iQ) \left(\frac{dx}{dt} + i \frac{dy}{dt} \right) dt$$

The “length” of the curve AB will be defined as

$$\int_a^b \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt$$

This length obviously exists if $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are continuous. We have thus reduced the discussion of a complex integral to the discussion of four real integrals, namely

$$\int_a^b P \frac{dx}{dt} dt, \quad \int_a^b P \frac{dy}{dt} dt, \quad \int_a^b Q \frac{dx}{dt} dt, \quad \int_a^b Q \frac{dy}{dt} dt$$

By Section 5.1.3, Example 4, this definition is consistent with the definition of an integral when AB happens to be part of the real axis.

Examples. First, consider

$$\int_{z_0}^{z_1} f(z) dz = - \int_{z_1}^{z_0} f(z) dz$$

with the paths of integration being the same, but taken in opposite directions, in the two integrals.

Also, $\int_{z_0}^{z_1} dz = z_1 - z_0$.

Using the substitutions $z = x + iy, dz = \left(\frac{dx}{dt} + i \frac{dy}{dt} \right) dt$, we have

$$\int_{z_0}^{z_1} z dz = \int_a^b \left\{ \left(x \frac{dx}{dt} - y \frac{dy}{dt} \right) + i \left(x \frac{dy}{dt} + y \frac{dx}{dt} \right) \right\} dt$$

⁸³ A treatment of complex integration based on a different set of ideas, and not making so many assumptions concerning the curve AB , will be found in Watson's Complex Integration and Cauchy's Theorem.

⁸⁴ This assumption will be made throughout the subsequent work. Compare Section 5.1.3, Example 4.

$$= \left[\frac{1}{2}x^2 - \frac{1}{2}y^2 + ixy \right]_{t=a}^{t=b} = \frac{1}{2}(z_1^2 - z_0^2)$$

5.6.1 The Fundamental Theorem of Complex Integration

From Section 5.1.3, the reader will easily deduce the following theorem.

Let a sequence of points be taken on a simple curve from z_0 to z_1 . Let the first n of them, rearranged in order of magnitude of their parameters, be called $w_1^{(n)}, w_2^{(n)}, \dots, w_n^{(n)}$ where $w_0^{(n)} = z_0, w_{n+1}^{(n)} = z_1$

Let their parameters be $t_1^{(n)}, t_2^{(n)}, \dots, t_n^{(n)}$ with $t_0^{(n)} = a, t_{n+1}^{(n)} = b$ and let the sequence be such that, given any positive number δ , we can find N such that, when $n > N$, $t_{r+1}^{(n)} - t_r^{(n)} < \delta$ for $r = 0, 1, 2, \dots, n$.

Let $\zeta_r^{(n)}$ be any point whose parameter lies between $t_r^{(n)}$ and $t_{r+1}^{(n)}$. Then we can make

$$\left| \sum_{r=0}^n (w_{r+1}^{(n)} - w_r^{(n)}) f(\zeta_r^{(n)}) - \int_{z_0}^{z_1} f(z) dz \right|$$

arbitrarily small by taking n sufficiently large.

5.6.2 An Upper Bound for a Complex Integral

Let M be the upper bound of the continuous function $|f(z)|$.

Then $\left| \int_{z_0}^{z_1} f(z) dz \right| \leq \int_a^b |f(z)| \left| \frac{dx}{dt} + i \frac{dy}{dt} \right| dt$. But $\left| \frac{dx}{dt} + i \frac{dy}{dt} \right| = \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2}$

Therefore, $\left| \int_{z_0}^{z_1} f(z) dz \right| \leq \int_a^b M \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt$

Hence $\left| \int_{z_0}^{z_1} f(z) dz \right| \leq M\ell$ where ℓ is the length of the curve from z_0 to z_1 .

That is to say, $\left| \int_{z_0}^{z_1} f(z) dz \right|$ cannot exceed $M\ell$.

5.7 Integration of Infinite Series

We shall now show that if $S(z) = u_1(z) + u_2(z) + \dots$ is a uniformly convergent series of continuous functions of z , for values of z contained within some region, then the series

$$\int_C u_1(z) dz + \int_C u_2(z) dz + \dots$$

where all the integrals are taken along some path C in the region, is convergent, and has for its sum $\int_C S(z) dz$.

Proof: For, writing $S(z) = u_1(z) + u_2(z) + \cdots + u_n(z) + R_n(z)$, we have

$$\int_C S(z) dz = \int_C u_1(z) dz + \cdots + \int_C u_n(z) dz + \int_C R_n(z) dz$$

Now, since the series is uniformly convergent, to every positive number ε there corresponds a number r , independent of z , such that when $n > r$, we have $|R_n(z)| < \varepsilon$ for all values of z in the region considered.

Therefore, if ℓ is the length of the path of integration, we have, by Section 5.6.2, $|\int_C R_n(z) dz| < \varepsilon \ell$

Therefore, the modulus of the difference between $\int_C S(z) dz$ and $\sum_{m=1}^n \int_C u_m(z) dz$ can be made less than any positive number by giving n any sufficiently large value. This proves both that the series $\sum_{m=1}^{\infty} \int_C u_m(z) dz$ is convergent, and that its sum is $\int_C S(z) dz$. ■

Corollary. As in the corollary to Section 5.4.4, it may be shown that⁸⁵

$$\frac{d}{dz} \sum_{n=0}^{\infty} u_n(z) = \sum_{n=0}^{\infty} \frac{d}{dz} u_n(z)$$

if the series on the right converges uniformly and the series on the left is convergent.

Example 1. Consider the series

$$\sum_{n=1}^{\infty} \frac{2x\{n(n+1)\sin^2(x^2) - 1\}\cos(x^2)}{\{1 + n^2\sin^2(x^2)\}\{1 + (n+1)^2\sin^2(x^2)\}}$$

in which x is real.

The n th term is

$$\frac{2xn \cos(x^2)}{1 + n^2\sin^2(x^2)} - \frac{2x(n+1)\cos(x^2)}{1 + (n+1)^2\sin^2(x^2)}$$

and the sum of n terms is therefore

$$\frac{2x \cos(x^2)}{1 + \sin^2(x^2)} - \frac{2x(n+1)\cos(x^2)}{1 + (n+1)^2\sin^2(x^2)}$$

Hence, the series is absolutely convergent for all real values of x except $\pm\sqrt{m\pi}, m = 1, 2, \dots$ but

$$R_n(x) = \frac{2x(n+1)\cos(x^2)}{1 + (n+1)^2\sin^2(x^2)}$$

and if n is any integer, by taking $x = (n+1)^{-1}$ this has the limit 2 as $n \rightarrow \infty$. The series is therefore non-uniformly convergent near $x = 0$.

⁸⁵ Here, $\frac{df(z)}{dz}$ means $\lim_{h \rightarrow 0} \frac{f(z+h)-f(z)}{h}$ where $h \rightarrow 0$ along a definite simple curve.

Now the sum to infinity of the series is $\frac{2x \cos(x^2)}{1 + \sin^2(x^2)}$ and so, the integral from 0 to x of the sum of the series is $\arctan(\sin(x^2))$.

On the other hand, the sum of the integrals from 0 to x of the first n terms of the series is

$$\arctan(\sin(x^2)) - \arctan((n+1)\sin(x^2))$$

and as $n \rightarrow \infty$, this tends to $\arctan(\sin(x^2)) - \frac{1}{2}\pi$.

Therefore, the integral of the sum of the series differs from the sum of the integrals of the terms by $\frac{1}{2}\pi$.

Example 2. Discuss, in a similar manner, the series

$$\sum_{n=1}^{\infty} \frac{2e^n x \{1 - n(e-1) + e^{n+1}x^2\}}{n^2(n+1)^2(1+e^n x^2)(1+e^{n+1}x^2)}$$

for real values of x .

Example 3. Discuss the series $u_1 + u_2 + u_3 + \dots$ where $u_1 = ze^{-z^2}$ and, for $n \geq 2$,

$$u_n = nze^{-nz^2} - (n-1)ze^{-(n-1)z^2}$$

for real values of z .

Let $A_n(z) = nze^{-nz^2}$.

Then $u_1 = A_1$ and, for $n \geq 2$, $u_n = A_n - A_{n-1}$.

Hence, the series telescopes. The sum of the first n terms is $u_1 + u_2 + \dots + u_n = A_n(z)$ that is,

$$u_1 + u_2 + \dots + u_n = nze^{-nz^2}$$

For each fixed real value of z , we have $nze^{-nz^2} \rightarrow 0$ as $n \rightarrow \infty$. Therefore, the sum to infinity is 0 for all real values of z .

Since the terms u_n are real and ultimately all of the same sign, the convergence is absolute.

Now consider the series obtained by integrating the terms:

$$\int_0^z u_1 dz + \int_0^z u_2 dz + \int_0^z u_3 dz + \dots$$

The sum of the first n terms in this integrated series is

$$\int_0^z u_1 dz + \int_0^z u_2 dz + \dots + \int_0^z u_n dz$$

For a finite sum, we may combine the integrals:

$$\int_0^z u_1 dz + \int_0^z u_2 dz + \dots + \int_0^z u_n dz = \int_0^z (u_1 + u_2 + \dots + u_n) dz$$

Using the telescoping sum, $u_1 + u_2 + \dots + u_n = nze^{-nz^2}$, we get

$$\int_0^z (u_1 + u_2 + \dots + u_n) dz = \int_0^z nze^{-nz^2} dz$$

To avoid confusion between the endpoint z and the variable of integration, write the integral with a dummy variable t : $\int_0^z nt e^{-nt^2} dt$.

Since $\frac{d}{dt} e^{-nt^2} = -2nte^{-nt^2}$ we have

$$\int_0^z nt e^{-nt^2} dt = -\frac{1}{2} [e^{-nt^2}]_0^z$$

Therefore,

$$\int_0^z nt e^{-nt^2} dt = -\frac{1}{2} (e^{-nz^2} - 1)$$

and so,

$$\int_0^z nt e^{-nt^2} dt = \frac{1}{2} (1 - e^{-nz^2})$$

Thus, the sum of the first n integrated terms is $\frac{1}{2} (1 - e^{-nz^2})$.

For fixed $z \neq 0$, $e^{-nz^2} \rightarrow 0$ as $n \rightarrow \infty$. Hence, $\frac{1}{2} (1 - e^{-nz^2}) \rightarrow \frac{1}{2}$.

But the sum of the original series is 0, so the integral from 0 to z of the sum of the series is

$$\int_0^z 0 dt = 0$$

Thus, for $z \neq 0$, the sum of the integrals is $\frac{1}{2}$ whereas the integral of the sum is 0. This is the discrepancy that Whittaker and Watson are pointing out.

The explanation of the discrepancy is the non-uniformity of the convergence near $z = 0$. The remainder after n terms is $R_n(z) = -nze^{-nz^2}$.

If we take $z = \frac{1}{n}$ then $R_n\left(\frac{1}{n}\right) = -e^{-1/n}$ and hence, $\left|R_n\left(\frac{1}{n}\right)\right| = e^{-1/n}$. This tends to 1, not to 0.

Therefore, the remainder cannot be made arbitrarily small uniformly for z near 0. The series is therefore non-uniformly convergent near $z = 0$.

Example 4. Compare the values of $\int_0^z (\sum_{n=1}^{\infty} u_n) dz$ and $\sum_{n=1}^{\infty} \int_0^z u_n dz$ where

$$u_n = \frac{2n^2 z}{(1 + n^2 z^2) \log(n+1)} - \frac{2(n+1)^2 z}{\{1 + (n+1)^2 z^2\} \log(n+2)}$$

(Trinity, 1903)

As printed in Whittaker & Watson (4th ed., p. 79), the two terms of u_n have unequal degrees: the first denominator is $1 + n^4 z^4$ while the second is $1 + (n + 1)^2 z^2$. With this mismatch the series does not telescope and in fact diverges. The second term behaves like $-\frac{2}{z \log n}$ for large n , and $\sum \frac{1}{\log n}$ diverges. So, the proposed comparison is not well posed. The intended form has both denominators of degree two; that is, the first denominator should read $1 + n^2 z^2$. With that correction $u_n = a_n - a_{n+1}$ with $a_n = \frac{2n^2 z}{(1+n^2 z^2) \log(n+1)}$, and the comparison below proceeds as intended.

Hint. Write $a_n = \frac{2n^2 z}{(1+n^2 z^2) \log(n+1)}$, so that $u_n = a_n - a_{n+1}$; the series telescopes. Since $a_{n+1} \rightarrow 0$ for each fixed z , the sum of the series is

$$\sum_{n=1}^{\infty} u_n = a_1 = \frac{2z}{(1+z^2) \log 2} \quad (\text{and } 0 \text{ at } z = 0).$$

Integral of the sum: With the substitution $w = t^2$,

$$\int_0^z \frac{2t}{(1+t^2) \log 2} dt = \frac{\log(1+z^2)}{\log 2}.$$

Sum of the integrals: The substitution $w = n^2 t^2$ gives

$$A_n(z) = \int_0^z a_n(t) dt = \frac{\log(1+n^2 z^2)}{\log(n+1)},$$

and so, $\int_0^z u_n dt = A_n - A_{n+1}$. Thus, the sum of the first m integrated terms telescopes to $A_1(z) - A_{m+1}(z)$. The decisive point is that A_{m+1} does not tend to zero: because $\log(1 + (m+1)^2 z^2) \sim 2 \log(m+1)$ matches the $\log(m+2)$ in the denominator,

$$A_{m+1}(z) = \frac{\log(1 + (m+1)^2 z^2)}{\log(m+2)} \rightarrow 2 \quad \text{as } m \rightarrow \infty.$$

Therefore

$$\sum_{n=1}^{\infty} \int_0^z u_n dt = \frac{\log(1+z^2)}{\log 2} - 2.$$

Conclusion: The two values differ by exactly 2:

$$\int_0^z \left(\sum_{n=1}^{\infty} u_n \right) dz - \sum_{n=1}^{\infty} \int_0^z u_n dt = 2.$$

The discrepancy is a consequence of the non-uniform convergence of the series near $z = 0$: the remainder $a_{n+1}(z)$ attains $\frac{n+1}{2\log(n+2)}$, which tends to ∞ , at the point $z = \frac{1}{n+1}$, so it cannot be made uniformly small.

Miscellaneous Examples

1. Show that the integrals

$$\int_0^{\infty} \sin(x^2) dx, \quad \int_0^{\infty} \cos(x^2) dx, \quad \int_0^{\infty} x \exp\{-x^6 \sin^2 x\} dx$$

converge. (Dirichlet and Du Bois-Reymond)

Solution: First consider $\int_0^{\infty} \sin x^2 dx$.

(I) The integral over $[0,1]$ is an ordinary proper integral, so there is no difficulty there. For the tail, use the substitution $u = x^2$.

Then $du = 2x dx$ and $dx = \frac{du}{2\sqrt{u}}$

Hence,

$$\int_1^{\infty} \sin x^2 dx = \frac{1}{2} \int_1^{\infty} u^{-1/2} \sin u du$$

The function $u^{-1/2}$ is positive, decreasing, and tends to 0. Also, the indefinite integrals of $\sin u$ are bounded, since $\int \sin u du = -\cos u$.

Therefore, by Dirichlet's test, $\int_1^{\infty} u^{-1/2} \sin u du$ converges.

Thus, $\int_0^{\infty} \sin(x^2) dx$ converges.

(II) The proof for $\int_0^{\infty} \cos(x^2) dx$ is the same.

Again using $u = x^2$, we obtain

$$\int_1^{\infty} \cos(x^2) dx = \frac{1}{2} \int_1^{\infty} u^{-1/2} \cos u du$$

The function $u^{-1/2}$ decreases to 0, and the indefinite integrals of $\cos u$ are bounded, since $\int \cos u du = \sin u$.

Thus, Dirichlet's test gives convergence.

(III) It remains to consider

$$\int_0^{\infty} x \exp\{-x^6 \sin^2 x\} dx$$

Hint. The only possible difficulty occurs near the zeros of $\sin x$, namely near $x = k\pi$. On the interval $k\pi - \frac{\pi}{2} \leq x \leq k\pi + \frac{\pi}{2}$ write $x = k\pi + t$.

Then x is comparable to k , and $|\sin x| = |\sin t| \geq C|t|$ for $|t| \leq \pi/2$.

Hence, the contribution from this interval is bounded by a constant multiple of

$$k \int_{-\pi/2}^{\pi/2} \exp\{-C k^6 t^2\} dt$$

Extending the integral to the whole real line gives a Gaussian integral of order k^{-3} , and so, the contribution from the interval around $k\pi$ is of order $k \cdot k^{-3} = k^{-2}$

Since $\sum_{k=1}^{\infty} k^{-2}$ converges, the integral converges.

2. If a is real, the integral

$$\int_0^{\infty} \frac{\cos(ax)}{1+x^2} dx$$

is a continuous function of a . (Stokes)

Hint: Split the integral into two, i.e., from 0 to A and from A to ∞ .

On $[0, A]$, continuity in a is ordinary continuity of a proper integral depending on a parameter.

On $[A, \infty)$, use the estimate

$$\left| \frac{\cos ax}{1+x^2} \right| \leq \frac{1}{1+x^2}$$

and make the tail uniformly small.

3. Discuss the uniformity of the convergence of

$$\int_0^{\infty} x \sin(x^3 - ax) dx$$

Hint, from the original text,

$$\begin{aligned} 3 \int x \sin(x^3 - ax) dx &= -\left(\frac{1}{x} + \frac{a}{3x^3}\right) \cos(x^3 - ax) \\ &- \int \left(\frac{1}{x^2} + \frac{a}{x^4}\right) \cos(x^3 - ax) dx + \frac{1}{3} a^2 \int \frac{\sin(x^3 - ax)}{x^3} dx \end{aligned}$$

(de la Vallée Poussin)

4. Show that

$$\int_0^{\infty} \exp\{-e^{ia}(x^3 - nx)\} dx$$

converges uniformly in the range $-\frac{1}{2}\pi < a < \frac{1}{2}\pi$ of values of a . (Stokes)

Hint: Use the real part of e^{ia} . Since $\operatorname{Re}(e^{ia}) = \cos a$ the exponential has decay when a stays in the interval $-\frac{1}{2}\pi < a < \frac{1}{2}\pi$. Split the integral into a finite part and a tail. For the tail, compare with an exponentially decaying function.

5. Discuss the convergence of

$$\int_0^{\infty} \frac{x^{\mu} dx}{1+x^{\nu}|\sin x|^p}$$

when μ, ν, p are positive. (Hardy, Messenger, XXXI (1902), p. 177)

Hint: To study

$$\int_0^\infty \frac{x^\mu dx}{1 + x^\nu |\sin x|^p}$$

there is no difficulty near $x = 0$, since $\mu > 0$. The main issue is the behavior near the zeros of $\sin x$ as $x \rightarrow \infty$.

Near $x = k\pi$, write $x = k\pi + t$.

Then x is comparable to k , and $|\sin x|$ is comparable to $|t|$.

So, the contribution from one period is comparable to

$$k^\mu \int_{-\pi/2}^{\pi/2} \frac{dt}{1 + k^\nu |t|^p}$$

Thus, the problem reduces to estimating

$$\int_0^{\pi/2} \frac{dt}{1 + k^\nu t^p}$$

as $k \rightarrow \infty$.

Use the substitution $u = k^{\nu/p} t$ and distinguish the cases $0 < p < 1, p = 1, p > 1$

This gives the required convergence condition by comparing the resulting series over k .

6. Examine the convergence of the integrals

$$\int_0^\infty \left(\frac{1}{x} - \frac{1}{2} e^{-x} + \frac{1}{1 - e^x} \right) \frac{dx}{x}, \quad \int_0^\infty \frac{\sin(x + x^2)}{x^n} dx$$

(Math. Trip. 1914)

Hint: For the first integral, expand the expression near $x = 0$ to see the cancellation of the singular terms. Then check the behavior as $x \rightarrow \infty$.

For the second integral, analyze separately the behavior near 0 and the oscillatory behavior as $x \rightarrow \infty$.

7. Show that $\int_\pi^\infty \frac{dx}{x^2 (\sin x)^{2/3}}$ exists.

Hint: The possible difficulties occur near the zeros of $\sin x$, namely near multiples of π . Use

$|\sin x| \sim |x - k\pi|$ near $x = k\pi$. Since $|x - k\pi|^{-\frac{2}{3}}$ is locally integrable, the singularities are harmless. The factor x^{-2} then ensures convergence over the infinitely many intervals.

8. Show that $\int_a^\infty x^{-n} e^{\sin x} \sin 2x dx$ converges if $a > 0, n > 0$ (Math. Trip. 1908)

Hint: Use the identity $\sin 2x = 2 \sin x \cos x$ and observe that the oscillatory factor has a bounded primitive. Then integrate by parts, using the fact that $x^{-n} \rightarrow 0$ when $n > 0$.

9. If a series

$$g(z) = \sum_{v=0}^{\infty} (c_v - c_{v+1}) \sin (2v + 1)\pi z$$

in which $c_0 = 0$ converges uniformly in an interval, show that $g(z) \frac{\pi}{\sin \pi z}$ is the derivative of the series

$$f(z) = \sum_{v=1}^{\infty} \frac{c_v}{v} \sin 2v\pi z$$

(Lerch, Ann. de l'Éc. norm. sup. (3), XII (1895), p. 351)

Hint: Use the trigonometric identity relating

$$\frac{\sin(2v + 1)\pi z}{\sin \pi z}$$

to a finite sum of cosines. This converts $g(z)\pi / \sin \pi z$ into the formally differentiated series for $f(z)$. Uniform convergence justifies passing to the limit.

10. Show that

$$\int_0^{\infty} \int_0^{\infty} \cdots \int_0^{\infty} \frac{dx_1 dx_2 \cdots dx_n}{(x_1^2 + x_2^2 + \cdots + x_n^2)^a}, \quad \int_0^{\infty} \int_0^{\infty} \cdots \int_0^{\infty} \frac{dx_1 dx_2 \cdots dx_n}{x_1^{\alpha} + x_2^{\beta} + \cdots + x_n^{\lambda}}$$

converge when $a > \frac{1}{2}n$ and $\alpha^{-1} + \beta^{-1} + \cdots + \lambda^{-1} < 1$, respectively. (Math. Trip. 1904)

Hint: For the first integral, use polar coordinates in n dimensions. The integrand behaves like r^{-2a} and the volume element contributes a factor r^{n-1} .

For the second integral, use scaling. The condition $\alpha^{-1} + \beta^{-1} + \cdots + \lambda^{-1} < 1$ is the condition that the corresponding volume growth is small enough for convergence.

11. If $f(x, y)$ is a continuous function of both x and y in the ranges $a \leq x \leq b, a \leq y \leq b$ except that it has ordinary discontinuities at points on a finite number of curves, with continuously turning tangents, each of which meets any line parallel to the coordinate axes only a finite number of times, then $\int_a^b f(x, y) dx$ is a continuous function of y .

Hint, from the original text: Consider

$$\int_a^{a_1 - \delta_1} + \int_{a_1 + \varepsilon_1}^{a_2 - \delta_2} + \cdots + \int_{a_n + \varepsilon_n}^b \{f(x, y + h) - f(x, y)\} dx$$

where the numbers $\delta_1, \delta_2, \dots, \varepsilon_1, \varepsilon_2, \dots$ are so chosen as to exclude the discontinuities of $f(x, y + h)$ from the range of integration, and $a_1, a_2, \dots$ are the discontinuities of $f(x, y)$. (Bôcher)

Hint: Exclude small intervals around the discontinuities (as shown above). On the remaining intervals, f is uniformly continuous. The contribution from the excluded intervals can be made small because their total length can be made small, while the remaining contribution is small by uniform continuity.

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Appendix: The Modern Extension (The Lebesgue Integral)

I don't know what's the matter with people: they don't learn by understanding; they learn by some other way—by rote or something. Their knowledge is so fragile! —Richard Feynman

The indoctrination begins. Students learn that mathematics is not something you do, but something that is done to you. ... The holy tablets, or 'Math Books,' are handed out, and the students learn to address the church elders as 'they' (as in 'What do they want here?').

—Paul Lockhart

The preceding section developed the theory of the Riemann integral. This theory gives a rigorous meaning to the idea of area under a curve by subdividing the domain of a function into intervals and forming upper and lower sums. For continuous functions, and for many functions with only a limited number of discontinuities, the Riemann integral is both natural and effective.

Nevertheless, the Riemann integral has limitations. Some functions are too discontinuous to be handled well by Riemann's method, even though they may be quite simple from another point of view, e.g., see Example 0 in Section 5.1.2. In addition, the Riemann integral does not always behave well when one passes to limits of sequences of functions, e.g., see the example in Section 5.4.4.1. Since much of modern analysis depends on limiting processes, this limitation is significant.

The Lebesgue integral was developed to overcome these difficulties. It does not contradict the Riemann integral. Rather, it extends it. For the functions usually encountered in elementary calculus, the two theories give the same answer. The advantage of the Lebesgue theory is that it applies to a wider class of functions and has better analytical properties.

Motivation

The difference between the Riemann and the Lebesgue integral can be put in a single sentence: Riemann partitions the domain of the function, while Lebesgue partitions its range. The rest of this section unpacks that sentence.

Riemann's method divides the interval $[a, b]$ into small subintervals

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

picks a sample point t_i in each subinterval $[x_{i-1}, x_i]$, and approximates the area under the graph by a sum of rectangles

$$\sum_{i=1}^n f(t_i) (x_i - x_{i-1})$$

Each rectangle stands on a piece of the x -axis: its width comes from the domain, and its height is whatever value the function happens to take at the sample point (see Figure 7). The method succeeds precisely when the choice of t_i hardly matters, that is, when f is nearly constant on each small subinterval. For continuous functions this is guaranteed. But if f oscillates wildly within every interval, however small, then the sample value $f(t_i)$ says nothing reliable about the rest of the subinterval, and the rectangles approximate nothing in particular.

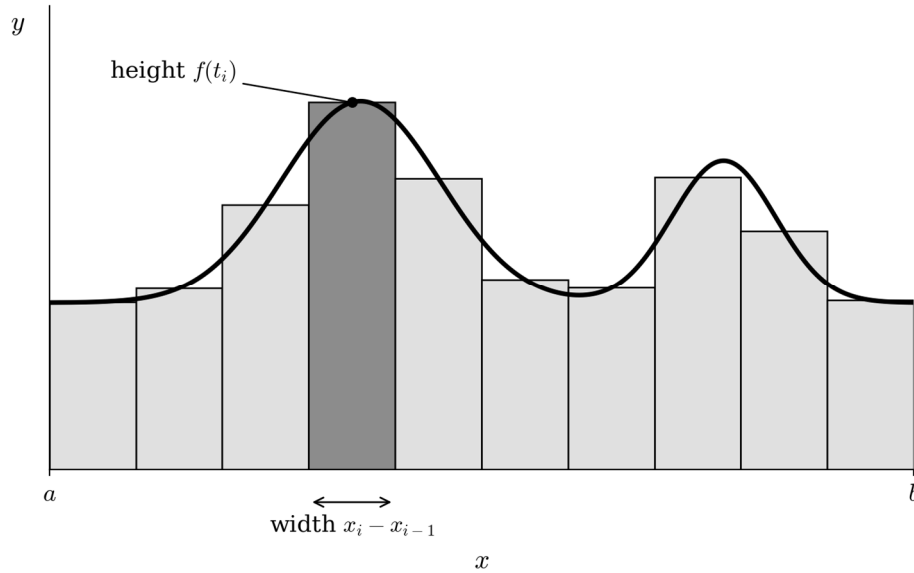


Figure 7. The Riemann viewpoint: partition the domain into subintervals

In the figure, each rectangle stands on a subinterval of the x -axis; its height is the value of f at a sample point t_i (highlighted strip).

Lebesgue's method turns this construction on its side. Instead of dividing the x -axis, it divides the y -axis into small slabs

$$0 = y_0 < y_1 < y_2 < \cdots < y_m$$

(taking $f \geq 0$ for simplicity). For each slab it asks: at which points x does $f(x)$ lie between y_{j-1} and y_j ? Those points form the set

$$E_j = \{x: y_{j-1} \leq f(x) < y_j\}$$

and the integral is approximated by the sum

$$\sum_{j=1}^m y_{j-1} \ell(E_j)$$

where $\ell(E_j)$ denotes the total length of the set E_j .

Two features distinguish this from the Riemann sum. First, the set E_j need not be an interval. In Figure 8, it is a union of three separated pieces; for a more complicated function it might be scattered through the domain in a way no picture could show. The Lebesgue method does not care, provided each E_j can be assigned a total length. Making "total length" precise for such sets is the subject of the next section, where it is given its proper name, the measure of E_j , written $\mu(E_j)$.

Second, and this is the decisive point, the quality of the approximation is now automatic. On the set E_j the values of f vary by less than $y_j - y_{j-1}$, simply by the definition of E_j . There is no analogue of a badly chosen sample point. However wildly f jumps from one point to a neighboring one, each value of f is filed into the correct slab. The burden of the construction thus shifts entirely from the smoothness of f to the measurability of the sets E_j .

In short:

- Riemann asks: on this piece of the domain, what values does f take?
- Lebesgue asks: at which points does f take values of this size, and how large is that set of points?

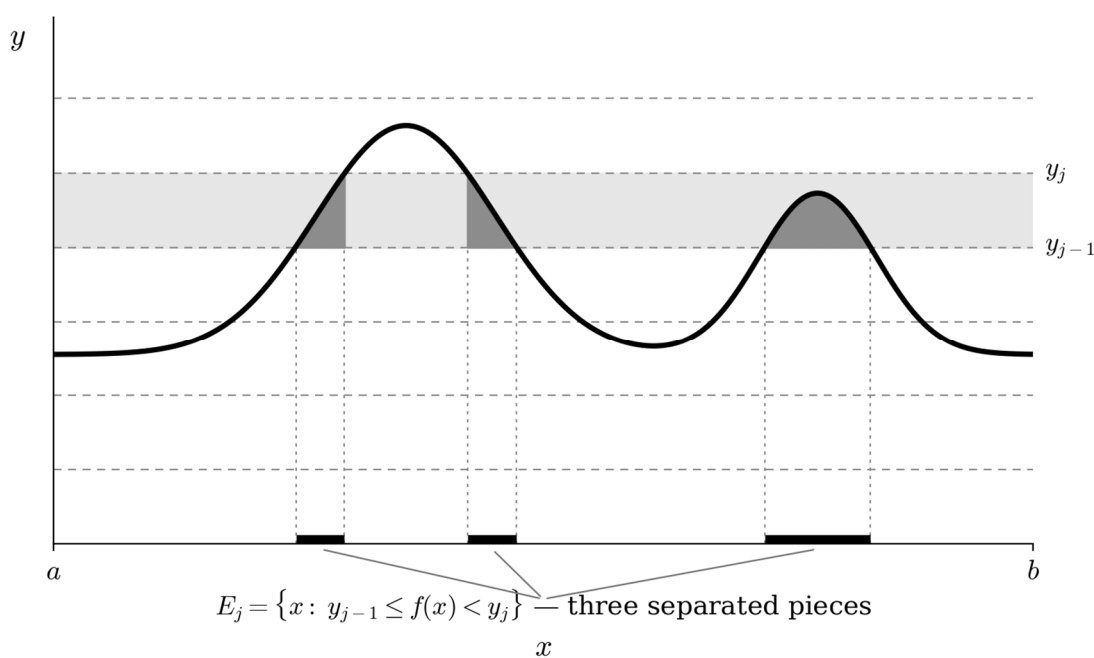


Figure 8. The Lebesgue viewpoint: partition the range into slabs.

In the figure, the slab between y_{j-1} and y_j is attained on the set E_j , here a union of three separated pieces of the x -axis (marked in black). Its contribution to the approximating sum is $y_{j-1} \ell(E_j)$.

An analogy, due to Lebesgue himself, may make the contrast vivid. Suppose a pile of coins is to be counted. One method takes the coins one at a time, in whatever order they happen to lie, adding each coin's value to a running total. This is the Riemann method: proceed across the domain and record the values as they come. The other method first sorts the coins by denomination, all the pennies in one pile, all the nickels in another, all the dimes in a third, and then counts each pile and multiplies: so many pennies times one cent, plus so many nickels times five cents, and so on. This is the Lebesgue method: group by value first, then measure how much of each value occurs. Both procedures give the same total, but the second is indifferent to how the coins were scattered; it needs to know only how many there are of each kind.

For ordinary well-behaved functions the two approaches lead to the same result. The advantage of the Lebesgue viewpoint appears for functions whose values are distributed across the domain in a complicated way. Riemann's vertical strips cannot cope with such functions, but Lebesgue's question, "how large is the set on which f is about this big?", still has a perfectly good answer.

Measure

To make the Lebesgue approach precise, one needs a way to measure the size of sets of points. For intervals this is familiar: the interval $[a, b]$ has length $b - a$. The problem is to extend this notion of length to sets like E_j above, which may be far from intervals.

What properties should such an extension have? Writing $\mu(E)$ for the measure of a set E , one asks at least the following. The measure of an interval should be its length, so that the new notion agrees with the old one where both apply. The measure should be countably additive: if E is split into a finite or countably infinite collection of pieces, no two of which overlap, the measure of E should be the sum of the measures of the pieces. And measure should be monotone: a subset of E should have measure no larger than that of E . Lebesgue's achievement was to construct such a μ , defined on a very large class of sets, the measurable sets. Intervals are measurable; complements, countable unions, and countable intersections of measurable sets are measurable; and consequently, every set obtained from intervals by ordinary limiting operations is measurable.

It is natural to ask whether every set of real numbers is measurable. Remarkably, the answer is no, but the known examples (the first was given by Vitali in 1905) cannot be exhibited concretely: their construction requires the axiom of choice. Non-measurable sets never arise in ordinary analysis, and we shall not meet one in this book.

A particularly important idea is that of a set of measure zero. A set E is said to have measure zero if it can be covered by countably many intervals whose total length is as small as desired. More precisely, E has measure zero if, for every $\varepsilon > 0$, there exists a countable collection of intervals $I_1, I_2, I_3, \dots$ such that

$$E \subseteq \bigcup_{n=1}^{\infty} I_n$$

and

$$\sum_{n=1}^{\infty} |I_n| < \varepsilon$$

where $|I_n|$ denotes the length of the interval I_n .

The following examples are fundamental.

- A single point has measure zero: it lies inside an interval of length ε , for every $\varepsilon > 0$.
- Any finite set has measure zero: cover each of its k points by an interval of length ε/k .
- Any countable set has measure zero. The proof is a pleasant trick with the geometric series: list the points as $p_1, p_2, p_3, \dots$ and cover p_n by an interval of length $\varepsilon/2^n$. The total length of the covering is then at most $\sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon$.
- In particular, the rational numbers in $[0,1]$ form a countable set, and therefore have measure zero.

This last fact is striking. The rationals are dense in $[0,1]$: every subinterval, however short, contains infinitely many of them. Nevertheless, from the point of view of length they occupy no space at all. They are everywhere, and yet they have measure zero. Density is a statement about where points are located; measure is a statement about how much room they take up. The two notions are

independent, and keeping them separate is one of the first habits of thought the Lebesgue theory requires.

One should not conclude that measure zero is merely a synonym for countable. There exist uncountable sets of measure zero; the standard example is Cantor's middle-thirds set, obtained from $[0,1]$ by repeatedly deleting the open middle third of every remaining interval. Countability implies measure zero, but not conversely.

Measurable Functions

The Lebesgue integral applies to measurable functions. A real-valued function f is called measurable if the set

$$\{x: f(x) > a\}$$

is a measurable set for every real number a .

At first sight this condition may seem arbitrary, but it is exactly what the program of the Motivation section requires. From sets of the form $\{x: f(x) > a\}$ one obtains, by taking complements and intersections,

$$\{x: y_{j-1} \leq f(x) < y_j\}$$

and these are precisely the sets E_j whose measures enter the Lebesgue sums. The definition asks for the least that makes the construction possible: every slab of values must be attained on a set that can be measured.

The class of measurable functions is very wide. Continuous functions are measurable, since for them $\{x: f(x) > a\}$ is an open set, and open sets are measurable. Step functions and piecewise continuous functions are measurable. Sums, products, and quotients of measurable functions are measurable. Most importantly of all, the pointwise limit of a sequence of measurable functions is again measurable. This last property has no analogue in the Riemann theory, where the limit of integrable functions can fail to be integrable, and it is the ultimate source of the convergence theorems described later in this appendix.

For the purposes of this appendix, the full technical development of measurable sets and functions will not be needed. The important point is that the Lebesgue integral is built on measuring sets of points, rather than merely partitioning intervals, and that measurability survives every limiting operation that arises in practice.

Simple Functions

The Lebesgue integral is first defined for simple functions, and then extended to general functions by approximation.

A simple function is a measurable function that takes only finitely many values. If the distinct values are $a_1, a_2, \dots, a_n$, and E_i is the set on which the value a_i is taken, then the function may be written

$$s(x) = a_1 \chi_{E_1}(x) + a_2 \chi_{E_2}(x) + \dots + a_n \chi_{E_n}(x)$$

where χ_E denotes the characteristic function of the set E :

$$\chi_E(x) = \begin{cases} 1, & x \in E \\ 0, & x \notin E \end{cases}$$

Every step function is simple: there the sets E_i are intervals. But a simple function is more general, because the sets E_i need only be measurable. They may be unions of scattered pieces, or sets as irregular as the rationals. Indeed, the function that equals 1 at every rational point and 0 at every irrational point, which will reappear below as the Dirichlet function, is a simple function: it is χ_E with E the set of rationals.

The integral of the simple function s is defined by

$$\int s \, d\mu = a_1 \mu(E_1) + a_2 \mu(E_2) + \cdots + a_n \mu(E_n)$$

Here $\mu(E_i)$ denotes the measure of the set E_i . This definition is the formal version of the grouping-by-height idea: each value of the function is multiplied by the size of the set on which that value occurs. It is the coin-counting method, written in symbols.

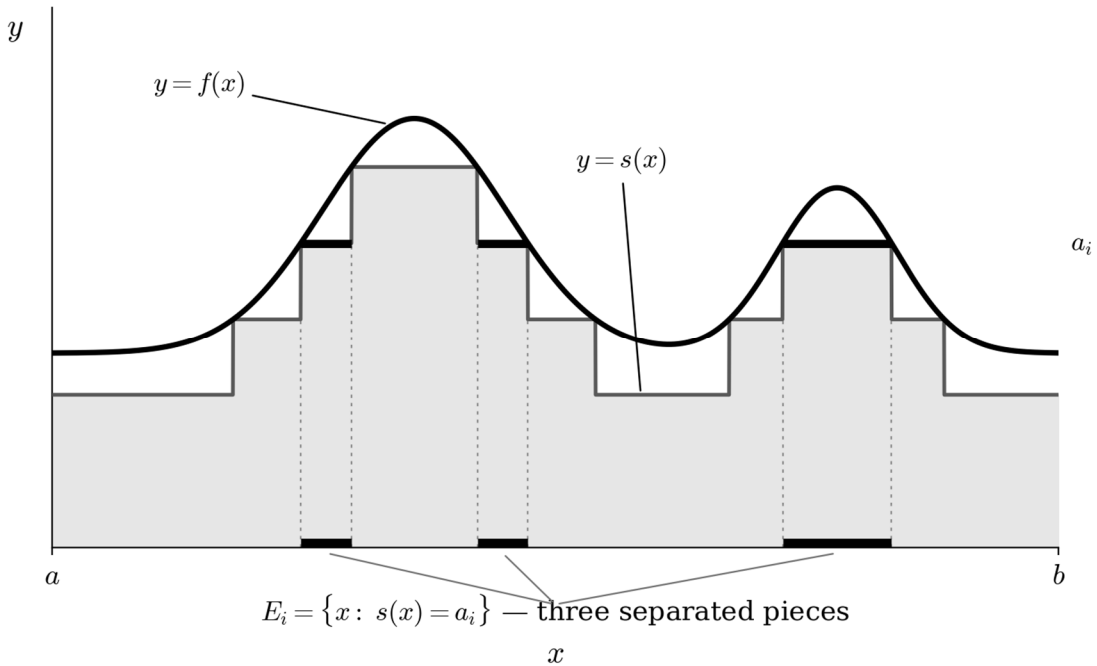


Figure 9. A simple function s approximating f from below

In the figure, the value a_i is taken on the set E_i , which here consists of three separated pieces of the x -axis (marked in black); its contribution to the integral is $a_i \mu(E_i)$.

Nonnegative Measurable Functions

The next step is to define the integral of an arbitrary nonnegative measurable function f . This is done by approximating f from below by simple functions:

$$\int f \, d\mu = \sup \int s \, d\mu$$

where the supremum is taken over all simple functions s satisfying

$$0 \leq s(x) \leq f(x)$$

Good approximating functions are easy to produce, and the construction is exactly the slab construction of the Motivation section. Cut the range into slabs of height 2^{-n} , and let $s_n(x)$ be the bottom of the slab in which $f(x)$ falls, that is, $s_n(x) = (k-1)/2^n$ on the set where $(k-1)/2^n \leq$

$f(x) < k/2^n$. Each s_n is a simple function, $s_n(x)$ never exceeds $f(x)$, and as n increases the slabs become finer, so that $s_n(x)$ climbs upward to $f(x)$ at every point. Figure 9 shows one stage of such an approximation.

This procedure resembles the construction of lower sums in the Riemann theory, but with the crucial difference already noted: the approximating rectangles of the Riemann theory stand on intervals of the domain, while the approximating simple functions of the Lebesgue theory stand on measurable sets on which the function has specified values. The geometric intuition of filling up the region below the graph is preserved; only the bookkeeping has changed, and the new bookkeeping never depends on the smoothness of f .

The supremum above may be infinite, in which case one writes $\int f d\mu = \infty$. The function f is said to be integrable when the integral is finite.

Functions with Positive and Negative Values

If a measurable function f takes both positive and negative values, it is separated into its positive and negative parts. Define

$$f^+(x) = \max(f(x), 0)$$

and

$$f^-(x) = \max(-f(x), 0)$$

Both are nonnegative measurable functions, and

$$f = f^+ - f^-, \quad |f| = f^+ + f^-$$

The Lebesgue integral of f is defined by

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

provided the two integrals on the right are not both infinite. A function is called Lebesgue integrable if $\int |f| d\mu < \infty$ in which case both integrals on the right are finite and the integral of f is well-defined.

Notice that Lebesgue integrability is by its nature absolute: f is integrable exactly when $|f|$ is. In this respect the Lebesgue integral is the analogue of an absolutely convergent series, while certain improper Riemann integrals behave like conditionally convergent series. For example, the improper integral

$$\int_0^\infty \frac{\sin x}{x} dx$$

converges (to $\pi/2$) by virtue of the cancellation between its positive and negative lobes, yet

$$\int_0^\infty \frac{|\sin x|}{x} dx = \infty$$

so the integrand is not Lebesgue integrable on $(0, \infty)$. Just as absolutely convergent series are the ones that may be rearranged and manipulated freely, Lebesgue integrable functions are the ones for which the theory's strong theorems hold; integrals that exist only by cancellation must still be handled as limits, in either theory.

Comparison with the Riemann Integral

The Lebesgue integral extends the Riemann integral: if a bounded function is Riemann integrable on a closed bounded interval, then it is also Lebesgue integrable, and the two integrals have the same value. For continuous functions, piecewise continuous functions, and the familiar functions of elementary calculus, the two theories assign the same value to the integral, though they arrive at that value by entirely different constructions. Nothing already computed needs to be revised.

There is, moreover, an exact criterion, due to Lebesgue, that settles which functions the Riemann theory can handle. A bounded function on a closed bounded interval is Riemann integrable if and only if its set of discontinuities has measure zero. It is a pleasing turn of events: the definitive theorem about Riemann's integral is stated in the language of Lebesgue's theory.

This criterion explains several familiar facts at a stroke. A continuous function is Riemann integrable because its set of discontinuities is empty. A function with finitely many jumps is Riemann integrable because a finite set has measure zero. A function may even have countably many discontinuities and remain Riemann integrable, because a countable set has measure zero; a standard example is the function on $[0,1]$ that equals $\frac{1}{q}$ at each rational $\frac{p}{q}$ in lowest terms, and 0 at each irrational, which is discontinuous at every rational and continuous at every irrational. What a Riemann integrable function cannot have is a set of discontinuities of positive measure.

The Dirichlet Function

A standard example illustrating the difference between the two theories is the Dirichlet function on $[0,1]$:

$$f(x) = \begin{cases} 1, & x \text{ is rational} \\ 0, & x \text{ is irrational} \end{cases}$$

This function is not Riemann integrable, as the criterion above predicts: it is discontinuous at every point, and $[0,1]$ certainly does not have measure zero. One can also see the failure directly. Every subinterval contains both rational and irrational numbers, so on every subinterval the supremum of f is 1 and the infimum is 0. Hence every upper sum equals 1, every lower sum equals 0, and the upper and lower integrals never meet.

From the Lebesgue point of view the function is not merely manageable but trivial, for it is a simple function: $f = \chi_E$, where E is the set of rationals in $[0,1]$. The definition of the integral of a simple function gives, immediately,

$$\int_{[0,1]} f \, d\mu = 1 \cdot \mu(E) + 0 \cdot \mu([0,1] \setminus E) = 1 \cdot 0 + 0 \cdot 1 = 0$$

The function that defeats the Riemann integral entirely is, for the Lebesgue integral, one of the very first functions whose integral is defined. This reversal is worth dwelling on: the Dirichlet function is complicated only from the pointwise, domain-first perspective. Sorted by value, in the manner of the coins, it could hardly be plainer; it takes one value on a set of measure zero and another value everywhere else.

A Function Changed on a Set of Measure Zero

Another important feature of the Lebesgue theory is that changing a function on a set of measure zero does not change its integral. The Dirichlet function differs from the zero function only on the

rational numbers, a set of measure zero, and accordingly the two functions have the same Lebesgue integral.

Two functions that are equal except on a set of measure zero are said to be equal almost everywhere. More generally, a property is said to hold almost everywhere if it holds except on a set of measure zero. From the point of view of integration, functions equal almost everywhere are indistinguishable, and the Lebesgue theory systematically treats them as the same function.

Probability theory, which will be seen at the end of this appendix to be a branch of measure theory, offers a vivid illustration. Suppose a number is drawn at random from $[0,1]$, uniformly, so that the probability of landing in any subinterval equals the length of that subinterval. The probability of drawing one particular point is then zero, and since the rationals in $[0,1]$ have measure zero, the probability of drawing a rational number is also zero, even though the rationals are dense and the draw must thread its way past infinitely many of them in every subinterval. The number drawn is irrational with probability 1. Probabilists say the outcome is irrational almost surely; “almost surely” is the probabilist’s name for “almost everywhere,” the exceptional set now being an event of probability zero. An event of probability zero is not an impossible event, since some point is certainly drawn and every point had probability zero of being it; it is an event too small to affect any probabilistic accounting.

The statement about integrals has a probabilistic counterpart as well. Consider a wager that pays 1 if the number drawn is rational and nothing otherwise; the payoff is the Dirichlet function, now wearing the costume of a random gain. Modifying a payoff on a set of outcomes of probability zero cannot change what the wager is worth, so this wager is worth exactly what the zero payoff is worth, namely nothing. Its expected value is 0, which is precisely the Lebesgue integral computed in the preceding section. Two payoffs that agree almost surely have the same expectation, just as two functions that agree almost everywhere have the same integral.

This concept is used constantly in modern analysis. It allows one to ignore exceptional sets that are too small to affect integration, to speak of limits that exist almost everywhere, and to state theorems in their natural generality. The hypotheses of the convergence theorems below, for instance, need only hold almost everywhere.

Better Behavior Under Limits

One of the chief advantages of the Lebesgue integral is its behavior under limiting processes. In analysis, one constantly studies a sequence of functions

$$f_1, f_2, f_3, \dots$$

and asks whether the limit may be passed through the integral, that is, whether

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int \lim_{n \rightarrow \infty} f_n d\mu$$

For the Riemann integral the difficulty begins earlier than one might expect: the right-hand side may fail to exist at all. Here is the classical example. List the rationals in $[0,1]$ as $r_1, r_2, r_3, \dots$ and let

$$f_n(x) = \begin{cases} 1, & x \in \{r_1, \dots, r_n\} \\ 0, & \text{otherwise} \end{cases}$$

Each f_n is Riemann integrable, since it has only finitely many discontinuities, and its integral is 0. The sequence increases at every point to a limit function, but that limit is the Dirichlet function, which is not Riemann integrable. The limit of the integrals exists and equals 0, while the Riemann

integral of the limit is meaningless. The class of Riemann integrable functions is not closed under pointwise limits, and no theorem about interchanging limits can repair that.

Within the Riemann theory the standard remedy is uniform convergence: if $f_n \rightarrow f$ uniformly on a closed bounded interval, then f is integrable and the limit passes through the integral. But uniform convergence is a strong hypothesis, and it fails in quite ordinary situations.

The Lebesgue theory does better in both respects. The pointwise limit of measurable functions is measurable, so the limit function never escapes the theory; and the integral and the limit may be interchanged under far weaker hypotheses than uniformity. The two fundamental results are the following.

The **Monotone Convergence Theorem** states that if

$$0 \leq f_1(x) \leq f_2(x) \leq f_3(x) \leq \dots$$

and $f_n(x) \rightarrow f(x)$ almost everywhere, then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

Applied to the example above, the theorem gives at once that the Dirichlet function has integral $\lim_{n \rightarrow \infty} 0 = 0$, in agreement with the direct computation made earlier.

The **Dominated Convergence Theorem** is even more widely used. It states that if $f_n(x) \rightarrow f(x)$ almost everywhere, and if there is a single integrable function g such that

$$|f_n(x)| \leq g(x)$$

for all n and almost every x , then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

The function g is called a dominating function. The point is that the f_n are allowed to converge merely pointwise, with no uniformity whatever, provided they are all controlled by one integrable function. A simple illustration: on $[0,1]$ the functions $f_n(x) = x^n$ converge to 0 almost everywhere (everywhere except at $x = 1$), and the convergence is not uniform; but $|f_n| \leq 1$, the constant function 1 is integrable on $[0,1]$, and the theorem gives

$$\lim_{n \rightarrow \infty} \int_{[0,1]} x^n d\mu = 0$$

as may be confirmed directly, since each integral equals $\frac{1}{n+1}$. The example is elementary, but the pattern it displays, pointwise convergence plus a single dominating bound, is one of the most frequently verified pairs of hypotheses in all of analysis.

A careful reader should ask why a dominating function is needed at all. The following example shows what can happen without one. Let $f_n = n$ on the interval $(0, \frac{1}{n})$ and 0 elsewhere. Then $\int f_n d\mu = n \cdot (1/n) = 1$ for every n ; but at every point x the values $f_n(x)$ are eventually 0, so the limit function is 0 and its integral is 0. The limit of the integrals is 1, the integral of the limit is 0. The mass of each f_n , squeezed into an ever-taller spike on an ever-thinner interval, has escaped in passing to the limit. The third classical result of the theory, **Fatou's lemma**, asserts that for nonnegative measurable functions this is the only thing that can go wrong:

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu$$

In words: in passing to a limit, mass may be lost, but it can never be created. The dominating function of the Dominated Convergence Theorem is precisely the hypothesis that blocks the escape. In the example, a function g dominating every f_n would have to exceed n on $(0, \frac{1}{n})$ for every n , forcing $g(x) \geq 1/x$ near 0, and no such g is integrable. When an integrable g does exist, the bound $|f_n| \leq g$ confines the mass on both sides, no escape is possible in either direction, and the inequality of Fatou's lemma sharpens to the equality of the Dominated Convergence Theorem.

These convergence theorems are the principal reason the Lebesgue integral became central in modern analysis. They permit limits to be passed through integrals under conditions that arise naturally in practice, rather than conditions imposed to make a proof work.

Beyond the Real Line: Measure Spaces

The Lebesgue integral has one further advantage: it generalizes naturally beyond subsets of the real line. An examination of the construction shows that only three ingredients were ever used:

- A set X ;
- A collection of subsets of X , the measurable sets;
- A measure μ that assigns a size to each measurable set.

Together these form a measure space. Nothing in the definition of the integral refers to lengths, intervals, or the geometry of the line; everything proceeds from simple functions, whose integrals are values times measures of sets, through approximation from below, exactly as before.

Therefore, the integral $\int f d\mu$ makes sense for measurable functions on any measure space.

This abstraction is one of the great strengths of the theory, and two examples show its reach.

First, let X be the set of positive integers and let $\mu(E)$ be the number of elements of E . This is called the counting measure. Here, in contrast with the situation on the real line, no question of measurability arises: the number of elements of a subset is well-defined for every subset, and so all subsets of X are measurable. A function on X is simply a sequence $f(1), f(2), f(3), \dots$, and its integral with respect to the counting measure is

$$\int f d\mu = \sum_{n=1}^{\infty} f(n)$$

Thus, the theory of infinite series is absorbed into the theory of integration: a series is an integral over the integers, absolute convergence is Lebesgue integrability, and the convergence theorems for integrals yield, as special cases, classical theorems on series. The parallel drawn earlier between absolutely convergent series and integrable functions is, from this vantage point, not an analogy but an identity.

Second, probability theory. There the underlying set is a sample space, the measurable sets are the events, and the measure is a probability measure P , a measure whose total mass is 1. A random variable X is a measurable function on the sample space, and its expected value is

$$\mathbb{E}[X] = \int X dP$$

Expectation is a form of Lebesgue integration. This connection is not merely notational: the modern foundation of probability, given by Kolmogorov in 1933, consists precisely in taking the measure-theoretic framework as the definition of the subject. The Dominated Convergence Theorem, for

example, is used throughout probability because it tells one when limits and expectations may be interchanged.

Integration, in the Lebesgue formulation, is thus revealed to be something more general than the computation of areas. It is a universal method of accumulating or averaging values against a measure, and the same theorems serve real analysis, the theory of series, probability, and beyond.

Why the Lebesgue Integral Matters

The Lebesgue integral matters for three main reasons.

First, it applies to a wider class of functions than the Riemann integral. Functions with complicated sets of discontinuities, of which the Dirichlet function is the simplest specimen, may still be Lebesgue integrable, and the exact boundary of the Riemann theory (discontinuities of measure zero) is itself a theorem of the Lebesgue theory.

Second, it has better analytical properties. The class of measurable functions is closed under pointwise limits, and the Monotone and Dominated Convergence Theorems allow limits to be passed through integrals under hypotheses weak enough to be verified in everyday practice. For a subject built on limiting processes, this is the decisive virtue.

Third, it generalizes to arbitrary measure spaces, so that one theory of integration serves real analysis, infinite series, Fourier analysis, functional analysis, and probability simultaneously.

The Riemann integral remains the natural starting point. It captures the geometric idea of area directly, and for continuous functions on closed bounded intervals it is entirely adequate and agrees with the Lebesgue integral. The Lebesgue integral does not displace it so much as complete it: it supplies the framework in which the functions, limits, and spaces of modern analysis can be handled without apology.

Summary

The Riemann integral divides the domain of a function into intervals and approximates the area under the graph by rectangles. The Lebesgue integral instead divides the range, and measures the sets on which the function takes its values. The cost of the new viewpoint is the need for a theory of measure; the rewards are an integral defined for a far wider class of functions, the systematic neglect of sets of measure zero, convergence theorems requiring only pointwise limits and mild control, and a definition that transfers without change to general measure spaces, including those of probability theory.

For elementary purposes the Riemann integral remains sufficient. But for the analysis that developed after Whittaker and Watson, above all in connection with limiting processes and probability, the Lebesgue integral became the natural and powerful tool, and it is the integral that modern analysis means when it writes $\int f \, d\mu$.

Acronyms and Symbols

Imagine music lessons in which students worked through hundreds of hours of sheet music, adjusting the notes on the page, receiving checks and crosses from the teachers, but never playing the music. Students would not continue with the subject because they would never experience what music is. Yet this is the situation that continues... in mathematics classes.”

— Paul Lockhart (A Mathematician's Lament)

$\equiv$ - indicates a definition (rather than equality)

$\forall$ - for every or for all

$\exists$ - there exists

$\ni$ - such that

$\arg(z)$ – represents the argument of a complex number

$\mathbb{C}$ - the complex numbers

$\liminf s_n$ – represents the limit inferior of the sequence s_n

$\limsup s_n$ – represents the limit superior of the sequence s_n

MVT – Mean Value Theorem

$\mathbb{R}$ - the real numbers

$|x|$ – the absolute value of a real number

$|z|$ – the modulus of a complex number

$\mathbb{Z}$ - the integers

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